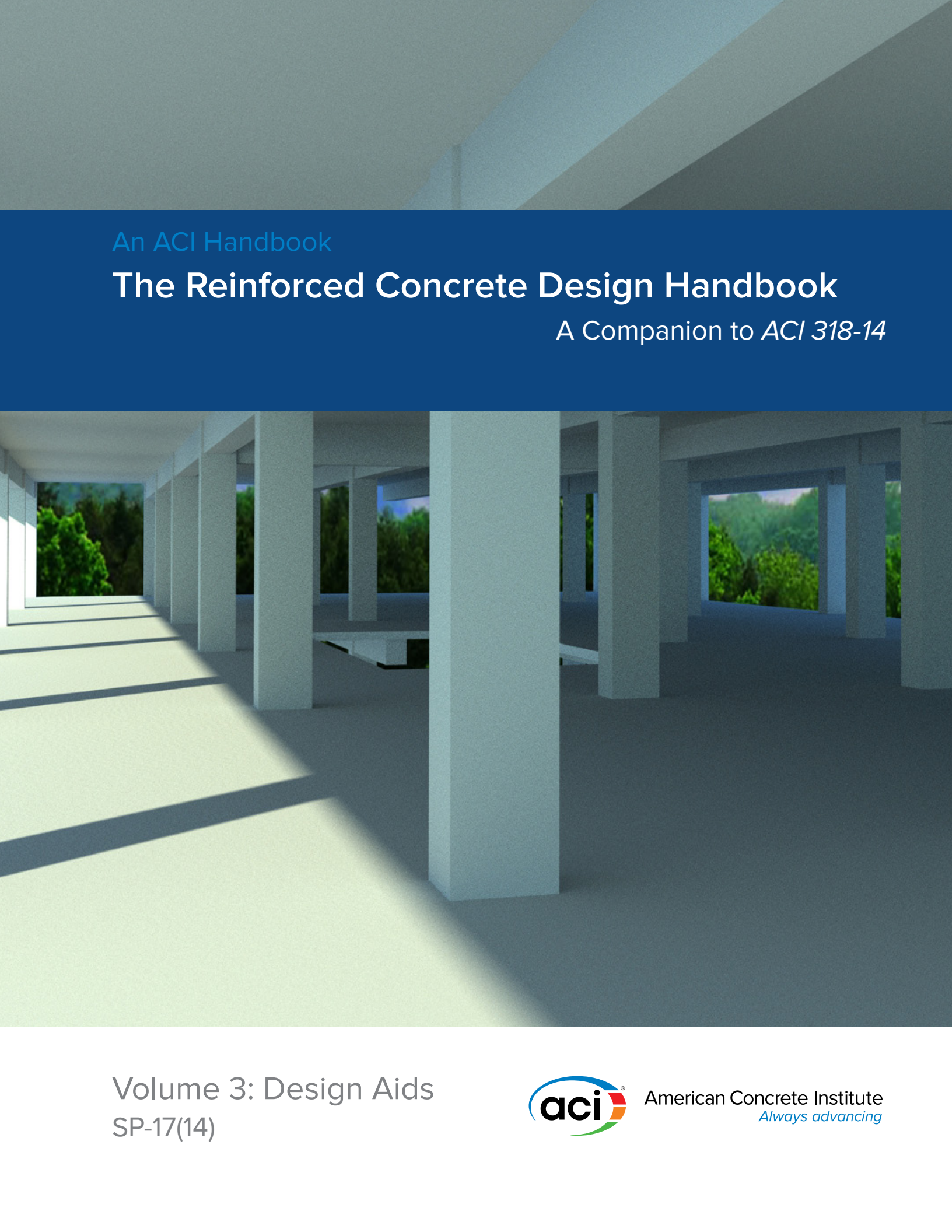




An ACI Handbook

The Reinforced Concrete Design Handbook

A Companion to *ACI 318-14*

Volume 3: Design Aids
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**ACI SP-17(14)
Volume 3**

THE REINFORCED CONCRETE DESIGN HANDBOOK

A Companion to *ACI 318-14*

Editors:
Andrew Taylor
Trey Hamilton III
Antonio Nanni



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THE REINFORCED CONCRETE DESIGN HANDBOOK Volume 3 ~ Ninth Edition

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American Concrete Institute
38800 Country Club Drive
Farmington Hills, MI 48331 USA
+1.248.848.3700

Managing Editor: Khaled Nahlawi
Staff Engineers: Daniel W. Falconer, Matthew R. Senecal, Gregory M. Zeisler, and Jerzy Z. Zemajtis
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APPENDIX A—REFERENCE TABLES

Table A-1—Nominal cross section area, weight, and nominal diameter of ASTM standard reinforcing bars

Bar size designation	Nominal cross section area, in. ²	Weight, lb/ft	Nominal diameter, in.	Nominal perimeter, in.
No. 3	0.11	0.376	0.375	1.18
No. 4	0.20	0.668	0.500	1.57
No. 5	0.31	1.043	0.625	1.96
No. 6	0.44	1.502	0.750	2.36
No. 7	0.60	2.044	0.875	2.75
No. 8	0.79	2.670	1.000	3.14
No. 9	1.00	3.400	1.128	3.54
No. 10	1.27	4.303	1.270	3.99
No. 11	1.56	5.313	1.410	4.43
No. 14	2.25	7.650	1.693	5.32
No. 18	4.00	13.600	2.257	7.09

Note: The nominal dimensions of a deformed bar are equivalent to those of a plain bar having the same mass per foot as the deformed bars.

Table A-2—Area of bars in a section 1 ft wide

Spacing, in.	Cross section area of bar A_s (or A'_s), in. ²											Spacing, in.
	Bar size											
	#3	#4	#5	#6	#7	#8	#9	#10	#11	#14	#18	
4.0	0.33	0.60	0.93	1.32	1.80	2.37	3.00	3.81	4.68	—	—	4.0
4.5	0.29	0.53	0.83	1.17	1.60	2.11	2.67	3.39	4.16	6.00	—	4.5
5.0	0.26	0.48	0.74	1.06	1.44	1.90	2.40	3.05	3.74	5.40	9.60	5.0
5.5	0.24	0.44	0.68	0.96	1.31	1.72	2.18	2.77	3.40	4.91	8.73	5.5
6.0	0.22	0.40	0.62	0.88	1.20	1.58	2.00	2.54	3.12	4.50	8.00	6.0
6.5	0.20	0.37	0.57	0.81	1.11	1.46	1.85	2.34	2.88	4.15	7.38	6.5
7.0	0.19	0.34	0.53	0.75	1.03	1.35	1.71	2.18	2.67	3.86	6.86	7.0
7.5	0.18	0.32	0.50	0.70	0.96	1.26	1.60	2.03	2.50	3.60	6.40	7.5
8.0	0.17	0.30	0.47	0.66	0.90	1.19	1.50	1.91	2.34	3.38	6.00	8.0
8.5	0.16	0.28	0.44	0.62	0.85	1.12	1.41	1.79	2.20	3.18	5.65	8.5
9.0	0.15	0.27	0.41	0.59	0.80	1.05	1.33	1.69	2.08	3.00	5.33	9.0
9.5	0.14	0.25	0.39	0.56	0.76	1.00	1.26	1.60	1.97	2.84	5.05	9.5
10.0	0.13	0.24	0.37	0.53	0.72	0.95	1.20	1.52	1.87	2.70	4.80	10.0
10.5	0.13	0.23	0.35	0.50	0.69	0.90	1.14	1.45	1.78	2.57	4.57	10.5
11.0	0.12	0.22	0.34	0.48	0.65	0.86	1.09	1.39	1.70	2.45	4.36	11.0
11.5	0.11	0.21	0.32	0.46	0.63	0.82	1.04	1.33	1.63	2.35	4.17	11.5
12.0	0.11	0.20	0.31	0.44	0.60	0.79	1.00	1.27	1.56	2.25	4.00	12.0
13.0	0.10	0.18	0.29	0.41	0.55	0.73	0.92	1.17	1.44	2.08	3.69	13.0
14.0	0.09	0.17	0.27	0.38	0.51	0.68	0.86	1.09	1.34	1.93	3.43	14.0
15.0	0.09	0.16	0.25	0.35	0.48	0.63	0.80	1.02	1.25	1.80	3.20	15.0
16.0	0.08	0.15	0.23	0.33	0.45	0.59	0.75	0.95	1.17	1.69	3.00	16.0
17.0	0.08	0.14	0.22	0.31	0.42	0.56	0.71	0.90	1.10	1.59	2.82	17.0
18.0	0.07	0.13	0.21	0.29	0.40	0.53	0.67	0.85	1.04	1.50	2.67	18.0

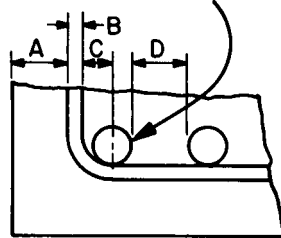
Example: #9 bar spaced 7-1/2 in. apart provides 1.60 in.²/ft of section width.

Table A-3—Minimum beam web widths required for two or more bars in one layer for cast-in-place non-prestressed concrete

Reference: ACI 318-11, Sections 7.2.2, 7.6.1, 7.7.1(c), and AASHTO Standard Specifications for Highway Bridges (17th edition, 2002) Division I, Sections 8.17.3.1, 8.21.1, 8.22.1, 8.23.2.2, and Table 8.23.2.1. For use of this Design Aid, see Flexure Example 1.

Minimum beam width = $2(A + B + C) + (n - 1)(D + d_b)$ where $A + B + C - 1/2 d_b \geq 2.0$ in. cover required for longitudinal bars and these assumptions are made:

assumed position of bar
nearest side face of beam



for ACI 318

$$B = 0.375 \text{ in. for \#3 stirrups}$$

$$= 0.500 \text{ in. for \#4 stirrups}$$

$$D = 1 d_b$$

$$\geq 1 \text{ in.}$$

$$\geq 1\text{-}1/3 \text{ nominal aggregate size}$$

For both ACI and AASHTO

$$A = 1\text{-}1/2 \text{ in. concrete cover to stirrup}$$

$$B = 0.635 \text{ in. for \#5 stirrups}$$

$$= 0.750 \text{ in. for \#6 stirrups}$$

$$C = \text{stirrup bend radius of 2 stirrup}$$

$$\text{bar diameter for \#5 and smaller stirrups}$$

$$= \text{stirrup bend radius of 3 stirrup}$$

$$\text{bar diameters for \#6 stirrups}$$

$$\geq 1/2 d_b \text{ of longitudinal bars}$$

For AASHTO

$$B = 0.375 \text{ in. for \#3 stirrups (minimum stirrup size for \#10 and smaller longitudinal bars)}$$

$$= 0.500 \text{ in. for \#4 stirrups (minimum stirrup size for \#11 and larger longitudinal bars)}$$

$$D = 1\text{-}1/2 d_b$$

$$\geq 1\text{-}1/2 \text{ in.}$$

$$\geq 1\text{-}1/2 \text{ nominal aggregate size}$$

Bar size	ACI 318-11 3/4-in. aggregate interior exposure #3 stirrups		ACI 318-11 1-in. aggregate interior exposure #3 stirrups		AASHTO requirements cast-in-place concrete 1-in. aggregate exposed to earth or weather	
	Minimum web width for 2 bars, in.	Increment for each added bar, in.	Minimum web width for 2 bars, in.	Increment for each added bar, in.	Minimum web width for 2 bars, in.	Increment for each added bar, in.
#4	6 3/4	1 1/2	7 1/8	1 7/8	7.25	2.000
#5	6 7/8	1 5/8	7 1/4	2	7.37	2.125
#6	7	1 3/4	7 3/8	2 1/8	7.50	2.250
#7	7 1/8	1 7/8	7 1/2	2 1/4	7.62	2.375
#8	7 1/4	2	7 5/8	2 3/8	7.75	2.500
#9	7 1/2	2 1/4	7 3/4	2 1/2	8.32	2.820
#10	7 7/8	2 1/2	7 7/8	2 5/8	8.68	3.175
#11	8 1/8	2 7/8	8 1/8	2 7/8	9.52	3.525
#14	8 7/8	3 3/8	8 7/8	3 3/8	10.23	4.232
#18	10 1/2	4 1/2	10 1/2	4 1/2	11.90	5.642

Notes

1. **Stirrups:** For stirrups larger than those used for table above, increase web width by the following amounts (in.):

Source	Main reinforcement size	#4 stirrup	#5 stirrup	#6 stirrup
ACI requirements	#4 through #11	3/4	1 1/2	2 1/4
	#14	1/2	1 1/4	2
	#18	1/4	3/4	1 1/2
AASHTO requirements	#4 through #10	0.75	1.50	2.25
	#11 through #14	—	0.75	1.50
	#18	—	0.49	1.24

2. **ACI cover requirements:** For exterior exposure with use of #6 or larger stirrups, add 1 in. to web width.

3. **AASHTO cover requirements:** For interior exposure, 1/2 in. may be deducted from beam widths.

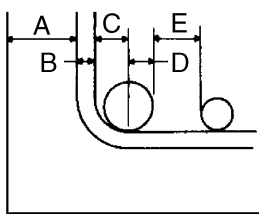
4. **Bars of different sizes:** For beams with bars of two or more sizes, determine from table the beam web width required for the given number of largest size bars; then add the indicated increments for each smaller bar.

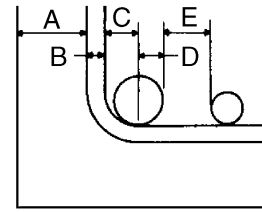
5. **Example:** Find the minimum web width for a beam reinforced with two #8 bars; a beam reinforced with three #8 bars; a beam reinforced with three #9 and two #6 bars.

	2 #8	3 #8	3 #9 + 2 #6
ACI (3/4 in. aggregate)	7 1/4	9 1/4	13 1/4
ACI (1 in. aggregate)	7 5/8	10	14 1/2
AASHTO	7.75	10.25	15.64

Table A-4—Minimum beam web widths for various bar combinations (interior exposure)

Reference: ACI 318-11 Sections 7.2.2, 7.6.1, and 7.7.1(c)

No. of bars	ACI min. b_w , in.		Columns at left headed 1-5 and 6-10 bars are for minimum web width b_w of beam having bars of one size only. Remaining columns are for combinations of 1 to 5 bars of each of two sizes. Calculated values of beam web width b_w rounded upward to nearest half inch. Where bars of two sizes are used, larger bar(s) assumed to be placed along outside face(s) of beam. Aggregate size assumed $\leq 3/4$ in.												
	Bar size	1 to 5 bars	6 to 10 bars	Size of smaller bars	ACI min. b_w , in.					Size of smaller bars	ACI min. b_w , in.				
					No. of smaller bars							No. of smaller bars			
				1	2	3	4	5		1	2	3	4	5	
1	#3	5.5	12.5	Size of smaller bars	ACI min. b_w , in.					Size of smaller bars	ACI min. b_w , in.				
2		7.0	13.5												
3		8.0	15.0												
4		9.5	16.5												
5		11.0	18.0												
1	#4	5.5	13.0	#3	7.0	8.5	9.5	11.0	12.5	Size of smaller bars	ACI min. b_w , in.				
2		7.0	14.5		8.5	9.5	11.0	12.5	14.0						
3		8.5	16.0		10.0	11.0	12.5	14.0	15.5						
4		10.0	17.5		11.5	12.5	14.0	15.5	17.0						
5		11.5	19.0		13.0	14.0	15.5	17.0	18.5						
1	#5	5.5	13.5	#4	7.0	8.5	10.0	11.5	13.0	#3	ACI min. b_w , in.				
2		7.0	15.0		8.5	10.0	11.5	13.0	14.5						
3		8.5	17.0		10.0	11.5	13.0	14.5	16.0						
4		10.5	18.5		12.0	13.5	15.0	16.5	18.0						
5		12.0	20.0		13.5	15.0	16.5	18.0	19.5						
1	#6	5.5	14.0	#5	7.0	9.0	10.5	12.0	13.5	#4	ACI min. b_w , in.				
2		7.0	16.0		9.0	10.5	12.0	13.5	15.5						
3		9.0	17.5		10.5	12.0	14.0	15.5	17.0						
4		10.5	19.5		12.5	14.0	15.5	17.0	19.0						
5		12.5	21.0		14.0	15.5	17.5	19.0	20.5						
1	#7	5.5	15.0	#6	7.5	9.0	11.0	12.5	14.5	#5	ACI min. b_w , in.				
2		7.5	16.5		9.0	11.0	12.5	14.5	16.0						
3		9.0	18.5		11.0	12.5	14.5	16.0	18.0						
4		11.0	20.5		13.0	14.5	16.5	18.0	20.0						
5		13.0	22.5		14.5	16.5	18.0	20.0	21.5						
1	#8	5.5	15.5	#7	7.5	9.5	11.0	13.0	15.0	#6	ACI min. b_w , in.				
2		7.5	17.5		9.5	11.0	13.0	15.0	17.0						
3		9.5	19.5		11.5	13.0	15.0	17.0	19.0						
4		11.5	21.5		13.5	15.0	17.0	19.0	21.0						
5		13.5	23.5		15.5	17.0	19.0	21.0	23.0						
1	#9	5.5	17.0	#8	7.5	9.5	11.5	13.5	15.5	#7	ACI min. b_w , in.				
2		8.0	19.0		10.0	12.0	14.0	16.0	18.0						
3		10.0	21.5		12.0	14.0	16.0	18.0	20.0						
4		12.5	23.5		14.5	16.5	18.5	20.5	22.5						
5		14.5	26.0		16.5	18.5	20.5	22.5	24.5						
1	#10	5.5	18.0	#9	8.0	10.0	12.5	14.5	17.0	#8	ACI min. b_w , in.				
2		8.0	20.5		10.5	12.5	15.0	17.0	19.5						
3		10.5	23.5		13.0	15.0	17.5	19.5	22.0						
4		13.0	26.0		15.5	17.5	20.0	22.0	24.5						
5		15.5	28.5		18.0	20.0	22.5	24.5	27.0						
1	#11	5.5	19.5	#10	8.0	10.5	13.0	15.5	18.0	#9	ACI min. b_w , in.				
2		8.5	22.5		11.0	13.5	16.0	18.5	21.0						
3		11.0	25.0		13.5	16.0	19.0	21.5	24.0						
4		14.0	28.0		16.5	19.0	21.5	24.0	26.5						
5		17.0	31.0		19.5	22.0	24.5	27.0	29.5						
1	#14	5.5	22.5	#11	8.5	11.5	14.5	17.0	20.0	#10	ACI min. b_w , in.				
2		9.0	26.0		12.0	14.5	17.5	20.5	23.0						
3		12.5	29.5		15.5	18.0	21.0	23.5	26.5						
4		16.0	33.0		18.5	21.5	24.5	27.0	30.0						
5		19.0	36.0		22.0	25.0	27.5	30.5	33.5						
1	#18	6.5	29.0	#14	10.0	13.5	16.5	20.0	23.5	#11	ACI min. b_w , in.				
2		11.0	33.5		14.0	17.5	21.0	24.5	27.5						
3		15.5	38.0		18.5	22.0	25.5	29.0	32.0						
4		20.0	42.5		23.0	26.5	30.0	33.5	36.5						
5		24.5	47.0		27.5	31.0	34.5	38.0	41.0						



A = clear cover of 1-

Table A-5—Properties of bundled bars

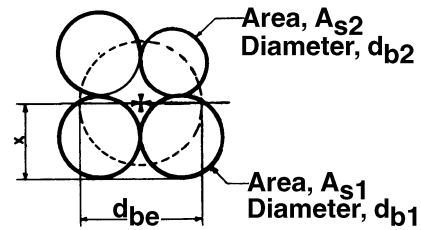
Reference: ACI 318-11 Section 7.6.6.5.

$$\text{Equivalent diameter, } d_{be} = \sqrt{\frac{4}{\pi} A_s}$$

Centroidal distance, x

For the bundled bars configuration shown here, the centroidal distance is calculated by the following equation:

$$x = \frac{\sum A_{si} d_{bi} + A_{s2} (d_{b1} + d_{b2}/2)}{\sum A_{si}}$$



Bars				Combination of bars							
Equivalent diameters d_{be} , in.											
#8	1.42	1.74	2.01	#7	#6	1.15	1.37	1.45	1.56	1.63	1.69
9	1.60	1.95	2.26	7	5	1.08	1.25	1.39	1.40	1.52	1.64
10	1.80	2.20	2.54	8	7	1.33	1.59	1.67	1.82	1.88	1.94
11	1.99	2.44	2.82	8	6	1.25	1.46	1.60	1.64	1.77	1.89
				9	8	1.51	1.81	1.88	2.07	2.13	2.20
				9	7	1.43	1.67	1.82	1.89	2.02	2.14
				10	9	1.70	2.04	2.12	2.33	2.40	2.47
				10	8	1.62	1.90	2.06	2.15	2.29	2.42
				11	10	1.90	2.28	2.36	2.61	2.68	2.75
				11	9	1.81	2.13	2.29	2.41	2.55	2.69
Centroidal distance x , from bottom of bundle, in.											
#4	0.25	0.39	0.50	#4	#3	0.23	0.38	0.33	0.43	0.40	0.46
5	0.31	0.49	0.62	5	4	0.29	0.47	0.43	0.55	0.53	0.58
6	0.37	0.59	0.75	5	3	0.28	0.46	0.37	0.49	0.44	0.55
				6	5	0.35	0.57	0.53	0.67	0.66	0.71
				6	4	0.34	0.55	0.47	0.61	0.57	0.67
7	0.44	0.69	0.87	7	6	0.41	0.67	0.62	0.80	0.78	0.83
8	0.50	0.79	1.00	7	5	0.39	0.65	0.56	0.73	0.69	0.79
9	0.56	0.89	1.13	8	7	0.47	0.77	0.72	0.93	0.90	0.95
				8	6	0.46	0.75	0.66	0.86	0.81	0.92
				9	8	0.54	0.86	0.82	1.05	1.03	1.08
				9	7	0.52	0.85	0.75	0.98	0.94	1.04
				10	9	0.60	0.98	0.92	1.19	1.16	1.22
				10	8	0.58	0.95	0.86	1.11	1.07	1.18
10	0.63	1.00	1.27	11	10	0.67	1.08	1.03	1.32	1.31	1.36
11	0.70	1.11	1.41	11	9	0.65	1.06	0.96	1.24	1.20	1.31

Example: Find the equivalent diameter of a single bar for 4 #9 bars. For 4 #9 bars, read $d_{be} = 2.26$ in., and the centroidal distance x equals 1.13 in.

Table A-6—Minimum beam web widths b_w for various combinations of bundled bars (interior exposure)

Reference: ACI 318-11 Sections 7.2.2, 7.6.6.1, 7.6.6.2, 7.6.6.3, 7.6.6.5, and 7.7.1.1

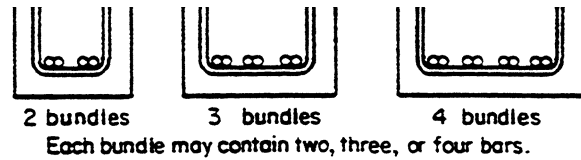
Calculated values of beam width b_w rounded upward to nearest half-inch.

Assumptions:

Aggregate size: $\leq 3/4$ in.

Clear cover of 1-1/2 in.

No. 3 stirrups



Minimum beam web width b_w , in.*			
Bar size	∞	⊞	⊞⊞
Two bundles			
#8	10.0	10.0	10.5
#9	10.5	11.0	11.0
#10	11.0	11.5	12.0
#11	11.5	12.0	12.5
Three bundles			
#8	13.5	14.0	14.5
#9	14.5	15.0	15.5
#10	15.5	16.0	17.0
#11	16.5	17.5	18.0
Four bundles			
#8	17.0	17.5	18.5
#9	18.0	19.0	20.0
#10	20.0	21.0	22.0
#11	21.5	22.5	24.0
*For beams conforming to AASHTO specifications, add 1 in. to tabulated beam web width. For example, for two bundles of three #10, minimum $b_w = 11.5$ in.			

Table A-7—Basic development length ratios of bars in tension

Reference: ACI 318-11 Sections 12.2.2 and 12.2.4

Development length ratios:

$$\frac{\ell_d}{d_b} = \alpha \frac{\psi_t \psi_e}{\lambda} \frac{f_y}{\sqrt{f'_c}}$$

For: $\psi_t = 1.0$; $\psi_e = 1.0$; and $\lambda = 1.0$ (see notes below).

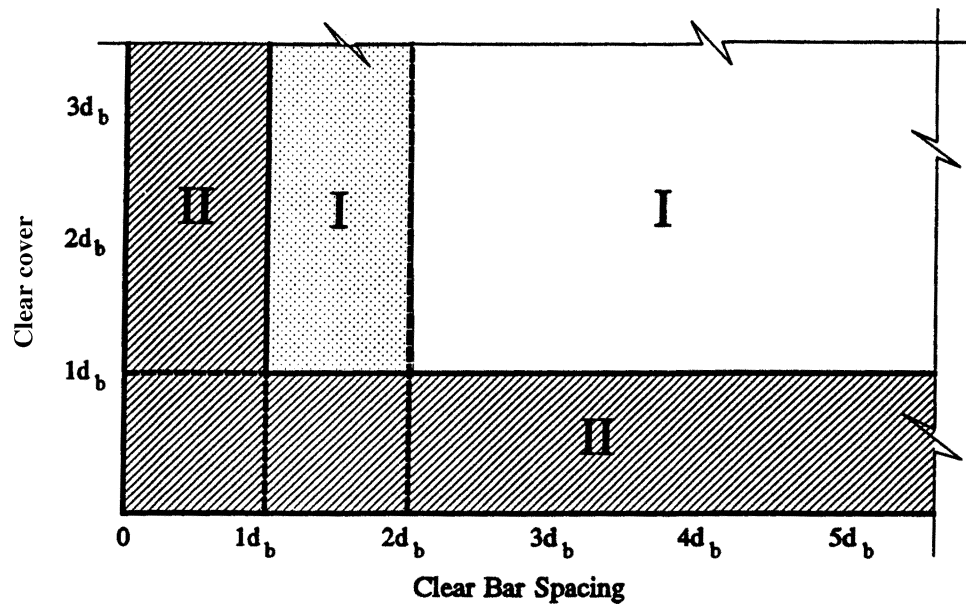
			Basic development length ratios of bars in tension															
Bar size	Category	f_y	40,000 psi				60,000 psi						80,000 psi					
		f'_c α	3000 psi	4000 psi	5000 psi	6000 psi	3000 psi	4000 psi	5000 psi	6000 psi	8000 psi	10,000 psi	3000 psi	4000 psi	5000 psi	6000 psi	8000 psi	10,000 psi
#3 to #6	I	1/25	29	25	23	21	44	38	34	31	27	24	58	51	45	41	36	32
	II	3/50	44	38	34	31	66	57	51	46	40	36	88	76	68	62	54	48
#7 to #18	I	1/20	37	32	28	26	55	47	42	39	34	30	73	63	57	52	45	40
	II	3/40	55	47	42	39	82	71	64	58	50	45	110	95	85	77	67	60

Notes:

1. See category chart for Categories I and II.

2. ψ_t = bar location factor (1.3 for bars placed such that more than 12 in. of fresh concrete is cast below the development length or splice; 1.0 for other bars). ψ_e = coating factor (1.5 = epoxy-coated reinforcement with cover $< 3d_b$ or clear spacing $< 6d_b$; 1.2 = all other epoxy-coated reinforcement; and 1.0 = uncoated and zinc-coated [galvanized] reinforcement). λ = lightweight-aggregate concrete factor (0.75 for lightweight concrete, and 1.0 for normalweight concrete).3. Minimum development length $\ell_d \geq 12$ in.

Table A-7—Basic development length ratios of bars in tension (cont.)



CATEGORY CHART

Category I: $\left\{ \begin{array}{l} \text{Clear spacing} \geq d_b \\ \text{Clear cover} \geq d_b \\ \text{Code minimum stirrups or ties throughout } \ell_d \end{array} \right.$

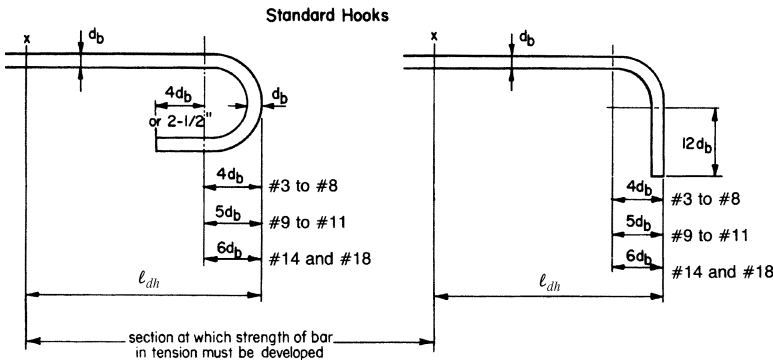
or

$\left\{ \begin{array}{l} \text{Clear spacing} \geq 2d_b \\ \text{Clear cover} \geq d_b \end{array} \right.$

Category II: All other cases

Table A-8—Basic development length ℓ_{dh} of standard hooks in tension

Reference: ACI 318-11 Sections 7.1 and 12.5.1 to 12.5.3



Development length, $\ell_{dh}^* = \alpha \ell_{dh} \geq 8d_b$, less than 6 in., where α represents modifiers from Note 1 below and ℓ_{dh} is basic development length of standard hooks in tension

$$\ell_{dh} = \frac{0.02 \psi_e f_y}{\lambda \sqrt{f'_c}} d_b$$

This table is calculated with $\psi_e = 1.0$ and $\lambda = 1.0$.

		Basic development length ℓ_{dh} , in., of standard hooks in tension												
	f_y	60,000 psi						80,000 psi						
	f_c'													
Bar size	d_b , in.	3000 psi	4000 psi	5000 psi	6000 psi	8000 psi	10,000 psi	3000 psi	4000 psi	5000 psi	6000 psi	8000 psi	10,000 psi	$8d_b$, in.
#3	0.375	8.2	7.1	6.4	5.8	5.0	4.5	11.0	9.5	8.5	7.7	6.7	6.0	3
#4	0.5	11.0	9.5	8.5	7.7	6.7	6.0	14.6	12.6	11.3	10.3	8.9	8.0	4
#5	0.625	13.7	11.9	10.6	9.7	8.4	7.5	18.3	15.8	14.1	12.9	11.2	10.0	5
#6	0.75	16.4	14.2	12.7	11.6	10.1	9.0	21.9	19.0	17.0	15.5	13.4	12.0	6
#7	0.875	19.2	16.6	14.8	13.6	11.7	10.5	25.6	22.1	19.8	18.1	15.7	14.0	7
#8	1	21.9	19.0	17.0	15.5	13.4	12.0	29.2	25.3	22.6	20.7	17.9	16.0	8
#9	1.128	24.7	21.4	19.1	17.5	15.1	13.5	33.0	28.5	25.5	23.3	20.2	18.0	9
#10	1.27	27.8	24.1	21.6	19.7	17.0	15.2	37.1	32.1	28.7	26.2	22.7	20.3	10
#11	1.41	30.9	26.8	23.9	21.8	18.9	16.9	41.2	35.7	31.9	29.1	25.2	22.6	11
#14	1.693	37.1	32.1	28.7	26.2	22.7	20.3	49.5	42.8	38.3	35.0	30.3	27.1	14
#18	2.257	49.5	42.8	38.3	35.0	30.3	27.1	65.9	57.1	51.1	46.6	40.4	36.1	18

Note 1: To compute development length ℓ_{dh}^* for a standard hook in tension, multiply basic development length ℓ_{dh} from table above by applicable modification factors:

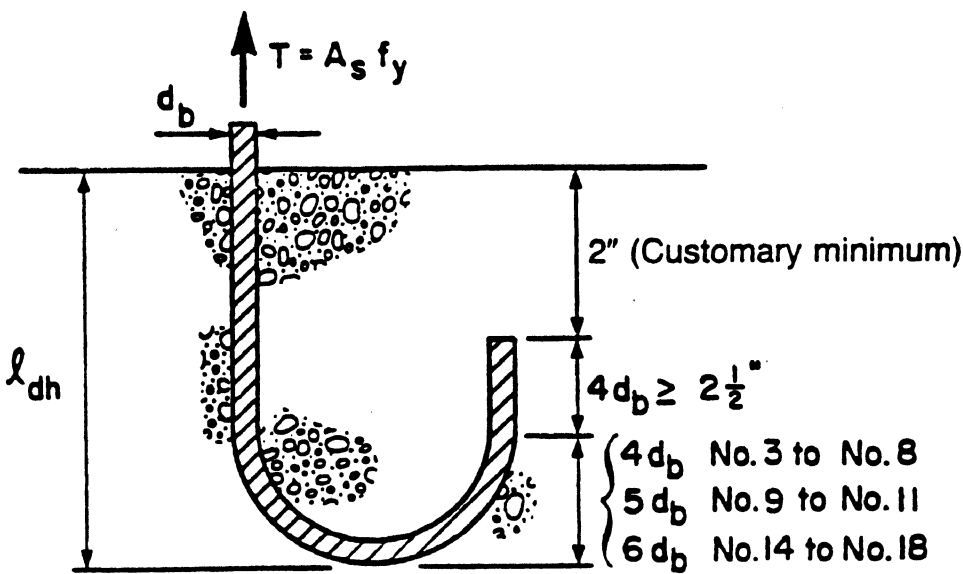
- $\alpha = 0.7$ for #11 and smaller bars with side cover normal to plane of hook not less than 2-1/2 in.; and for 90-degree hook, cover on bar extension beyond hook not less than 2 in.
- $\alpha = 0.8$ for #11 and smaller bars with a 90-degree hook enclosed vertically or horizontally within ties or stirrup ties spaced along the full development length not greater than $3d_b$, where d_b is diameter of hooked bar.
- $\alpha = 0.8$ for #11 and smaller bars with 180-degree hook that are enclosed within ties or stirrups perpendicular to the bar being developed, spaced not greater than $3d_b$ along ℓ_{dh} .
- $\alpha = A_s \text{ required} / A_s \text{ provided}$

Note 2: Values of basic development length ℓ_{dh} above the heavy line are less than the minimum development length of 6 in. Development length ℓ_{dh} shall not be less than $8d_b$, nor less than 6 in., whichever is greater.

REF.
TABLES

Table A-8—Basic development length ℓ_{dh} of standard hooks in tension (cont.)

Reference: ACI 318-11 Sections 7.1 and 7.21



Bar size	#3	#4	#5	#6	#7	#8	#9	#10	#11	#14	#18
ℓ_{dh} , in.	6	7	7	8	9	10	13	14	15	19	25

Example: Find minimum embedment depth ℓ_{dh} that will provide 2 in. cover over the tail of a standard 180-degree end hook in a #8 bar. For #8 bar, read $\ell_{dh} = 10$ in.

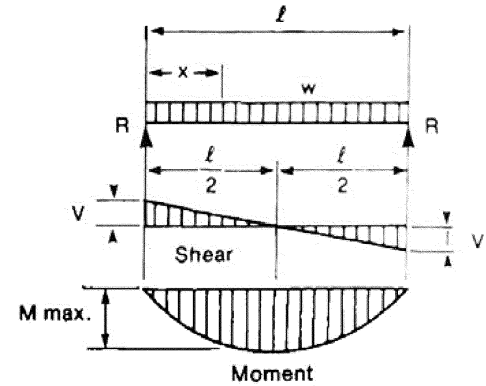
APPENDIX B—ANALYSIS TABLES

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Table B-1 Beam Diagrams

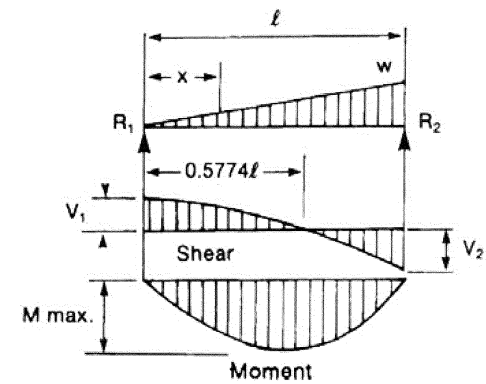
Simple Beam — uniformly distributed load

$$\begin{aligned}
 R_1 = V_1 & \dots\dots\dots = \frac{w\ell}{2} \\
 V_x & \dots\dots\dots = w\left(\frac{\ell}{2} - x\right) \\
 M_{\text{max. (at centre)}} & \dots\dots\dots = \frac{w\ell^2}{8} \\
 M_x & \dots\dots\dots = \frac{wx}{2}(\ell - x) \\
 \Delta_{\text{max. (at centre)}} & \dots\dots\dots = \frac{5w\ell^4}{384EI} \\
 \Delta_x & \dots\dots\dots = \frac{wx}{24EI}(\ell^3 - 2\ell x^2 + x^3)
 \end{aligned}$$



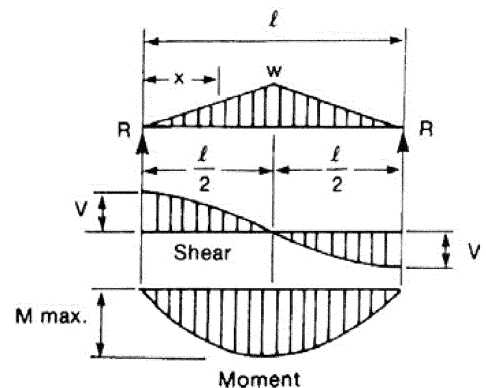
Simple Beam — load increasing uniformly to one end

$$\begin{aligned}
 R_1 = V_1 & \dots\dots\dots = \frac{w\ell}{6} \\
 R_2 = V_2 & \dots\dots\dots = \frac{w\ell}{3} \\
 V_x & \dots\dots\dots = \frac{w\ell}{3} - \frac{wx^2}{2\ell} \\
 M_{\text{max. (at } x = \sqrt{\frac{\ell}{3}} = 0.5774\ell)} & \dots\dots\dots = \frac{w\ell^2}{9\sqrt{3}} = 0.06415w\ell^2 \\
 M_x & \dots\dots\dots = \frac{wx}{6\ell}(\ell^2 - x^2) \\
 \Delta_{\text{max. (at } x = \ell\sqrt{1 - \sqrt{\frac{8}{15}}} = 0.5193\ell)} & \dots\dots\dots = 0.00652 \frac{w\ell^4}{EI} \\
 \Delta_x & \dots\dots\dots = \frac{wx}{360EI\ell}(3x^4 - 10\ell^2x^2 + 7\ell^4)
 \end{aligned}$$



Simple Beam — load increasing uniformly to centre

$$\begin{aligned}
 R = V & \dots\dots\dots = \frac{w\ell}{4} \\
 V_x \left(\text{when } x < \frac{\ell}{2}\right) & \dots\dots\dots = \frac{w}{4\ell}(\ell^2 - 4x^2) \\
 M_{\text{max. (at centre)}} & \dots\dots\dots = \frac{w\ell^2}{12} \\
 M_x \left(\text{when } x < \frac{\ell}{2}\right) & \dots\dots\dots = \frac{w\ell x}{2}\left(\frac{1}{2} - \frac{2x^2}{3\ell^2}\right) \\
 \Delta_{\text{max. (at centre)}} & \dots\dots\dots = \frac{w\ell^4}{120EI} \\
 \Delta_x & \dots\dots\dots = \frac{wx}{960EI\ell}(5\ell^2 - 4x^2)^2
 \end{aligned}$$



ANALYSIS
TABLES

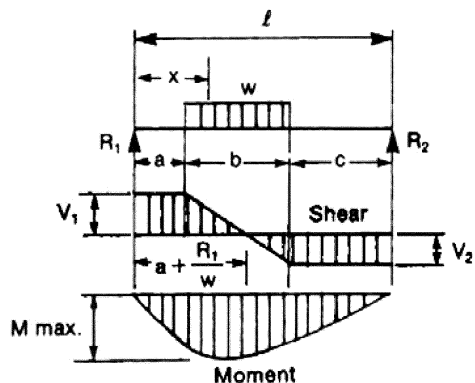
Note:

w: Distributed load per unit length. In the case of triangular distribution, w represents the maximum intensity of load per unit length.
P: Concentrated load.

Table B-1 Beam Diagrams (Cont'd)

Simple Beam — uniform load partially distributed

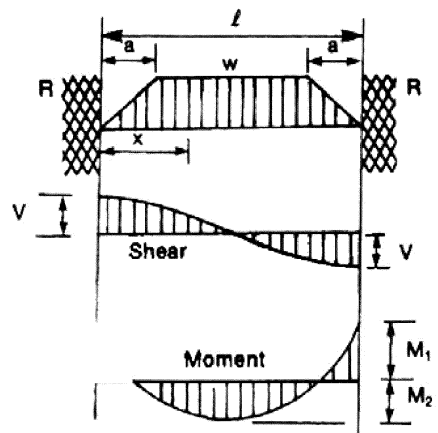
$$\begin{aligned}
 R_1 = V_1 \text{ (max. when } a < c) & \dots\dots\dots = \frac{wb}{2\ell} (2c + b) \\
 R_2 = V_2 \text{ (max. when } a > c) & \dots\dots\dots = \frac{wb}{2\ell} (2a + b) \\
 V_x \text{ (when } x > a \text{ and } < (a + b)) & \dots\dots\dots = R_1 - w(x - a) \\
 M \text{ max. (at } x = a + \frac{R_1}{w}) & \dots\dots\dots = R_1 \left(a + \frac{R_1}{2w} \right) \\
 M_x \text{ (when } x < a) & \dots\dots\dots = R_1 x \\
 M_x \text{ (when } x > a \text{ and } < (a + b)) & \dots\dots\dots = R_1 x - \frac{w}{2} (x - a)^2 \\
 M_x \text{ (when } x > (a + b)) & \dots\dots\dots = R_2 (\ell - x)
 \end{aligned}$$



Beam fixed at both ends — symmetrical trapezoidal load

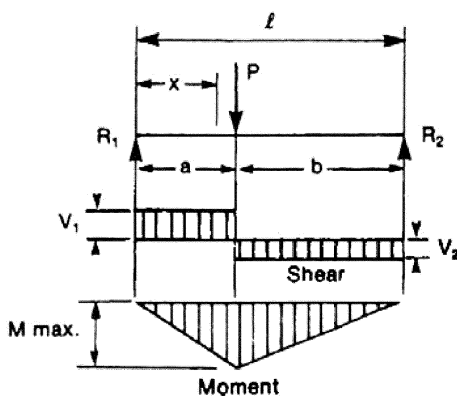
$$\begin{aligned}
 R = V & \dots\dots\dots = \frac{wl}{2} \left(1 - \frac{a}{\ell} \right) \\
 M_1 & \dots\dots\dots = -\frac{wl^2}{12} \left(1 - 2\frac{a^2}{\ell^2} + \frac{a^3}{\ell^3} \right) \\
 M_2 & \dots\dots\dots = \frac{wl^2}{24} \left(1 - 2\frac{a^3}{\ell^3} \right) \\
 M_x \text{ (when } x < a) & \dots\dots\dots = M_1 + R_1 x - \frac{wx^3}{6a} \\
 M_x \text{ (when } a < x < \ell - a) & \dots\dots\dots = M_1 + R_1 x - \frac{wa}{2} \left(x - \frac{2}{3}a \right) \\
 & \dots\dots\dots - \frac{w}{2} (x - a)^2
 \end{aligned}$$

Note: When $a = \ell/2$ loading is triangular



Simple Beam — concentrated load at any point

$$\begin{aligned}
 R_1 = V_1 \text{ (max. when } a < b) & \dots\dots\dots = \frac{Pb}{\ell} \\
 R_2 = V_2 \text{ (max. when } a > b) & \dots\dots\dots = \frac{Pa}{\ell} \\
 M \text{ max. (at point of load)} & \dots\dots\dots = \frac{Pab}{\ell} \\
 M_x \text{ (when } x < a) & \dots\dots\dots = \frac{Pbx}{\ell} \\
 \Delta \text{ max. (at } x = \sqrt{\frac{a(a+2b)}{3}} \text{ when } a > b) & \dots\dots\dots = \frac{Pab(a+2b)\sqrt{3a(a+2b)}}{27EI\ell} \\
 \Delta a \text{ (at point of load)} & \dots\dots\dots = \frac{Pa^2b^2}{3EI\ell} \\
 \Delta_x \text{ (when } x < a) & \dots\dots\dots = \frac{Pbx}{6EI\ell} (\ell^2 - b^2 - x^2)
 \end{aligned}$$



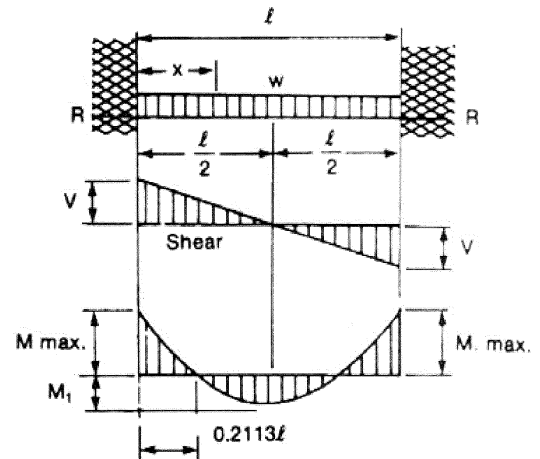
Note:

w: Distributed load per unit length. In the case of triangular distribution, w represents the maximum intensity of load per unit length.
P: Concentrated load.

Table B-1 Beam Diagrams (Cont'd)

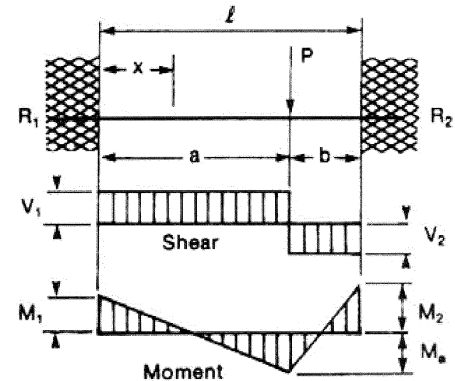
Beam fixed at both ends — uniformly distributed loads

$$\begin{aligned}
 R_1 = V_1 &= \frac{w\ell}{2} \\
 R_2 = V_2 &= w\left(\frac{\ell}{2} - x\right) \\
 M_1 \text{ (at ends)} &= -\frac{w\ell^2}{12} \\
 M_2 \text{ (at centre)} &= \frac{w\ell^2}{24} \\
 M_x &= \frac{w}{12}(6\ell x - \ell^2 - 6x^2) \\
 \Delta \text{ max. (at centre)} &= \frac{w\ell^4}{384 EI} \\
 \Delta_x &= \frac{wx^2}{24 EI}(\ell - x)^2
 \end{aligned}$$



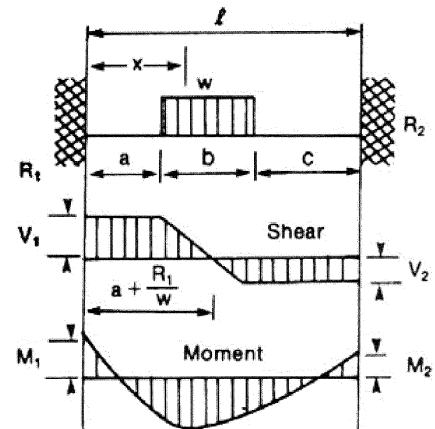
Beam fixed at both ends — concentrated load at any point

$$\begin{aligned}
 R_1 = V_1 \text{ (max. when } a < b) &= \frac{Pb^2}{\ell^3}(3a + b) \\
 R_2 = V_2 \text{ (max. when } a > b) &= \frac{Pa^2}{\ell^3}(a + 3b) \\
 M_1 \text{ (max. when } a < b) &= -\frac{Pab^2}{\ell^2} \\
 M_2 \text{ (max. when } a > b) &= -\frac{Pa^2b}{\ell^2} \\
 M_x \text{ (at point of load)} &= \frac{2Pa^2b^2}{\ell^3} \\
 M_x \text{ (when } x < a) &= R_1x - \frac{Pab^2}{\ell^2} \\
 \Delta \text{ max. (when } a > b \text{ at } x = \frac{2a\ell}{3a + b}) &= \frac{2Pa^3b^2}{3EI(3a + b)^2} \\
 \Delta_x \text{ (at point of load)} &= \frac{Pa^3b^3}{3EI\ell^3} \\
 \Delta_x \text{ (when } x < a) &= \frac{Pb^2x^2}{6EI\ell^3}(3a\ell - 3ax - bx)
 \end{aligned}$$



Beam fixed at both ends — uniform load partially distributed

$$\begin{aligned}
 M_1 &= \frac{w}{12\ell^2}[(\ell - a)^3(\ell + 3a) - c^3(4\ell - 3c)] \\
 M_2 &= \frac{w}{12\ell^2}[(\ell - c)^3(\ell + 3c) - a^3(4\ell - 3a)] \\
 R_1 = V_1 &= \frac{1}{\ell}[M_1 - M_2 + wc(c + \frac{b}{2})] \\
 R_2 = V_2 &= \frac{1}{\ell}[M_2 - M_1 + wc(a + \frac{b}{2})] \\
 M_x \text{ (when } x < a) &= R_1x - M_1 \\
 M_x \text{ (when } a < x < (a + c)) &= R_1x - M_1 - \frac{w}{2}(x - a)^2 \\
 M_x \text{ (when } x > (a + c)) &= R_2(\ell - x) - M_2
 \end{aligned}$$



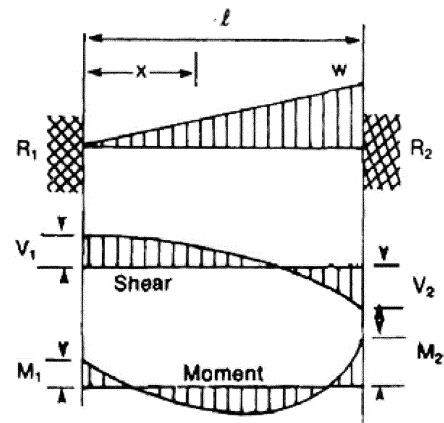
Note:

w: Distributed load per unit length. In the case of triangular distribution, w represents the maximum intensity of load per unit length.
 P: Concentrated load.

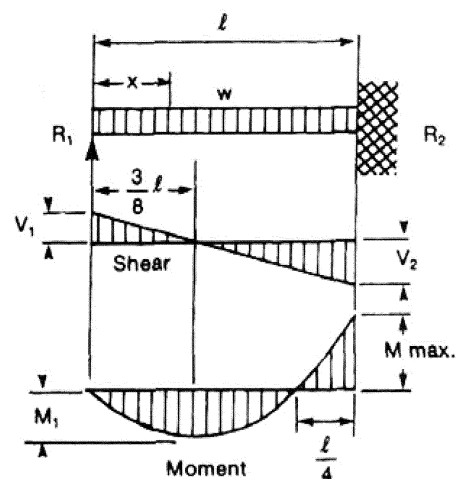
Table B-1 Beam Diagrams (Cont'd)

Beam fixed at both ends — load increasing uniformly to one end

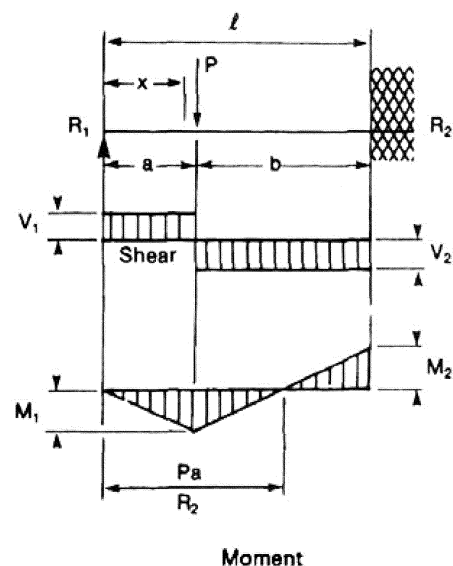
$$\begin{aligned}
 R_1 = V_1 & \dots\dots\dots = \frac{3wl}{20} \\
 R_2 = V_2 & \dots\dots\dots = \frac{7wl}{20} \\
 V_x & \dots\dots\dots = \frac{3wl}{20} - \frac{wx^2}{2l} \\
 M_1 & \dots\dots\dots = -\frac{wl^2}{30} \\
 M_2 & \dots\dots\dots = -\frac{wl^2}{20} \\
 M_x & \dots\dots\dots = \frac{3wlx}{20} - \frac{wl^2}{30} - \frac{wx^3}{6l}
 \end{aligned}$$

**Beam fixed at one end, supported at other — uniformly distributed load**

$$\begin{aligned}
 R_1 = V_1 & \dots\dots\dots = \frac{3wl}{8} \\
 R_2 = V_2 \text{ max.} & \dots\dots\dots = \frac{5wl}{8} \\
 V_x & \dots\dots\dots = R_1 - wx \\
 M \text{ max.} & \dots\dots\dots = -\frac{wl^2}{8} \\
 M_1 \left(\text{at } x = \frac{3}{8}l \right) & \dots\dots\dots = \frac{9}{128} wl^2 \\
 M_x & \dots\dots\dots = R_1 x - \frac{wx^2}{2} \\
 \Delta \text{ max.} \left(\text{at } x = \frac{l}{16} (1 + \sqrt{33}) = 0.4215l \right) & \dots\dots\dots = \frac{wl^4}{185 EI} \\
 \Delta_x & \dots\dots\dots = \frac{wx}{48 EI} (l^3 - 3lx^2 + 2x^3)
 \end{aligned}$$

**Beam fixed at one end, supported at other — concentrated load at any point**

$$\begin{aligned}
 R_1 = V_1 & \dots\dots\dots = \frac{Pb^2}{2l^3} (a + 2l) \\
 R_2 = V_2 & \dots\dots\dots = \frac{Pa}{2l^3} (3l^2 - a^2) \\
 M_1 \text{ (at point of load)} & \dots\dots\dots = R_1 a \\
 M_2 \text{ (at fixed end)} & \dots\dots\dots = -\frac{Pab}{2l^2} (a + l) \\
 M_x \text{ (when } x < a) & \dots\dots\dots = R_1 x \\
 M_x \text{ (when } x > a) & \dots\dots\dots = R_1 x - P(x - a) \\
 \Delta \text{ max.} \left(\text{when } a < 0.414l \text{ at } x = l \frac{l^2 + a^2}{3l^2 - a^2} \right) & \dots\dots\dots = \frac{Pa}{3EI} \frac{(l^2 - a^2)^3}{(3l^2 - a^2)^2} \\
 \Delta \text{ max.} \left(\text{when } a > 0.414l \text{ at } x = l \sqrt{\frac{a}{2l + a}} \right) & \dots\dots\dots = \frac{Pab^2}{6EI} \sqrt{\frac{a}{2l + a}} \\
 \Delta a \text{ (at point of load)} & \dots\dots\dots = \frac{Pa^2 b^3}{12EI l^3} (3l + a) \\
 \Delta x \text{ (when } x < a) & \dots\dots\dots = \frac{Pb^2 x}{12EI l^3} (3al^2 - 2lx^2 - ax^2) \\
 \Delta x \text{ (when } x > a) & \dots\dots\dots = \frac{Pa}{12EI l^3} (l - x)^2 (3l^2 x - a^2 x - 2a^2 l)
 \end{aligned}$$

**Note:**

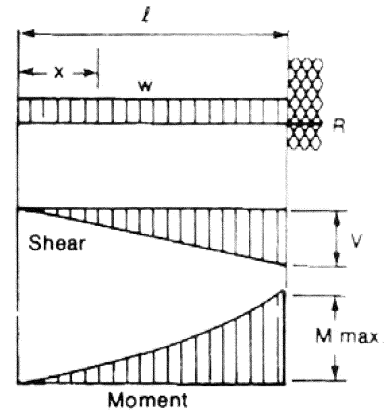
w: Distributed load per unit length. In the case of triangular distribution, w represents the maximum intensity of load per unit length.
P: Concentrated load.

Moments are positive if they cause compression in the top of the beam.

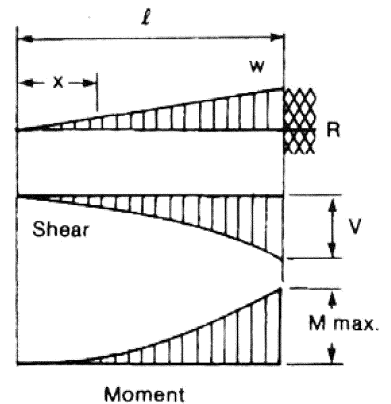
Table B-1 Beam Diagrams (Cont'd)

Cantilever Beam — uniformly distributed load

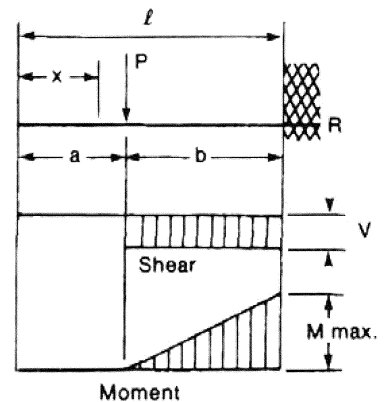
$$\begin{aligned}
 R = V & \dots\dots\dots = w\ell \\
 V_x & \dots\dots\dots = wx \\
 M \text{ max. (at fixed end)} & \dots\dots\dots = -\frac{w\ell^2}{2} \\
 M_x & \dots\dots\dots = -\frac{wx^2}{2} \\
 \Delta \text{ max. (at free end)} & \dots\dots\dots = \frac{w\ell^4}{8EI} \\
 \Delta_x & \dots\dots\dots = \frac{w}{24EI} (x^4 - 4\ell^3x + 3\ell^4)
 \end{aligned}$$

**Cantilever Beam — load increasing uniformly to fixed end**

$$\begin{aligned}
 R = V & \dots\dots\dots = \frac{w\ell}{2} \\
 V_x & \dots\dots\dots = \frac{wx^2}{2\ell} \\
 M \text{ max. (at fixed end)} & \dots\dots\dots = -\frac{w\ell^2}{6} \\
 M_x & \dots\dots\dots = -\frac{wx^3}{6\ell} \\
 \Delta \text{ max. (at free end)} & \dots\dots\dots = \frac{w\ell^4}{30EI} \\
 \Delta_x & \dots\dots\dots = \frac{w}{120EI\ell} (x^5 - 5\ell^4x + 4\ell^5)
 \end{aligned}$$

**Cantilever Beam — concentrated load at any point**

$$\begin{aligned}
 R = V & \dots\dots\dots = P \\
 M \text{ max. (at fixed end)} & \dots\dots\dots = -Pb \\
 M_x \text{ (when } x > a) & \dots\dots\dots = P(x - a) \\
 \Delta \text{ max. (at free end)} & \dots\dots\dots = \frac{Pb^2}{6EI} (3\ell - b) \\
 \Delta a \text{ (at point of load)} & \dots\dots\dots = \frac{Pb^3}{3EI} \\
 \Delta_x \text{ (when } x < a) & \dots\dots\dots = \frac{Pb^2}{6EI} (3\ell - 3x - b) \\
 \Delta_x \text{ (when } x > a) & \dots\dots\dots = \frac{P(\ell - x)^2}{6EI} (3b - \ell + x)
 \end{aligned}$$

**Note:**

w: Distributed load per unit length. In the case of triangular distribution, w represents the maximum intensity of load per unit length.

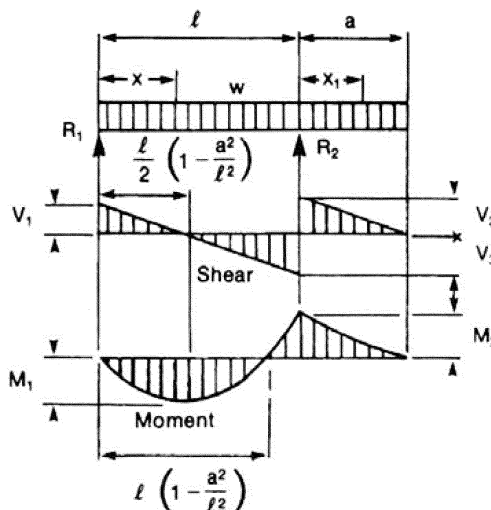
P: Concentrated load.

Moments are positive if they cause compression in the top of the beam.

Table B-1 Beam Diagrams (Cont'd)

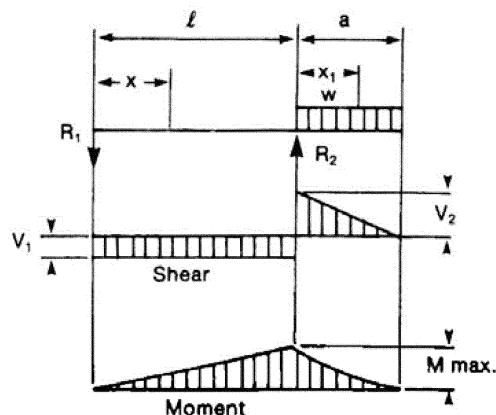
Beam overhanging one support — uniformly distributed load

$$\begin{aligned}
 R_1 = V_1 &= \frac{w}{2\ell} (\ell^2 - a^2) \\
 R_2 = V_2 + V_3 &= \frac{w}{2\ell} (\ell + a)^2 \\
 V_2 &= wa \\
 V_3 &= \frac{w}{2\ell} (\ell^2 + a^2) \\
 V_x \text{ (between supports)} &= R_1 - wx \\
 V_{x_1} \text{ (for overhang)} &= w(a - x_1) \\
 M_1 \left(\text{at } x = \frac{\ell}{2} \left[1 - \frac{a^2}{\ell^2} \right] \right) &= \frac{w}{8\ell^2} (\ell + a)^2 (\ell - a)^2 \\
 M_2 \text{ (at } R_2) &= \frac{wa^2}{2} \\
 M_x \text{ (between supports)} &= \frac{wx}{2\ell} (\ell^2 - a^2 - x\ell) \\
 M_{x_1} \text{ (for overhang)} &= \frac{w}{2} (a - x_1)^2 \\
 \Delta_x \text{ (between supports)} &= \frac{wx}{24EI} (\ell^4 - 2\ell^2 x^2 + \ell x^3 - 2a^2 \ell^2 + 2a^2 x^2) \\
 \Delta_{x_1} \text{ (for overhang)} &= \frac{wx_1}{24EI} (4a^2 \ell - \ell^3 + 6a^2 x_1 - 4ax_1^2 + x_1^3)
 \end{aligned}$$



Beam overhanging one support — uniformly distributed load on overhang

$$\begin{aligned}
 R_1 = V_1 &= \frac{wa^2}{2\ell} \\
 R_2 = V_1 + V_2 &= \frac{wa}{2\ell} (2\ell + a) \\
 V_2 &= wa \\
 V_{x_1} \text{ (for overhang)} &= w(a - x_1) \\
 M \text{ max. (at } R_2) &= \frac{wa^2}{2} \\
 M_x \text{ (between supports)} &= \frac{wa^2 x}{2\ell} \\
 M_{x_1} \text{ (for overhang)} &= \frac{w}{2} (a - x_1)^2 \\
 \Delta \text{ max. (between supports at } x = \frac{\ell}{\sqrt{3}}) &= \frac{wa^2 \ell^2}{18\sqrt{3}EI} = 0.03208 \frac{wa^2 \ell^2}{EI} \\
 \Delta \text{ max. (for overhang at } x_1 = a) &= \frac{wa^3}{24EI} (4\ell + 3a) \\
 \Delta_x \text{ (between supports)} &= \frac{wa^2 x}{12EI} (\ell^2 - x^2) \\
 \Delta_{x_1} \text{ (for overhang)} &= \frac{wx_1}{24EI} (4a^2 \ell + 6a^2 x_1 - 4ax_1^2 + x_1^3)
 \end{aligned}$$



Note:

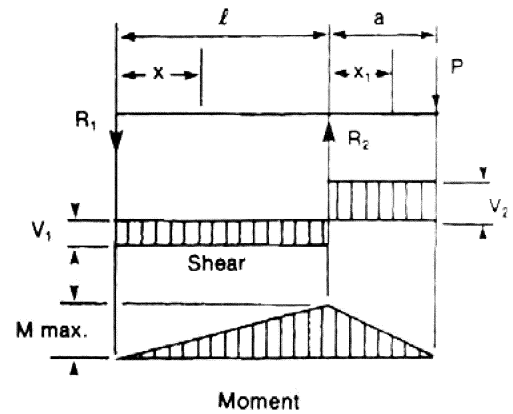
w: Distributed load per unit length. In the case of triangular distribution, w represents the maximum intensity of load per unit length.
P: Concentrated load.

Moments are positive if they cause compression in the top of the beam.

Table B-1 Beam Diagrams (Cont'd)

Beam overhanging one support — concentrated load at end of overhang

$$\begin{aligned}
 R_1 = V_1 &= \frac{Pa}{\ell} \\
 R_2 = V_1 + V_2 &= \frac{P}{\ell} (\ell + a) \\
 V_2 &= P \\
 M \text{ max. (at } R_2) &= Pa \\
 M_1 \text{ (between supports)} &= \frac{Pax}{\ell} \\
 M_{11} \text{ (for overhang)} &= P(a - x_1) \\
 \Delta \text{ max. (between supports at } x = \frac{\ell}{\sqrt{3}}) &= \frac{Pa\ell^2}{9\sqrt{3}EI} = 0.06415 \frac{Pa\ell^2}{EI} \\
 \Delta \text{ max. (for overhang at } x_1 = a) &= \frac{Pa^2}{3EI} (\ell + a) \\
 \Delta_1 \text{ (between supports)} &= \frac{Pax}{6EI} (\ell^2 - x^2) \\
 \Delta_{11} \text{ (for overhang)} &= \frac{Px_1}{6EI} (2a\ell + 3ax_1 - x_1^2)
 \end{aligned}$$

**Note:**

w : Distributed load per unit length. In the case of triangular distribution, w represents the maximum intensity of load per unit length.

P : Concentrated load.

Moments are positive if they cause compression in the top of the beam.

Table B-2 Moments and Reactions in Continuous Beams under Uniformly Distributed Loads

Moment : $\text{Coefficient} \times w\ell^2$ Reaction : $\text{Coefficient} \times w\ell$		w: Uniform load per unit length ℓ : Length of one span	
Two Equal Spans		Four Equal Spans	
Three Equal Spans			

Table B-3 Moments and Reactions in Continuous Beams under Central Point Loads

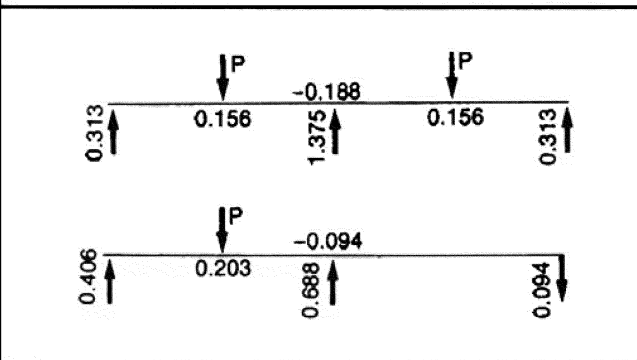
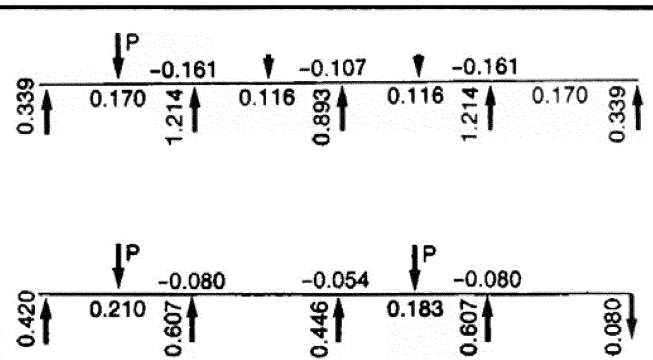
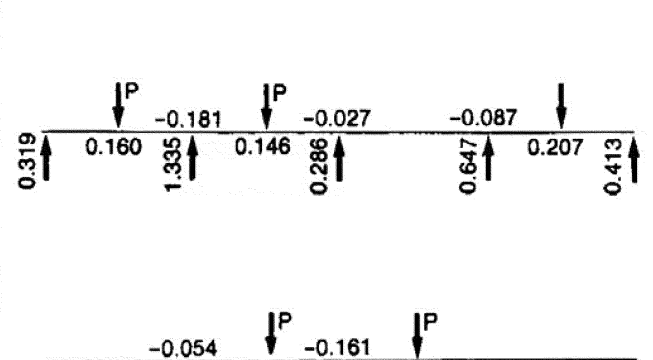
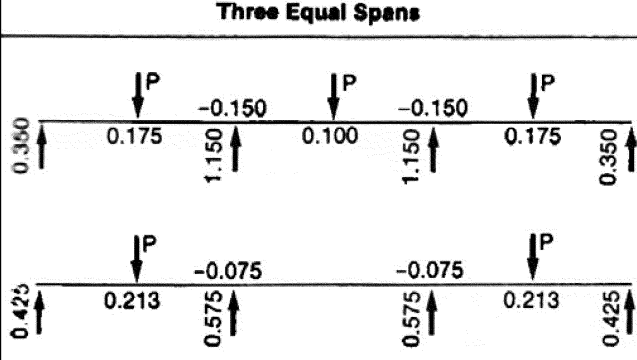
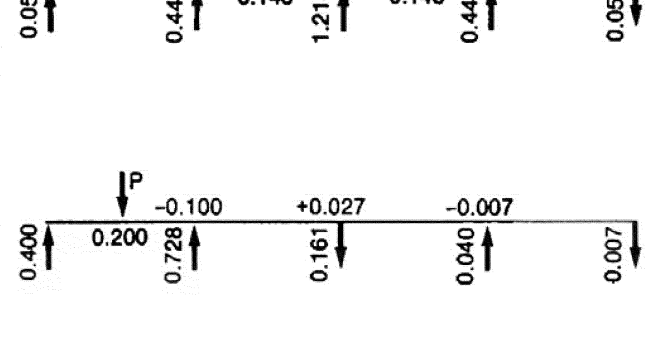
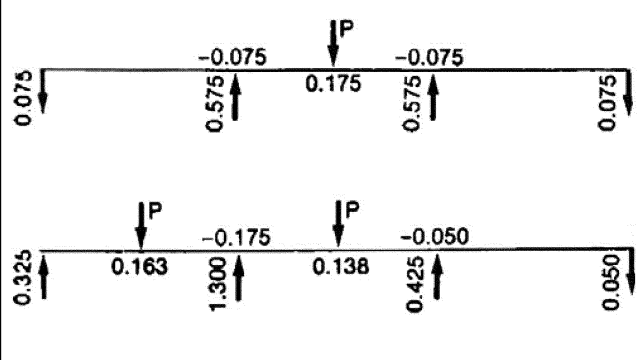
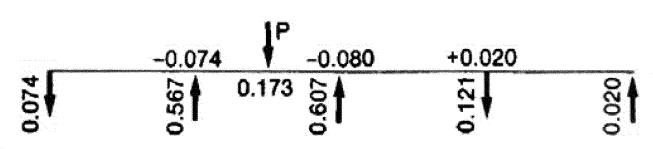
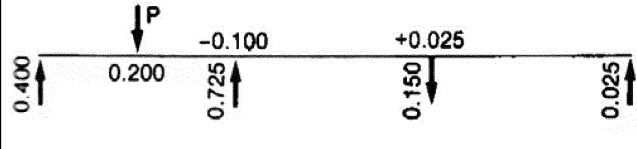
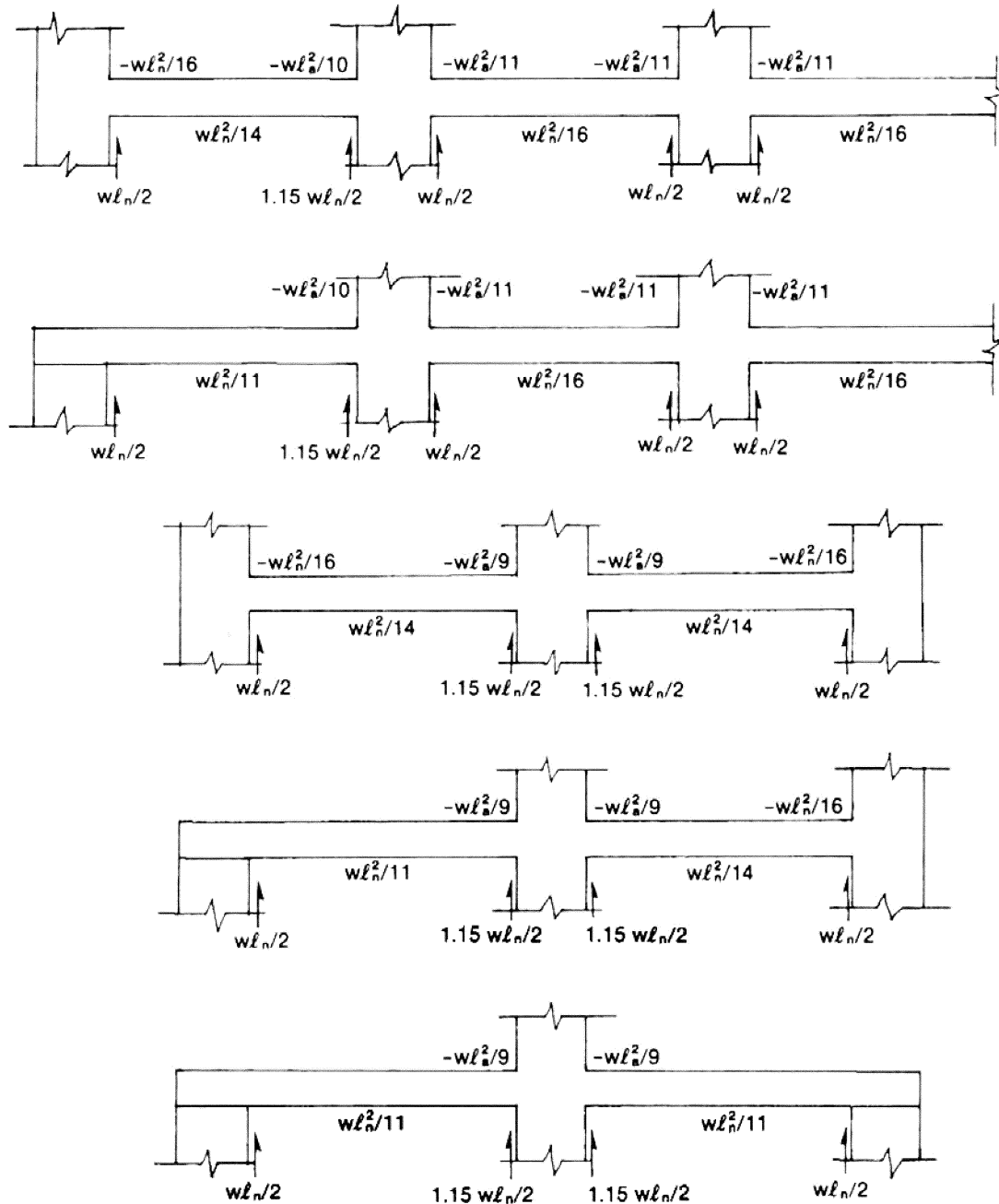
Moment : Coefficient $\times Pl$ Reaction: Coefficient $\times P$		P: Concentrated load l: Length of one span	
Two Equal Spans		Four Equal Spans	
			
Three Equal Spans			
			
			
			
			

Table B-4 Moments and Reactions in Continuous Beams, Point Loads at Third Points of Span

Moment : Coefficient $\times Pl$ Reaction: Coefficient $\times P$		P: Total concentrated load on one span l: Length of one span	
Two Equal Spans		Four Equal Spans	
Three Equal Spans			

Table B-5 Approximate Moments and Shears for Continuous Beams and One-way Slabs

w : Factored load per unit length
 l_n : Clear span length
 l_a : Average length of adjacent clear spans


Notes:

- 1 This figure is applicable to prismatic members loaded with uniformly distributed load where the ratio of factored live load to factored dead load is not greater than 3.0, and the span lengths approximately equal, with the longer of the two adjacent spans not greater than the shorter by more than 20 per cent.
- 2 If the exterior support is a spandrel or girder, the negative moment at the interior face of exterior support is $wl_n^2/24$.
- 3 For slabs with spans not exceeding 3 m or beams with the ratio of the sum of the column stiffnesses to the beam stiffness exceeds eight at each end, the negative moment at all supports can be taken equal to $wl_n^2/12$.

Table B-6 Beams with Prismatic Haunch at One End

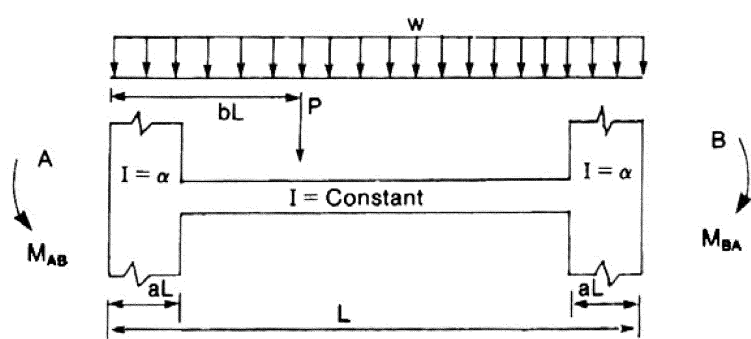
Right haunch	r	Carry-over factors		Stiffness factors		Unif. load		Concentrated load F.E.M. — coef. $\times PL$										Haunch load	
						F.E.M. coef. $\times WL^2$		b										F.E.M. coef. $\times WL^2$	
		C_{AB}	C_{BA}	k_{AB}	k_{BA}	M_{AB}	M_{BA}	0.1		0.3		0.5		0.7		0.9		M_{AB}	M_{BA}
0.1	0.4	0.593	0.491	4.24	5.12	0.0749	0.1016	0.0799	0.0113	0.1397	0.0788	0.1110	0.1553	0.0478	0.1798	0.0042	0.0911	0.0001	0.0047
	0.6	0.615	0.490	4.30	5.40	0.0727	0.1062	0.0797	0.0119	0.1378	0.0828	0.1074	0.1630	0.0439	0.1881	0.0029	0.0937	0.0001	0.0048
	1.0	0.639	0.488	4.37	5.72	0.0703	0.1114	0.0794	0.0125	0.1358	0.0873	0.1035	0.1716	0.0396	0.1974	0.0016	0.0966	0.0001	0.0049
	1.5	0.652	0.487	4.40	5.89	0.0690	0.1143	0.0792	0.0129	0.1346	0.0898	0.1012	0.1764	0.0373	0.2026	0.0008	0.0982	0.0000	0.0049
	2.0	0.658	0.487	4.42	5.97	0.0684	0.1156	0.0791	0.0131	0.1341	0.0910	0.1002	0.1786	0.0361	0.2050	0.0005	0.0990	0.0000	0.0050
0.2	0.4	0.677	0.469	4.42	6.37	0.0706	0.1126	0.0791	0.0134	0.1345	0.0925	0.1020	0.1788	0.0409	0.1975	0.0050	0.0890	0.0013	0.0171
	0.6	0.730	0.463	4.56	7.18	0.0664	0.1225	0.0785	0.0149	0.1302	0.1025	0.0942	0.1972	0.0335	0.2148	0.0037	0.0917	0.0010	0.0178
	1.0	0.793	0.458	4.74	8.22	0.0610	0.1353	0.0777	0.0168	0.1248	0.1154	0.0843	0.2207	0.0242	0.2368	0.0022	0.0951	0.0006	0.0187
	1.5	0.831	0.455	4.86	8.88	0.0576	0.1434	0.0772	0.0180	0.1214	0.1235	0.0781	0.2355	0.0182	0.2507	0.0012	0.0973	0.0003	0.0193
	2.0	0.849	0.453	4.91	9.20	0.0559	0.1473	0.0769	0.0186	0.1197	0.1276	0.0750	0.2429	0.0153	0.2576	0.0007	0.0984	0.0002	0.0196
0.3	0.4	0.741	0.439	4.52	7.63	0.0698	0.1155	0.0787	0.0149	0.1319	0.1013	0.0987	0.1899	0.0420	0.1929	0.0056	0.0868	0.0045	0.0338
	0.6	0.831	0.427	4.75	9.24	0.0642	0.1296	0.0777	0.0175	0.1255	0.1182	0.0877	0.2185	0.0338	0.2130	0.0045	0.0893	0.0036	0.0359
	1.0	0.954	0.415	5.09	11.69	0.0559	0.1511	0.0762	0.0215	0.1158	0.1440	0.0711	0.2621	0.0217	0.2436	0.0028	0.0930	0.0023	0.0391
	1.5	1.036	0.409	5.34	13.53	0.0497	0.1673	0.0751	0.0245	0.1085	0.1633	0.0587	0.2948	0.0128	0.2665	0.0017	0.0959	0.0014	0.0415
	2.0	1.078	0.407	5.48	14.54	0.0464	0.1762	0.0745	0.0262	0.1045	0.1740	0.0520	0.3129	0.0080	0.2792	0.0010	0.0974	0.0008	0.0448
0.4	0.4	0.774	0.405	4.55	8.70	0.0703	0.1117	0.0786	0.0156	0.1315	0.1035	0.0992	0.1855	0.0445	0.1773	0.0059	0.0849	0.0106	0.0509
	0.6	0.901	0.386	4.83	11.28	0.0646	0.1269	0.0774	0.0192	0.1240	0.1254	0.0875	0.2182	0.0377	0.1932	0.0049	0.0869	0.0089	0.0547
	1.0	1.102	0.367	5.33	16.03	0.0549	0.1548	0.0752	0.0257	0.1105	0.1658	0.0671	0.2780	0.0267	0.2222	0.0034	0.0904	0.0063	0.0616
	1.5	1.260	0.357	5.79	20.46	0.0462	0.1807	0.0732	0.0319	0.0982	0.2035	0.0485	0.3339	0.0173	0.2491	0.0022	0.0938	0.0037	0.0679
	2.0	1.349	0.352	6.09	23.32	0.0407	0.1975	0.0719	0.0358	0.0903	0.2278	0.0367	0.3699	0.0113	0.2664	0.0014	0.0959	0.0027	0.0720
0.5	0.4	0.768	0.371	4.56	9.45	0.0700	0.1048	0.0786	0.0154	0.1312	0.0993	0.0983	0.1679	0.0442	0.1663	0.0059	0.0836	0.0189	0.0656
	0.6	0.919	0.343	4.84	12.94	0.0651	0.1176	0.0774	0.0193	0.1240	0.1218	0.0884	0.1935	0.0386	0.1769	0.0051	0.0849	0.0167	0.0702
	1.0	1.200	0.316	5.42	20.61	0.0561	0.1451	0.0749	0.0280	0.1096	0.1709	0.0706	0.2486	0.0299	0.1993	0.0038	0.0877	0.0131	0.0802
	1.5	1.470	0.301	6.10	29.74	0.0466	0.1777	0.0720	0.0384	0.0934	0.2290	0.0516	0.3137	0.0215	0.2255	0.0027	0.0909	0.0094	0.0918
	2.0	1.647	0.295	6.63	37.04	0.0393	0.2036	0.0698	0.0466	0.0807	0.2755	0.0370	0.3655	0.0153	0.2463	0.0019	0.0934	0.0067	0.1011
0.6	0.4	0.726	0.341	4.62	9.84	0.0675	0.0986	0.0782	0.0146	0.1280	0.0916	0.0923	0.1519	0.0419	0.1603	0.0056	0.0829	0.0283	0.0769
	0.6	0.872	0.305	4.88	13.97	0.0630	0.1072	0.0771	0.0183	0.1214	0.1096	0.0835	0.1664	0.0368	0.1666	0.0048	0.0837	0.0254	0.0813
	1.0	1.196	0.267	5.43	24.35	0.0560	0.1277	0.0748	0.0274	0.1092	0.1537	0.0705	0.1999	0.0299	0.1804	0.0038	0.0854	0.0212	0.0913
	1.5	1.588	0.247	6.18	39.79	0.0482	0.1572	0.0718	0.0408	0.0939	0.2183	0.0572	0.2478	0.0237	0.1997	0.0030	0.0878	0.0171	0.1055
	2.0	1.905	0.237	6.92	55.51	0.0412	0.1870	0.0688	0.0544	0.0792	0.2839	0.0455	0.2960	0.0186	0.2189	0.0023	0.0901	0.0136	0.1197
0.7	0.4	0.657	0.321	4.86	9.96	0.0631	0.0954	0.0770	0.0138	0.1175	0.0846	0.0844	0.1461	0.0392	0.1582	0.0053	0.0827	0.0372	0.0854
	0.6	0.770	0.275	5.14	14.39	0.0580	0.1006	0.0758	0.0167	0.1097	0.0955	0.0745	0.1543	0.0335	0.1621	0.0045	0.0832	0.0330	0.0890
	1.0	1.056	0.224	5.62	26.45	0.0516	0.1122	0.0738	0.0243	0.0992	0.1213	0.0626	0.1710	0.0269	0.1694	0.0035	0.0841	0.0280	0.0965
	1.5	1.491	0.196	6.24	47.48	0.0463	0.1304	0.0714	0.0371	0.0890	0.1633	0.0537	0.1959	0.0223	0.1796	0.0028	0.0854	0.0241	0.1076
	2.0	1.944	0.183	6.95	73.85	0.0417	0.1523	0.0687	0.0530	0.0793	0.2149	0.0468	0.2255	0.0191	0.1915	0.0024	0.0869	0.0210	0.1210
0.8	0.4	0.583	0.319	5.46	9.97	0.0585	0.0951	0.0741	0.0137	0.1040	0.0837	0.0793	0.1456	0.0380	0.1580	0.0053	0.0826	0.0452	0.0917
	0.6	0.645	0.263	5.89	14.44	0.0516	0.0990	0.0721	0.0160	0.0921	0.0907	0.0667	0.1520	0.0311	0.1614	0.0043	0.0831	0.0388	0.0951
	1.0	0.818	0.196	6.47	27.06	0.0435	0.1053	0.0696	0.0211	0.0781	0.1025	0.0521	0.1615	0.0232	0.1660	0.0031	0.0838	0.0314	0.1004
	1.5	1.128	0.155	6.98	50.85	0.0385	0.1130	0.0676	0.0296	0.0692	0.1175	0.0432	0.1715	0.0184	0.1705	0.0024	0.0844	0.0268	0.1064
	2.0	1.533	0.135	7.47	84.60	0.0355	0.1222	0.0658	0.0412	0.0638	0.1357	0.0384	0.1824	0.0159	0.1750	0.0020	0.0849	0.0242	0.1133
0.9	0.4	0.524	0.356	6.87	10.10	0.0604	0.0948	0.0674	0.0157	0.1031	0.0835	0.0844	0.1439	0.0418	0.1568	0.0059	0.0824	0.0550	0.0942
	0.6	0.542	0.295	7.95	14.58	0.0497	0.0991	0.0623	0.0184	0.0866	0.0913	0.0691	0.1510	0.0339	0.1605	0.0048	0.0830	0.0460	0.0985
	1.0	0.594	0.206	9.44	27.16	0.0372	0.1052	0.0553	0.0226	0.0642	0.1023	0.0484	0.1609	0.0231	0.1656	0.0032	0.0837	0.0337	0.1044
	1.5	0.695	0.142	10.48	51.25	0.0289	0.1098	0.0506	0.0266	0.0492	0.1105	0.0346	0.1680	0.0159	0.1692	0.0021	0.0842	0.0255	0.1089
	2.0	0.842	0.107	11.07	86.80	0.0245	0.1147	0.0481	0.0306	0.0414	0.1159	0.0274	0.1723	0.0121	0.1714	0.0016	0.0845	0.0213	0.1117

Table B-7 Beams with Prismatic Haunch at Both Ends

The diagram illustrates a beam of total length L with prismatic haunches at both ends. The haunches have a length aL and a height rh . The central section of the beam has a length h . A uniformly distributed load W is applied over the entire length of the beam. A concentrated load P is applied at a distance bL from the left end. The beam is supported at both ends (A and B) with moments M_{AB} and M_{BA} .

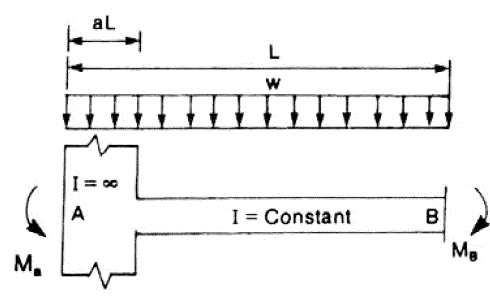
a	r	Carry-over factors $C_{AB} = C_{BA}$	Stiffness factors $k_{AB} = k_{BA}$	Unif. load	Concentrated load F.E.M. — coef. $\times PL$											Haunch load both haunches F.E.M. coef. $\times WL^2$	
				F.E.M. coef. $\times WL^2$	b												
					0.1		0.3		0.5		0.7		0.9				
				$M_{AB} = M_{BA}$	M_{AB}	M_{BA}	M_{AB}	M_{BA}	M_{AB}	M_{BA}	M_{AB}	M_{BA}	M_{AB}	M_{BA}	$M_{AB} = M_{BA}$		
0.1	0.4	0.583	5.49	0.0921	0.0905	0.0053	0.1727	0.0606	0.1396	0.1396	0.0606	0.1727	0.0053	0.0905	0.0049		
	0.6	0.603	5.93	0.0940	0.0932	0.0040	0.1796	0.0589	0.1428	0.1428	0.0589	0.1796	0.0040	0.0932	0.0049		
	1.0	0.624	6.45	0.0961	0.0962	0.0023	0.1873	0.0566	0.1462	0.1462	0.0566	0.1873	0.0023	0.0962	0.0050		
	1.5	0.636	6.75	0.0972	0.0980	0.0013	0.1918	0.0551	0.1480	0.1480	0.0551	0.1918	0.0013	0.0980	0.0050		
	2.0	0.641	6.90	0.0976	0.0988	0.0008	0.1939	0.0543	0.1489	0.1489	0.0543	0.1939	0.0008	0.0988	0.0050		
0.2	0.4	0.634	7.32	0.0970	0.0874	0.0079	0.1852	0.0623	0.1506	0.1506	0.0623	0.1852	0.0079	0.0874	0.0187		
	0.6	0.674	8.80	0.1007	0.0899	0.0066	0.1993	0.0584	0.1575	0.1575	0.0584	0.1993	0.0066	0.0899	0.0191		
	1.0	0.723	11.09	0.1049	0.0935	0.0046	0.2193	0.0499	0.1654	0.1654	0.0499	0.2193	0.0046	0.0935	0.0195		
	1.5	0.752	12.87	0.1073	0.0961	0.0029	0.2338	0.0420	0.1699	0.1699	0.0420	0.2338	0.0029	0.0961	0.0197		
	2.0	0.765	13.87	0.1084	0.0976	0.0018	0.2410	0.0372	0.1720	0.1720	0.0372	0.2410	0.0018	0.0976	0.0198		
0.3	0.4	0.642	9.02	0.0977	0.0845	0.0097	0.1763	0.0707	0.1558	0.1558	0.0707	0.1763	0.0097	0.0845	0.0397		
	0.6	0.697	12.09	0.1027	0.0861	0.0095	0.1898	0.0700	0.1665	0.1665	0.0700	0.1898	0.0095	0.0861	0.0410		
	1.0	0.775	18.68	0.1091	0.0890	0.0084	0.2136	0.0627	0.1803	0.1803	0.0627	0.2136	0.0084	0.0890	0.0426		
	1.5	0.828	26.49	0.1132	0.0920	0.0065	0.2376	0.0492	0.1891	0.1891	0.0492	0.2376	0.0065	0.0920	0.0437		
	2.0	0.855	32.77	0.1153	0.0943	0.0048	0.2555	0.0366	0.1934	0.1934	0.0366	0.2555	0.0048	0.0943	0.0442		
0.4	0.4	0.599	10.15	0.0937	0.0825	0.0101	0.1601	0.0732	0.1509	0.1509	0.0732	0.1601	0.0101	0.0825	0.0642		
	0.6	0.652	14.52	0.0966	0.0833	0.0106	0.1668	0.0776	0.1632	0.1632	0.0776	0.1668	0.0106	0.0833	0.0668		
	1.0	0.744	26.06	0.1067	0.0847	0.0112	0.1790	0.0835	0.1833	0.1833	0.0835	0.1790	0.0112	0.0847	0.0711		
	1.5	0.827	45.95	0.1131	0.0862	0.0113	0.1919	0.0852	0.1995	0.1995	0.0852	0.1919	0.0113	0.0862	0.0746		
	2.0	0.878	71.41	0.1169	0.0876	0.0108	0.2033	0.0822	0.2089	0.2089	0.0822	0.2033	0.0108	0.0876	0.0766		
0.5	0.0	0.500	4.00	0.0833	0.0810	0.0090	0.1470	0.0630	0.1250	0.1250	0.0630	0.1470	0.0090	0.0810	0.0833		

Table B-8 Prismatic Member with Equal Infinitely Stiff End Regions



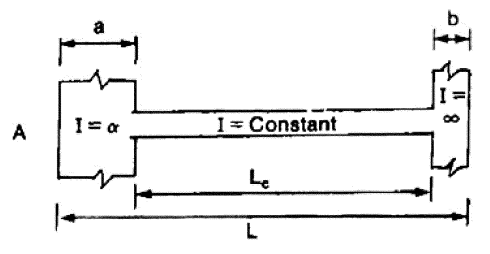
a	Carry-over Factors	Stiffness Factors	Unif. Load F.E.M. Coef. × wL ²	Concentrated Load F.E.M. — Coef. × PL									
				b									
				0.1		0.2		0.3		0.4		0.5	
				M _{AB}	M _{BA}	M _{AB}	M _{BA}	M _{AB}	M _{BA}	M _{AB}	M _{BA}	M _{AB}	M _{BA}
0.05	0.575	5.23	0.0913	0.0940	0.0030	0.1505	0.0245	0.1711	0.0595	0.1640	0.0999	0.1375	0.1375
0.10	0.648	7.11	0.0983	0.1000	0.0000	0.1722	0.0152	0.1968	0.0532	0.1856	0.1019	0.1500	0.1500
0.15	0.719	10.17	0.1046	0.1000	0.0000	0.1909	0.0056	0.2247	0.0431	0.2095	0.1013	0.1625	0.1625
0.20	0.786	15.56	0.1100	0.1000	0.0000	0.2000	0.0000	0.2546	0.0286	0.2369	0.0964	0.1750	0.1750
0.25	0.846	26.00	0.1146	0.1000	0.0000	0.2000	0.0000	0.2830	0.0118	0.2699	0.0851	0.1875	0.1875

Table B-9 Prismatic Member with Infinitely Stiff Region at One End



a	Carry-over Factors		Stiffness Factors		Unif. Load F.E.M. Coef. × WL ²	
	C _{AB}	C _{BA}	k _{AB}	k _{BA}	M _{AB}	M _{BA}
0.05	0.496	0.579	4.91	4.21	0.1002	0.0752
0.10	0.486	0.657	6.00	4.44	0.1175	0.0675
0.15	0.471	0.765	7.64	4.71	0.1352	0.0602
0.20	0.452	0.875	9.60	5.00	0.1533	0.0533
0.25	0.429	1.000	12.44	5.33	0.1719	0.0469

Table B-10 Prismatic Member with Unequal Infinitely Stiff End Regions



$$k' = \left[1 - \left[\frac{3(L_c + 2b)(L_c + 2a)}{L_c^2} \right] \right] \frac{L}{L_c}$$
$$k_{AB} = \left[1 + 3 \left[\frac{L_c + 2a}{L_c} \right]^2 \right] \frac{L}{L_c}$$
$$k_{BA} = \left[1 + 3 \left[\frac{L_c + 2b}{L_c} \right]^2 \right] \frac{L}{L_c}$$
$$C_{AB} = k'/k_{AB}$$
$$C_{BA} = k'/k_{BA}$$

APPENDIX C—SECTIONAL PROPERTIES

Table C-1—Properties of sections

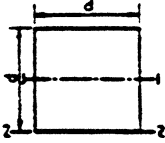
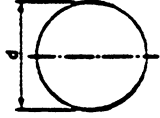
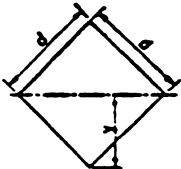
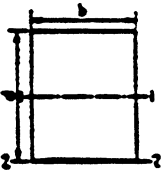
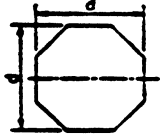
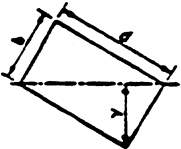
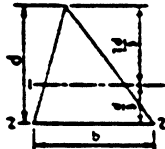
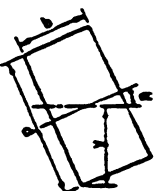
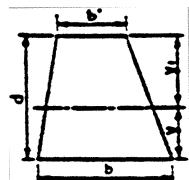
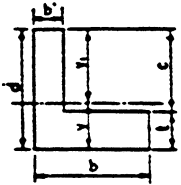
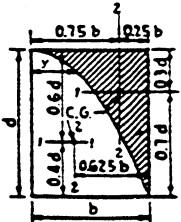
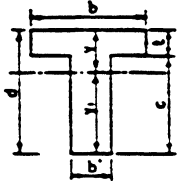
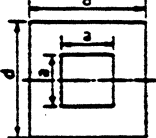
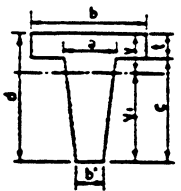
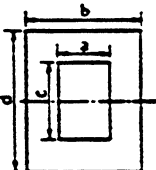
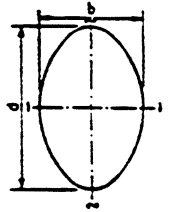
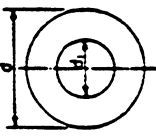
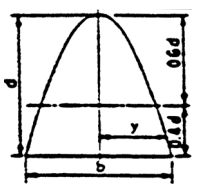
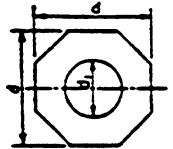
Dash-and-dot lines are drawn through centers of gravity A = area of section; I = moment of inertia; R = radius of gyration			
	$A = d^2$ $I_1 = \frac{d^4}{12}$ $I_2 = \frac{d^4}{3}$ $R_1 = 0.2887d$ $R_2 = 0.57774d$		$A = \frac{\pi d^2}{4} = 0.7854d^2$ $I = \frac{\pi d^4}{64} = 0.0491d^4$ $R = \frac{d}{4}$
	$A = d^2$ $y = 0.7071d$ $I = \frac{d^4}{12}$ $R = 0.2887d$		$A = 0.8660d^2$ $I = 0.060d^4$ $R = 0.264d$
	$A = bd$ $I_1 = \frac{bd^3}{12}$ $I_2 = \frac{bd^3}{3}$ $R_1 = 0.2887d$ $R_2 = 0.57774d$		$A = 0.8284d^2$ $I = 0.055d^4$ $R = 0.257d$
	$A = bd$ $y = \frac{bd}{\sqrt{b^2 + d^2}}$ $I = \frac{b^3 d^3}{6(b^2 + d^2)}$ $R = \frac{bd}{\sqrt{6(b^2 + d^2)}}$		$A = \frac{bd}{2}$ $I_1 = \frac{bd^3}{36}$ $I_2 = \frac{bd^3}{12}$ $R_1 = 0.236d$ $R_2 = 0.408d$
	$A = bd$ $y = \frac{b \sin \alpha + d \cos \alpha}{2}$ $R = \frac{\sqrt{b^2 \sin^2 \alpha + d^2 \cos^2 \alpha}}{12}$		$A = \frac{d}{2}(b + b')$ $y = \frac{d(2b + b')}{3(b + b')}$ $y = \frac{d(b + 2b')}{3(b + b')}$ $I = \frac{d^3(b^2 + 4bb' + b'^2)}{36(b + b')}$ $R = \frac{d}{6(b + b')} \sqrt{2(b^2 + 4bb' + b'^2)}$

Table C-2—Properties of sections

Dash-and-dot lines are drawn through centers of gravity A = area of section; I = moment of inertia; R = radius of gyration	
 $A = bt + b'_c$ $y = \frac{d^2 b' + t^2 (b - b')}{2(bt + b'_c)}$ $y_1 = d - y$ $I = \frac{b'y_1^3 + by^3 - (b - b')(y - t)^3}{3}$ $R = \sqrt{\frac{I}{A}}$	Section of parabola $y^2 = \frac{b^2}{d}x$ For parabola: For complement: $A = \frac{2bd}{3} \quad A = \frac{bd}{3}$ $I_1 = \frac{8}{175}bd^3 \quad I_1 = \frac{37}{2100}bd^3$ $I_2 = \frac{19}{480}b^3d \quad I_2 = \frac{1}{80}b^3d$ 
 $A = bt + b'_c$ $y = \frac{d^2 b' + t^2 (b - b')}{2(bt + b'_c)}$ $y_1 = d - y$ $I = \frac{b'y_1^3 + by^3 - (b - b')(y - t)^3}{3}$ $R = \sqrt{\frac{I}{A}}$	 $A = d^2 - a^2$ $I = \frac{d^4 - a^4}{12}$ $R = \sqrt{\frac{d^2 + a^2}{12}}$
 $A = bt + \frac{c(a + b')}{2}$ $y = \frac{3bt^2 + 3b'c(d + t) + c(a - b')(c + 3t)}{3[2bt + c(a + b')]}$ $y_1 = d - y$ $I = \frac{4bt^2 + c^3(3b' + a)}{12} - A(y - t)^2$ $R = \sqrt{\frac{I}{A}}$	 $A = bd - ac$ $I = \frac{bd^3 - ac^3}{12}$ $R = \sqrt{\frac{bd^3 - ac^3}{12(bd - ac)}}$
 Ellipse $A = 0.7854bd$ $I_1 = \frac{wb d^3}{64} = 0.0491bd^3$ $I_2 = \frac{wb^3 d}{64} = 0.0491b^3d$ $R_1 = \frac{d}{4}$ $R_2 = \frac{b}{4}$	 $A = \frac{\pi(d^2 - d_1^2)}{4} = 0.7854(d^2 - d_1^2)$ $I = \frac{\pi(d^4 - d_1^4)}{64} = 0.0491(d^4 - d_1^4)$ $R = 1/4 \sqrt{d^2 + d_1^2}$
 Equation: $y^2 = \frac{b^2}{4d^3}x$ $A = \frac{2bd}{3}$ Parabola	 $A = 0.8284d^2 - 0.7854d_1^2$ $= 0.7854(1.055d^2 - d_1^2)$ $I = 0.0547d^4 - 0.0491d_1^4$ $= 0.0491(1.115d^4 - d_1^4)$ $= 0.0491[(1.056)^2 d^4 - d_1^4]$ $R = 1/4 \sqrt{1.056d^2 + d_1^2}$

APPENDIX D—COLUMN INTERACTION DIAGRAMS (DESIGN AIDS)

D.1—Column interaction diagrams

The column axial load-bending moment interaction diagrams illustrated in D.5 conform to ACI 318-14. The equations used to generate data for plotting the interaction diagrams were originally developed for ACI SP-7 (Everard and Cohen 1964). In addition, complete derivations of the equations for square and circular columns having the steel arranged in a circle have been published in the *ACI Structural Journal* (Everard 1997). The interaction diagrams contained in SP-7 were subsequently published in SP-17A (ACI Committee 340 1970). The related equations were derived considering the following:

- (a) For rectangular and square columns having steel bars placed on the end faces only, reinforcement was assumed to consist of two equal thin strips parallel to the compression face of the section;
- (b) For rectangular and square columns having steel bars equally distributed along all four faces, the reinforcement was considered to consist of a thin rectangular or square tube; and
- (c) For square and circular sections having steel bars arranged in a circle, the reinforcement was considered to consist of a thin circular tube.

The interaction diagrams were developed using the rectangular stress block (ACI 318-14, Section 22.2.2.4). In all cases, for reinforcement within the compressed portion of the depth perpendicular to the compression face of the concrete ($a = \beta_1 c$), the compression stress in the steel was reduced by $0.85f'_c$ to account for the concrete area that is displaced by the reinforcing bars within the compression stress block.

The interaction diagrams were plotted in nondimensional form. The vertical coordinate [$K_n = P_n / (f'_c A_g)$] represents the nondimensional form of the nominal axial load strength of the section. The horizontal coordinate [$R_n = M_n / (f'_c A_g h)$] represents the nondimensional nominal bending moment strength of the section. The nondimensional forms were used so that the interaction diagrams could be used with any system of units (SI or in.-lb units). Because ACI 318-14 contains different ϕ factors in Chapter 21, the strength reduction factor ϕ was considered as 1.0 so that the nominal values in the interaction diagrams could be used with any set of ϕ factors.

Note that the ϕ factors provided in Chapter 21 of ACI 318-14 are based on strain values in the tension reinforcement farthest from the compression face of a member, or at the centroid of the tension reinforcement.

Also note the eccentricity ratios ($e/h = M/P$), sometimes included as diagonal lines on interaction diagrams, are not included in the interaction diagrams. Using the eccentricity ratio as a coordinate with K_n or R_n could lead to inaccuracies because the e/h lines converge rapidly at the lower ends of the diagrams. Straight lines for the tension steel stress ratios f_s/f_y have been plotted to assist in designing splices for the reinforcement. Further, the ratio $f_s/f_y = 1.0$ represents steel strain $\epsilon_y = f_y/E_s$, which is the boundary point for the compression control ϕ factor, and the beginning of the transition zone for linear increase of the ϕ factor to that for tension control.

To provide interpolation for the ϕ factor, other strain lines were plotted. The strain line for $\epsilon_t = 0.005$, the beginning of the zone for tension control, has been plotted on all diagrams. The intermediate strain line for $\epsilon_t = 0.0035$ has been plotted for steel yield strength 60.0 ksi. The intermediate strain line for $\epsilon_t = 0.0038$ has been plotted for steel yield strength 75.0 ksi. Note that all strains refer to the reinforcing bar or bars farthest from the compression face of the section. Discussions and tables related to the strength reduction factors are contained in two publications in *Concrete International* (Everard 2002a,b).

Straight lines for K_{max} are also provided on each interaction diagram. Here, K_{max} refers to the maximum permissible nominal axial load, $P_{n,max}$, on a column that is laterally reinforced with ties or hoops (ACI 318-14, Section 22.4.2.1).

$$P_{n,max} = 0.8 \left[0.85 f'_c (A_g - A_{st}) + f_y A_{st} \right] \quad (D.1a)$$

Then,

$$K_{max} = P_{max} / f'_c A_g \quad (D.1b)$$

For columns with spirals, values of K_{max} from the interaction diagrams are multiplied by 0.85/0.80 ratio.

The number of longitudinal reinforcing bars is not limited to the number shown on the interaction diagrams. The diagrams only illustrate the type of reinforcement patterns. However, for circular and square columns with steel arranged in a circle, and for rectangular or square columns with steel equally distributed along all four faces, 12 bars are preferred; using at least 8 bars is good practice. Although side steel was assumed to be 50 percent of the total steel for columns having longitudinal steel equally distributed along all four faces, accurate and conservative designs can result when side steel is only 30 percent of the total steel. The maximum number of bars used in any column cross section is limited by the maximum allowable steel ratio of 0.08, the cover, and spacing between bars.

Tension axial loads are not included in the interaction diagrams.

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- Parma, A. L.; Nieves, J. M.; and Gouwens, A., 1966, "Capacity of Reinforced Rectangular Columns Subject to Biaxial Bending," *ACI JOURNAL Proceedings*, V. 63, No. 9, Sept., pp. 911-923.

D.2—Columns subject to biaxial bending

D.2.1 General—Most columns are subjected to significant bending in one direction while subjected to relatively small bending moments in the orthogonal direction. These columns are designed using the interaction

diagrams discussed in the preceding section for uniaxial bending and, when required, checked for strength in the orthogonal direction. Other columns, such as corner columns, are subjected to equally significant bending moments in two orthogonal directions and, therefore, might have to be designed for biaxial bending.

A circular column subjected to moments about two axes may be designed as a uniaxial column acted upon by the resultant moment

$$M_x = \sqrt{M_{ux}^2 + M_{uy}^2} \leq \phi M_n = \phi \sqrt{M_{nx}^2 + M_{ny}^2} \quad (\text{D.2.1})$$

For the design of rectangular columns subjected to moments about two axes, this Handbook provides design aids for two methods: The reciprocal load ($1/P_i$) method (Bresler 1960) and the load contour method (Parme et al. 1966). The reciprocal load method is convenient for making a trial section analysis. The load contour method is suitable for selecting a column cross section. Both methods use a failure surface to reflect the interaction of three variables: the nominal axial load P_n and the nominal biaxial bending moments M_{nx} and M_{ny} . In combination, these variables cause failure strain at the extreme compression fiber; that is, the failure surface reflects the strength of short compression members subject to biaxial bending and compression. Figure D.2.1a illustrates the bending axes, eccentricities, and biaxial moments.

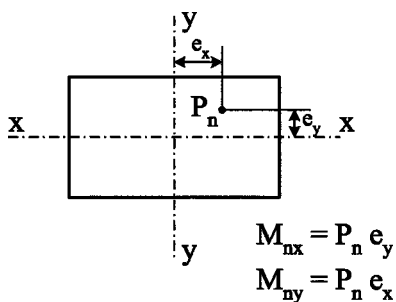


Fig. D.2.1a—Notation used for column sections subjected to biaxial bending.

A failure surface S_1 may be represented by P_n , e_x , and e_y , as in Fig. D.2.1b, or it may be represented by surface S_2 represented by P_n , M_{nx} , and M_{ny} , as shown in Fig. D.2.1c. Note that S_1 is a single curvature surface having no discontinuity at the balance point, whereas S_2 has discontinuity.

Note that when biaxial bending exists with a nominal axial force smaller than the lesser of P_b or $0.1 f'_c A_g$, it is sufficiently accurate and conservative to ignore the axial force and design the section for bending only.

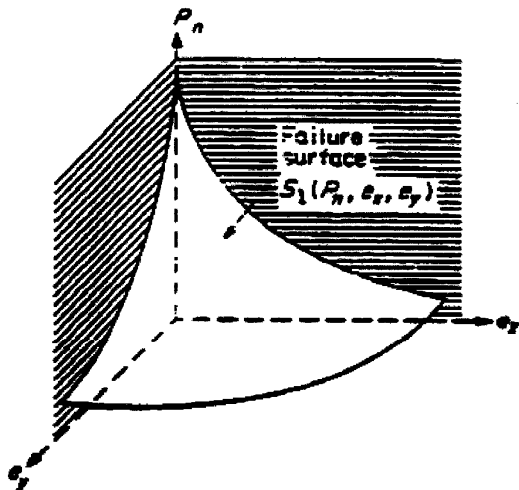


Fig. D.2.1b—Failure surface S_1 .

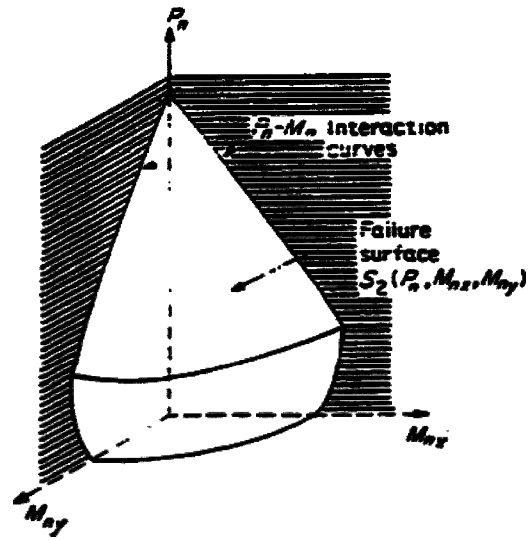


Fig. D.2.1c—Failure surface S_2 .

D.2.2 Reciprocal load method—In the reciprocal load method, S_1 is inverted by plotting $1/P_n$ as the vertical axis, giving S_3 , shown in Fig. D.2.2a. As Figure D.2.2b shows, a true point $(1/P_{ni}, e_{xA}, e_{yB})$ on this reciprocal failure surface can be approximated by a point $(1/P_{ni}, e_{xA}, e_{yB})$ on a plane S_3' passing through points A, B, and C. Each point is approximated by a different plane, that is, the entire failure surface is defined by an infinite number of planes.

Point A represents the nominal axial load strength P_{ny} when the load has an eccentricity of e_{xA} with $e_y = 0$. Point B represents the nominal axial load strength P_{nx} when the load has an eccentricity of e_{yB} with $e_x = 0$. Point C is based on the axial strength P_0 with zero eccentricity. The equation of the plane passing through the three points is

$$\frac{1}{P_{ni}} = \frac{1}{P_{nx}} + \frac{1}{P_{ny}} - \frac{1}{P_0} \quad (\text{D.2.2a})$$

where

P_{ni} = approximation of nominal axial load strength at eccentricities e_x and e_y

P_{nx} = nominal axial load strength for eccentricity e_y along the y-axis only (x-axis is axis of bending)

P_{ny} = nominal axial load strength for eccentricity e_x along the x-axis only (y-axis is axis of bending)

P_0 = nominal axial load strength for zero eccentricity

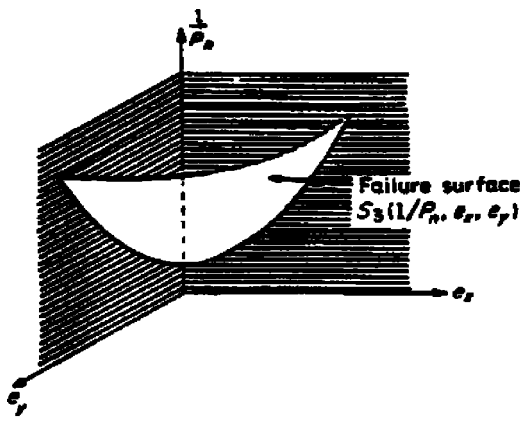


Fig. D.2.2a—Failure surface S_3 which is reciprocal of surface S_1 .

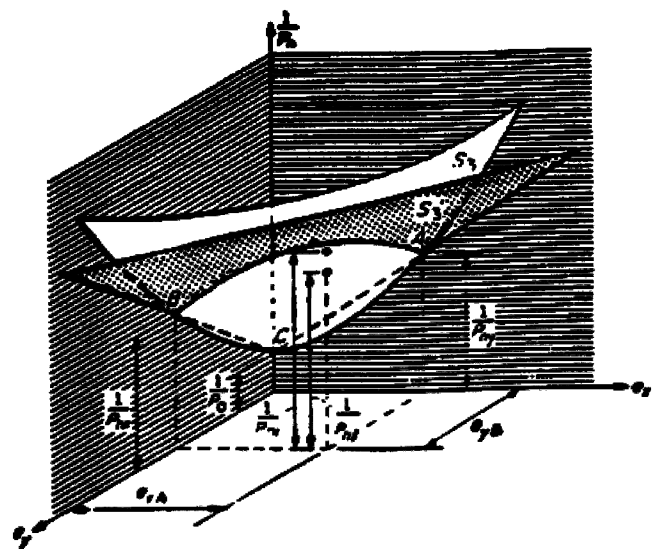


Fig. D.2.2b—Graphical representation of reciprocal load method.

For design purposes, when ϕ is constant, $1/P_{ni}$ in Eq. (D.2.2a) can be used. The variable $K_n = P_n / (f_c' A_g)$ can be used directly in the reciprocal equation, as follows (D.2.2b)

$$\frac{1}{K_{ni}} = \frac{1}{K_{nx}} + \frac{1}{K_{ny}} - \frac{1}{K_0} \quad (\text{D.2.2b})$$

where the K values refer to the corresponding P_n values as defined above.

Once a preliminary cross section with an estimated steel ratio ρ_g is selected, calculate R_{nx} and R_{ny} using the actual bending moments about the cross section x- and y-axes. Obtain the corresponding values of K_{nx} and K_{ny} from the interaction diagrams presented in this chapter as the intersection of R_n value and the assumed steel ratio curve for ρ_g . Then, obtain the theoretical compression axial load strength K_0 at the intersection of the steel ratio curve and the vertical axis for zero R_n .

D.2.3 Load contour method—The load contour method uses the failure surface S_2 (Fig. D.2.1c) and works with a load contour defined by a plane at a constant value of P_n (Fig. D.2.3a). The load contour defining the relationship between M_{nx} and M_{ny} for a constant P_n can be expressed nondimensionally as

$$\left(\frac{M_{nx}}{M_{nox}} \right)^\alpha + \left(\frac{M_{ny}}{M_{noy}} \right)^\alpha = 1 \quad (\text{D.2.3a})$$

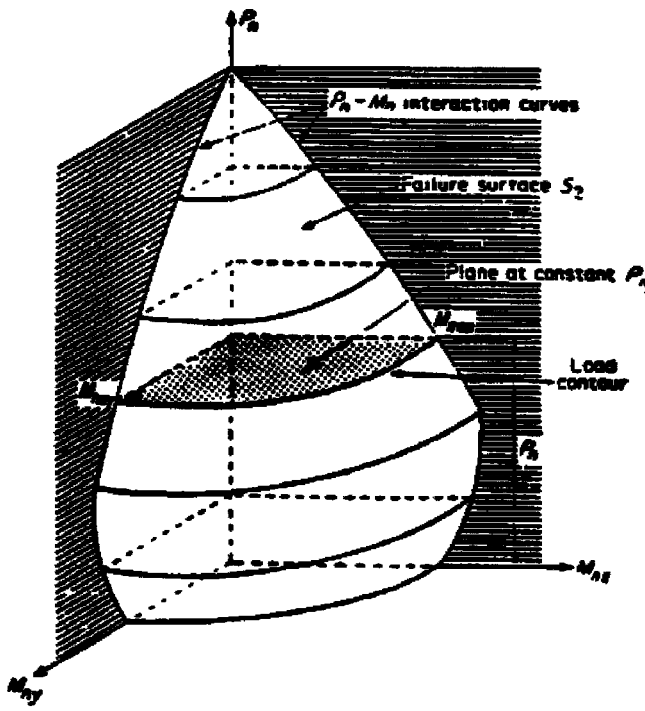


Fig. D.2.3a—Load contour constant P_n on failure surface.

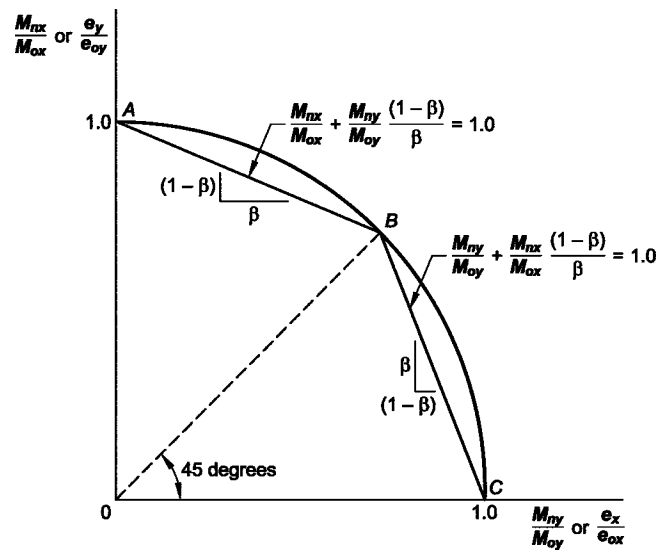


Fig. D.2.3b—Nondimensional load contour at constant P_n .

For design, when each term is multiplied by ϕ , the equation is unchanged. Thus, M_{ux} , M_{uy} , M_{ox} , and M_{oy} , which should correspond to ϕM_{nx} , ϕM_{ny} , ϕM_{nox} , and ϕM_{noy} , are used in the remainder of this section. To simplify the equation for application, a point on the nondimensional diagram (Fig. D.2.3b) is defined such that the biaxial moment capacities M_{nx} and M_{ny} are in the same ratio as the uniaxial moment capacities M_{ox} and M_{oy} ; thus

$$\frac{M_{nx}}{M_{ny}} = \frac{M_{ox}}{M_{oy}} \quad (\text{D.2.3b})$$

or

$$M_{nx} = \beta M_{ox} \text{ and } M_{ny} = \beta M_{oy} \quad (\text{D.2.3c})$$

In a physical sense, the ratio β is the constant portion of the uniaxial moment capacities that may act simultaneously on the column section. The actual value of β depends on the ratio P_n/P_{og} , as well as properties of the material and cross section. The usual range is 0.55 to 0.70 and an average value of 0.65 is suggested for design.

The load contour equation (Eq. (D.2.3a)) may be written in terms of β , as shown

$$\left(\frac{M_{nx}}{M_{nox}} \right)^{\log 0.5 / \log \beta} + \left(\frac{M_{ny}}{M_{noy}} \right)^{\log 0.5 / \log \beta} = 1 \quad (\text{D.2.3d})$$

Figure D.2.3b illustrates the relationship using β . The true relationship between Points A, B, and C is a curve, but for design purposes, it may be approximated by straight lines. The load contour equations as straight line approximation are

$$\text{For } \frac{M_{ny}}{M_{nx}} = \frac{M_{oy}}{M_{ox}}, \quad M_{oy} = M_{ny} + M_{nx} \left(\frac{M_{oy}}{M_{ox}} \right) \left(\frac{1-\beta}{\beta} \right) \quad (\text{D.2.3e})$$

$$\text{For } \frac{M_{ny}}{M_{nx}} \leq \frac{M_{oy}}{M_{ox}}, \quad M_{ox} = M_{nx} + M_{ny} \left(\frac{M_{ox}}{M_{oy}} \right) \left(\frac{1-\beta}{\beta} \right) \quad (\text{D.2.3f})$$

For rectangular sections with reinforcement equally distributed on all four faces, the above equations can be approximated by

$$M_{oy} = M_{ny} + M_{nx} \left(\frac{b}{h} \right) \left(\frac{1-\beta}{\beta} \right) \quad (\text{D.2.3g})$$

$$\text{For } \frac{M_{ny}}{M_{nx}} \leq \frac{M_{oy}}{M_{ox}} \text{ or } \frac{M_{ny}}{M_{nx}} \leq \frac{b}{h} \quad (\text{D.2.3f})$$

where b and h are dimensions of the rectangular column section parallel to x- and y-axes, respectively. Using the straight line approximation equations, the design problem can be solved by converting nominal moments into equivalent uniaxial moment capacities M_{ox} or M_{oy} . This is accomplished by:

- (a) Assuming a value for b/h
- (b) Estimating β as 0.65
- (c) Calculating the approximate equivalent uniaxial bending moment using Eq. (D.2.3e) or (D.2.3f)
- (d) Choosing the trial section and reinforcement using the methods for uniaxial bending and axial load.

The trial section should be verified using the load contour method or the reciprocal load method.

D.3—Column examples using interaction diagrams (D.5)

D.3.1 Column Example 1—Determination of required area of steel for a rectangular tied column with bars on four faces with slenderness ratio below critical value.

For a rectangular tied column with bars equally distributed along four faces, find steel area.

Given:

Loading—

$P_u = 560$ kip, and $M_u = 3920$ in.-kip

Assume $\phi = 0.70$

Nominal axial load $P_n = 560 \text{ kip} / 0.70 = 800$ kip

Nominal moment $M_n = 3920 \text{ in.-kip} / 0.70 = 5600$ in.-kip

Materials—

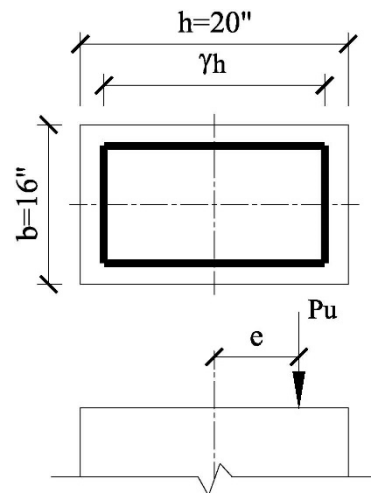
Compressive strength of concrete $f'_c = 4$ ksi

Yield strength of reinforcement $f_y = 60$ ksi

Normalized maximum size of aggregate is 1 in.

Design conditions—

Short column braced against sidesway



ACI 318-14 section	Discussion	Calculation	Design aid
	Determine column section size.	Given: $h = 20$ in. and $b = 16$ in.	
	Determine reinforcement ratio ρ_g using known values of variables on appropriate interaction diagrams and compute required cross section area A_{st} of longitudinal reinforcement.	$P_n = 800$ kip $M_n = 5600$ in.-kip $h = 20$ in. $b = 16$ in. $A_g = b \times h = 20 \times 16 = 320 \text{ in.}^2$	
	Compute $K_n = \frac{P_n}{f'_c A_g}$	$K_n = \frac{800}{(4)(320)} = 0.625$	
	Compute $R_n = \frac{M_n}{f'_c A_g h}$	$R_n = \frac{5600}{(4)(320)(20)} = 0.22$	
	Estimate $\gamma \approx \frac{h-5}{h}$	$\gamma \approx \frac{20-5}{20} = 0.75$	
10.5	Determine the appropriate interaction diagrams.	For a rectangular tied column with bars along four faces, $f'_c = 4$ ksi, $f_y = 60$ ksi, and an estimated γ of 0.75, enter diagram R4-60.7 and R4-60.8 with $K_n = 0.625$ and $R_n = 0.22$, respectively.	
	Read ρ_g for K_n and R_n values from appropriate interaction diagrams.	Read $\rho_g = 0.041$ for $\gamma = 0.7$ and $\rho_g = 0.039$ for $\gamma = 0.8$. Interpolating $\rho_g = 0.040$ for $\gamma = 0.75$	R4-60.7 R4-60.8
	Compute required A_{st} from $A_{st} = \rho_g A_g$.	Required $A_{st} = 0.040 \times 320 \text{ in.}^2 = 12.8 \text{ in.}^2$	

D.3.2 Column Example 2—For a specified reinforcement ratio, selection of a column section size for a rectangular tied column with bars on end faces only.

For minimum longitudinal reinforcement ($\rho_g = 0.01$) and column section dimension $h = 16$ in., select the column dimension b for a rectangular tied column with bars on end faces only.

Given:

Loading—

$P_u = 660$ kip and $M_u = 2790$ in.-kip

Assume $\phi = 0.70$

Nominal axial load $P_n = 660 \text{ kip}/0.70 = 943$ kip

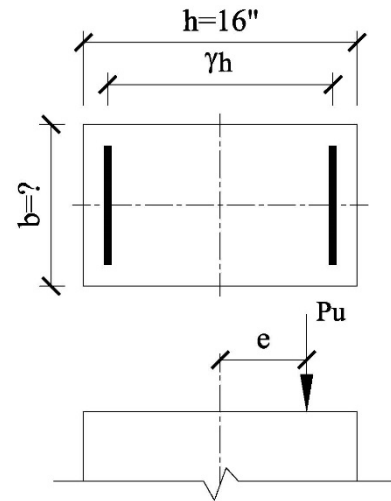
Nominal moment $M_n = 4200 \text{ in.-kip}/0.70 = 3986$ in.-kip

Materials—

Compressive strength of concrete $f'_c = 4$ ksi

Yield strength of reinforcement $f_y = 60$ ksi

Nominal maximum size of aggregate is 1 in.



Design conditions—

Slenderness effects may be neglected because $k\ell_u/h$ is known to be below critical value.

ACI 318-14 section	Discussion	Calculation			Design aid
	Determine trial column dimension b corresponding to known values of variables on appropriate interaction diagrams.	$P_n = 943$ kip $M_n = 3986$ in.-kip $h = 16$ in. $\rho_g = 0.039$			
	Assume a series of trial column sizes b , in inches, and compute $A_g = b \times h$, in. ²	b	24	26	28
		A_g	384	416	448
	Compute $K_n = \frac{P_n}{f_c' A_g}$	$\frac{943}{(4)(384)} = 0.61$	$\frac{943}{(4)(416)} = 0.57$	$\frac{943}{(4)(448)} = 0.53$	
	Compute $R_n = \frac{M_n}{f_c' A_g h}$	$\frac{3986}{(4)(384)(16)} = 0.16$	$\frac{3986}{(4)(416)(16)} = 0.15$	$\frac{3986}{(4)(448)(16)} = 0.14$	
	Estimate $\gamma \approx \frac{h-5}{h}$	0.7	0.7	0.7	
10.5	Determine the appropriate interaction diagrams.	For a rectangular tied column with bars along four faces, $f_c' = 4$ ksi, $f_y = 60$ ksi, and an estimated γ of 0.70, use diagram L4-60.7.			
	Read ρ_g for K_n and R_n values for $\gamma = 0.7$, select dimension corresponding to ρ_g nearest desired value of $\rho_g = 0.01$.	0.018	0.014	0.011	L4-60.7
		Therefore, try a 16 x 28-in. column			

D.3.3 Example 3—Selection of reinforcement for a square spiral column (slenderness ratio is below critical value).

For the square spiral column section shown, select reinforcement.

Given:

Loading—

$$P_u = 660 \text{ kip and } M_u = 2640 \text{ in.-kip}$$

Assume $\phi = 0.70$

$$\text{Nominal axial load } P_n = 660 \text{ kip}/0.70 = 943 \text{ kip}$$

$$\text{Nominal moment } M_n = 2640 \text{ in.-kip}/0.70 = 3771 \text{ in.-kip}$$

Materials—

Compressive strength of concrete $f'_c = 4 \text{ ksi}$

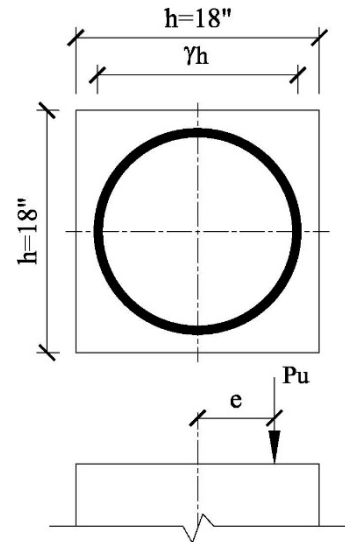
Yield strength of reinforcement $f_y = 60 \text{ ksi}$

Nominal maximum size of aggregate is 1 in.

Design conditions—

Column section size $h = b = 18 \text{ in}$

Slenderness effects may be neglected because $k\ell_u/h$ is known to be below critical value



ACI 318-14 section	Discussion	Calculation	Design aid
	Determine reinforcement ratio ρ_g using known values of variables on appropriate interaction diagram(s) and compute required cross section area A_{st} of longitudinal reinforcement.	$P_n = 943 \text{ kip}$ $M_n = 3771 \text{ in.-kip}$ $A_g = b \times h = 18 \times 18 = 324 \text{ in.}^2$	
	Compute $K_n = \frac{P_n}{f'_c A_g}$	$K_n = \frac{943}{(4)(324)} = 0.73$	
	Compute $R_n = \frac{M_n}{f'_c A_g h}$	$R_n = \frac{3771}{(4)(324)(18)} = 0.16$	
	Estimate $\gamma \approx \frac{h-5}{h}$	$\gamma \approx \frac{18-5}{18} = 0.72$	
10.5	Determine the appropriate interaction diagrams.	For a square spiral column, $f'_c = 4 \text{ ksi}$, $f_y = 60 \text{ ksi}$, and an estimated γ of 0.72, use diagram S4-60.7 and S4-60.8.	
	Read ρ_g for K_n and R_n .	For $K_n = 0.73$ and $R_n = 0.16$, and for: $\gamma = 0.70 \quad \rho_g = 0.035$ $\gamma = 0.80 \quad \rho_g = 0.031$ $\gamma = 0.72 \quad \rho_g = 0.034$ $A_{st} = 0.034 \times 324 \text{ in.}^2 = 11 \text{ in.}^2$	S4-60.7 S4-60.8

D.3.4 Example 4—Design of square column section subject to biaxial bending using resultant moment

Select column section size and reinforcement for a square column with $\rho_g \leq 0.04$ and bars equally distributed along four faces, subject to biaxial bending.

Given:

Loading—

$P_u = 193$ kip, $M_{ux} = 1917$ in.-kip, and $M_{uy} = 769$ in.-kip

Assume $\phi = 0.65$

Nominal axial load $P_n = 193 \text{ kip} / 0.65 = 297$ kip

Nominal moment about x-axis $M_{nx} = 1917 \text{ in.-kip} / 0.65 = 2949$ in.-kip

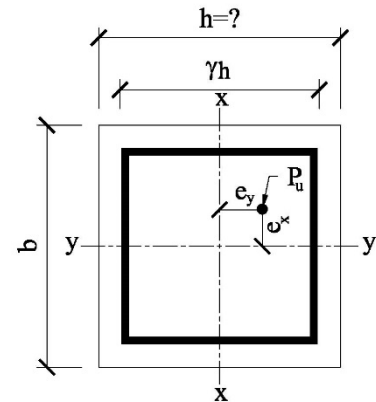
Nominal moment about y-axis $M_{ny} = 769 \text{ in.-kip} / 0.65 = 1183$ in.-kip

Materials—

Compressive strength of concrete $f'_c = 5$ ksi

Yield strength of reinforcement $f_y = 60$ ksi

Nominal maximum size of aggregate is 1 in.



ACI 318-14 section	Discussion	Calculation			Design aid
	Assume load contour curve at constant P_n is an ellipse, and determine resultant moment M_{nr} from $M_{nr} = \sqrt{M_{nx}^2 + M_{ny}^2}$	For a square column: $h = b$ $M_{nr} = \sqrt{2949^2 + 1183^2} = 3177 \text{ in.-kip}$			
	Assume a series of trial column sizes h , in inches.	14	16	18	
	Compute $A_g = h^2, \text{ in.}^2$	196	256	324	
	Compute $K_n = \frac{P_n}{f'_c A_g}$	$\frac{297}{(5)(196)} = 0.30$	$\frac{297}{(5)(256)} = 0.23$	$\frac{297}{(5)(324)} = 0.18$	
	Compute $R_n = \frac{M_n}{f'_c A_g h}$	$\frac{3177}{(5)(196)(14)} = 0.23$	$\frac{3177}{(5)(256)(16)} = 0.16$	$\frac{3177}{(5)(324)(18)} = 0.11$	
	Estimate $\gamma \approx \frac{h-5}{h}$	0.64	0.69	0.72	
10.5	Determine the appropriate interaction diagrams.	For a square tied column, $f'_c = 5$ ksi, $f_y = 60$ ksi, use diagram R5-60.6, R5-60.7 and R5-60.8.			
	Read ρ_g for K_n and R_n values for $\gamma = 0.60, 0.70$, and 0.08 .	0.064 0.048	0.030 0.026	0.012 0.011	R5-60.6 R5-60.7 R5-60.8
	Interpolate	0.058	0.026	0.012	
		Therefore, try $h = 15$ in.			
	Determine reinforcement ratio ρ_g using known values of variables on appropriate interaction diagrams and compute required cross section area A_{st} of longitudinal reinforcement.	$A_g = h^2 = 15^2 = 225 \text{ in.}^2$ $P_n = 297$ kip $M_n = 3177$ in.-kip			
	Compute $K_n = \frac{P_n}{f'_c A_g}$	$K_n = \frac{297}{(5)(225)} = 0.264$			

ACI 318-14 section	Discussion	Calculation	Design aid
	Compute $R_n = \frac{M_n}{f_c' A_g h}$	$R_n = \frac{3177}{(5)(225)(15)} = 0.188$	
	Estimate $\gamma \approx \frac{h-5}{h}$	$\gamma \approx \frac{15-5}{15} = 0.67$	
10.5	Determine the appropriate interaction diagrams.	For a square tied column, $f_c' = 5$ ksi, $f_y = 60$ ksi, use diagram R5-60.6 and R5-60.7.	
	Read ρ_g for K_n and R_n values for $\gamma = 0.60$ and 0.70 .	For $K_n = 0.264$ and $R_n = 0.188$: $\gamma = 0.60$ $\rho_g = 0.043$ $\gamma = 0.70$ $\rho_g = 0.034$ $\gamma = 0.37$ $\rho_g = 0.037$	R5-60.6 R5-60.7
	Compute required A_{st} from $A_{st} = \rho_g A_g$ and add about 15 percent for skew bending.	Required $A_{st} = 0.037 \times 225 \text{ in.}^2 = 8.32 \text{ in.}^2$ Use $A_{st} = 9.50 \text{ in.}^2$	

D.3.5 Example 5—Design of circular spiral column section subject to very small design moment

For a circular spiral column, select column section diameter h and choose reinforcement. Use relatively high proportion of longitudinal steel.

Given:

Loading—

$P_u = 940$ kip and $M_u = 480$ in.-kip

Assume $\phi = 0.70$

Nominal axial load $P_n = 940 \text{ kip} / 0.70 = 1343$ kip

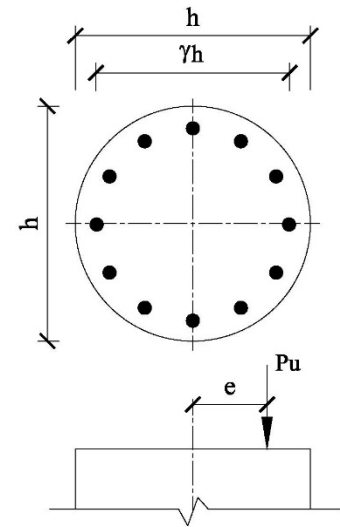
Nominal moment $M_n = 480 \text{ in.-kip} / 0.70 = 686$ in.-kip

Materials—

Compressive strength of concrete $f'_c = 5$ ksi

Yield strength of reinforcement $f_y = 60$ ksi

Nominal maximum size of aggregate is 1 in.



Design condition—

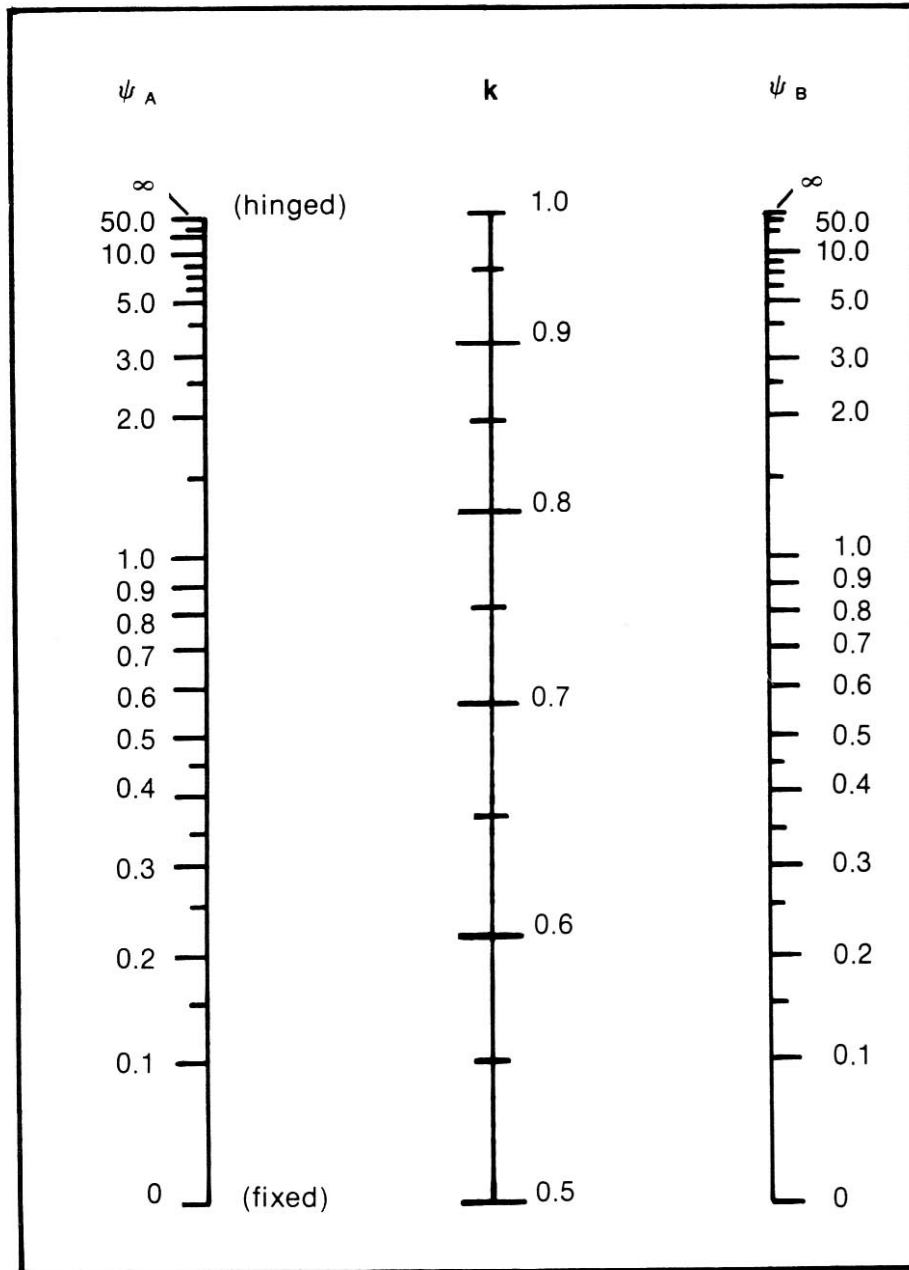
Slenderness effects may be neglected because $k\ell_u/h$ is known to be below critical value

ACI 318-14 section	Discussion	Calculation			Design aid
	Determine trial column dimension b corresponding to known values of variables on appropriate interaction diagrams.	$P_n = 1343$ kip $M_n = 686$ in.-kip $\rho_g = 0.04$			
	Assume a series of trial column sizes b , in inches.	12	16	20	
	Compute $A_g = h^2, \text{ in.}^2$	113	201	314	
	Compute $R_n = \frac{M_n}{f'_c A_g h}$	$\frac{686}{(5)(113)(12)} = 0.10$	$\frac{686}{(5)(201)(16)} = 0.04$	$\frac{686}{(5)(314)(20)} = 0.02$	
	Estimate $\gamma \approx \frac{h-5}{h}$	0.58	0.69	0.75	
10.5	Determine the appropriate interaction diagrams.	For a circular column, $f'_c = 5$ ksi, $f_y = 60$ ksi, use diagram C5-60.6, C5-60.7 and C5-60.8.			
	Read K_n and R_n and ρ_g values for $\gamma = 0.60, 0.70$, and 0.08 .	No chart 0.9	1.08 1.18	1.23 1.25	C5-60.6 C5-60.7 C5-60.8
	Interpolate	0.9	1.17	1.24	
	Compute $A_g = \frac{P_n}{f'_c K_n}$	298	229	217	
	Compute $h = 2\sqrt{\frac{A_g}{\pi}}$	19.5	17.1	16.6	
		Therefore, try 17 in. diameter column.			
	Determine reinforcement ratio ρ_g using known values of variables on appropriate interaction diagrams and compute required cross section area A_{st} of longitudinal reinforcement.	$A_g = \pi \left(\frac{17}{2} \right)^2 = 227 \text{ in.}^2$			

ACI 318-14 section	Discussion	Calculation	Design aid
	Compute $K_n = \frac{P_n}{f'_c A_g}$	$K_n = \frac{1343}{(5)(227)(17)} = 1.18$	
	Compute $R_n = \frac{M_n}{f'_c A_g h}$	$R_n = \frac{1343}{(5)(227)(17)} = 0.0356$	
	Estimate $\gamma \approx \frac{h-5}{h}$	$\gamma \approx \frac{17-5}{17} = 0.71$	
10.5	Determine the appropriate interaction diagrams.	For a circular column, $f'_c = 5$ ksi, $f_y = 60$ ksi, use diagram C5-60.7.	
	Read ρ_g for K_n and R_n values.	For $K_n = 1.18$ and $R_n = 0.0356$: $\gamma = 0.70$ $\rho_g = 0.040$	C5-60.7
	Compute required A_{st} from $A_{st} = \rho_g A_g$.	Required $A_{st} = 0.04 \times 227 \text{ in.}^2 = 9.08 \text{ in.}^2$	

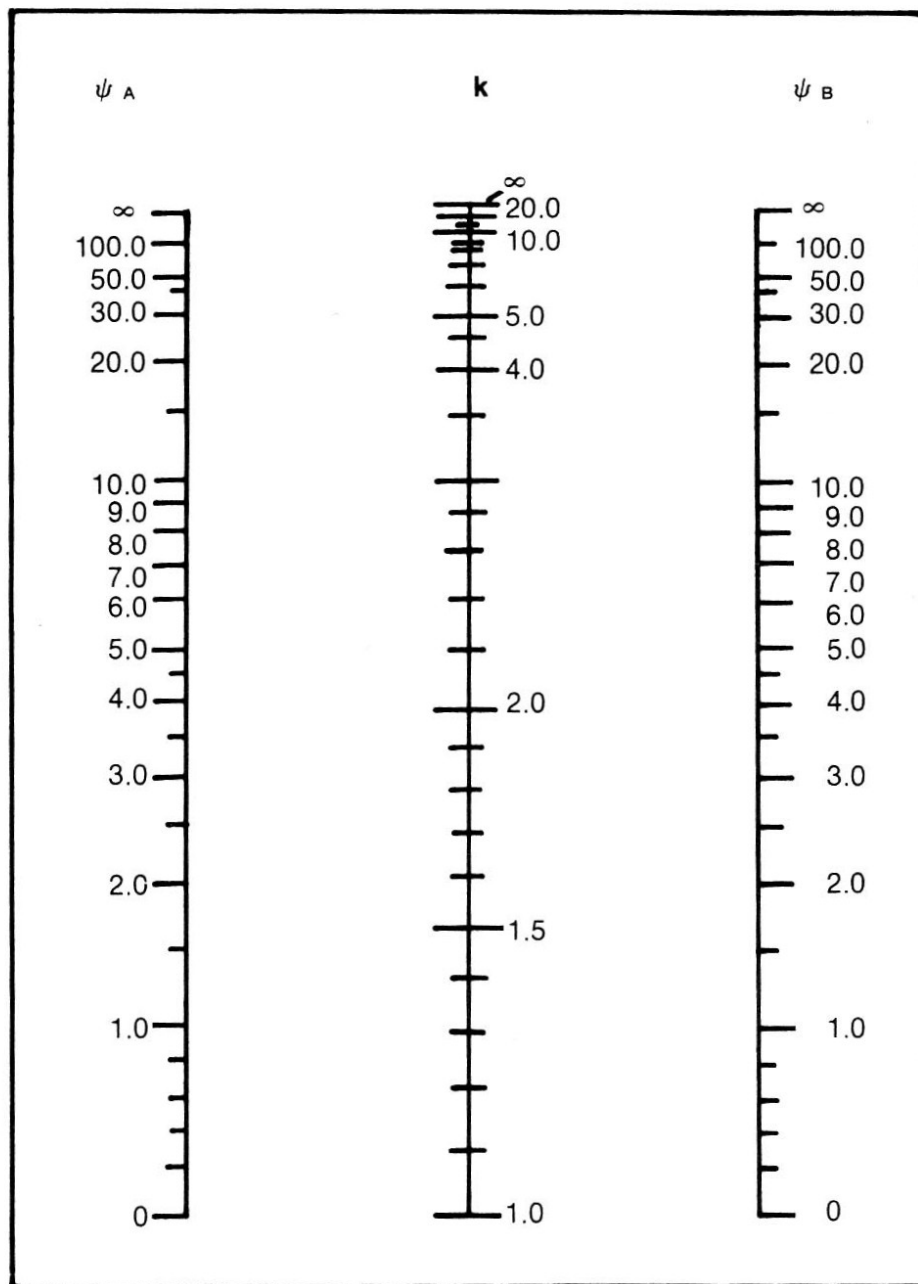
D.4—Column design aids

Table D.4.1—Effective Length Factor, Jackson and Moreland alignment chart for columns in braced (nonsway) frames (Column Research Council 1966)



$$\psi_i = \frac{(\sum EI/\ell_c)_{\text{columns}}}{(\sum EI/\ell)_{\text{beams}}} \quad \text{at end } i \text{ of column}$$

Table D.4.2—Effective Length Factor, Jackson and Moreland alignment chart for columns in unbraced (sway) frames (Column Research Council 1966)



$$\psi_i = \frac{(\sum EI/\ell_c)_{\text{columns}}}{(\sum EI/\ell)_{\text{beams}}} \quad \text{at end i of column}$$

Table D.4.3—Recommended flexural rigidities (EI) for use in first-order and second order analyses of frames for design of slender columns (ACI 318-14, Section 6.6.3.1.1.)

Second-order analysis of frames for design of slender columns

f'_c , ksi	3	4	5	6	7	8	9	10	
E_c , ksi	3120	3605	4031	4415	4769	5098	5407	5700	
$E_c I / I_g$, ksi									I/I_g
Beams	1092	1262	1411	1545	1669	1784	1892	1995	0.3
Columns	2184	2524	2822	3091	3338	3569	3785	3990	0.7
Walls (uncracked)	2184	2524	2822	3091	3338	3569	3785	3990	0.7
Walls (cracked)	1092	1262	1411	1545	1669	1784	1892	1995	0.3
Flat plates Flat slabs	780	901	1008	1104	1192	1275	1352	1425	0.2

Notes:

- Alternatively, the following more refined values of I can be used:
For columns:

$$I = \left(0.8 + 25 \frac{A_{st}}{A_g} \right) \left(1 - \frac{M_u}{P_u h} - 0.5 \frac{P_u}{P_o} \right) I_g \leq 0.5 I_g$$

Where P_u and M_u are for the particular load combinations under consideration, or the combination of P_u and M_u that leads to the smallest value of I , I need not be less than $0.35 I_g$.

For beams:

$$I = (0.10 + 25\rho) \left(1.2 - 0.2 \frac{b_w}{d} \right) I_g \leq 0.5 I_g$$

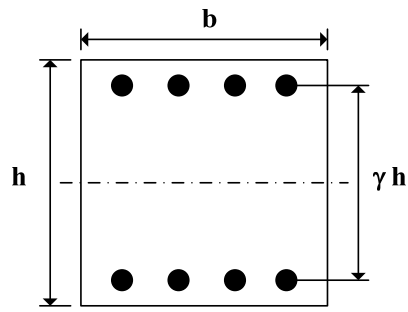
For continuous beams, I can be taken as the average of values for the critical positive and negative moment sections. I need not be less than $0.25 I_g$.

- When sustained lateral loads are applied, I for columns should be divided by $(1 + \beta_{ds})$, where β_{ds} is the ratio of maximum factored sustained shear within a story to the maximum factored story shear for the same load combination. β_{ds} shall not be taken greater than 1.0.
- The above values are applicable to normal-density concrete with w_c between 90 and 155 lb/ft³.
- The moment of inertia of a T-beam should be based on the effective flange width as shown in Flexure 6. It is generally accurate to take I_g of a T-beam as two times the I_g for the web.
- Member area will not be reduced for analysis.

Table D.4.4—Effective length factor *k* for columns in braced frames (Concrete Design Handbook 2005)

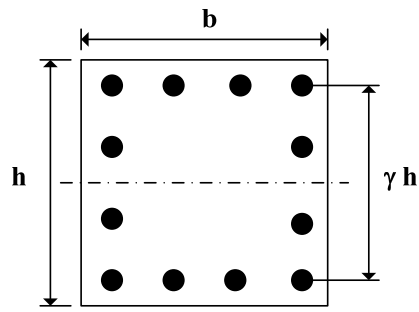
		k			
TOP	Hinged 	0.81	0.91	0.95	1.00
	Elastic 	0.77	0.86	0.90	0.95
	Elastic 	0.74	0.83	0.86	0.91
	Stiff 	0.67	0.74	0.77	0.81
					
		Stiff	Elastic	Elastic	Hinged
		BOTTOM			

Table D.4.5—Moment of inertia of reinforcement about sectional centroid (Based on Table 12-1, MacGregor 1997)



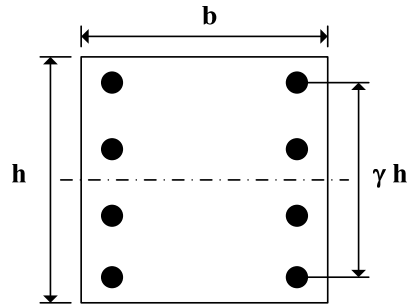
Bars in two end faces

$$I_{se} = 0.25 \rho_t b h^3 \gamma^2$$



Equal reinforcement on four sides

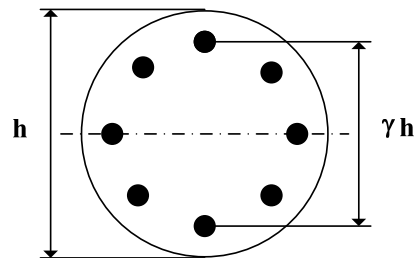
$$I_{se} = 0.18 \rho_t b h^3 \gamma^2$$



Bars in two side faces

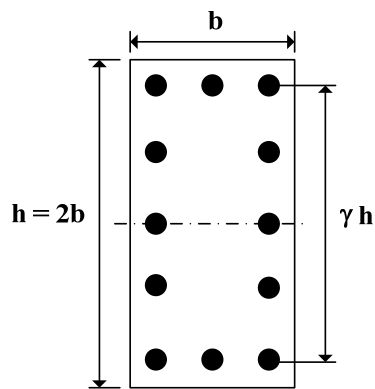
$$I_{se} = 0.17 \rho_t b h^3 \gamma^2 \quad (3 \text{ bars per face})$$

$$I_{se} = 0.12 \rho_t b h^3 \gamma^2 \quad (6 \text{ bars per face})$$



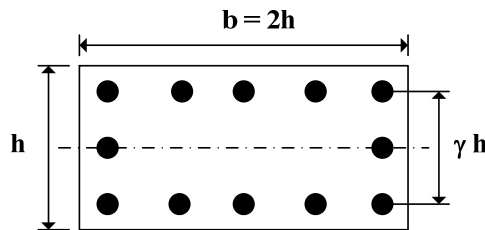
Uniformly distributed reinforcement

$$I_{se} = 0.10 \rho_t h^4 \gamma^2$$



Bars uniformly spaced on all sides

$$I_{se} = 0.13 \rho_t b h^3 \gamma^2$$

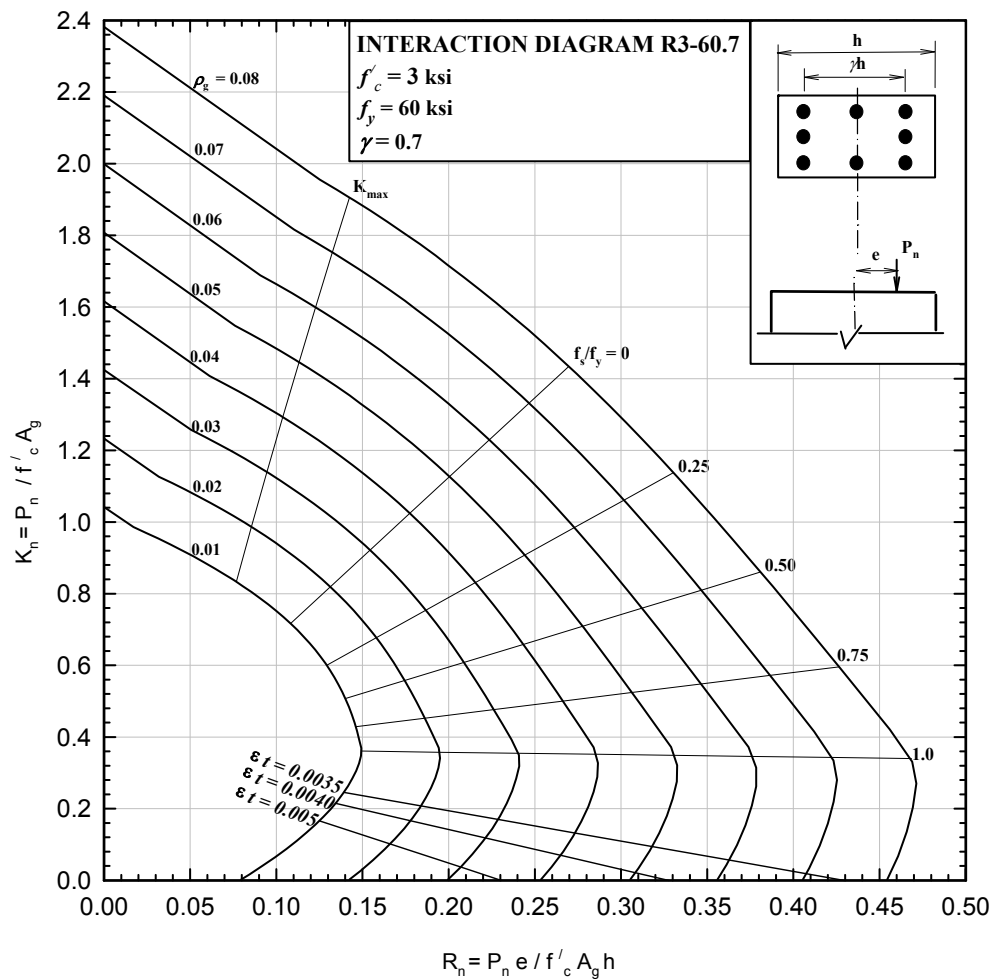
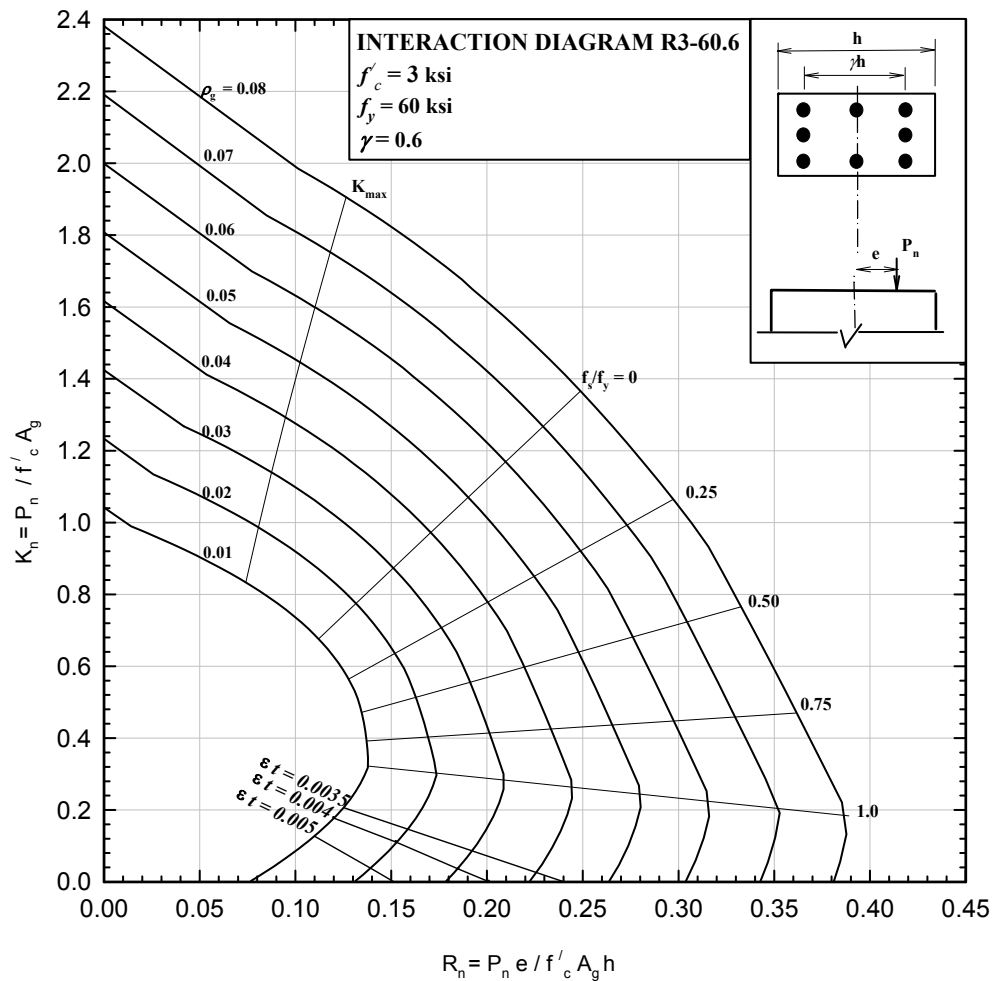


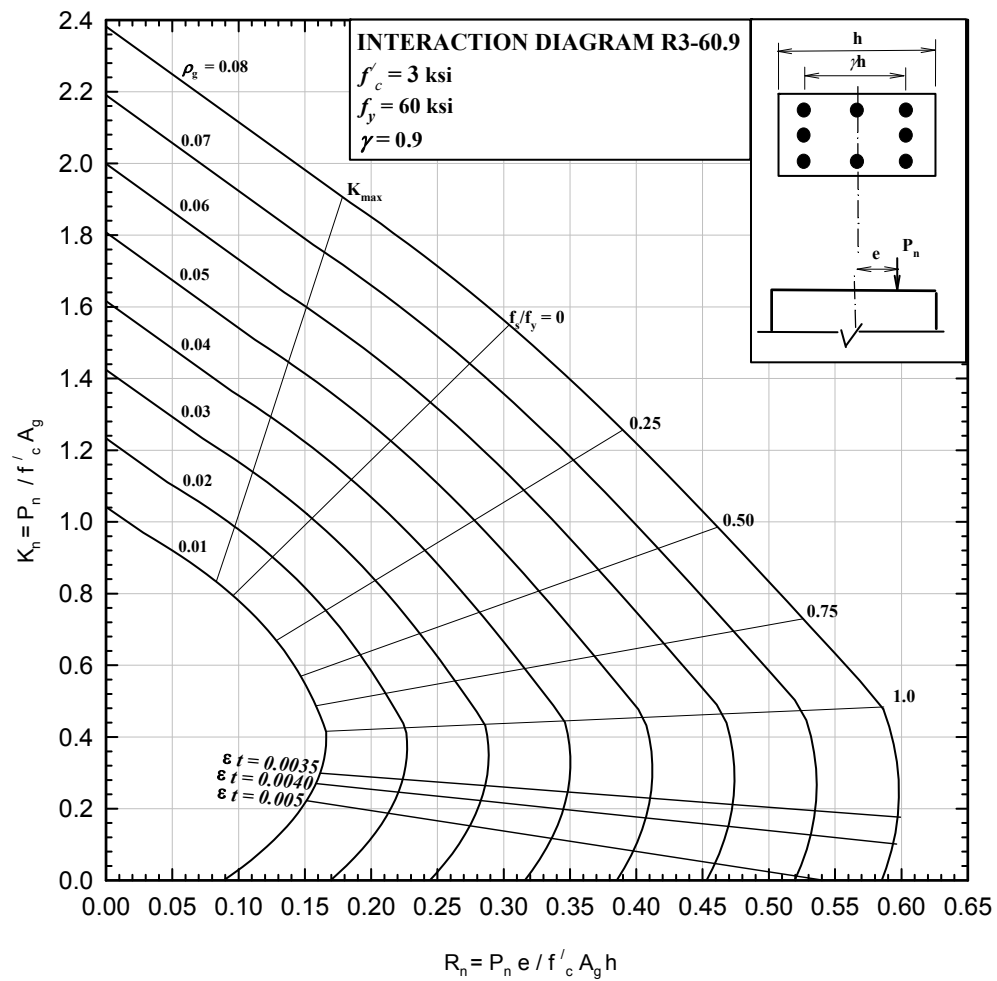
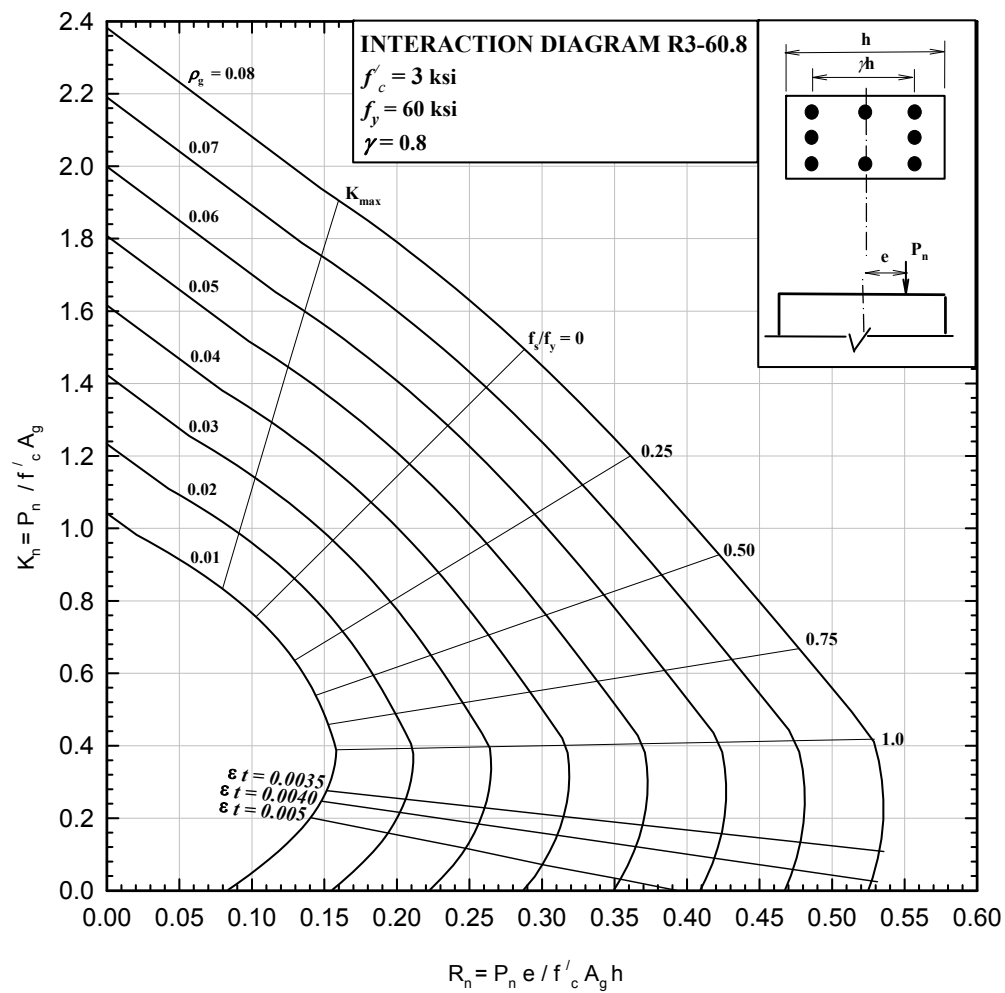
Bars uniformly spaced on all sides

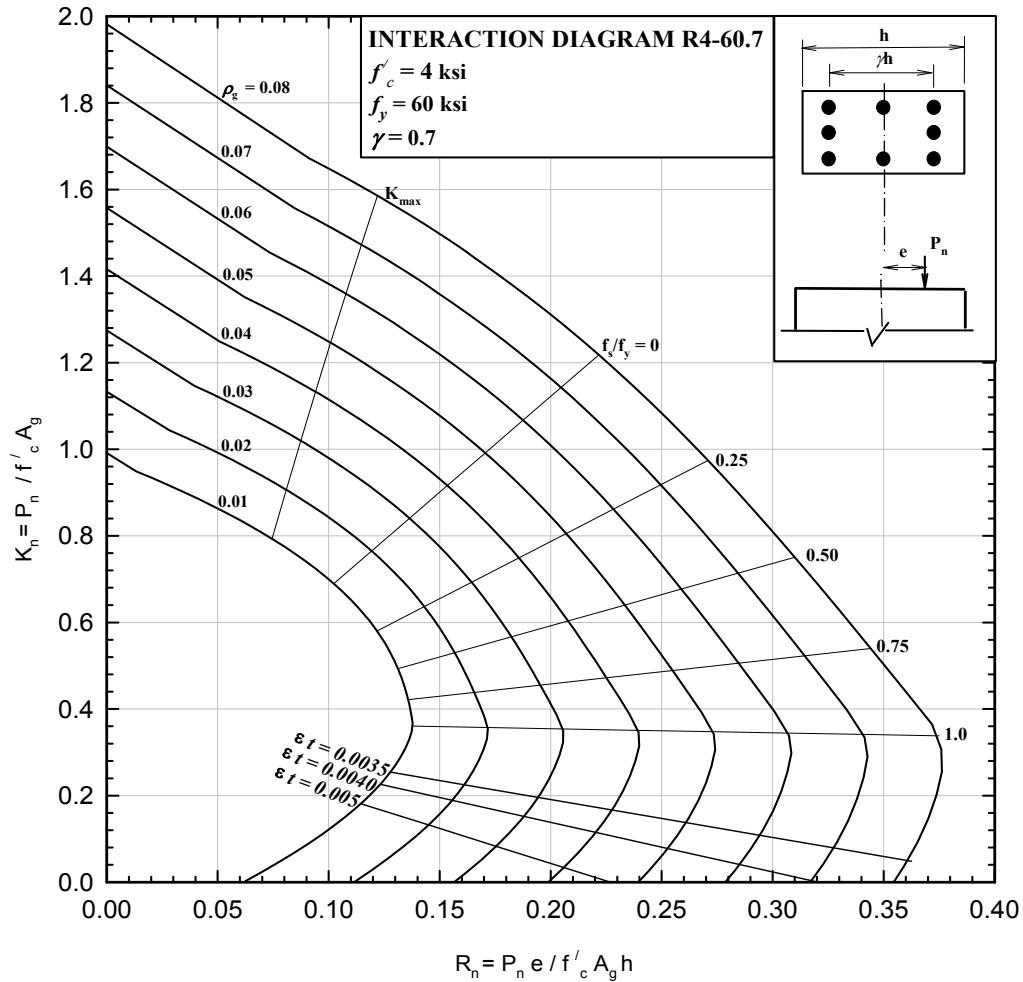
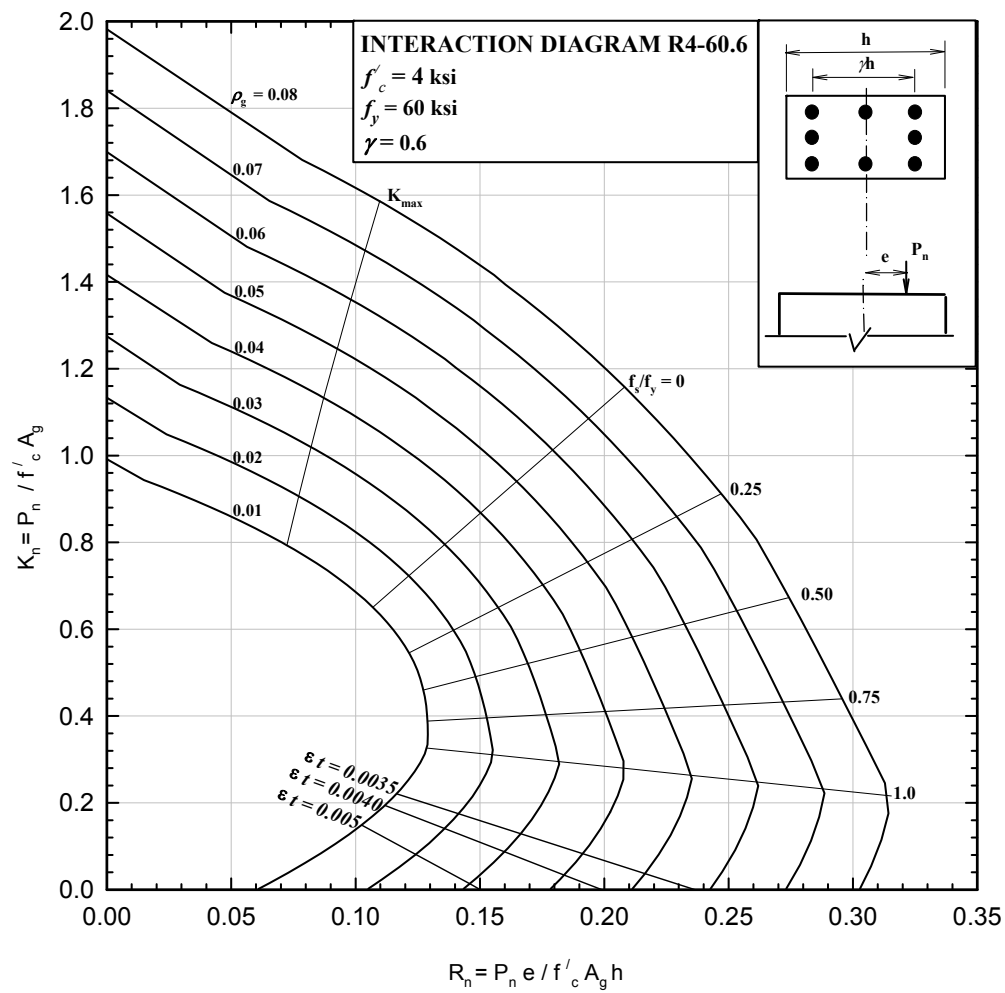
$$I_{se} = 0.22 \rho_t b h^3 \gamma^2$$

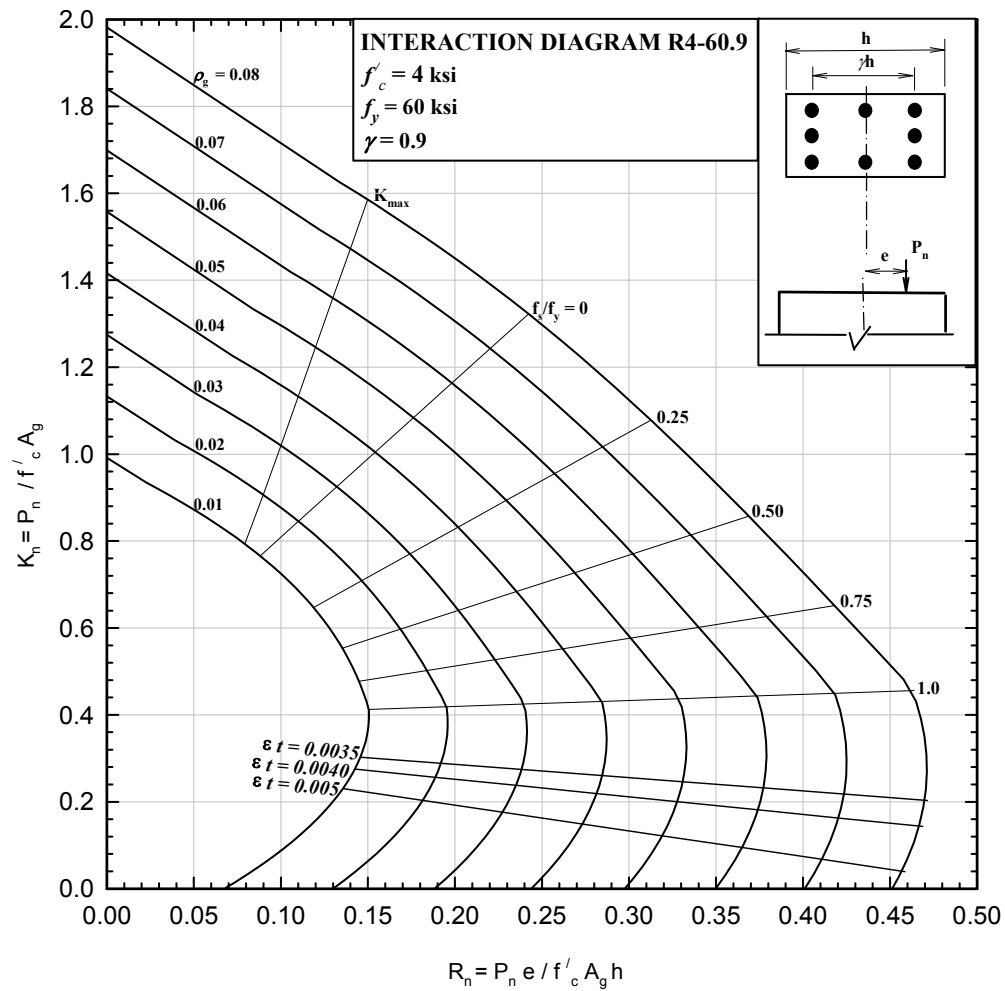
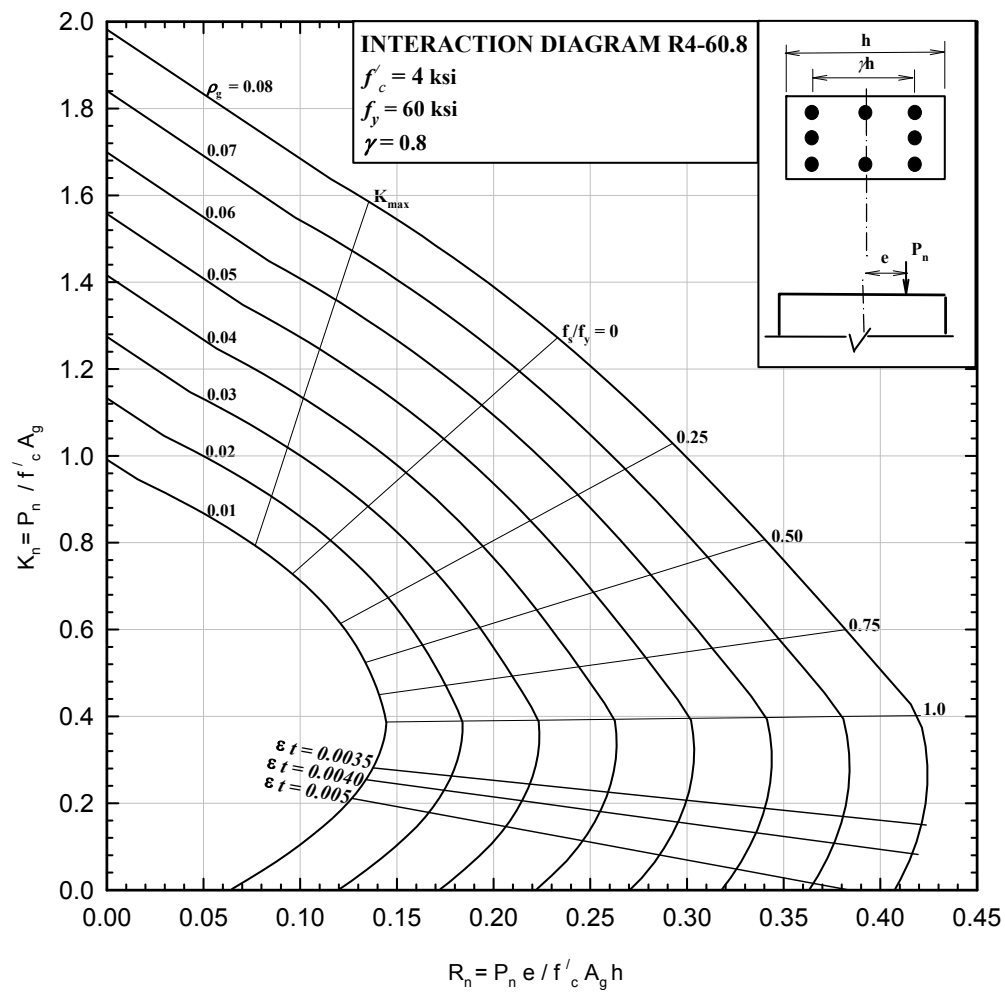
γ is the ratio of the distance between the centers of the outermost bars to the column dimension perpendicular to the axis of bending.

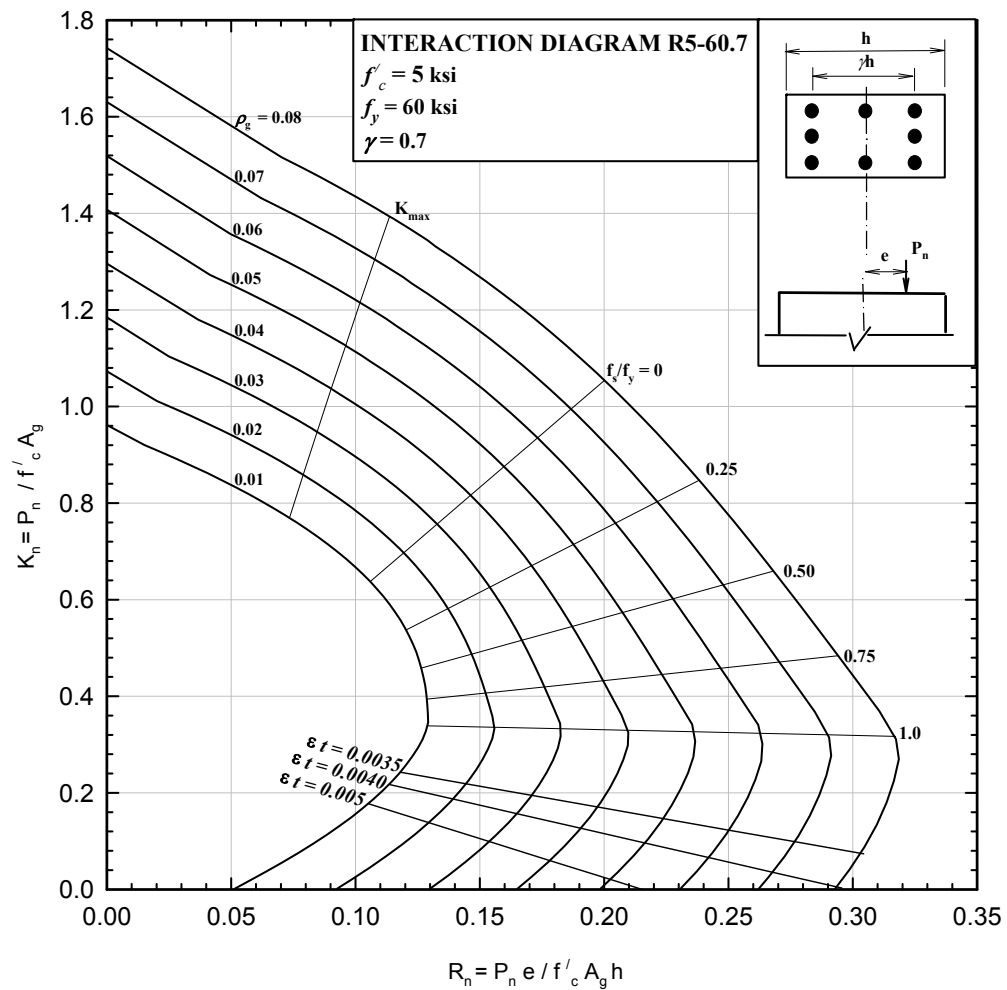
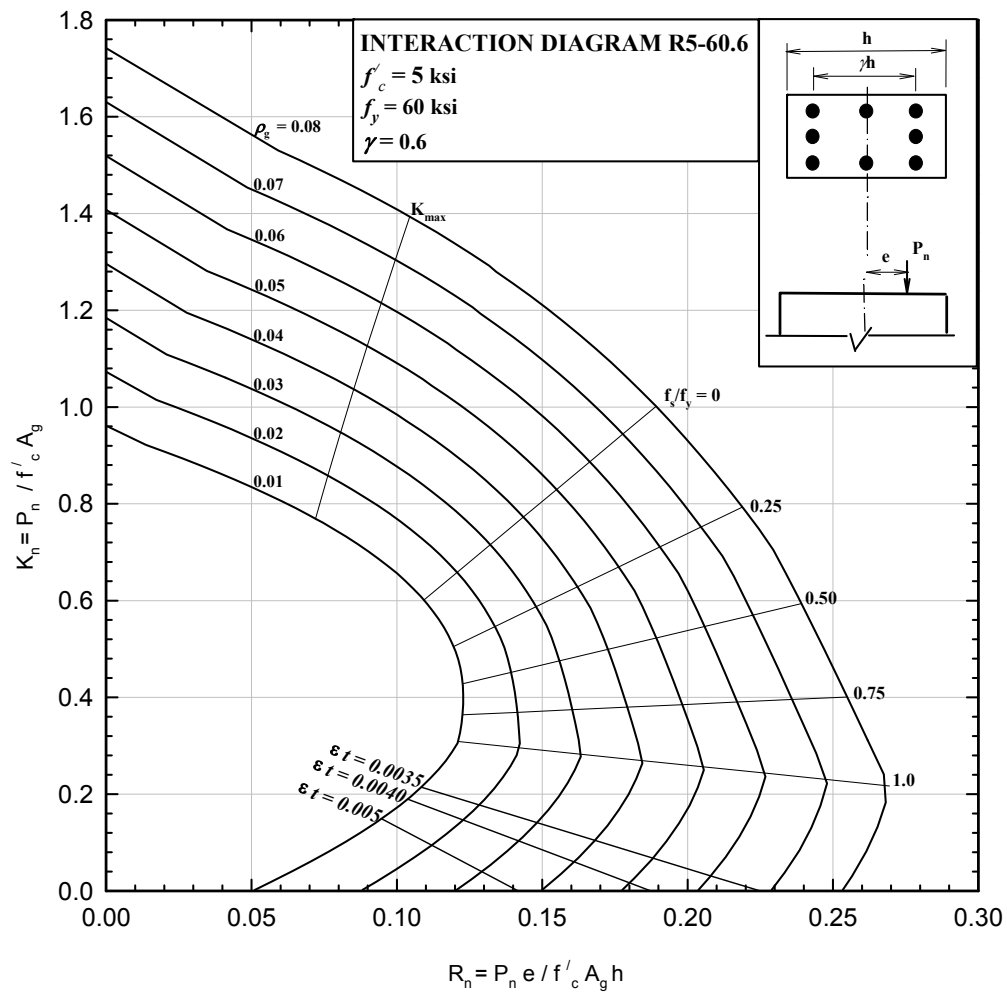
D.5—Interaction diagrams

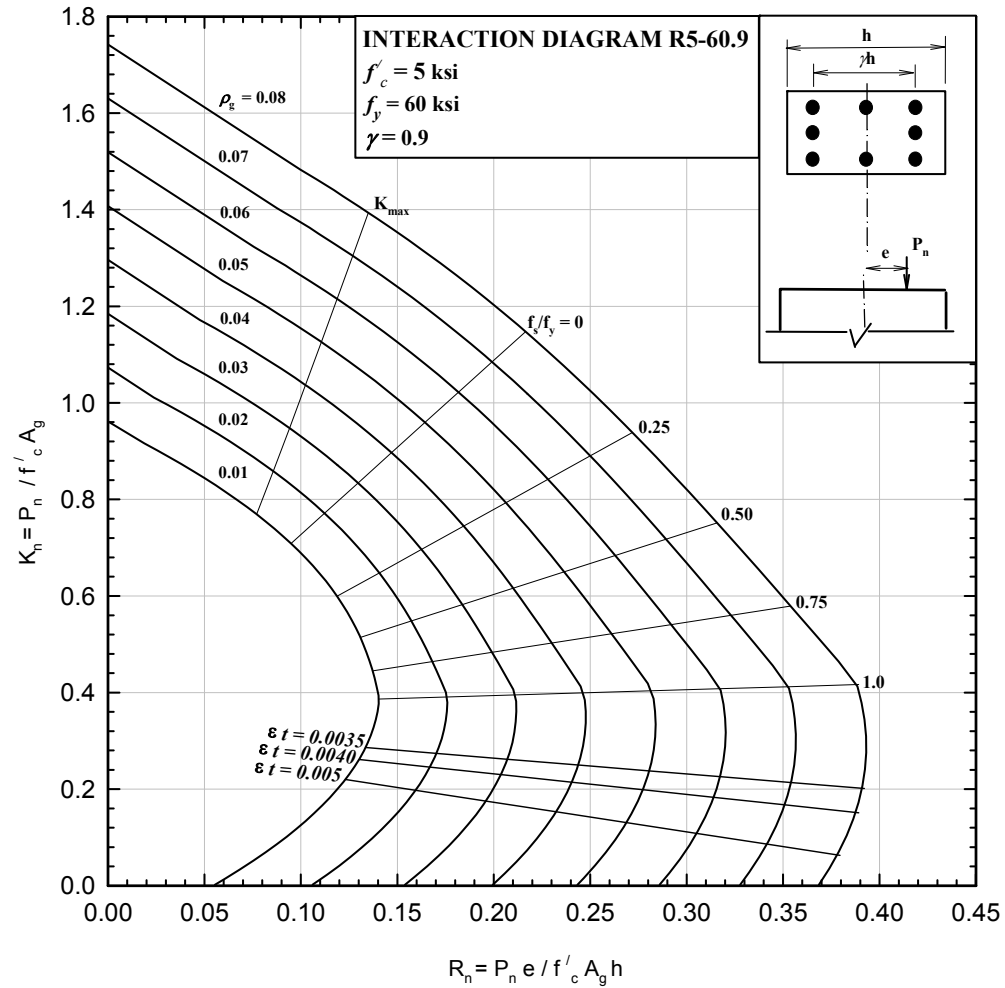
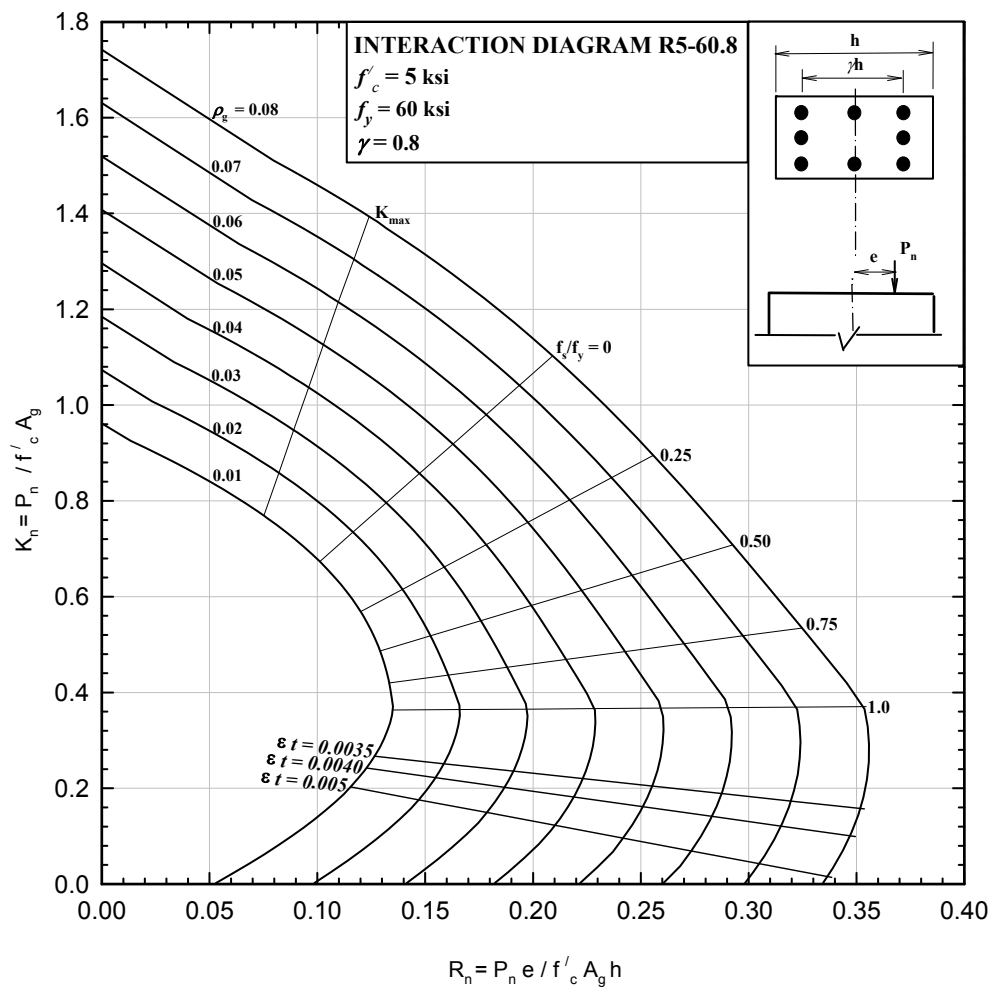


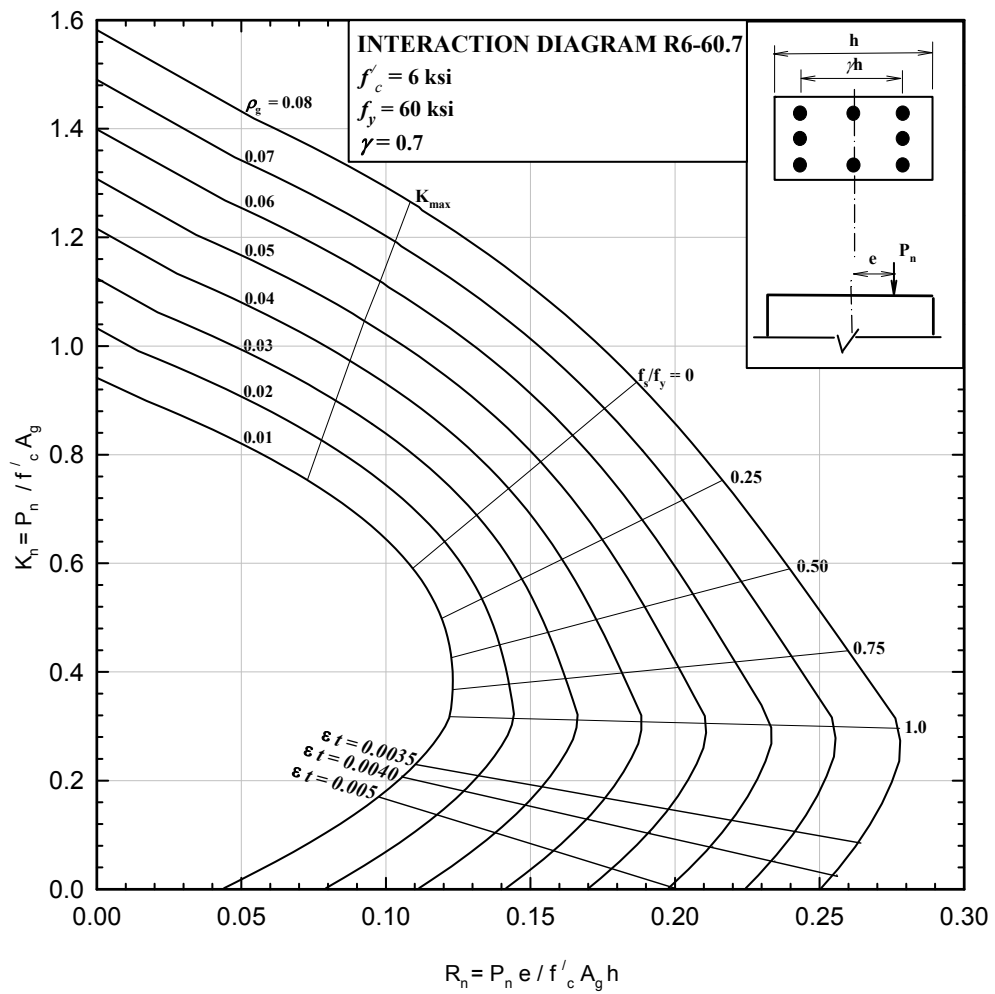
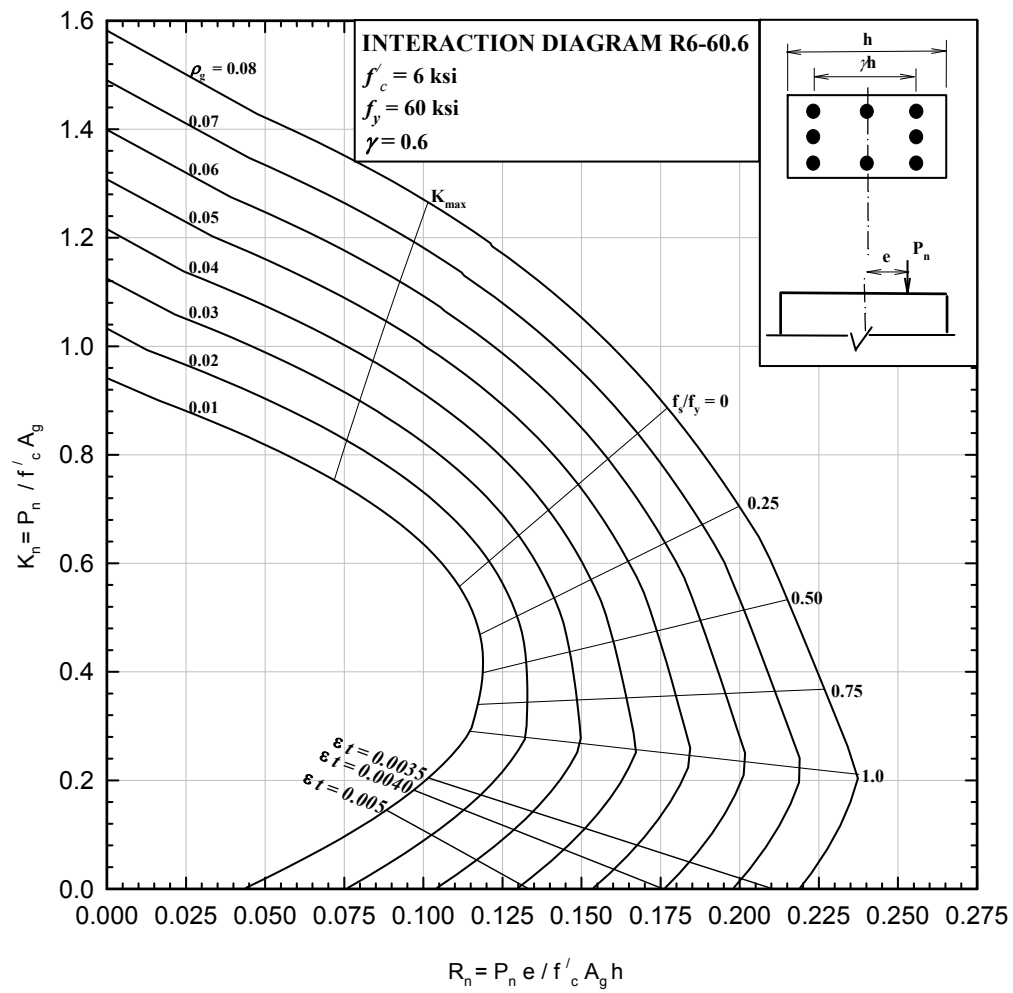


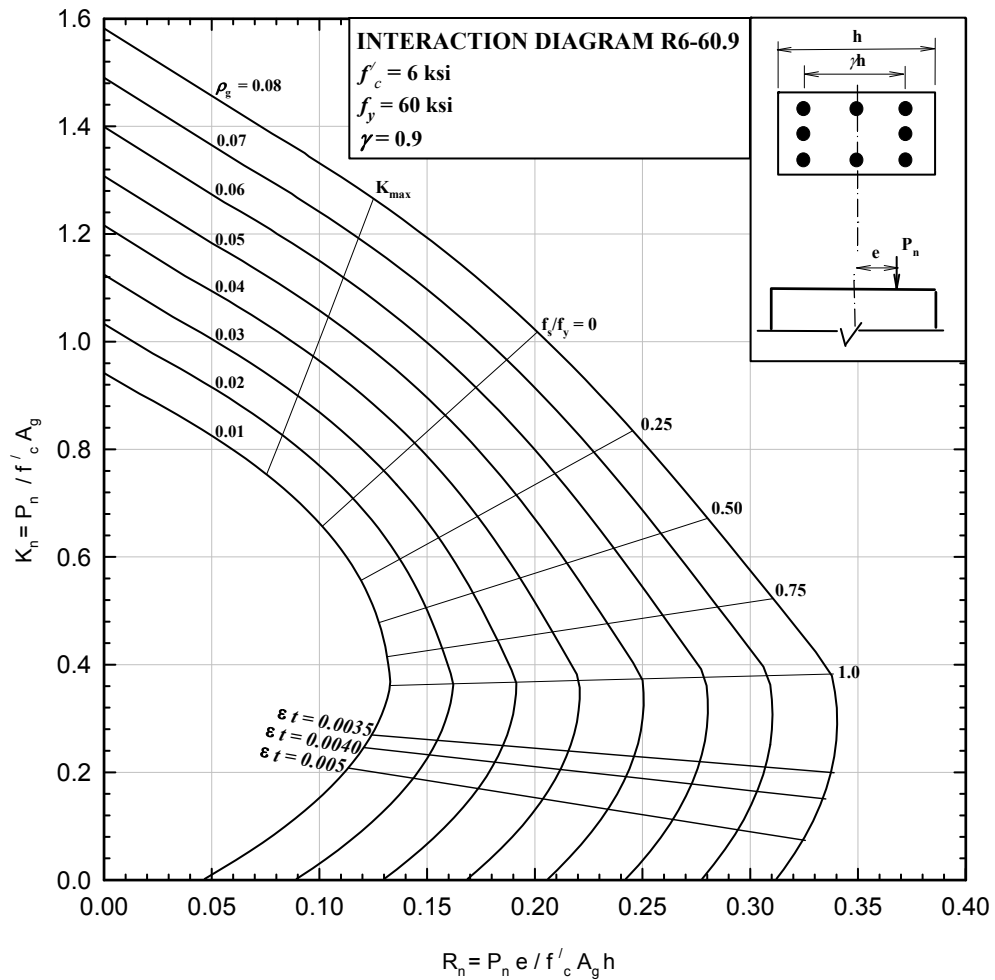
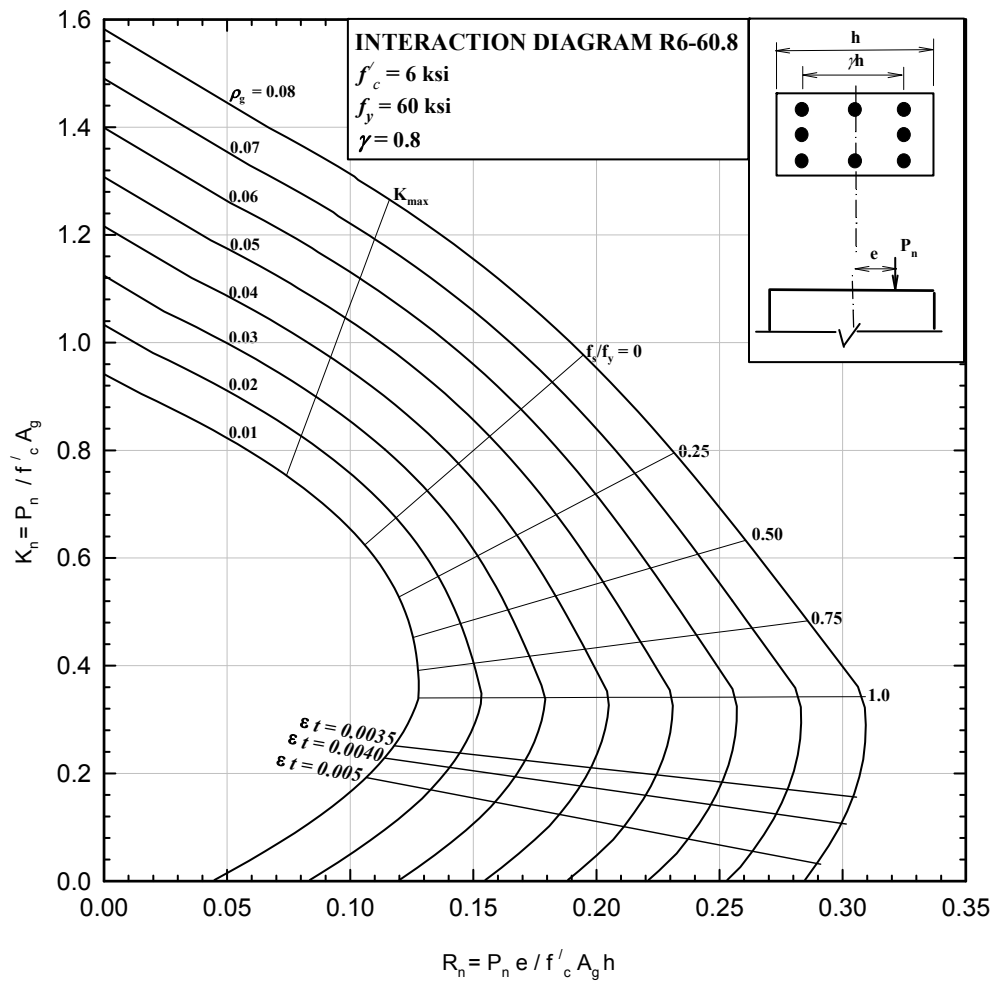


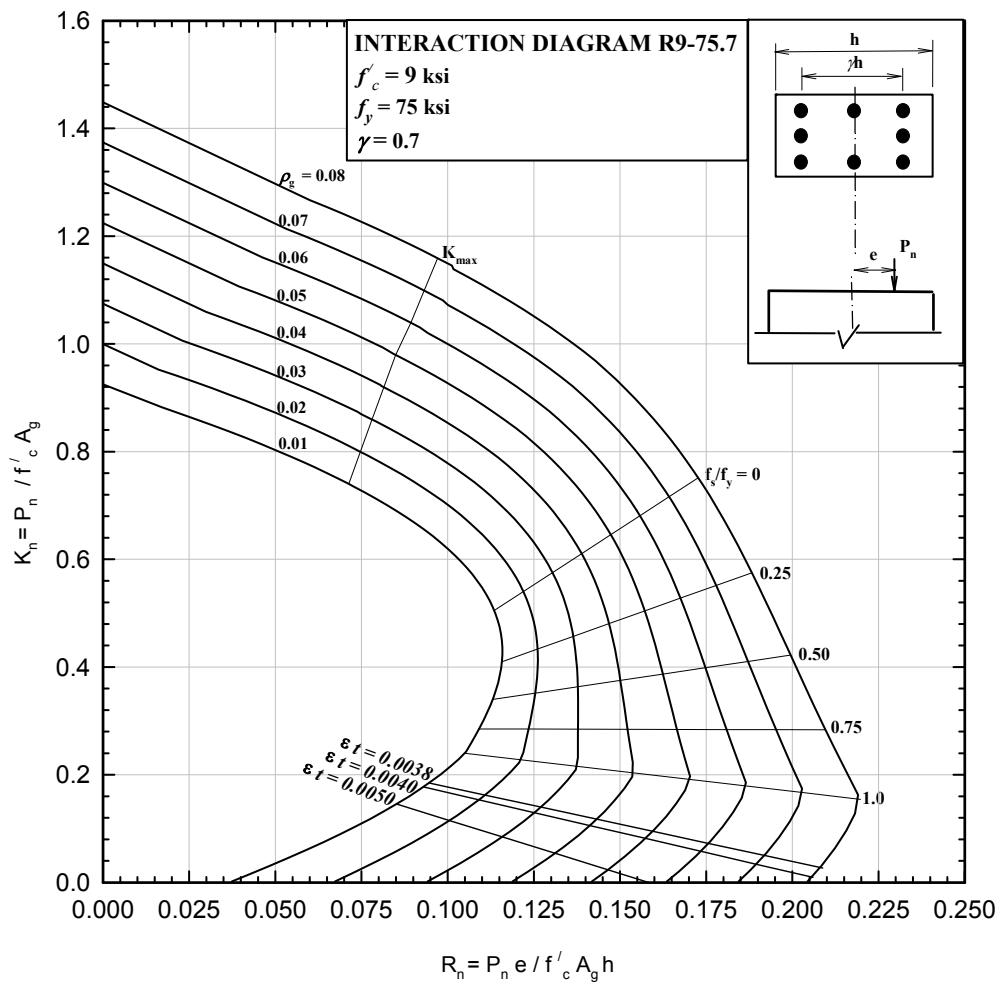
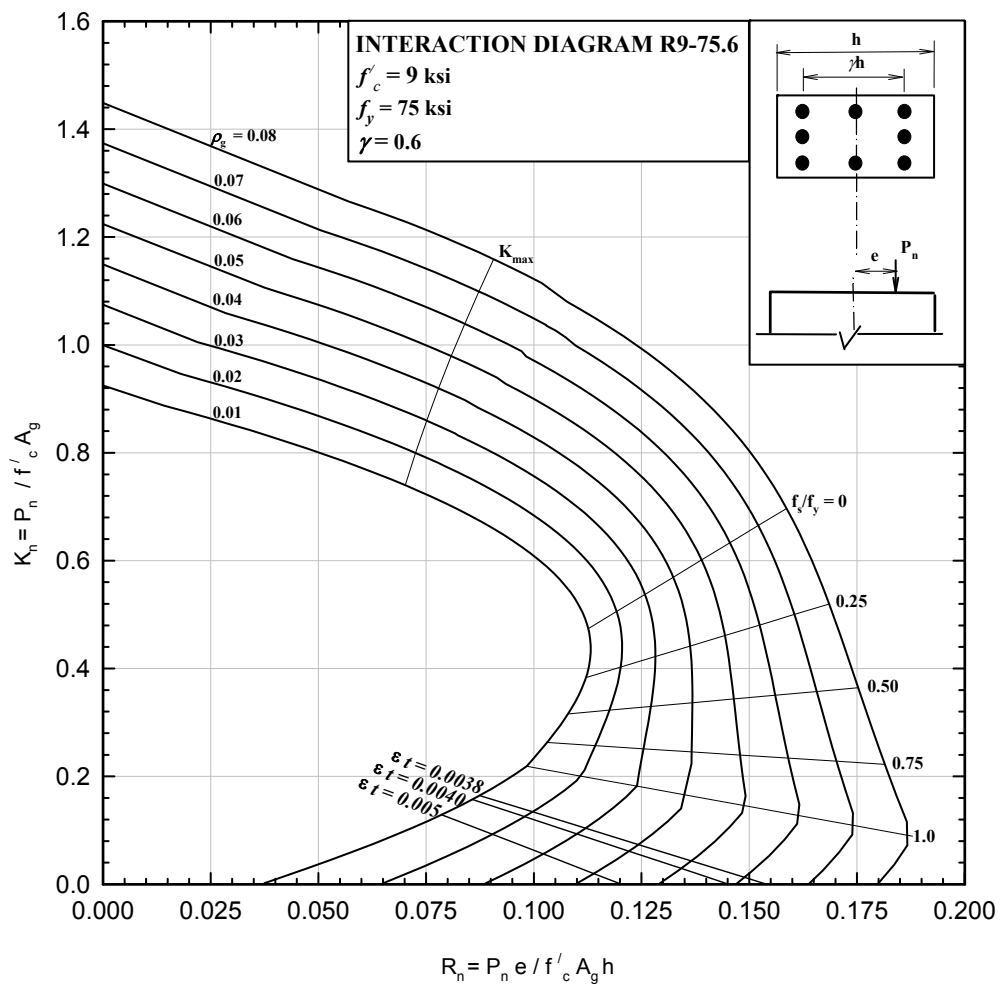


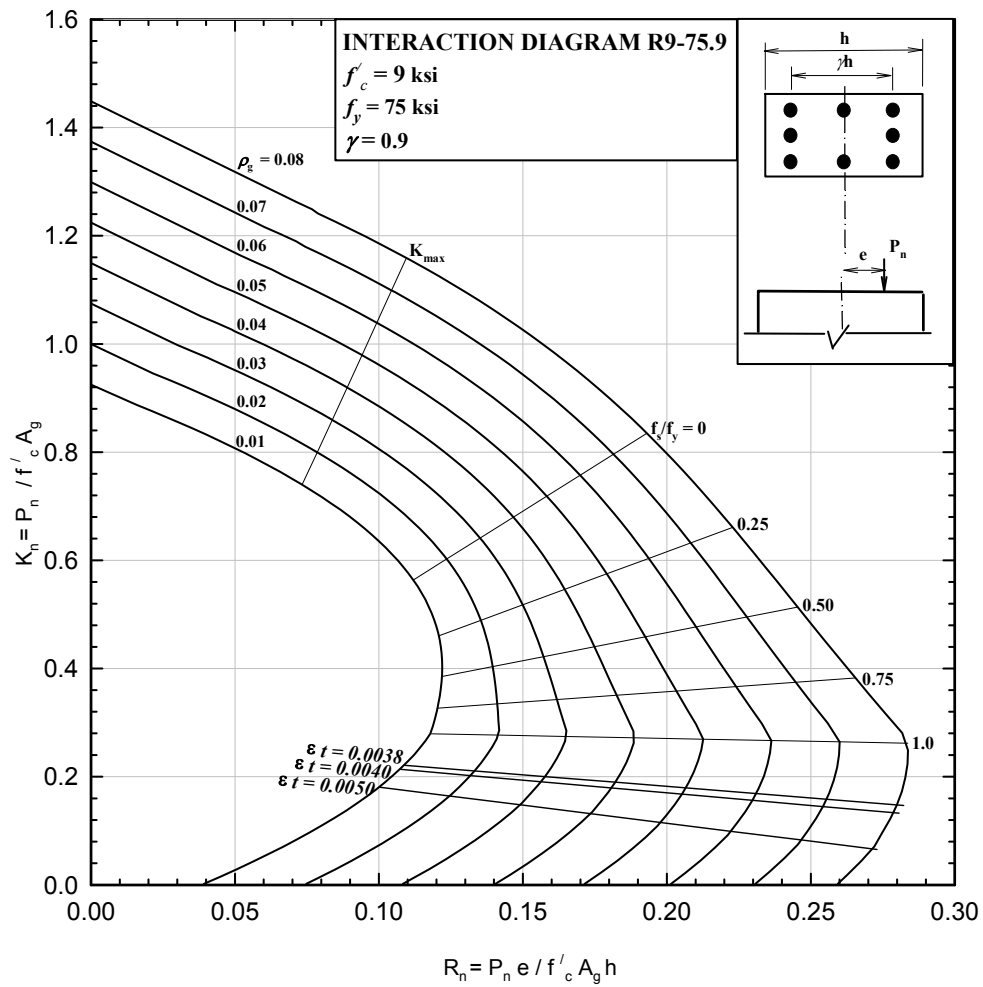
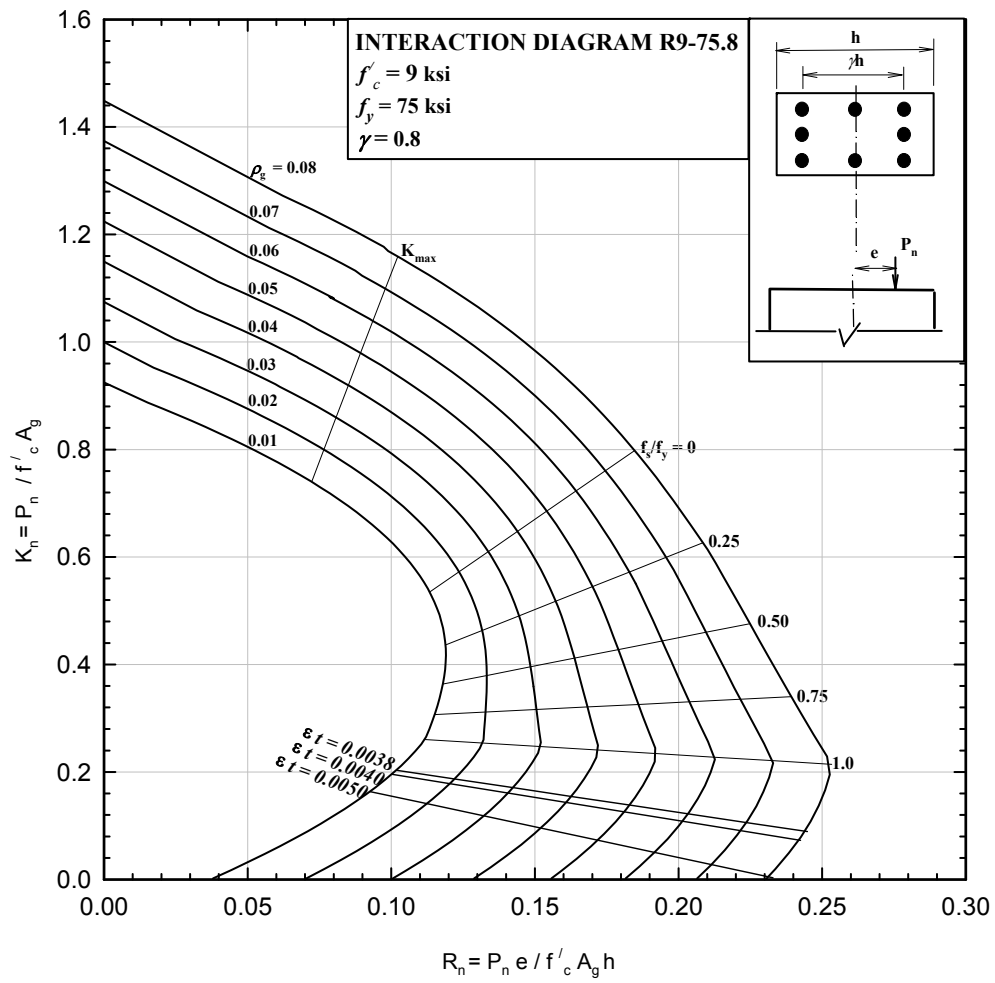


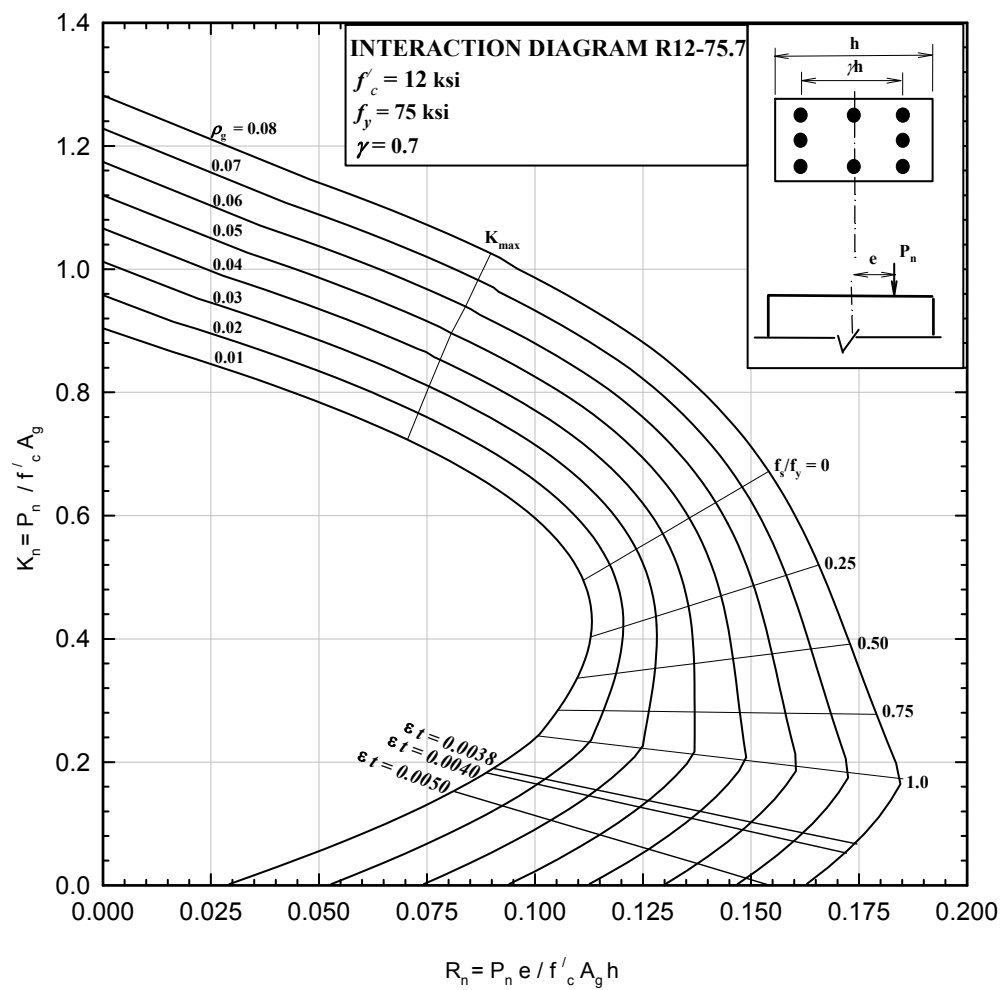
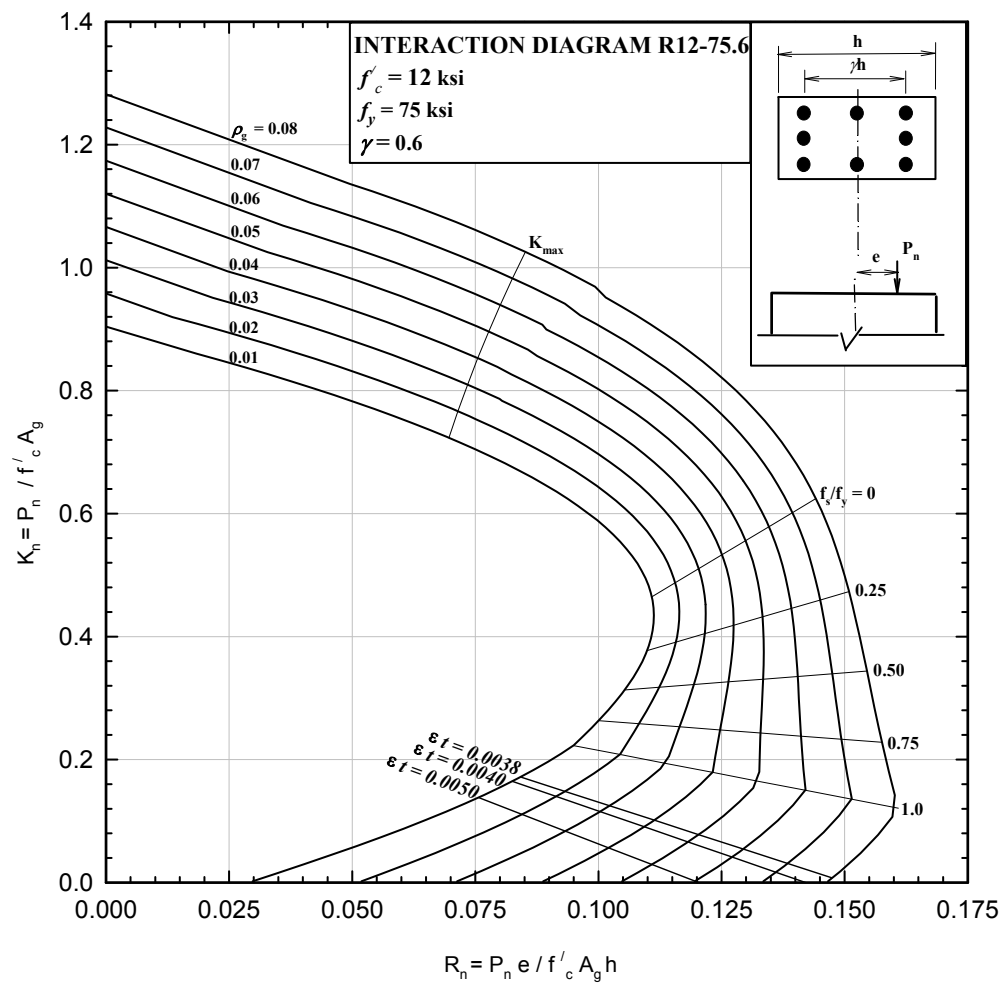


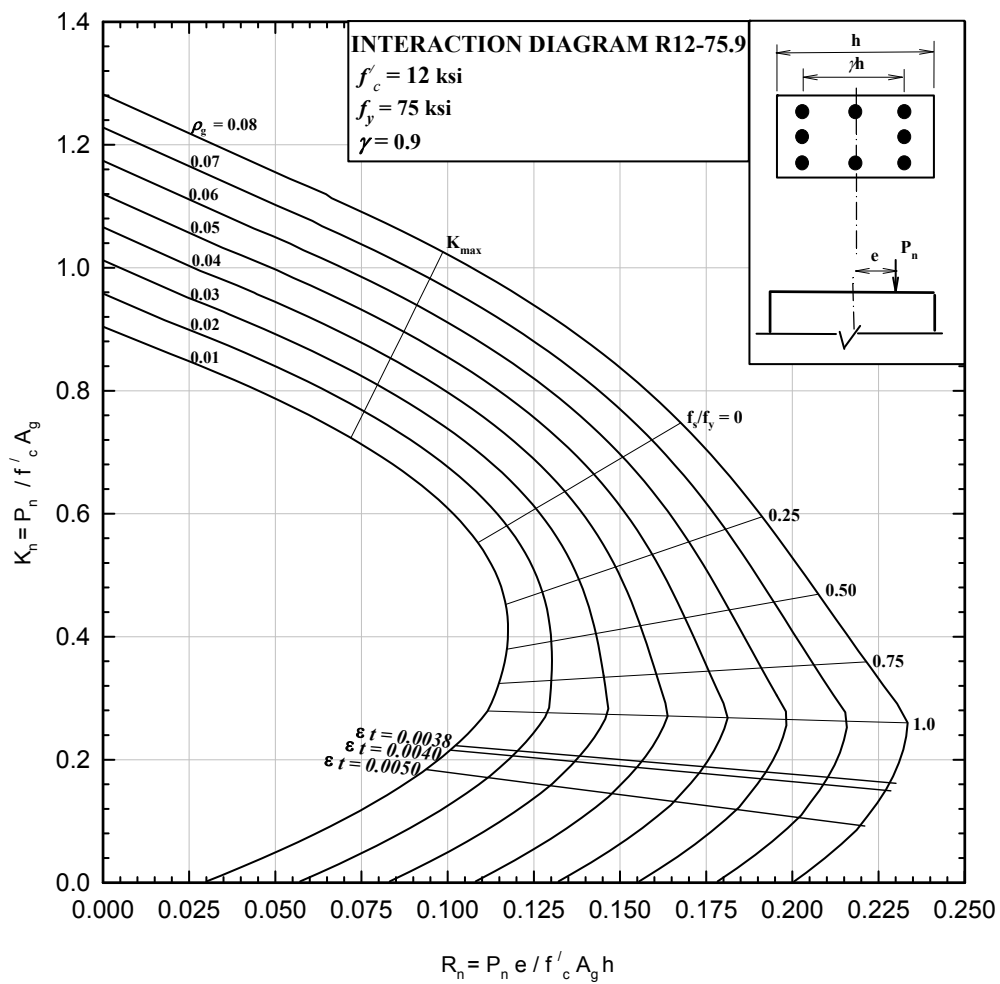
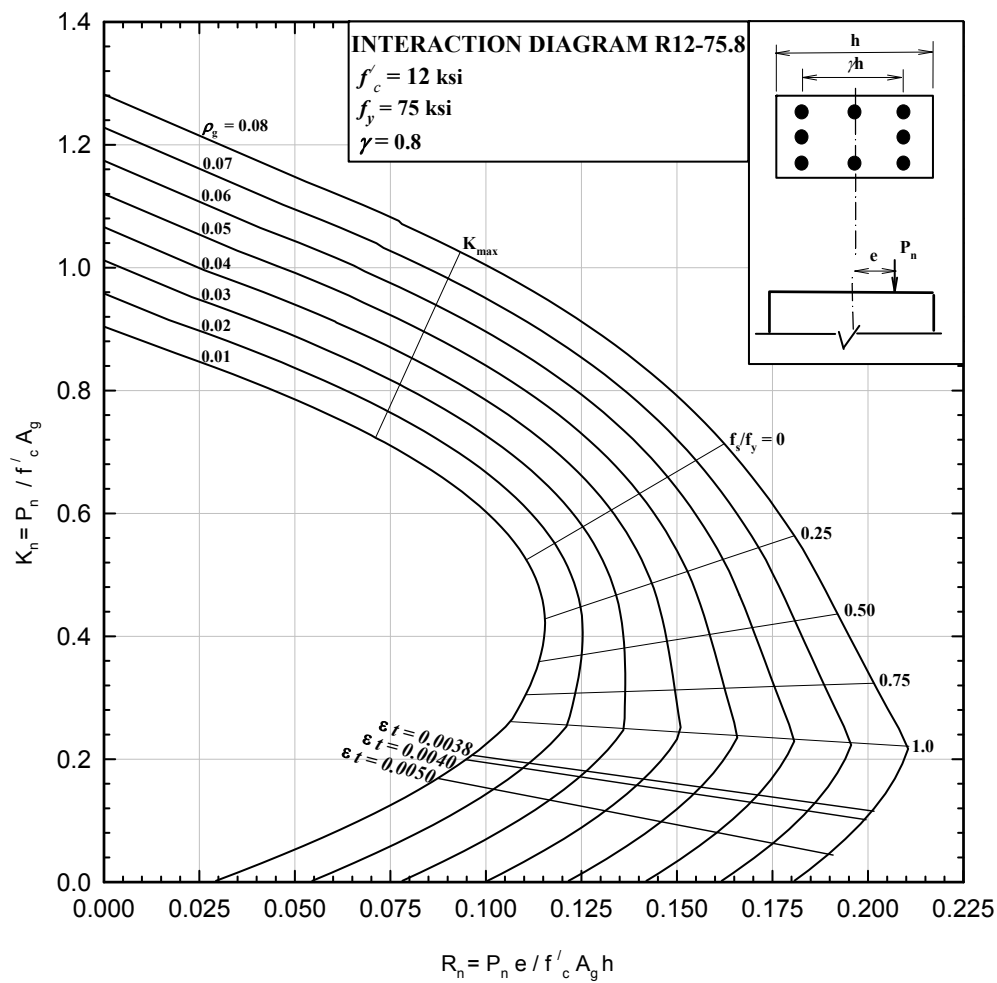


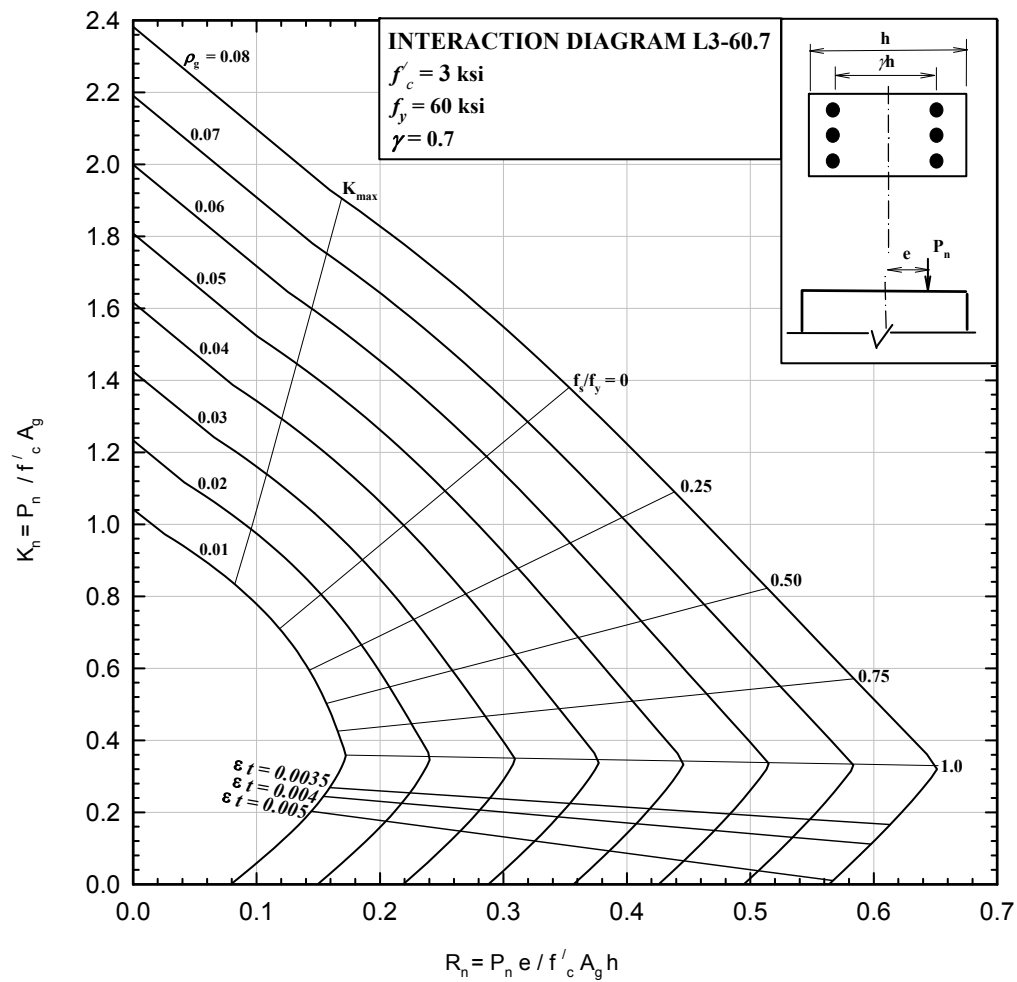
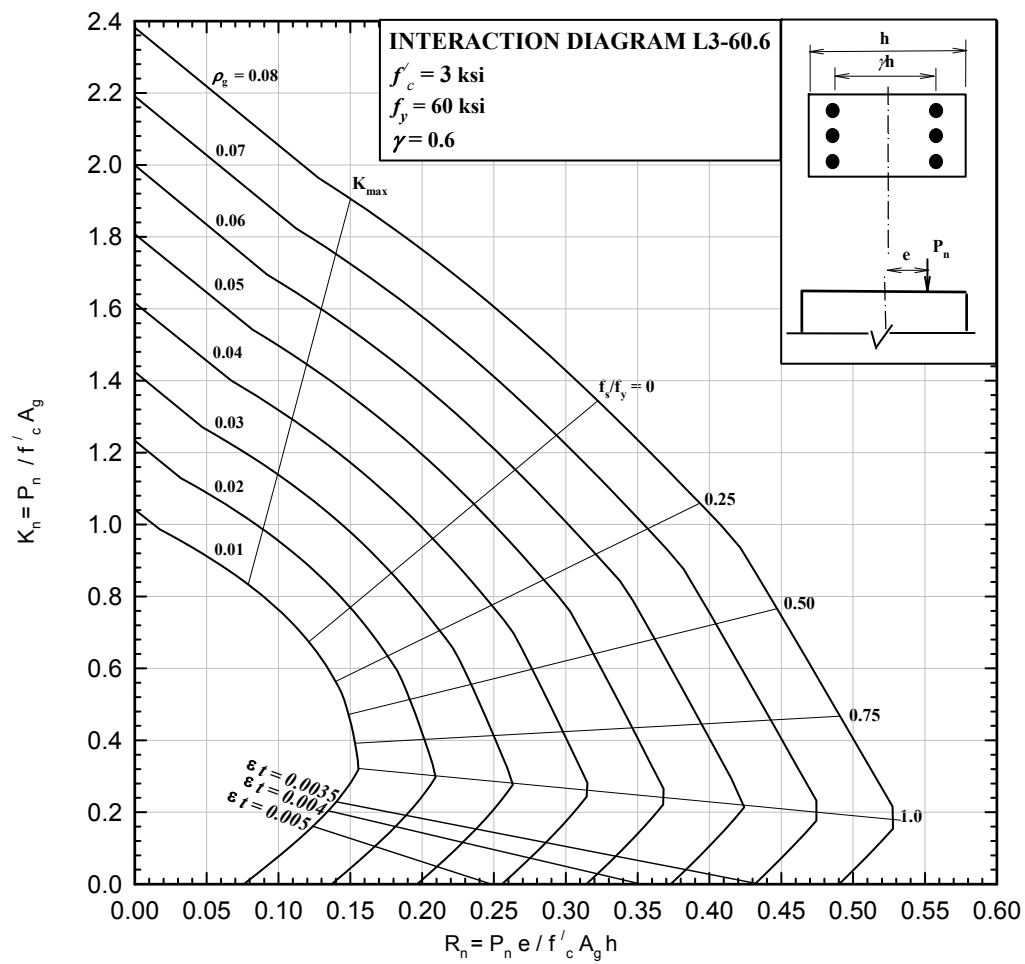


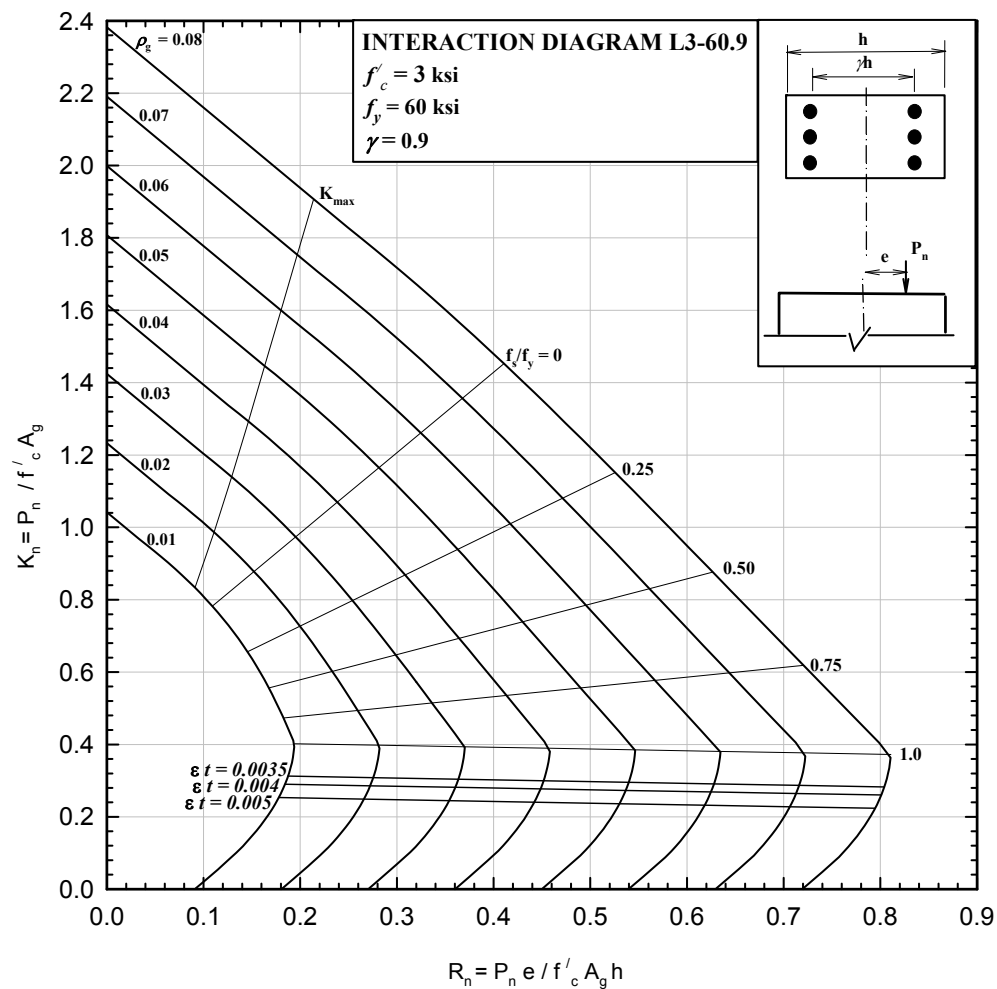
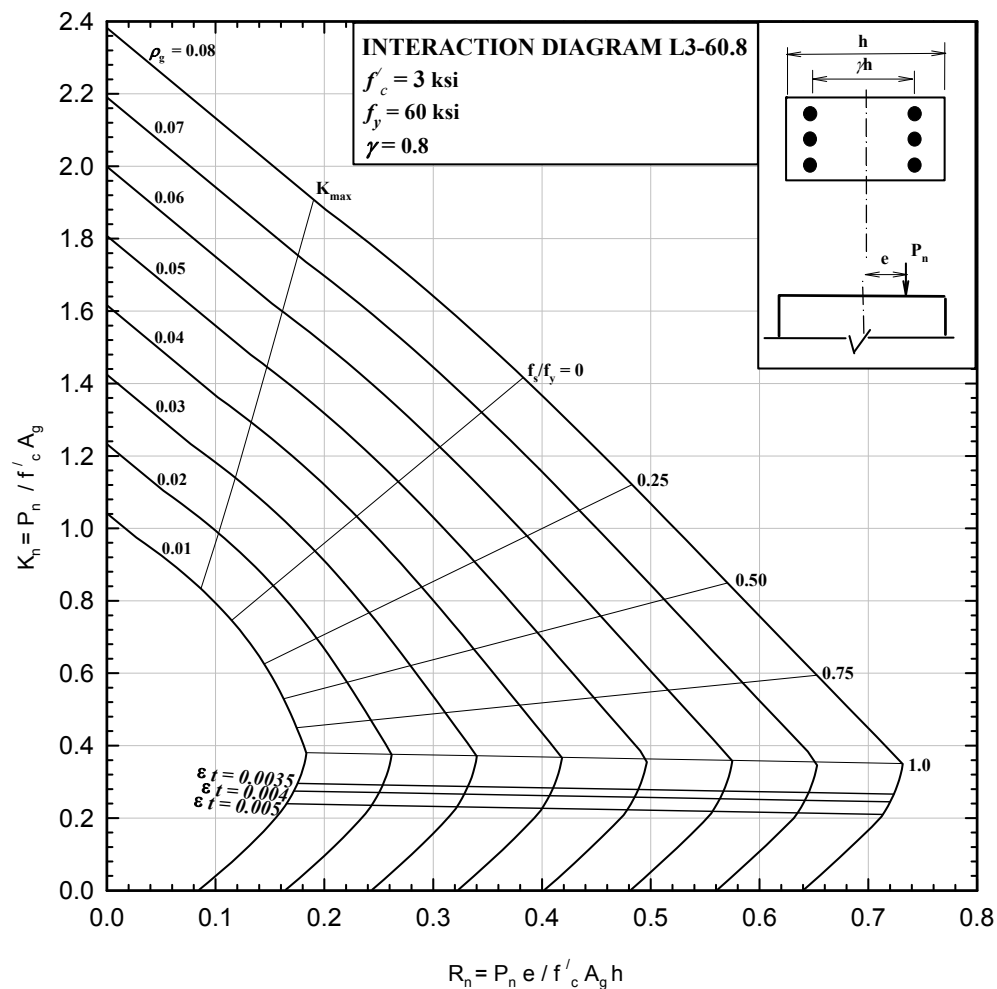


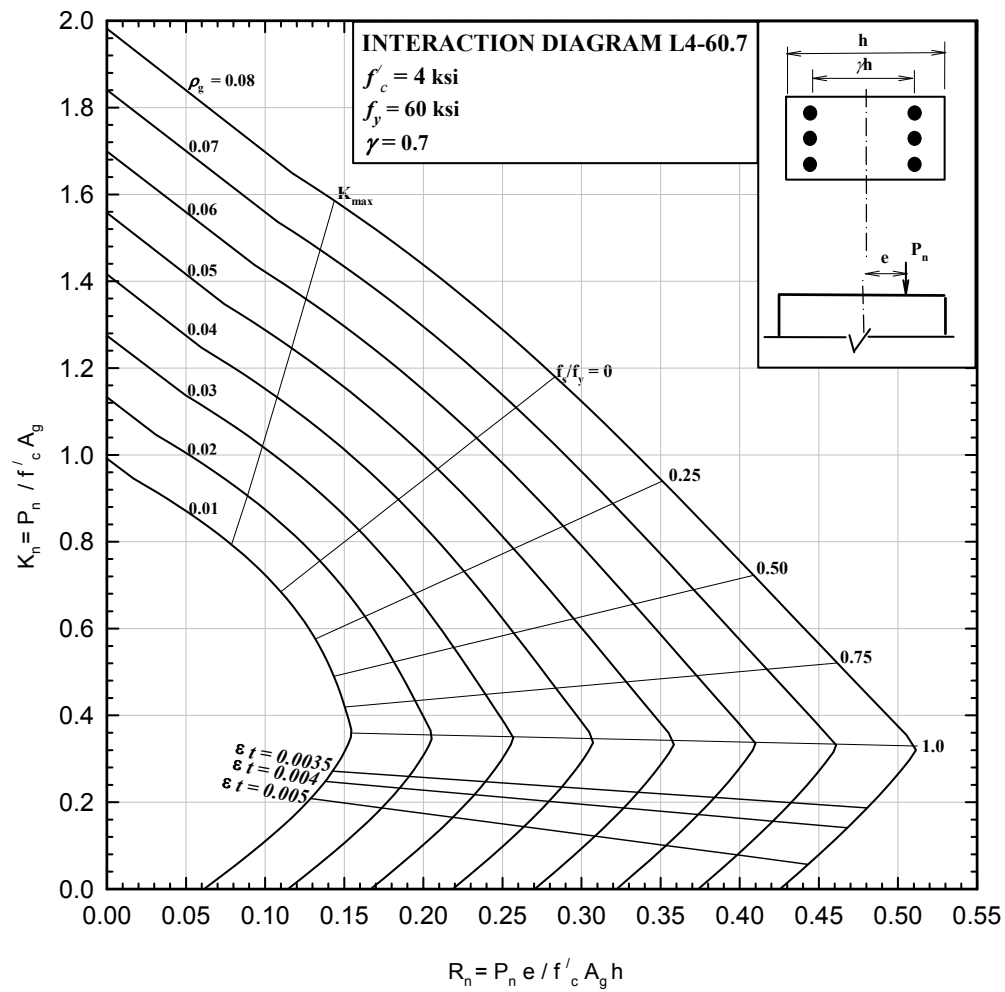
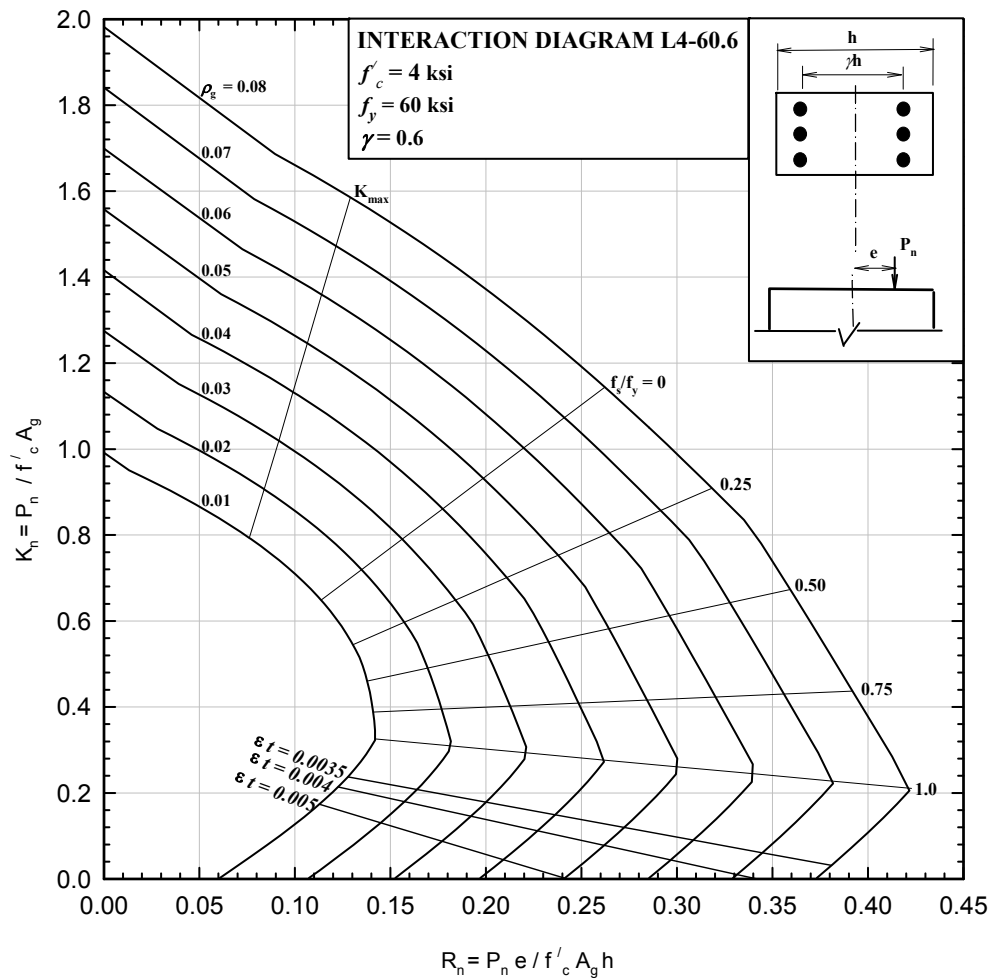


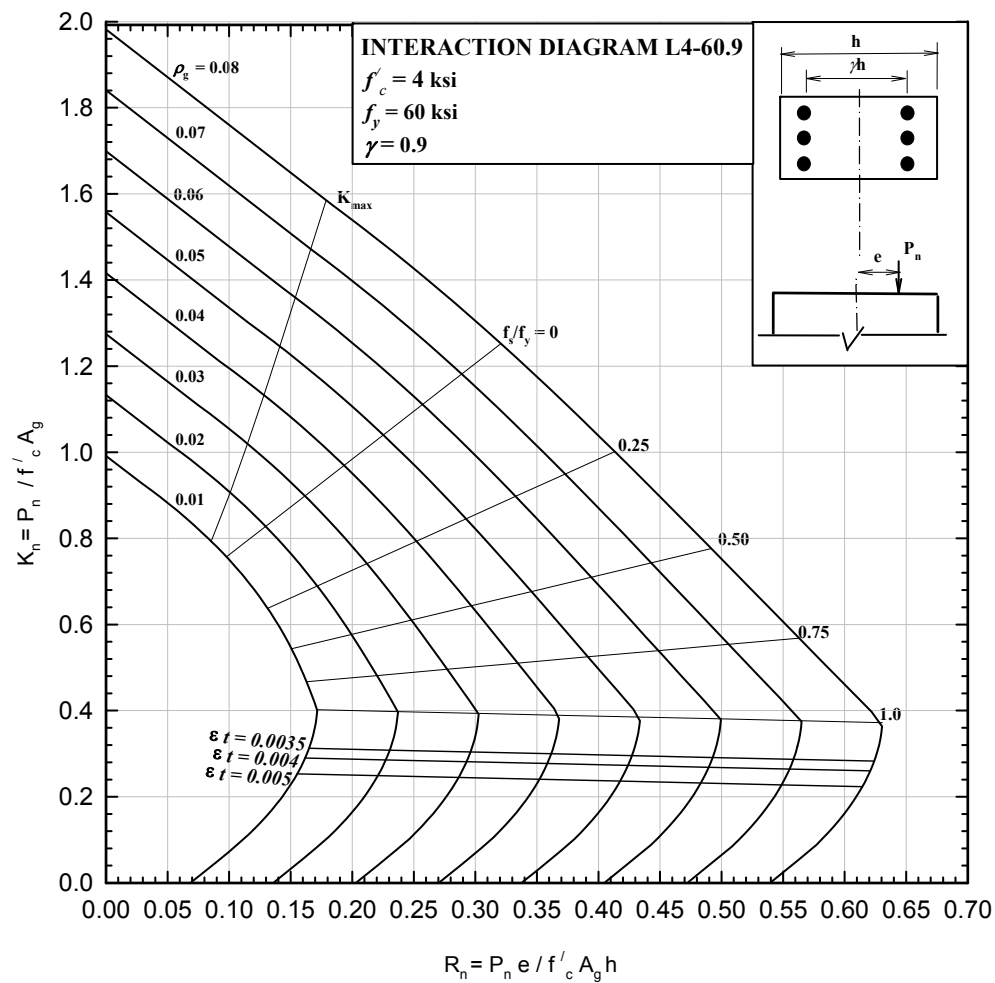
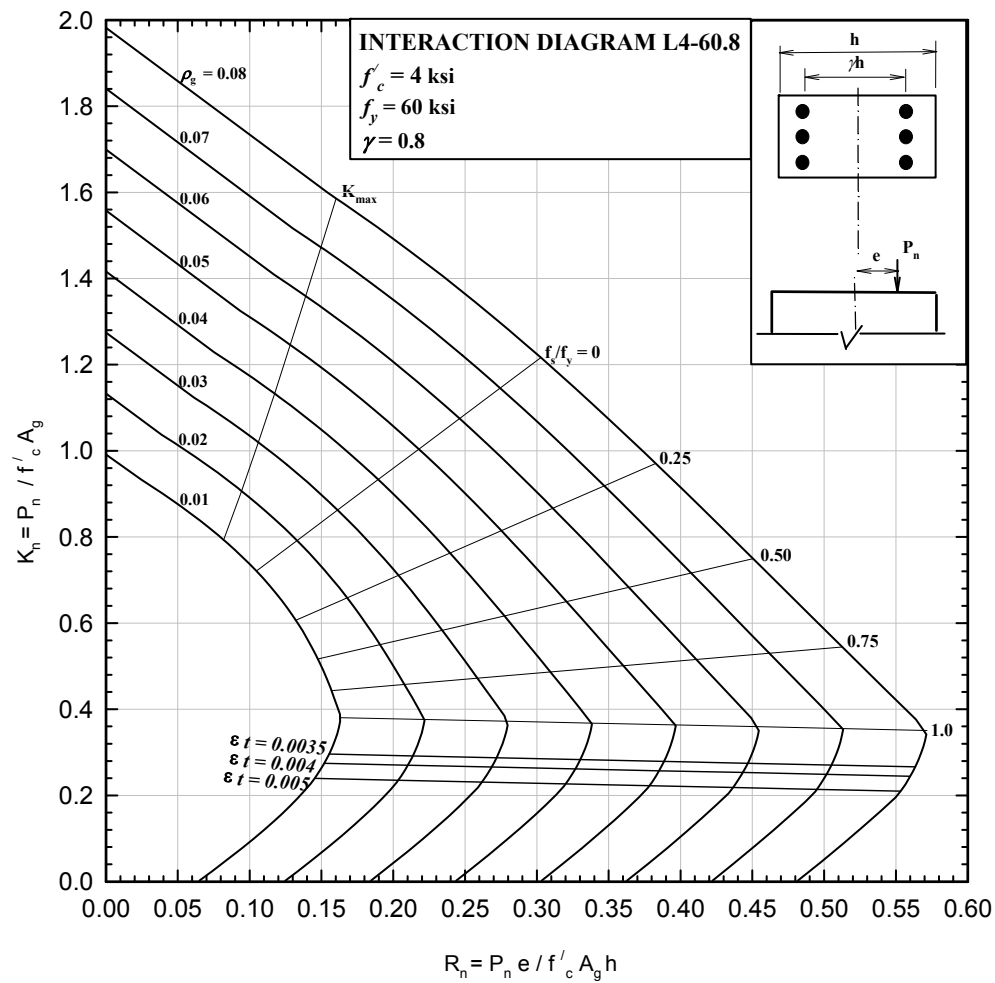


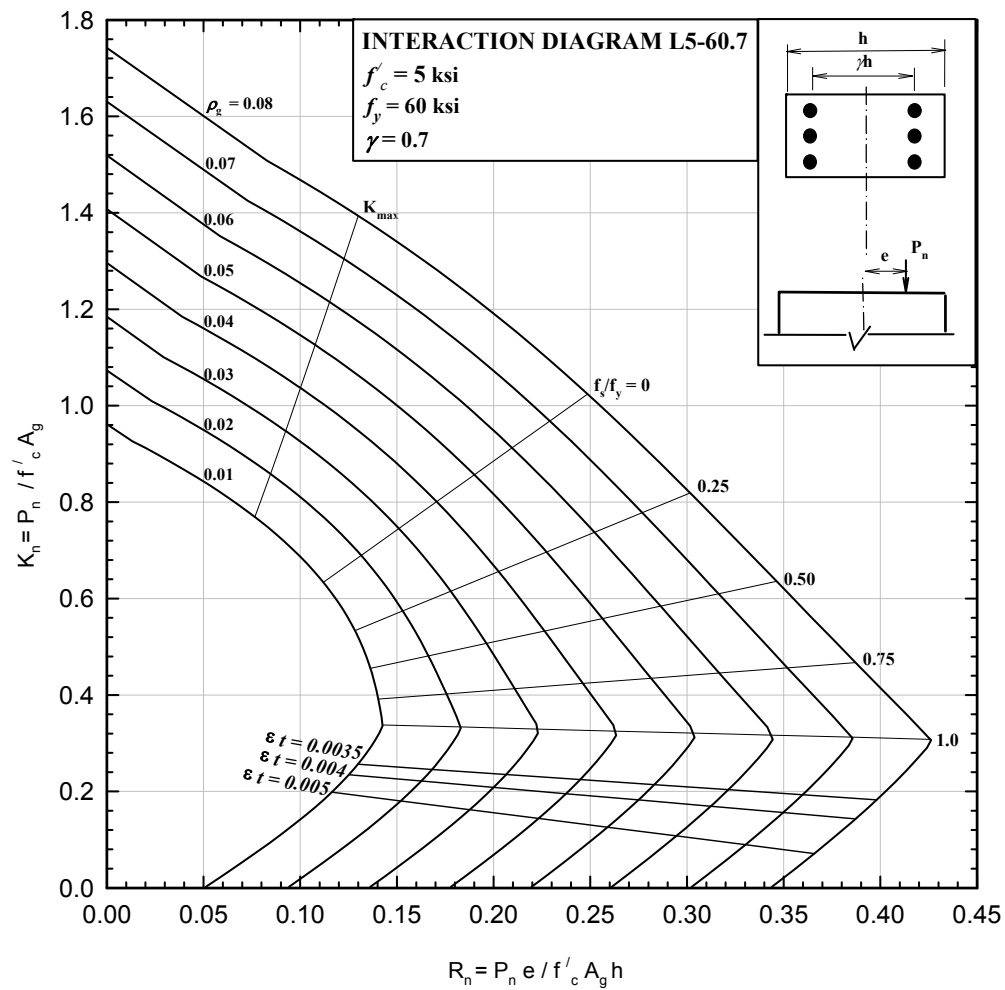
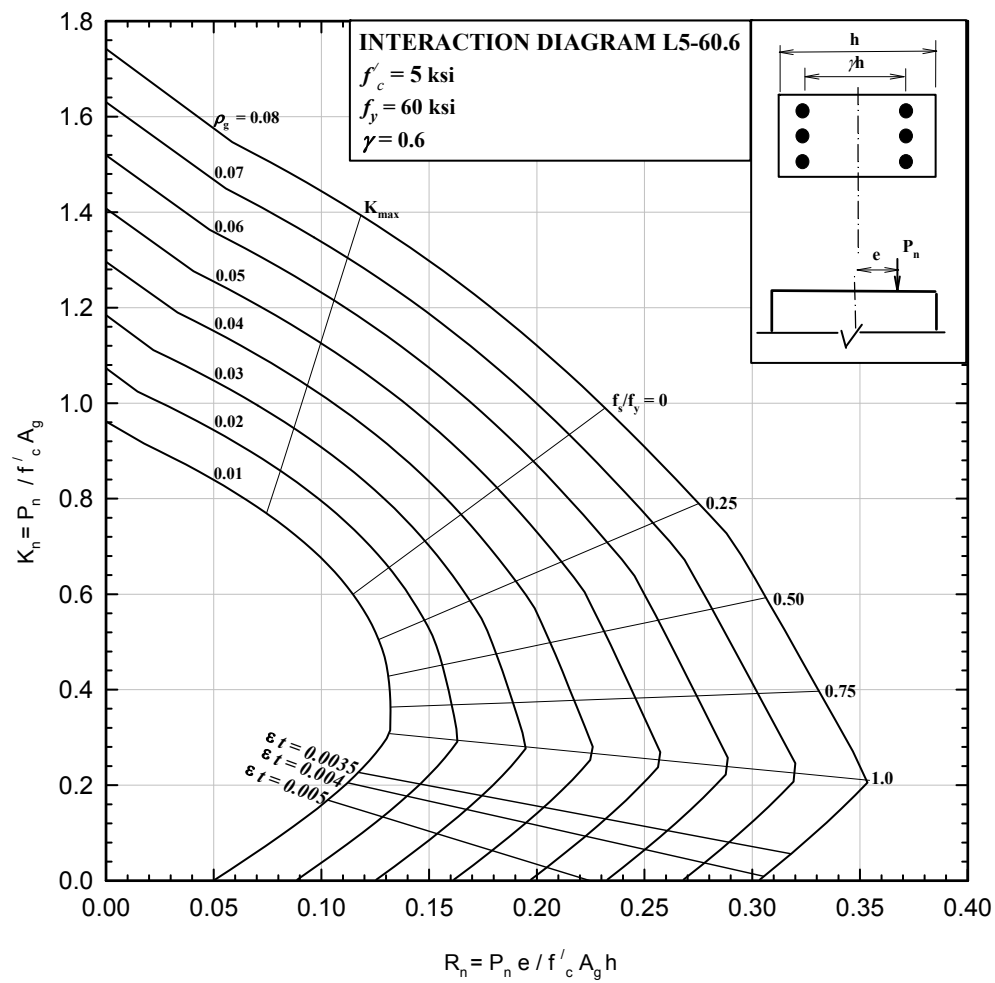


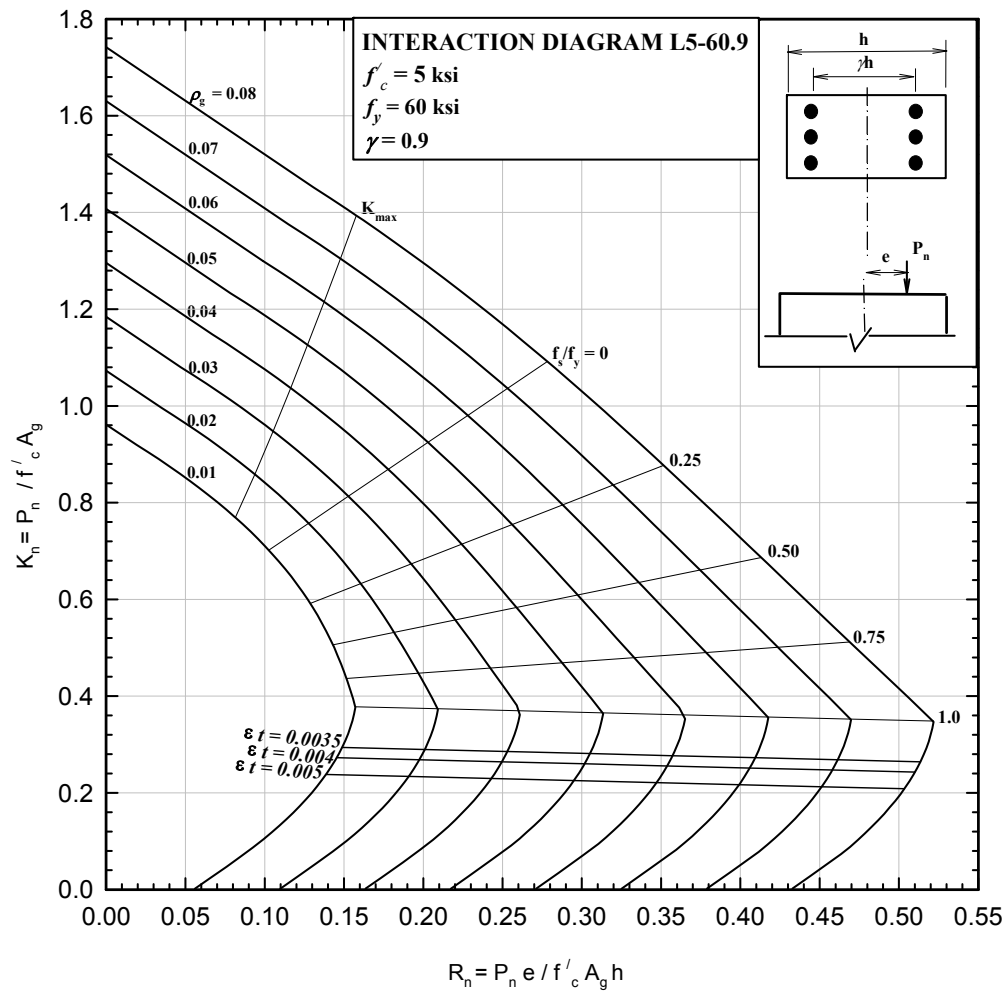
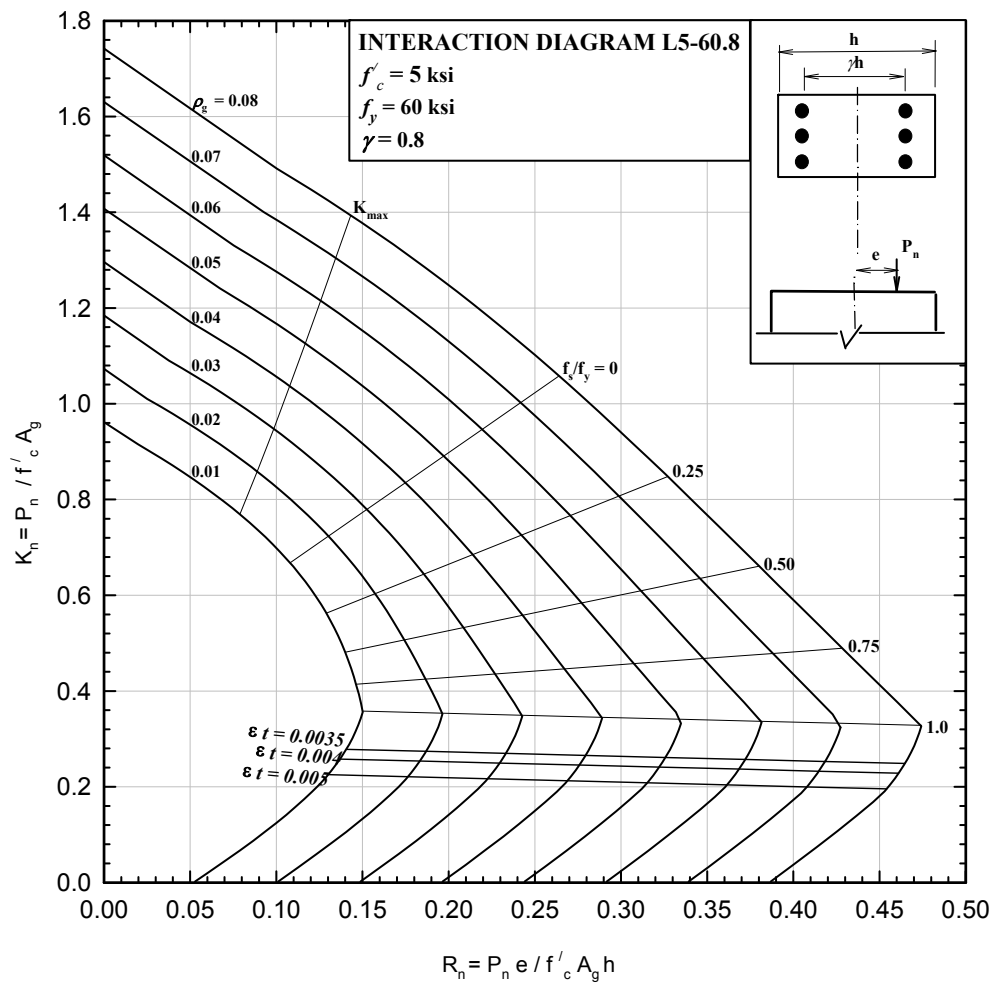


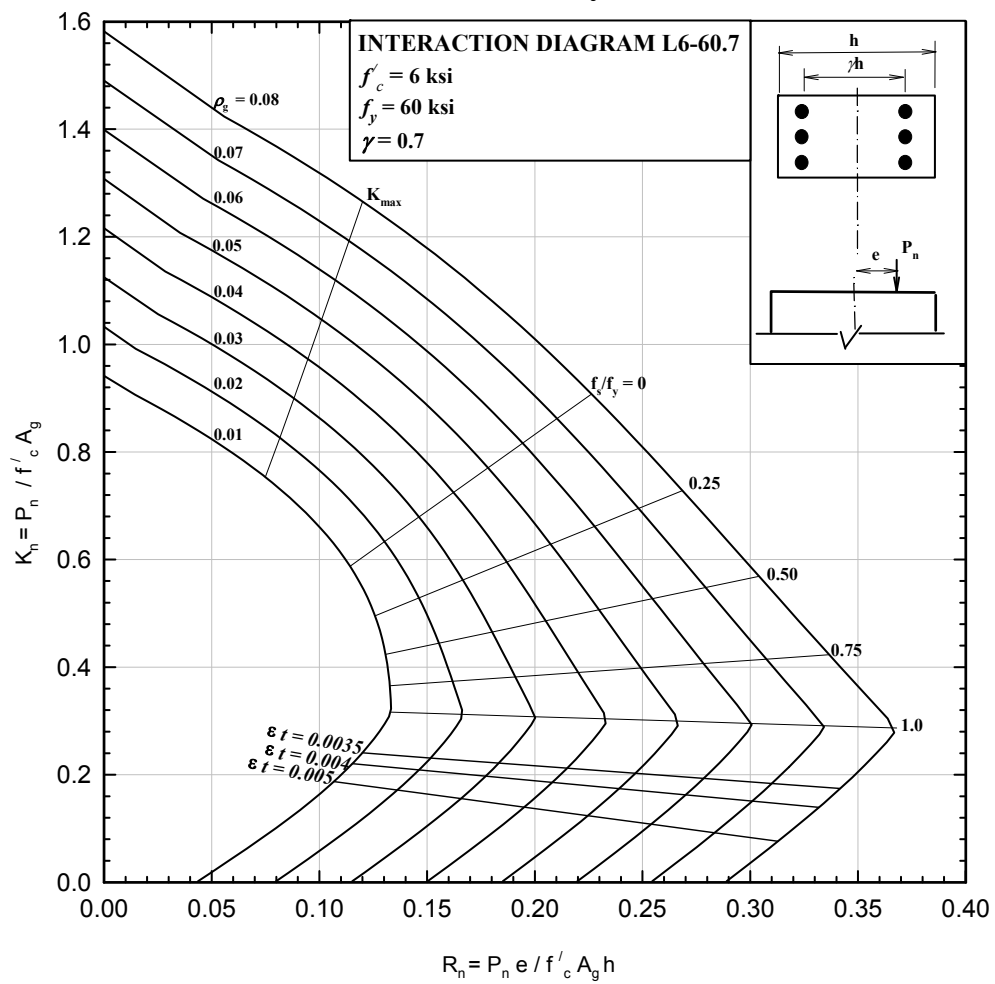
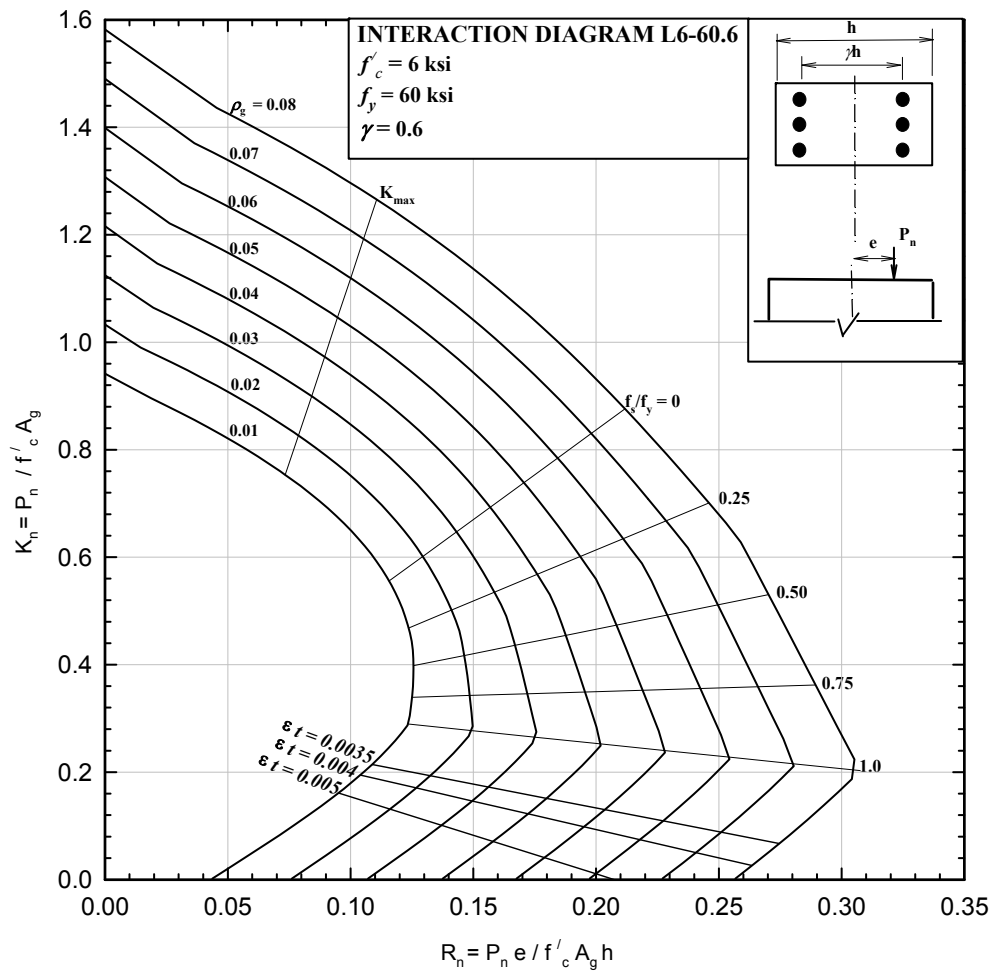


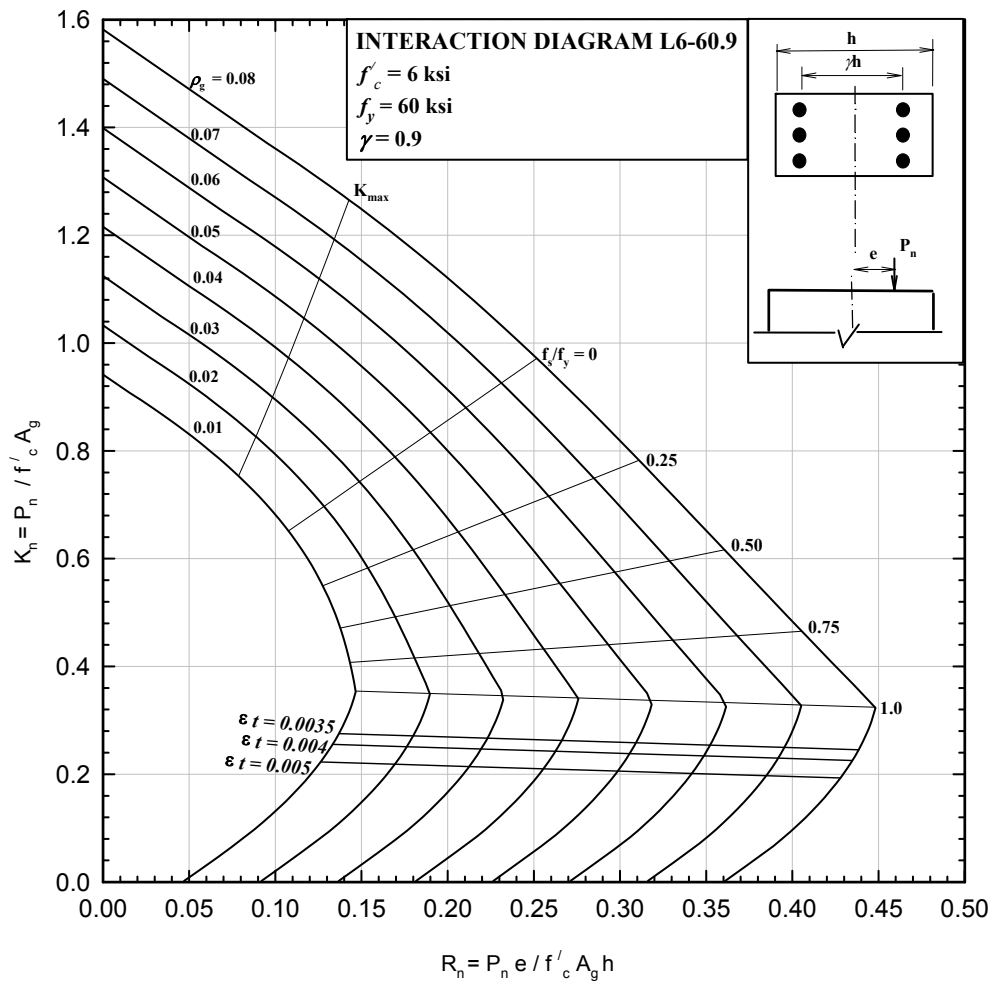
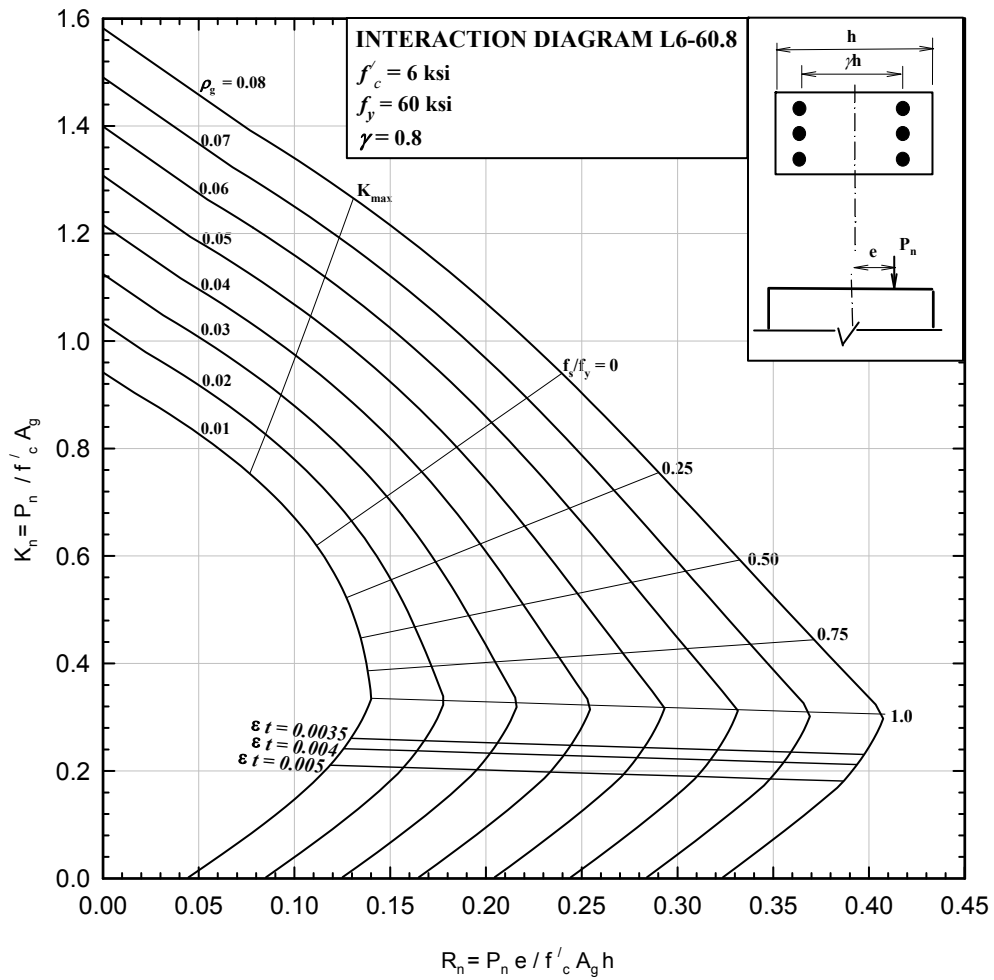


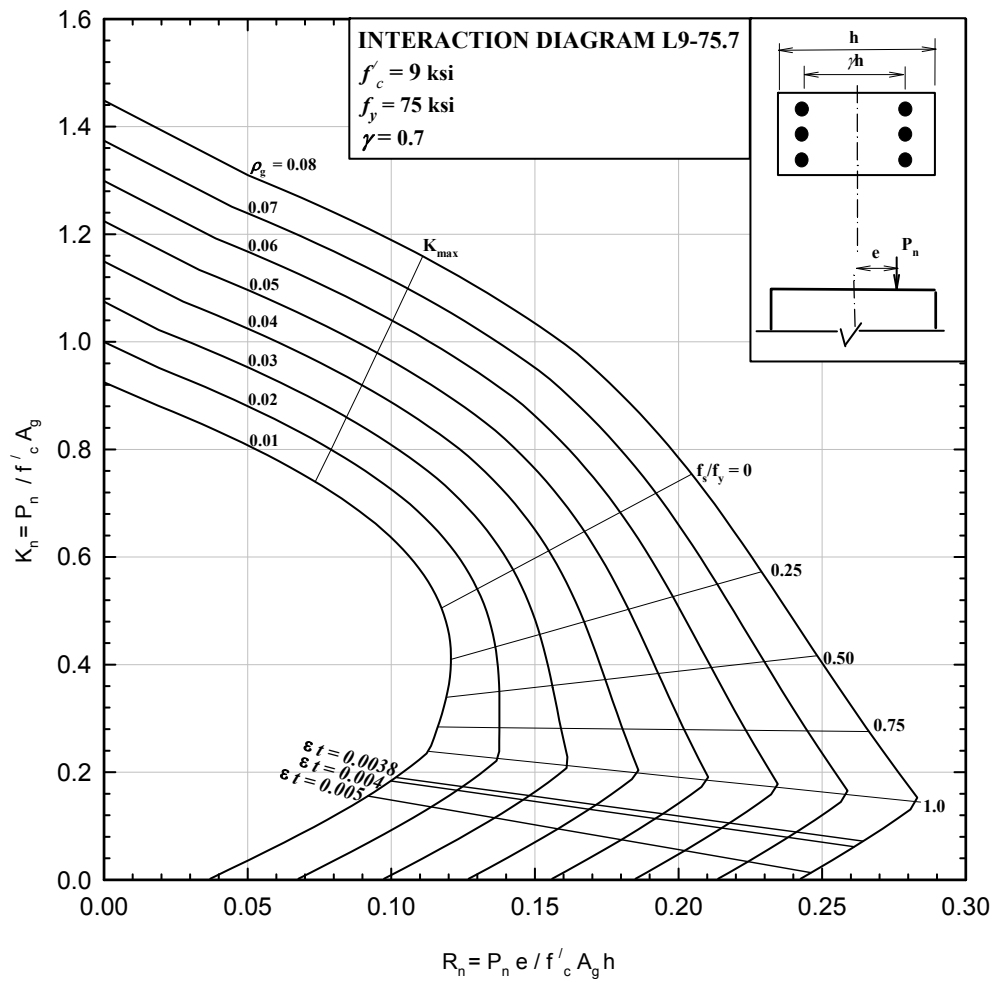
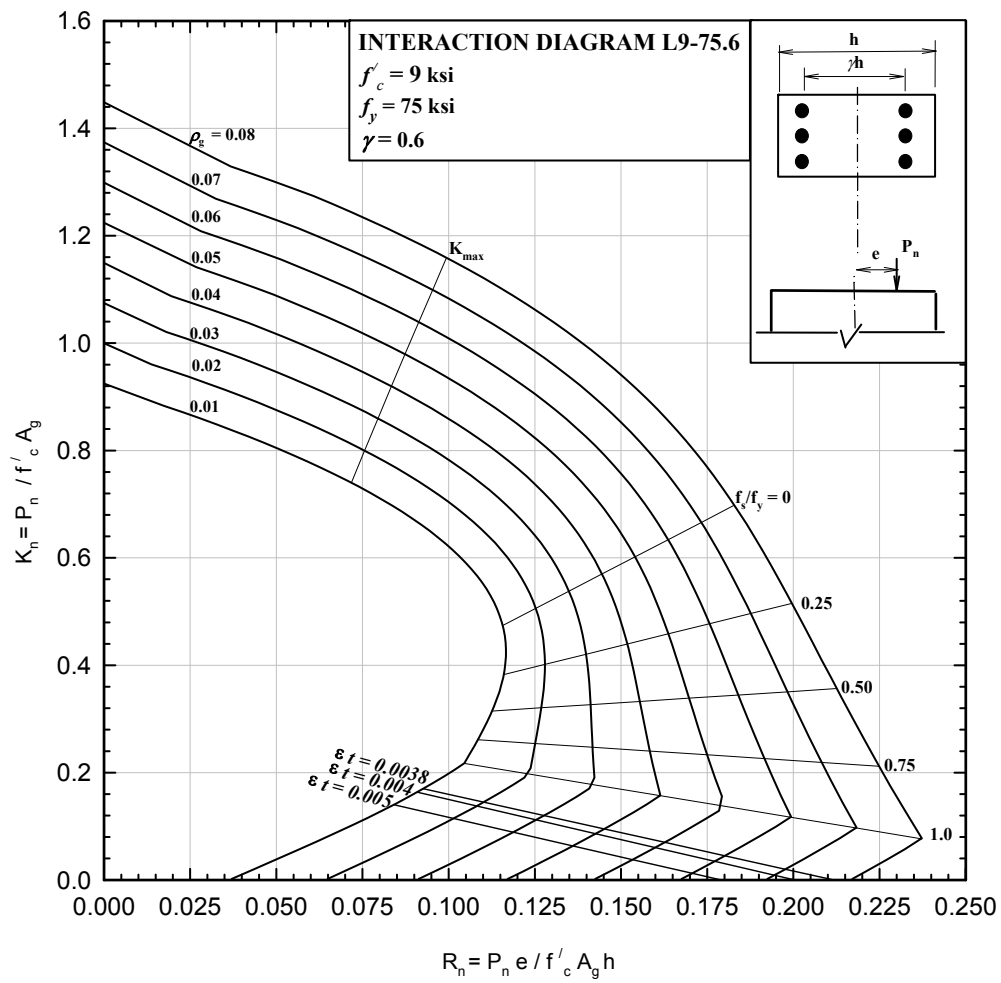


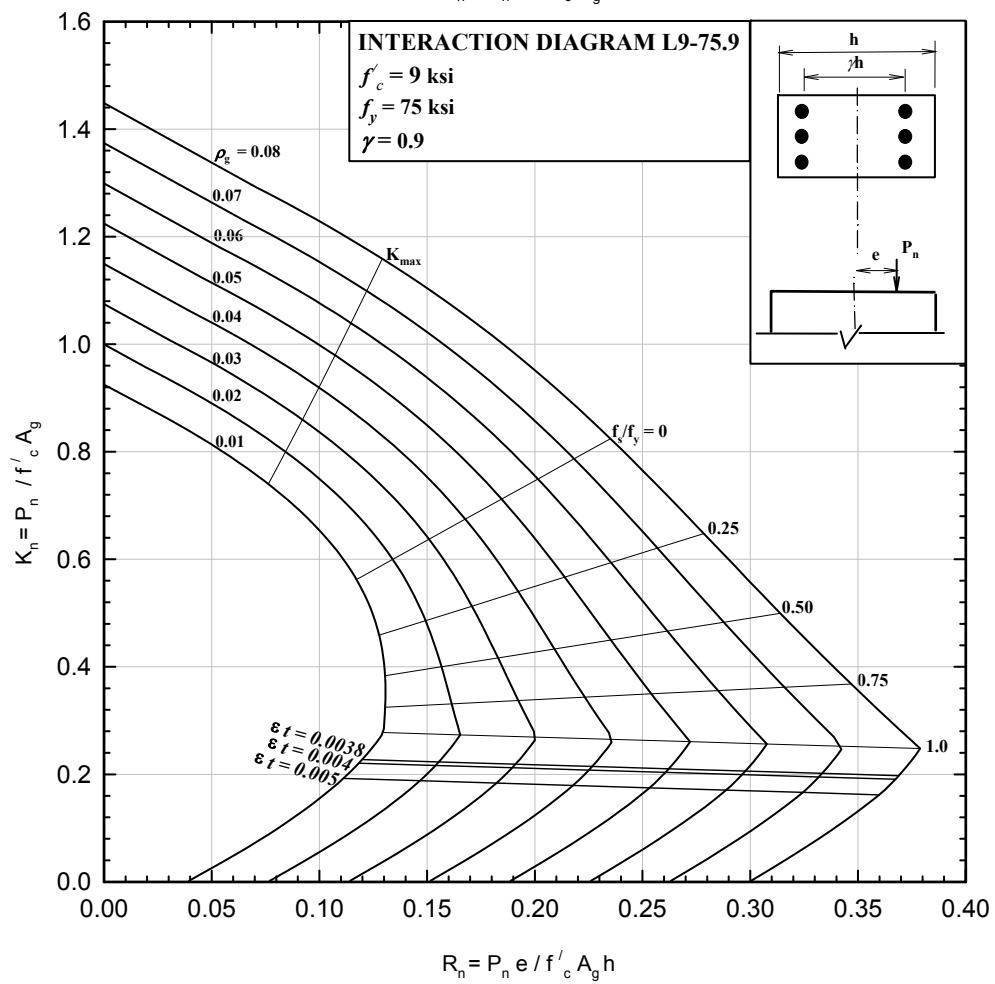
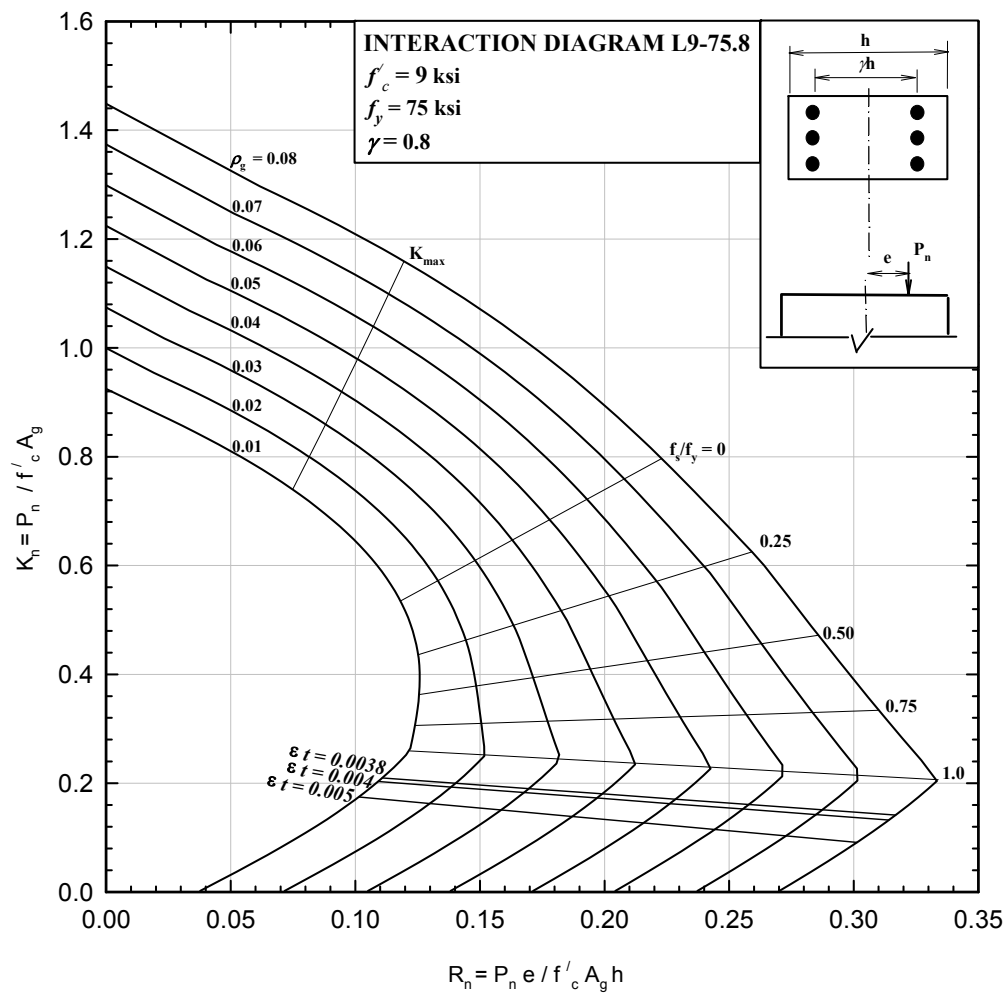


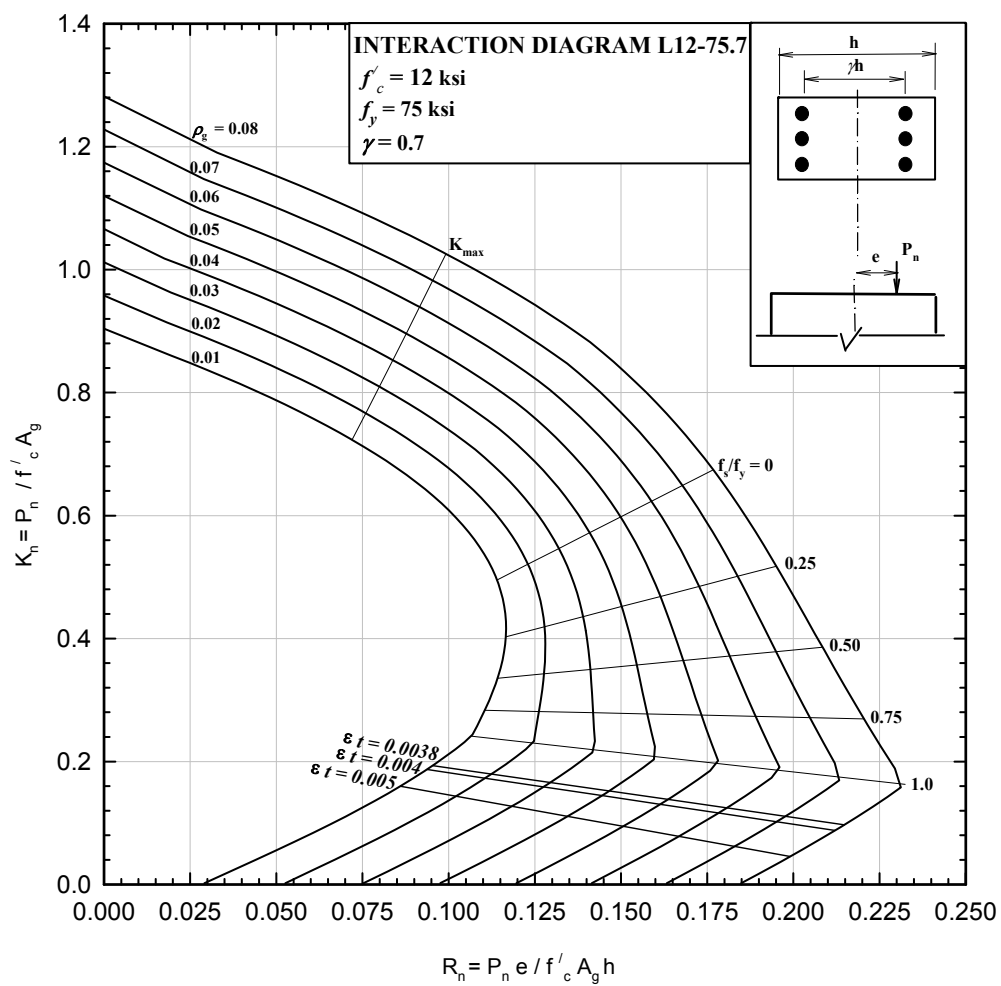
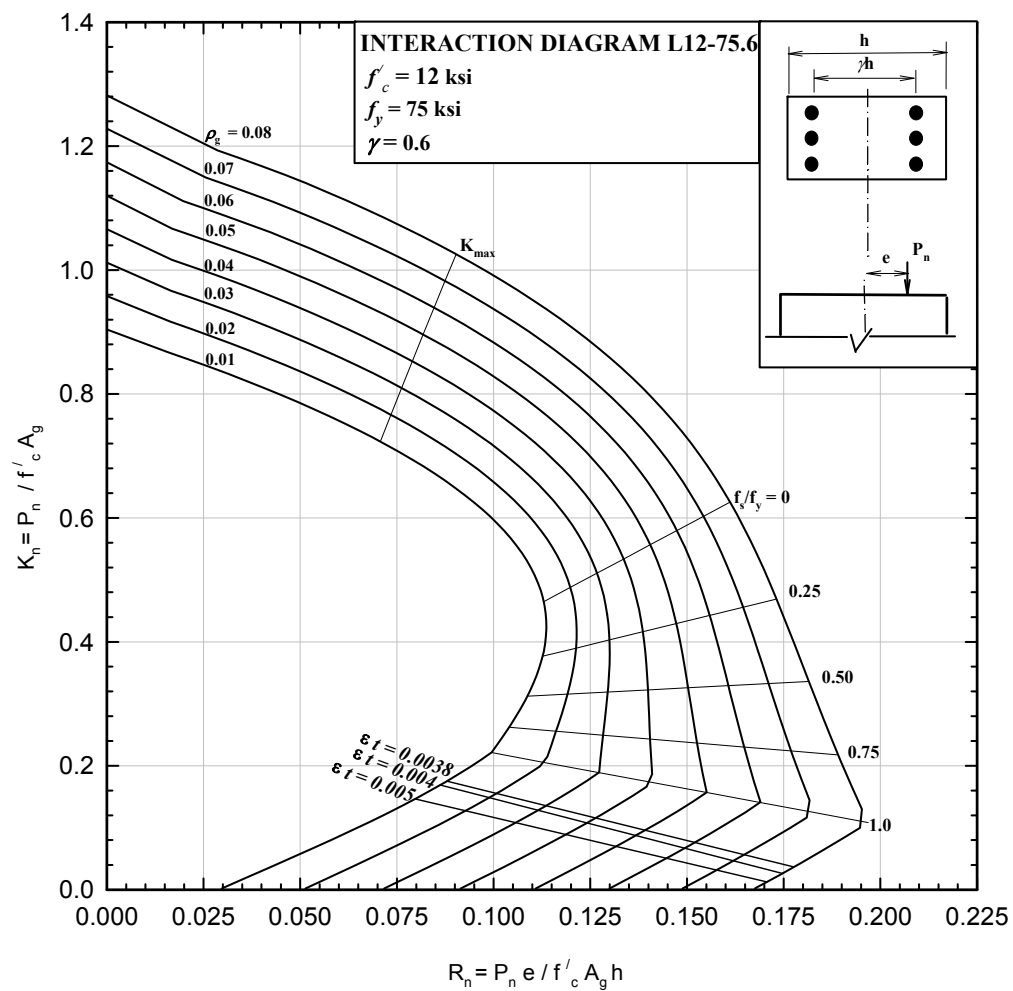


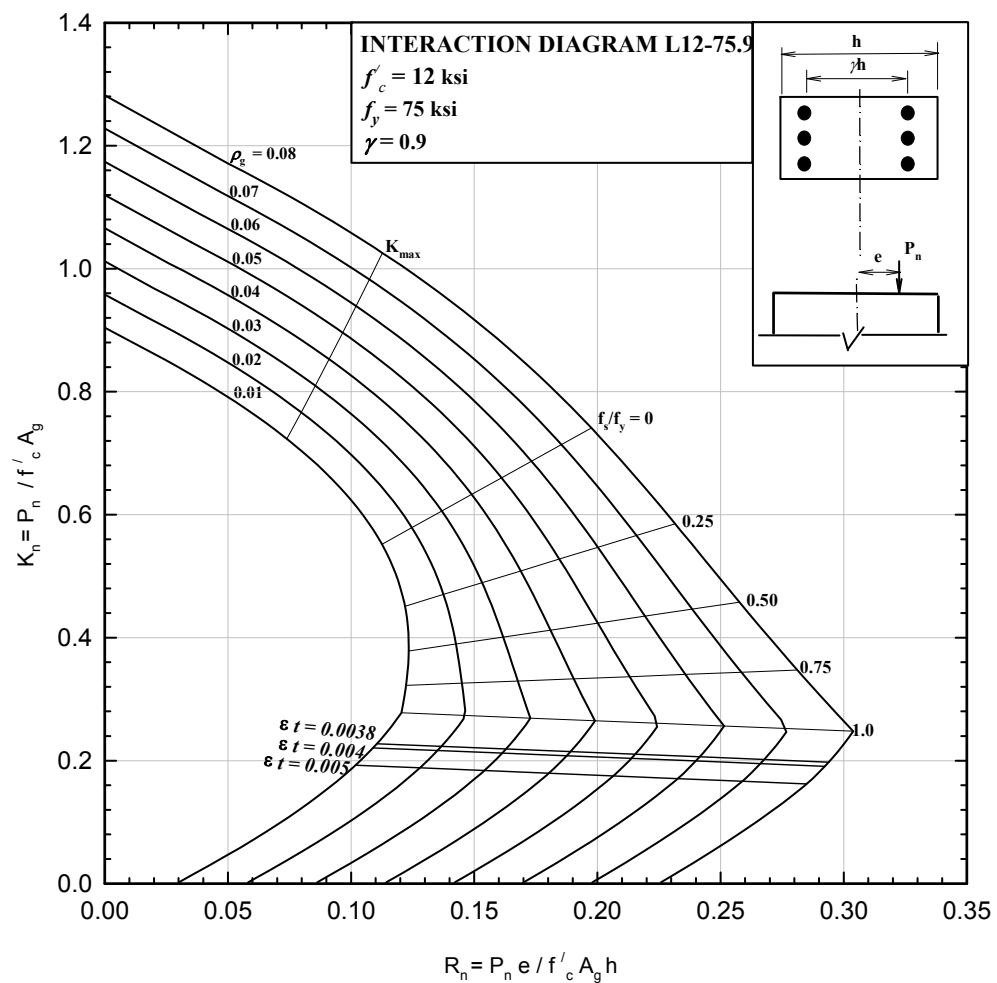
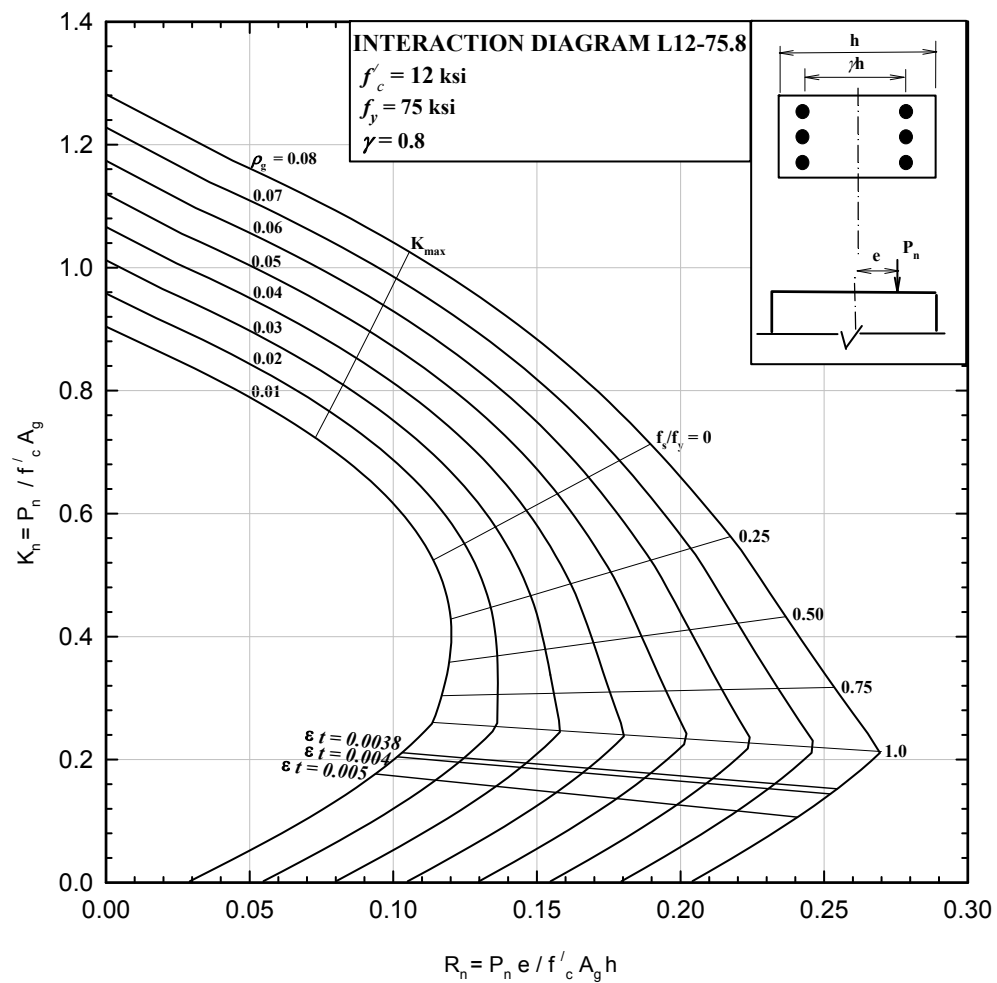


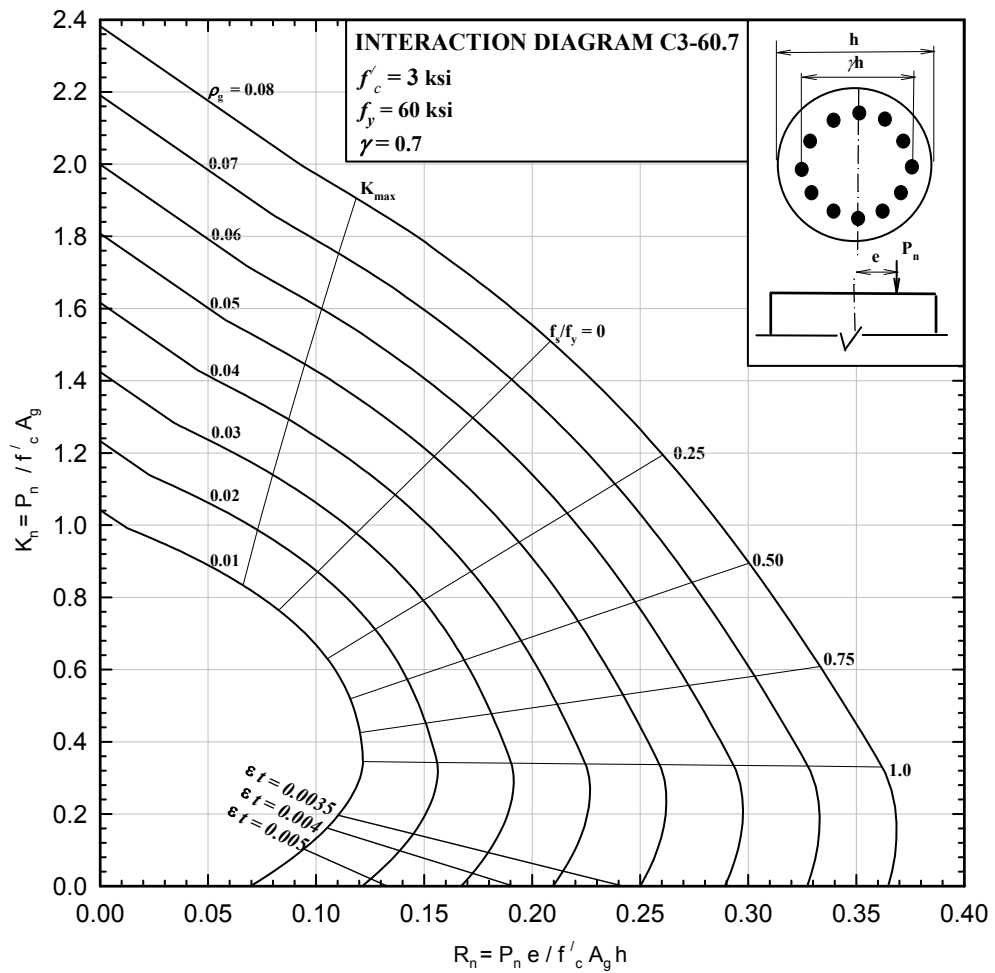
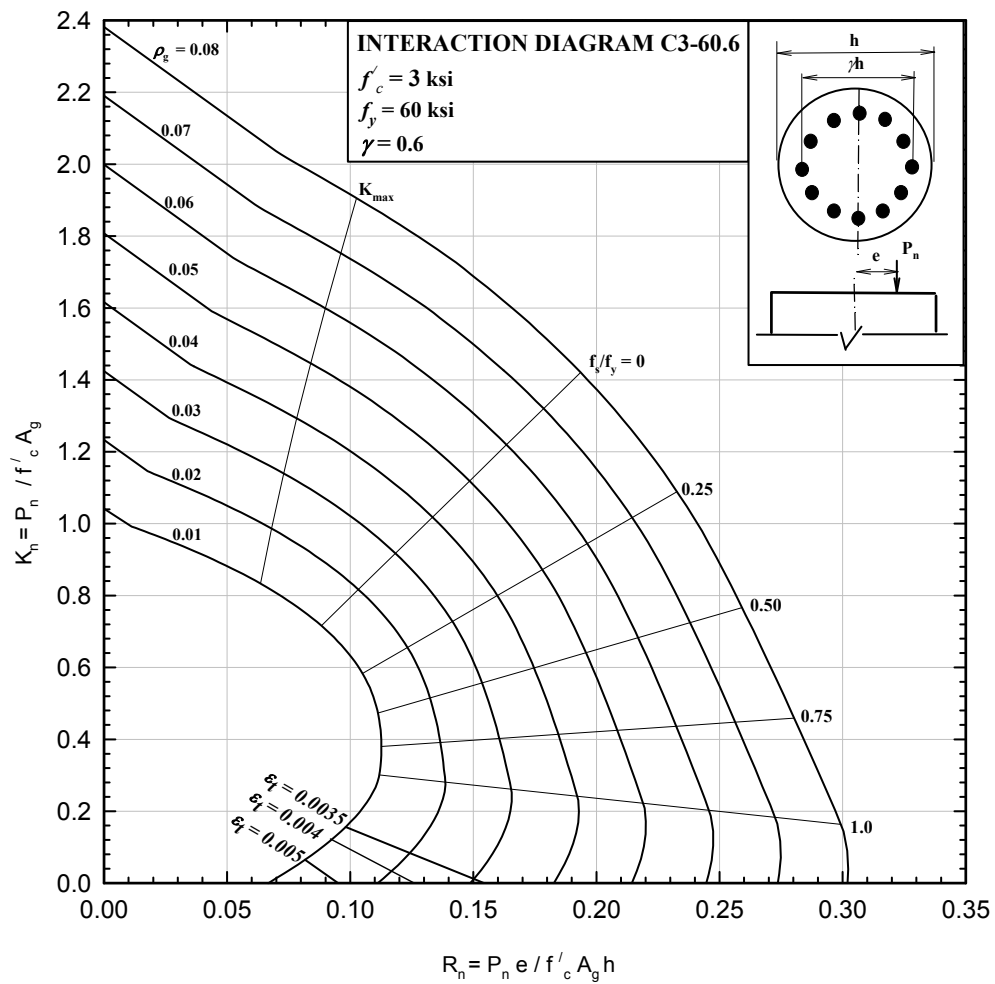


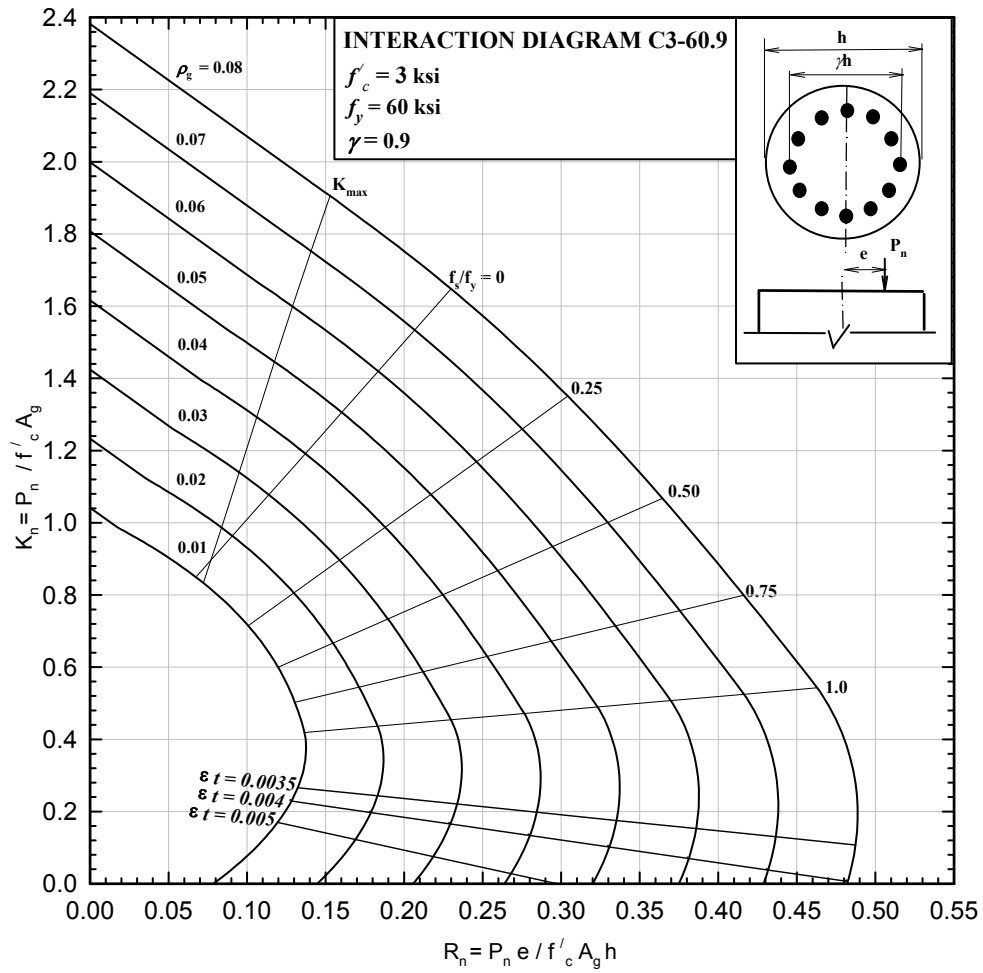
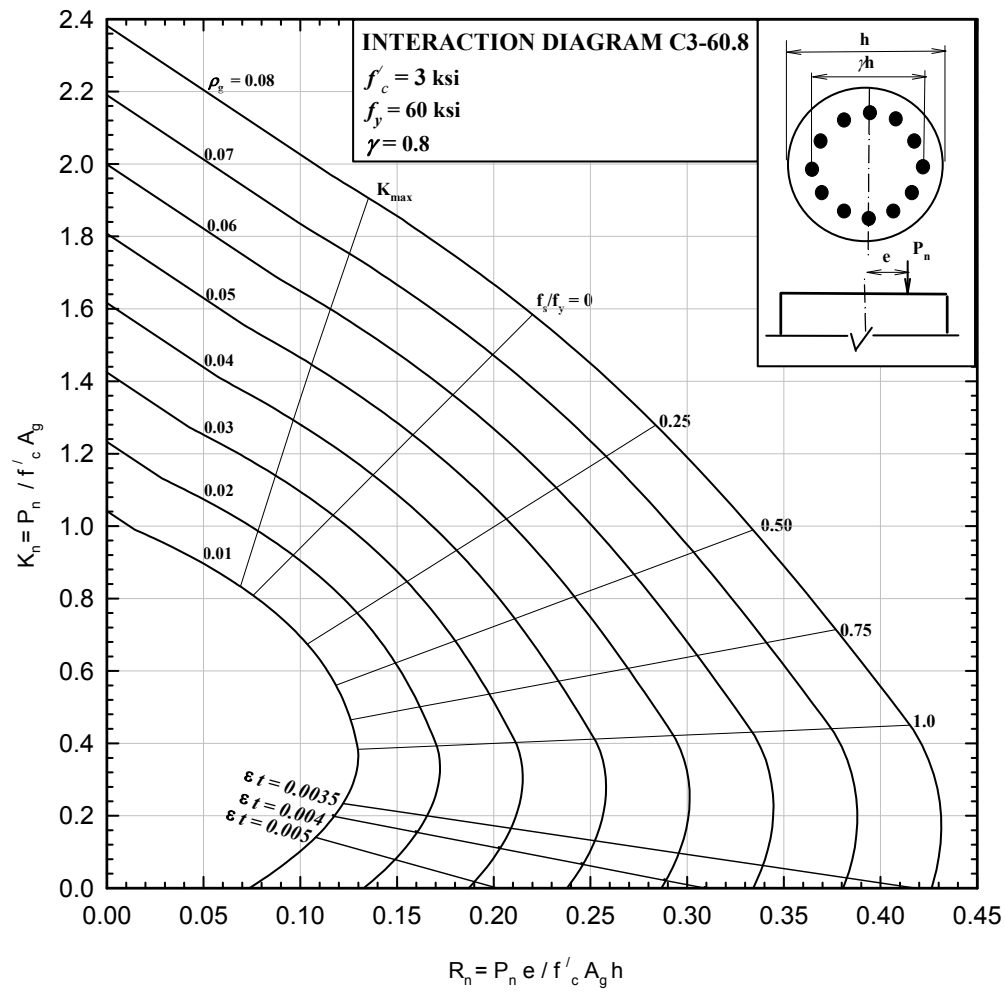


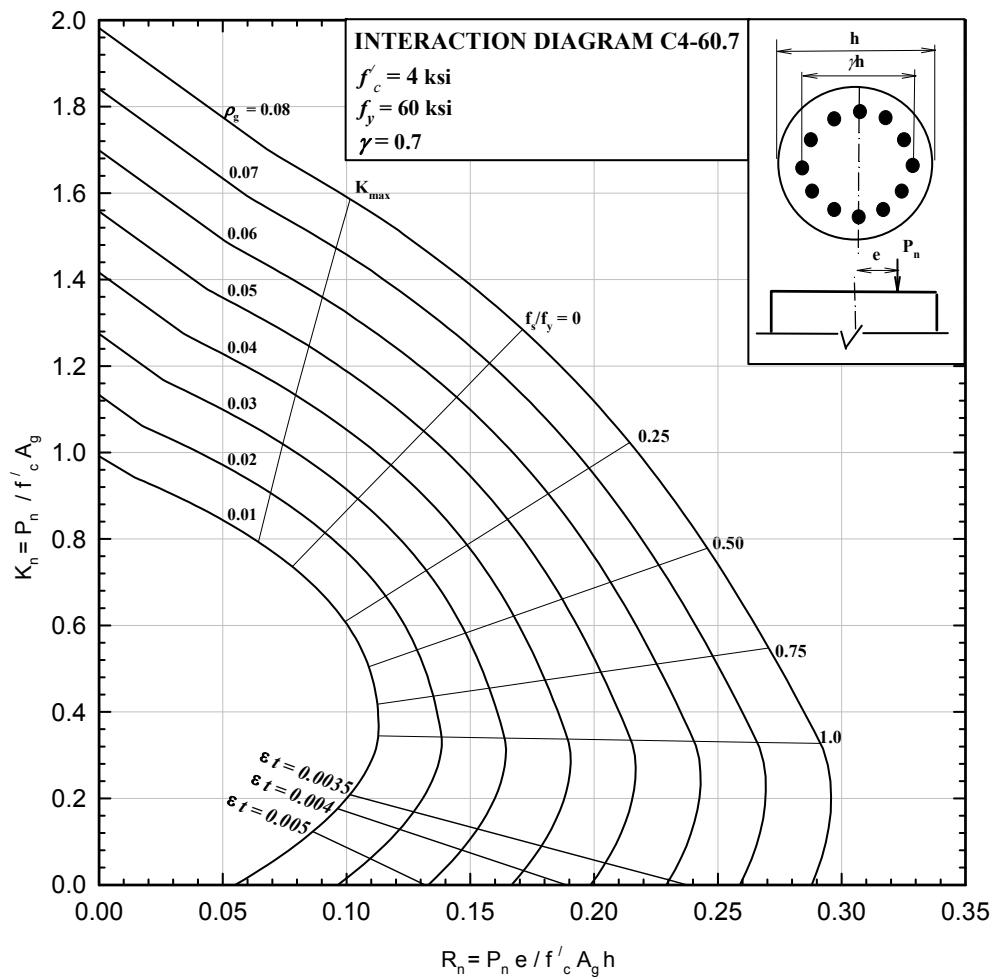
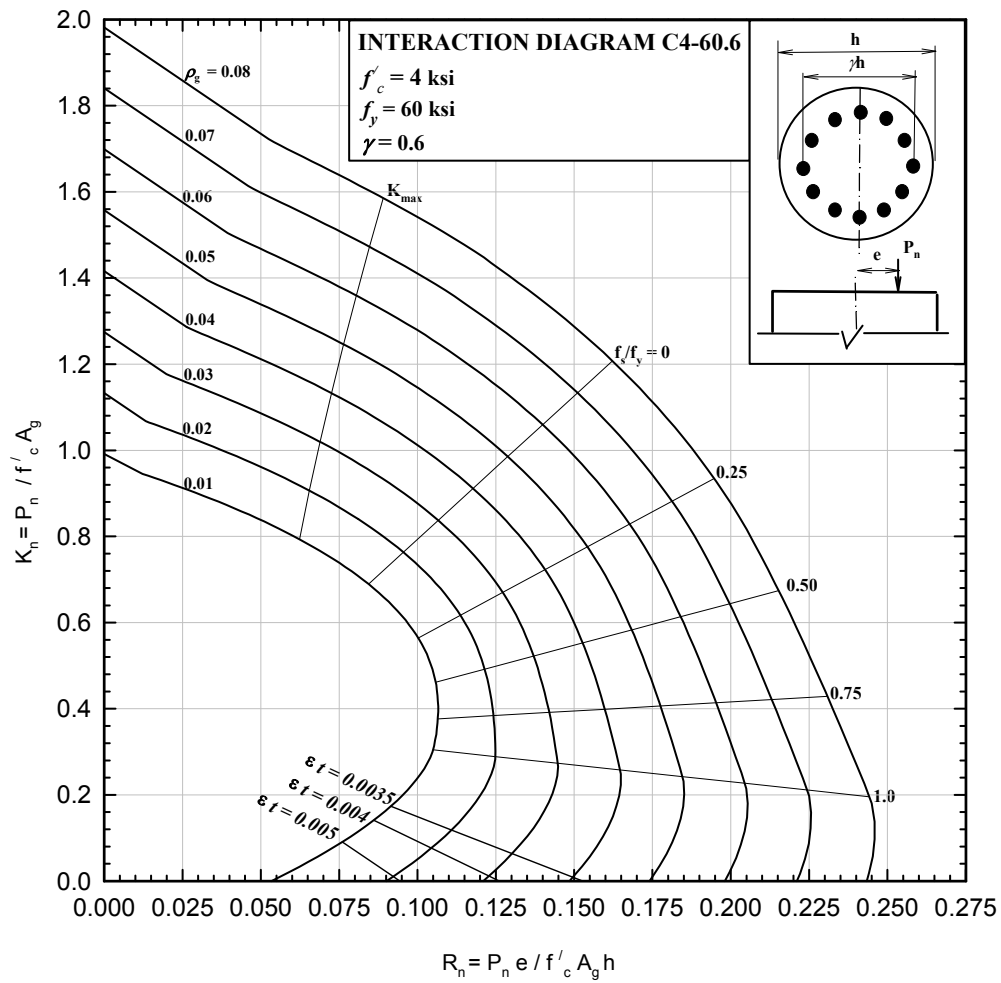


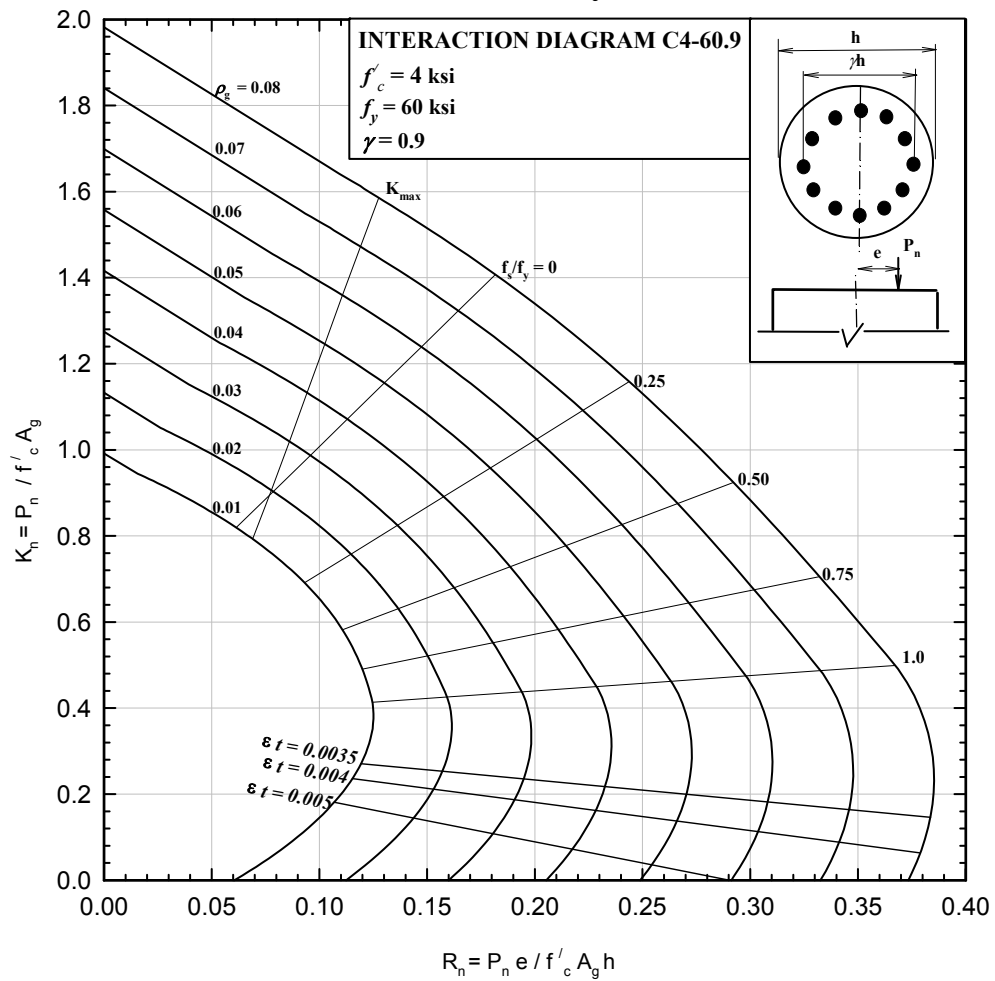
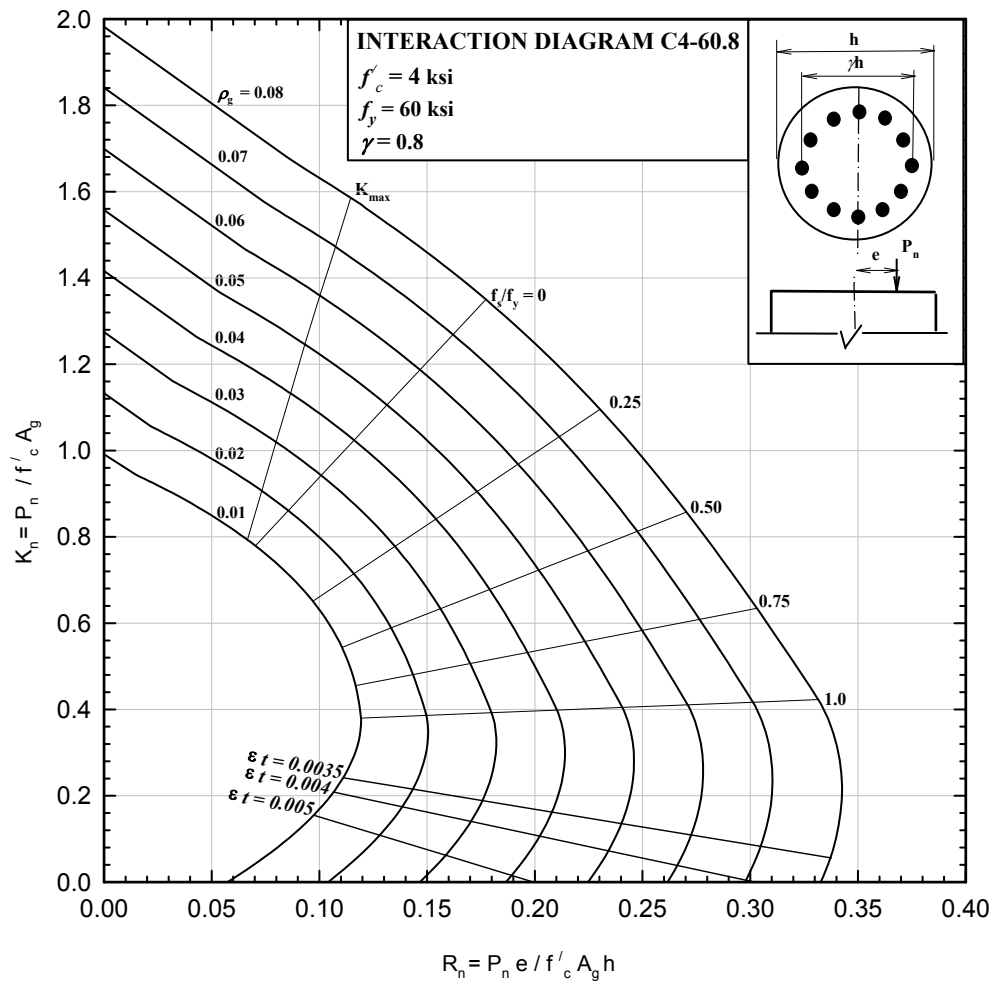


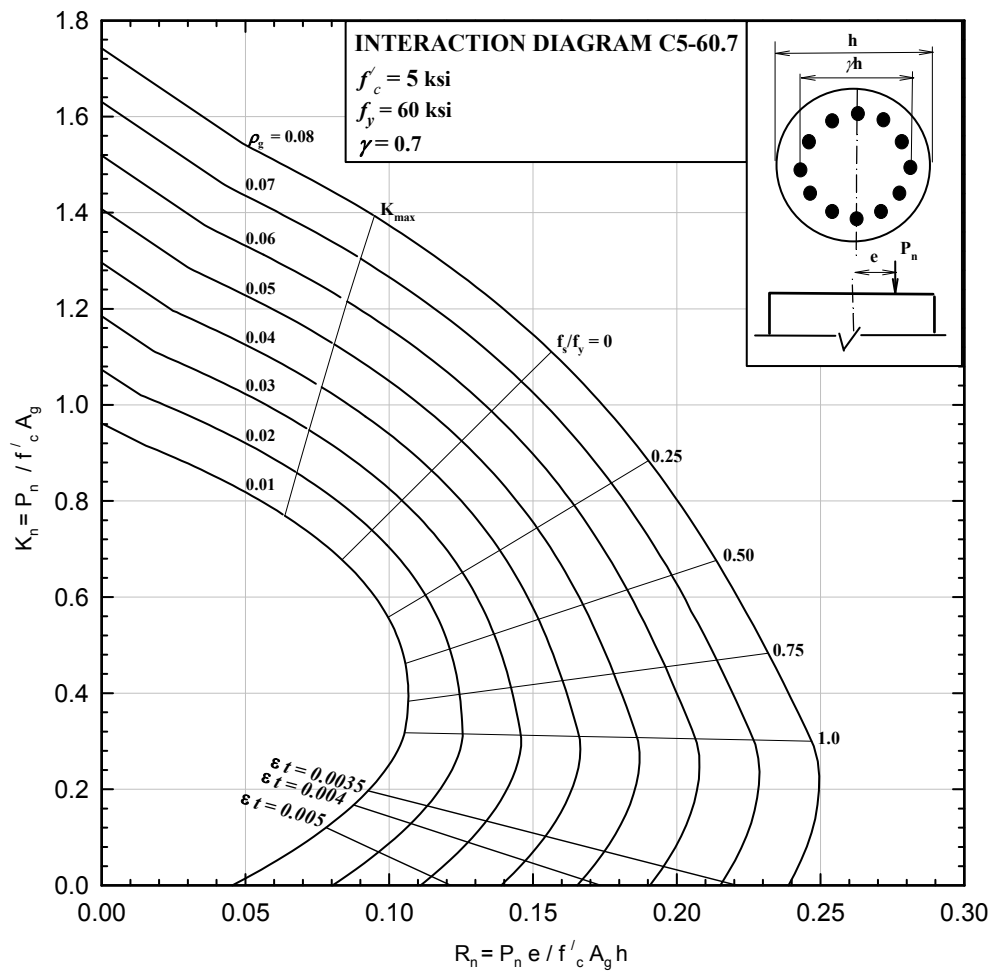
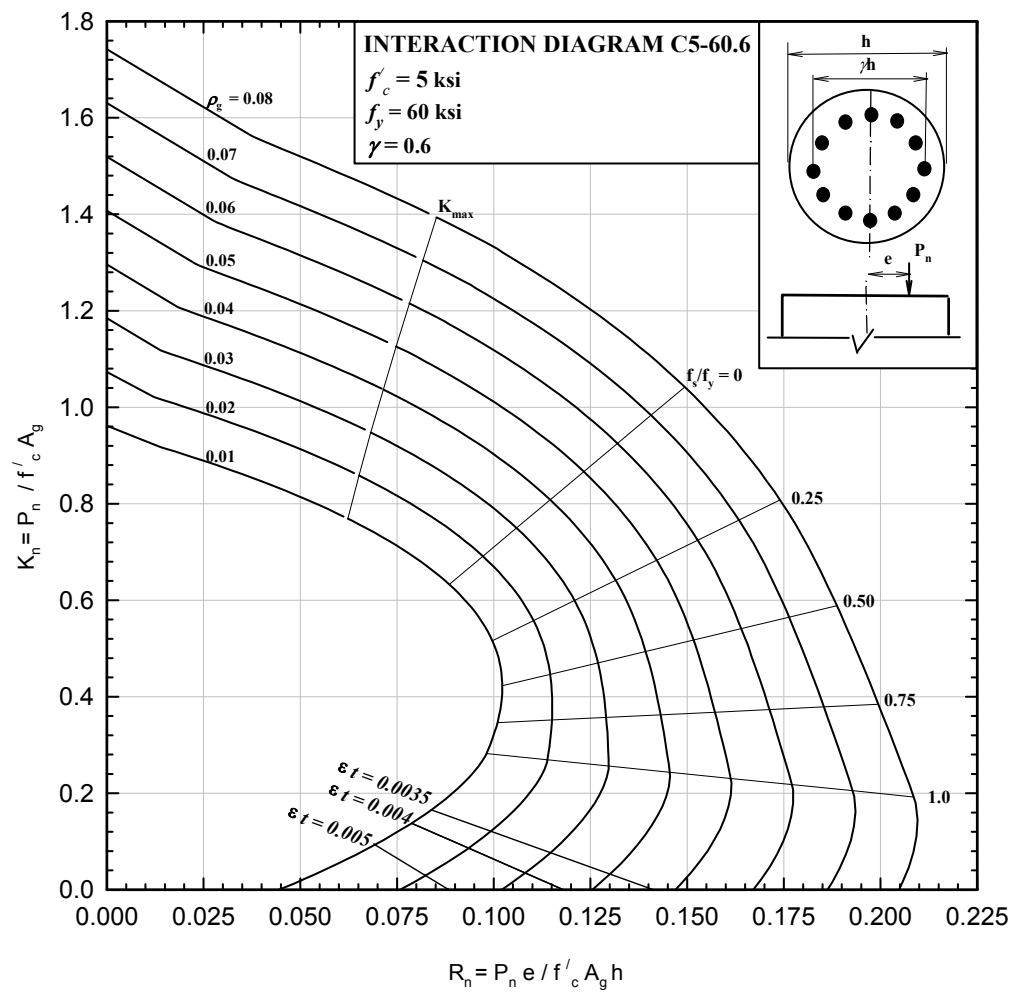


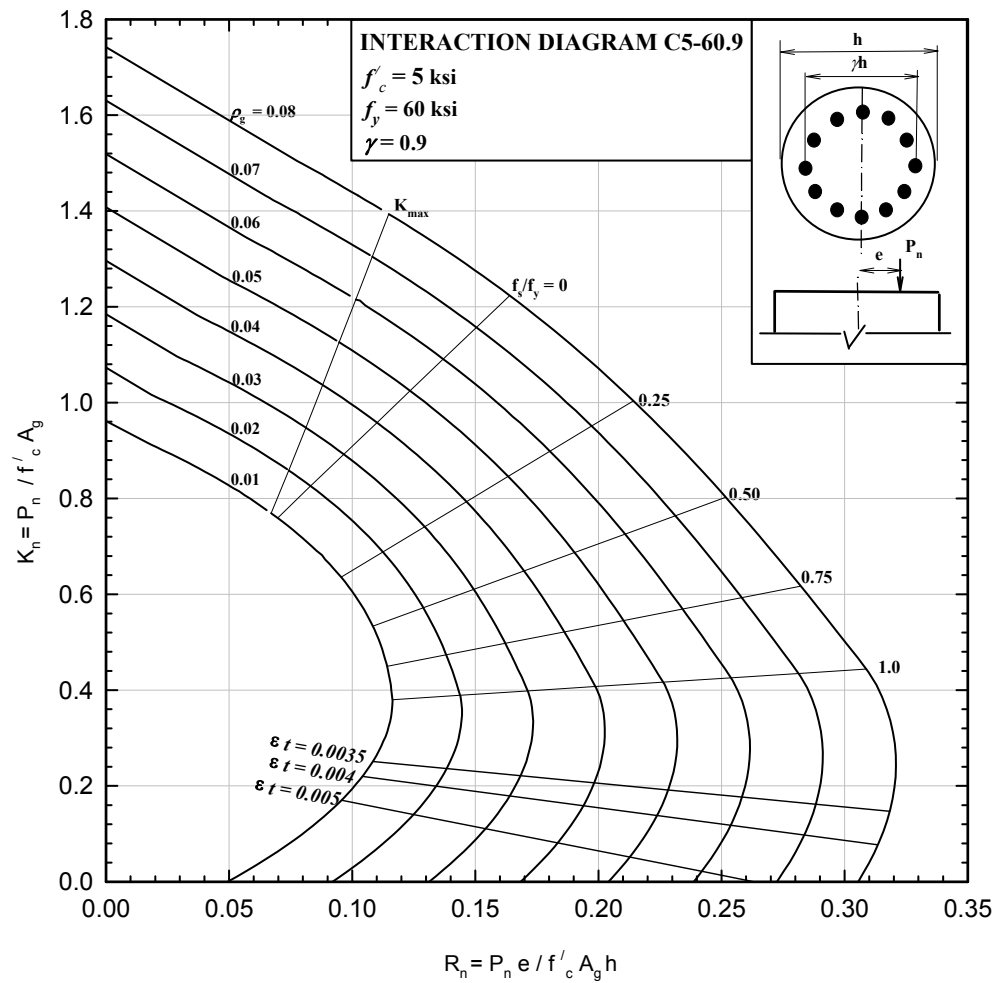
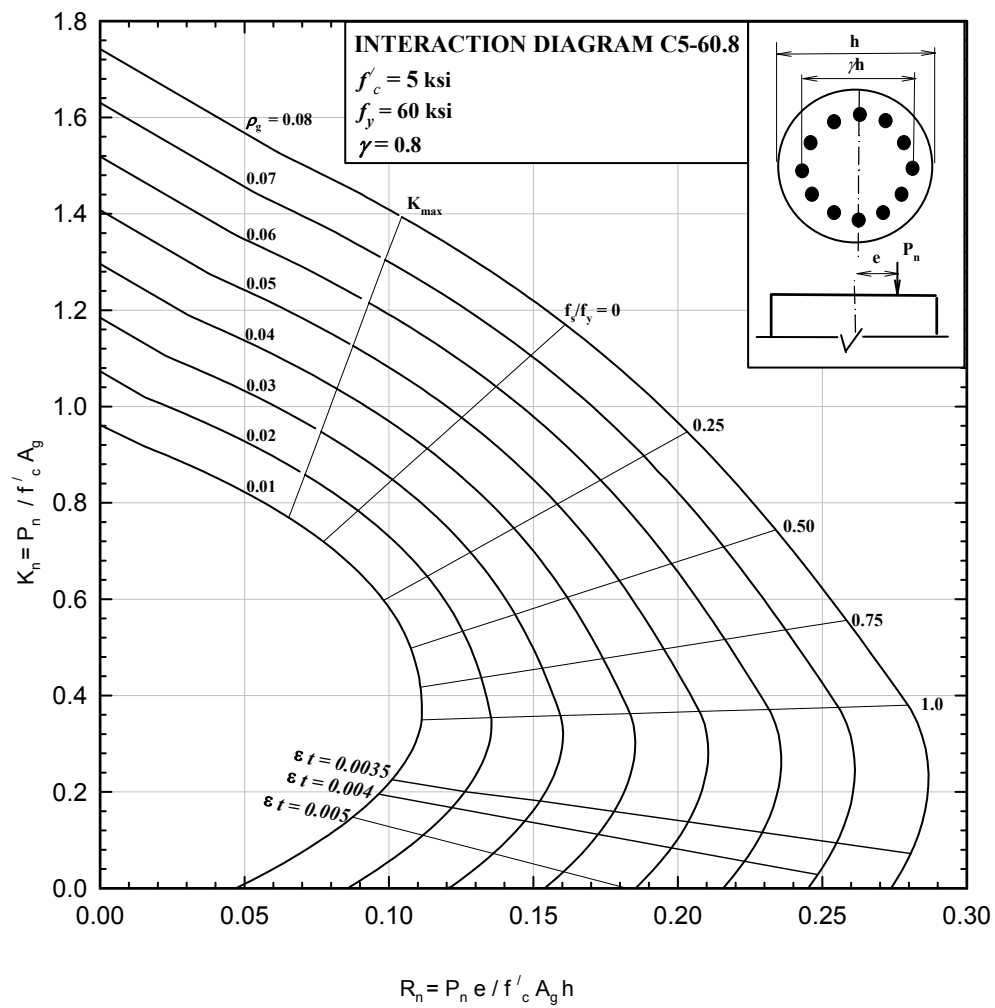


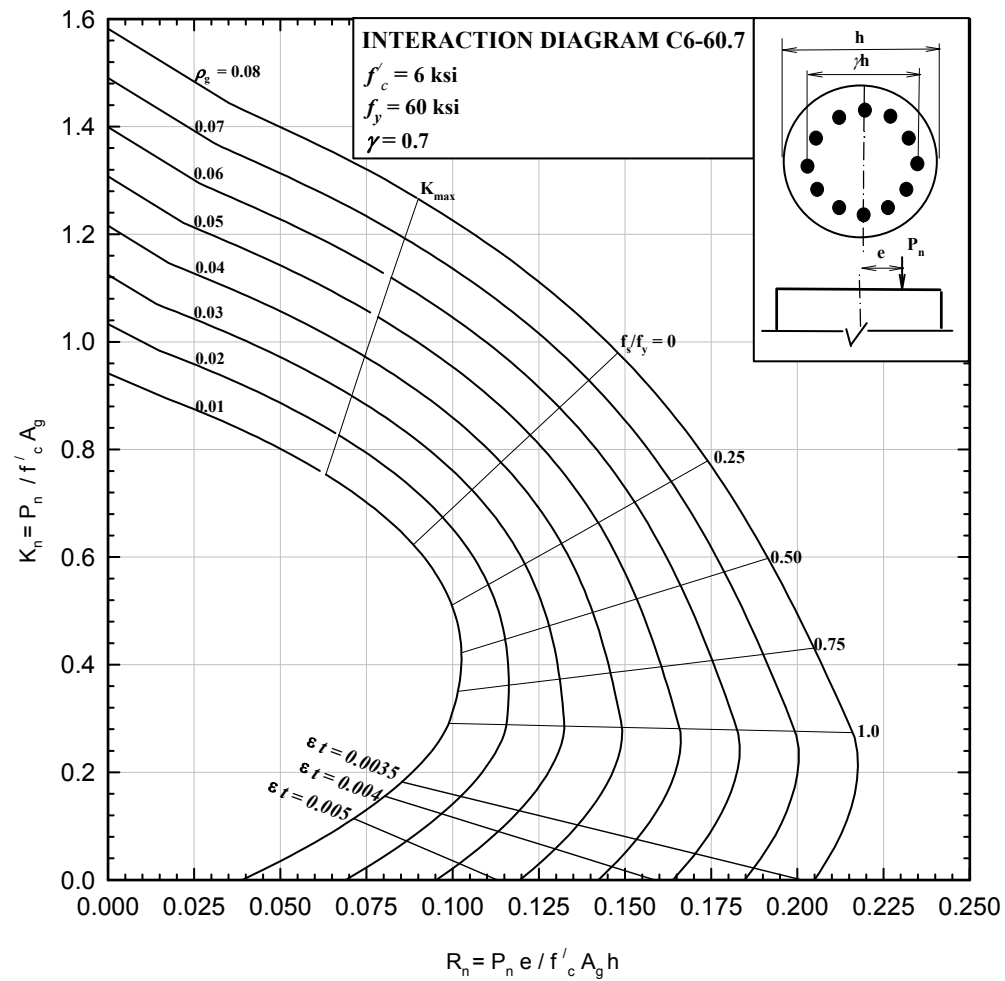
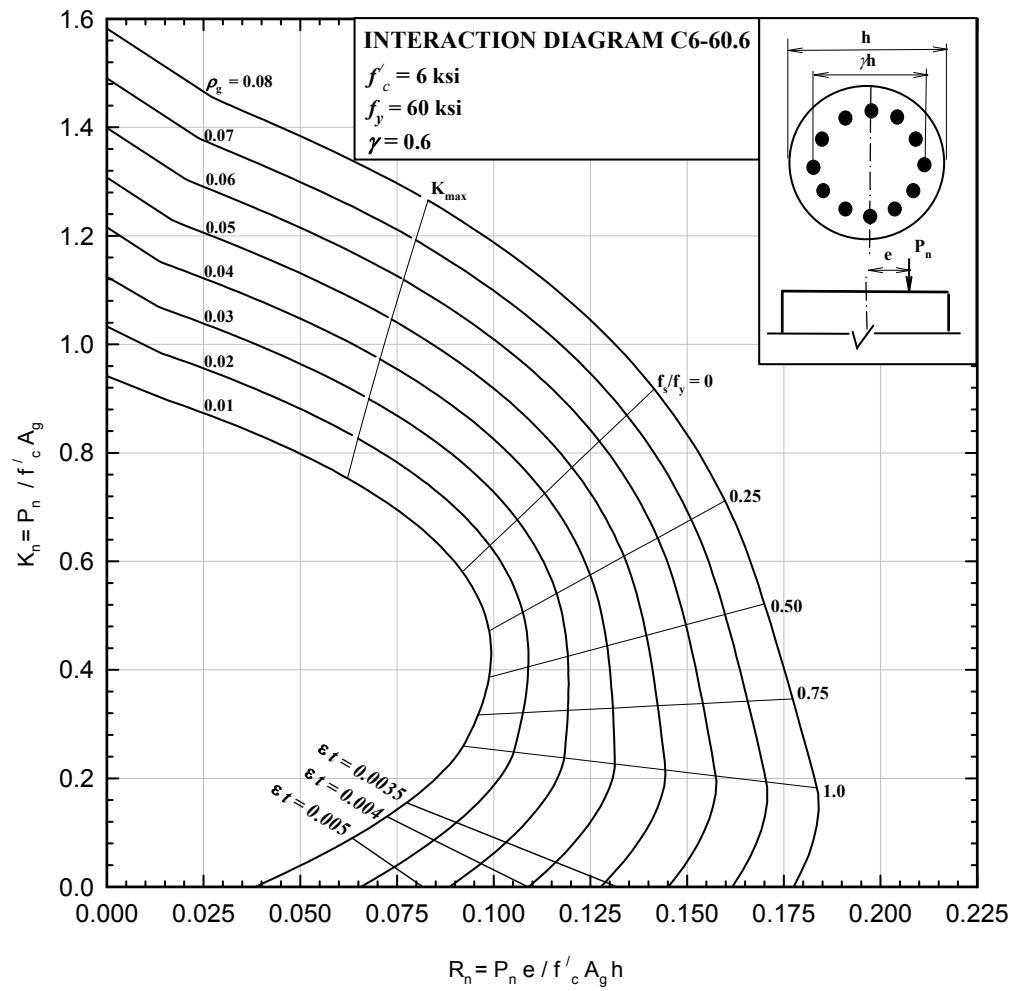


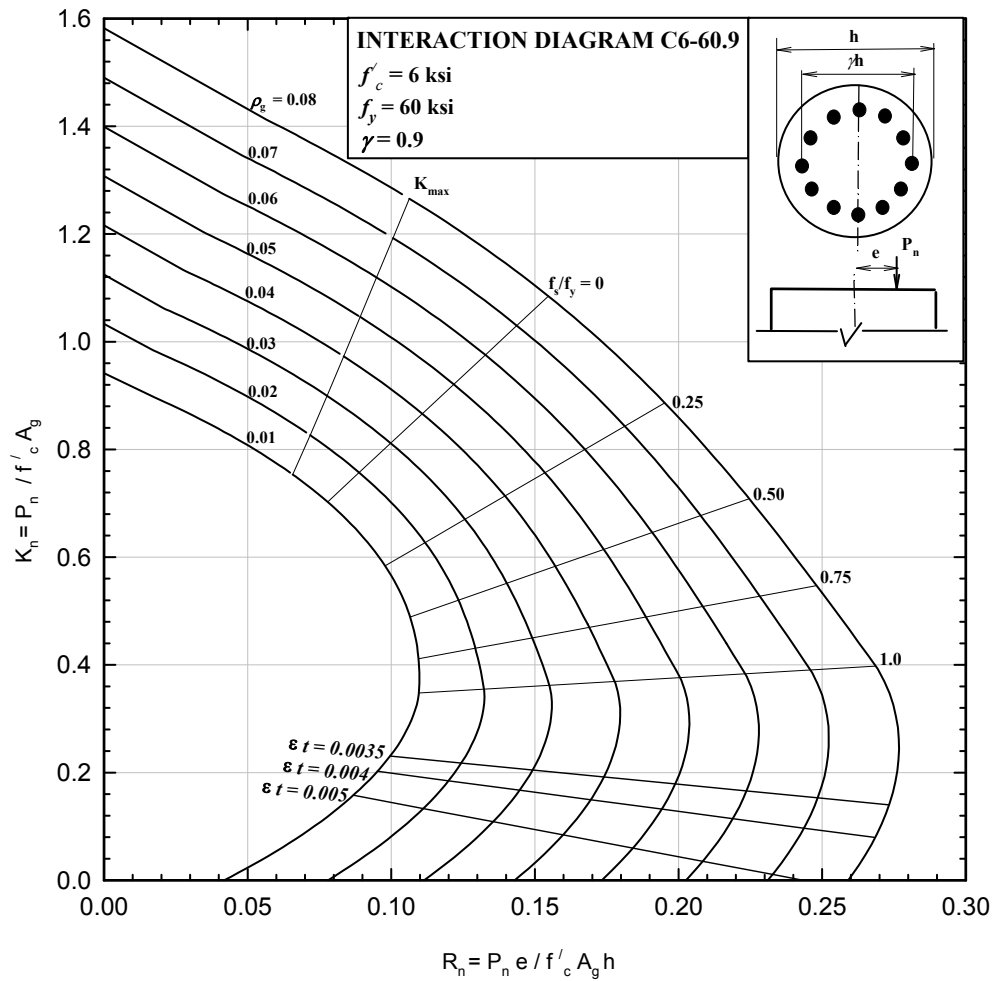
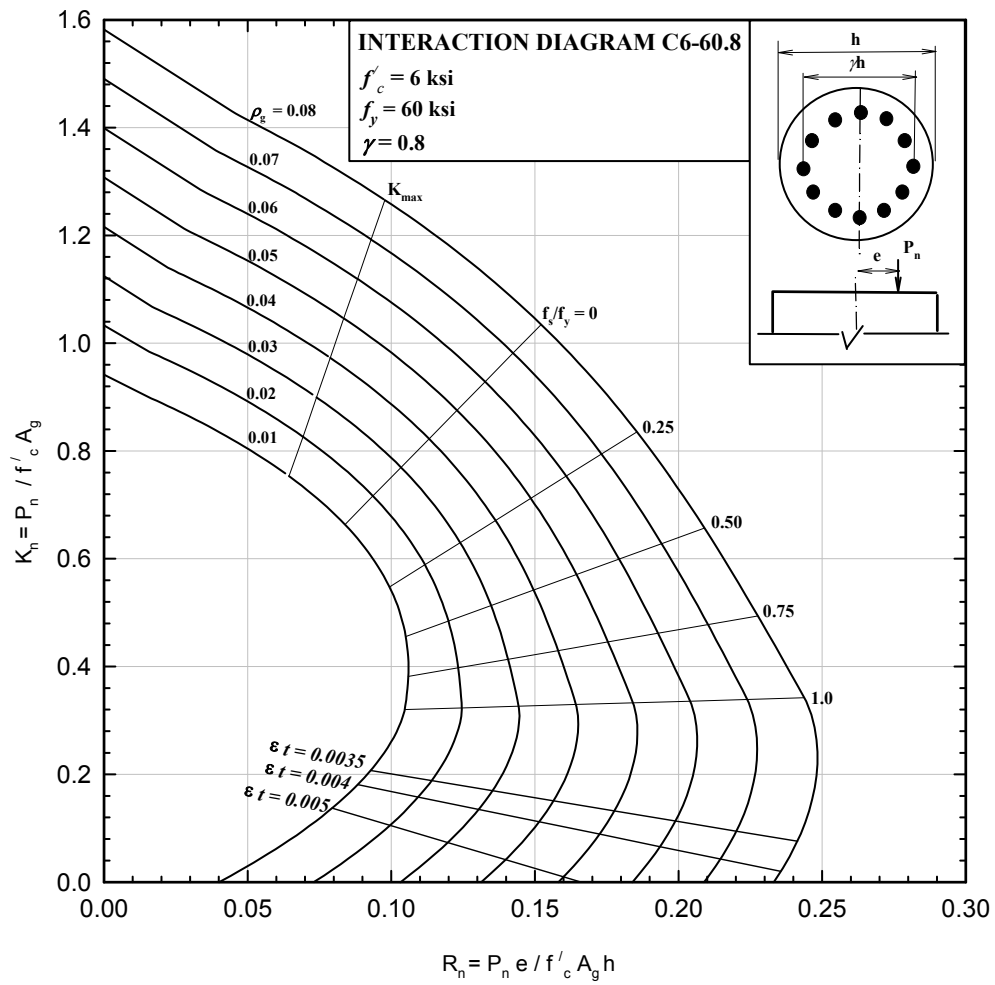


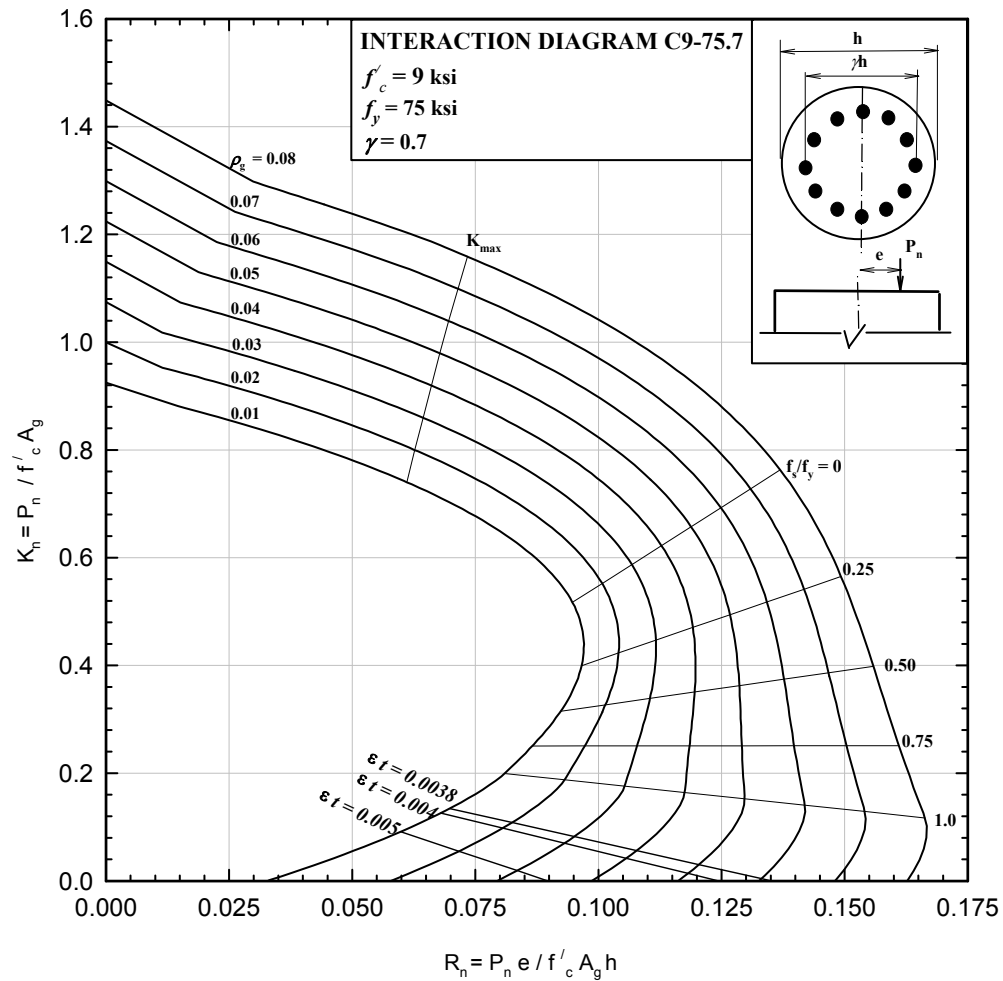
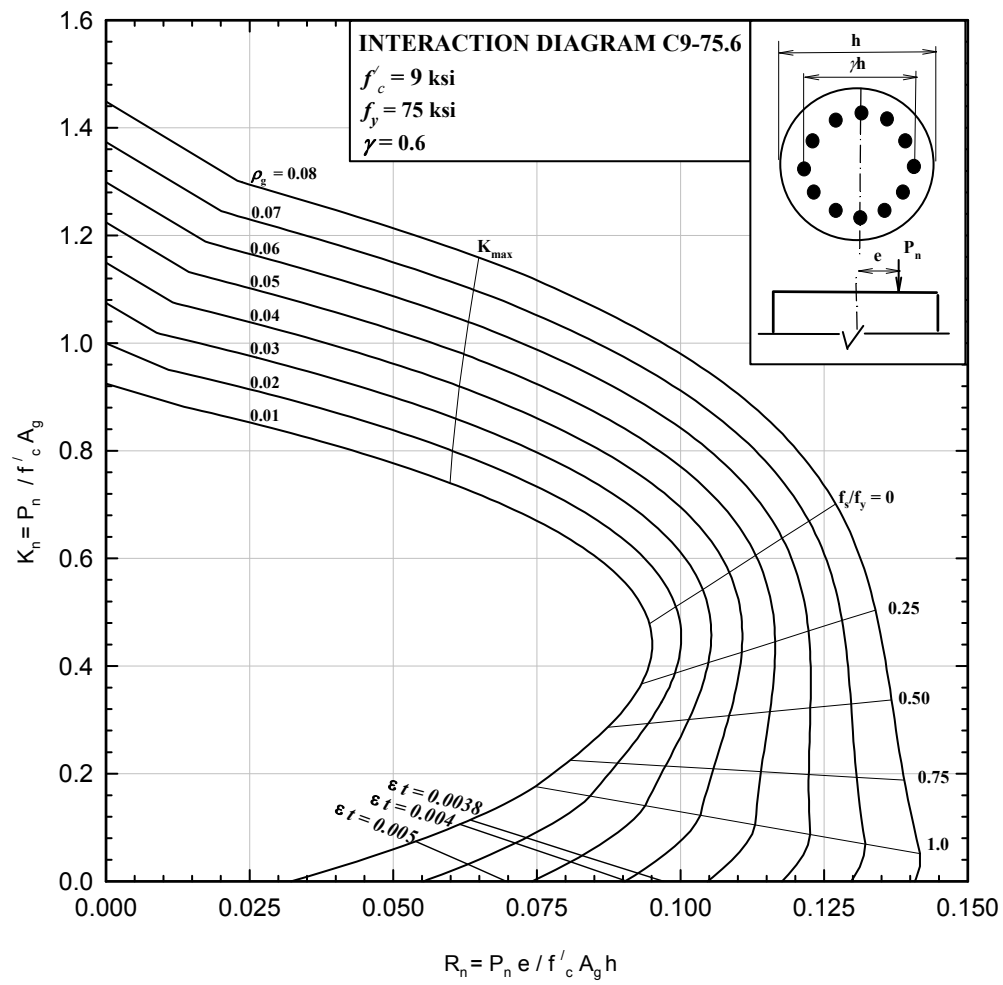


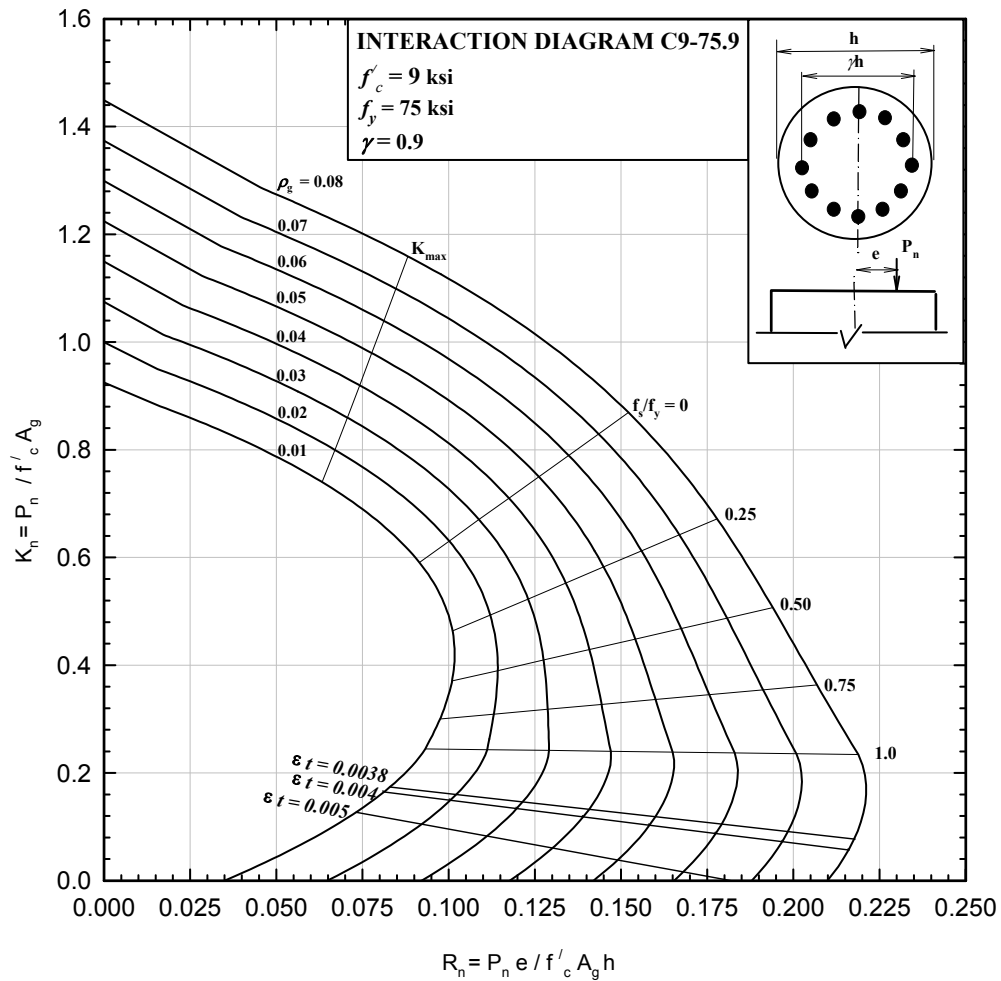
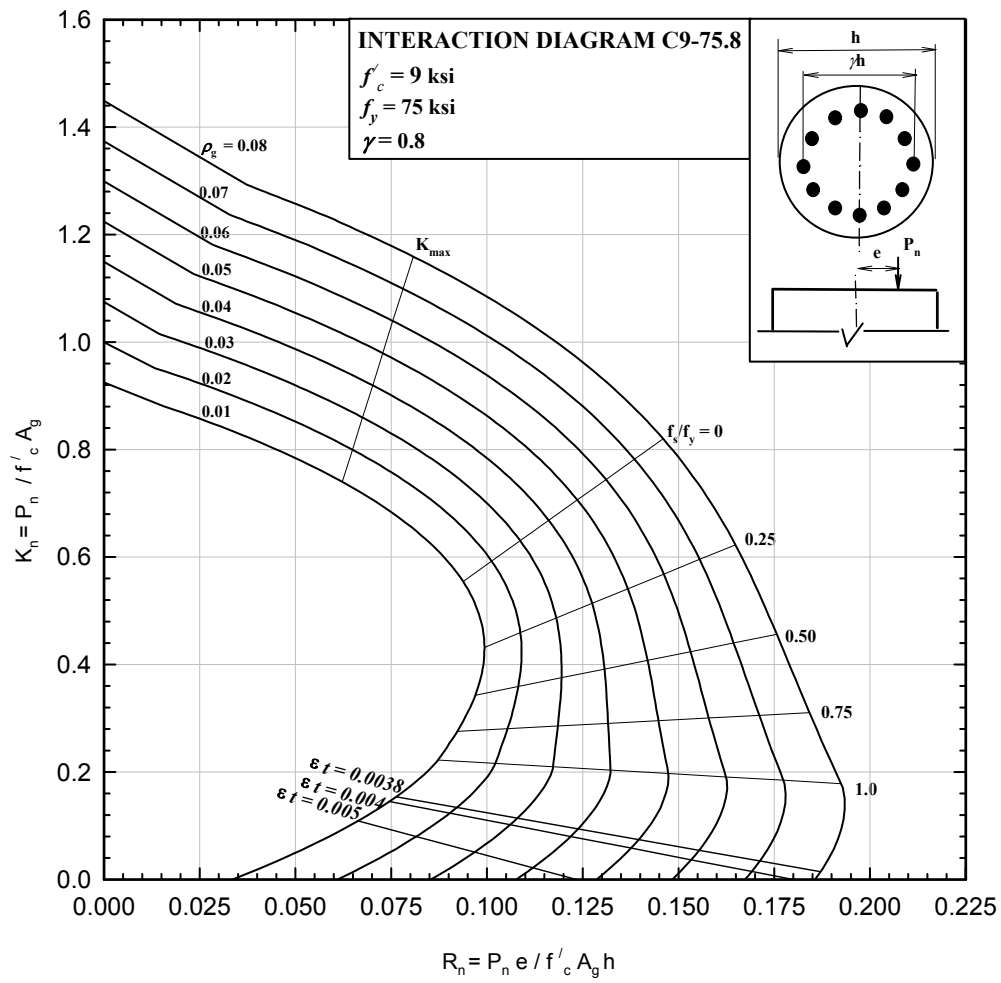


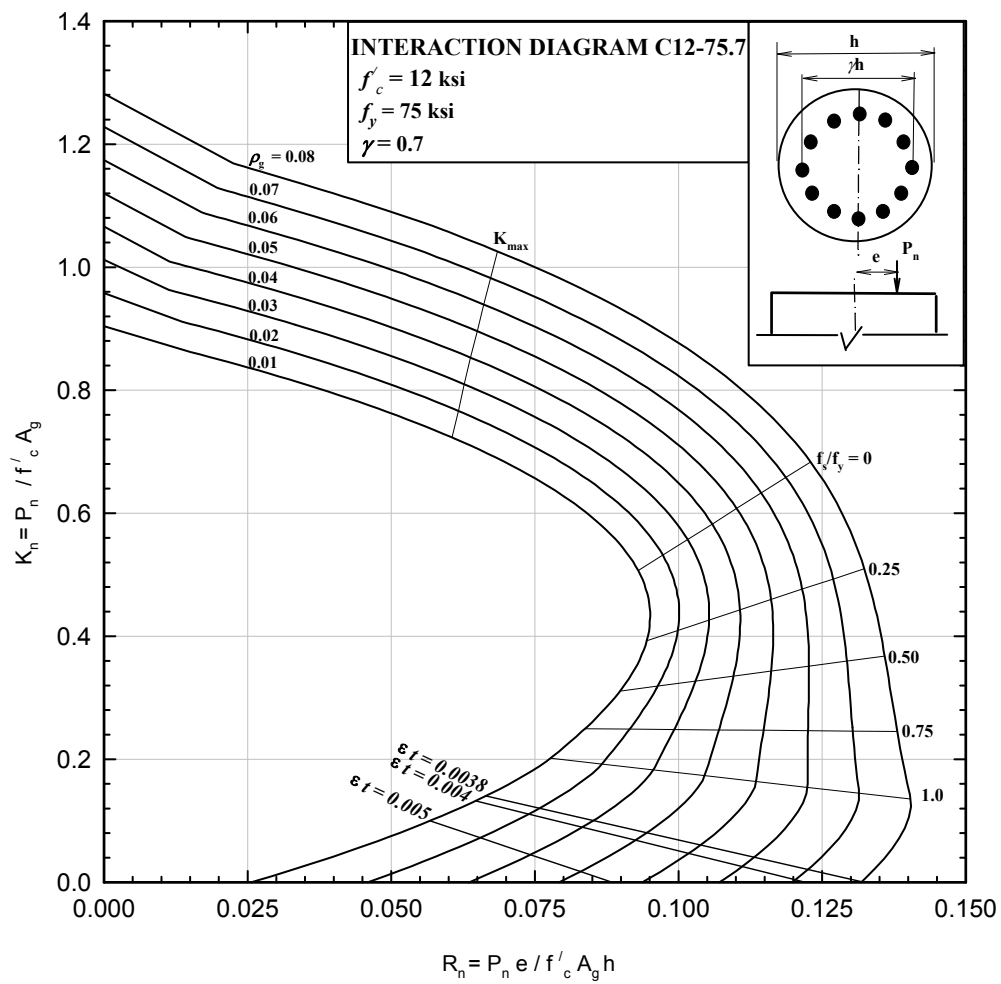
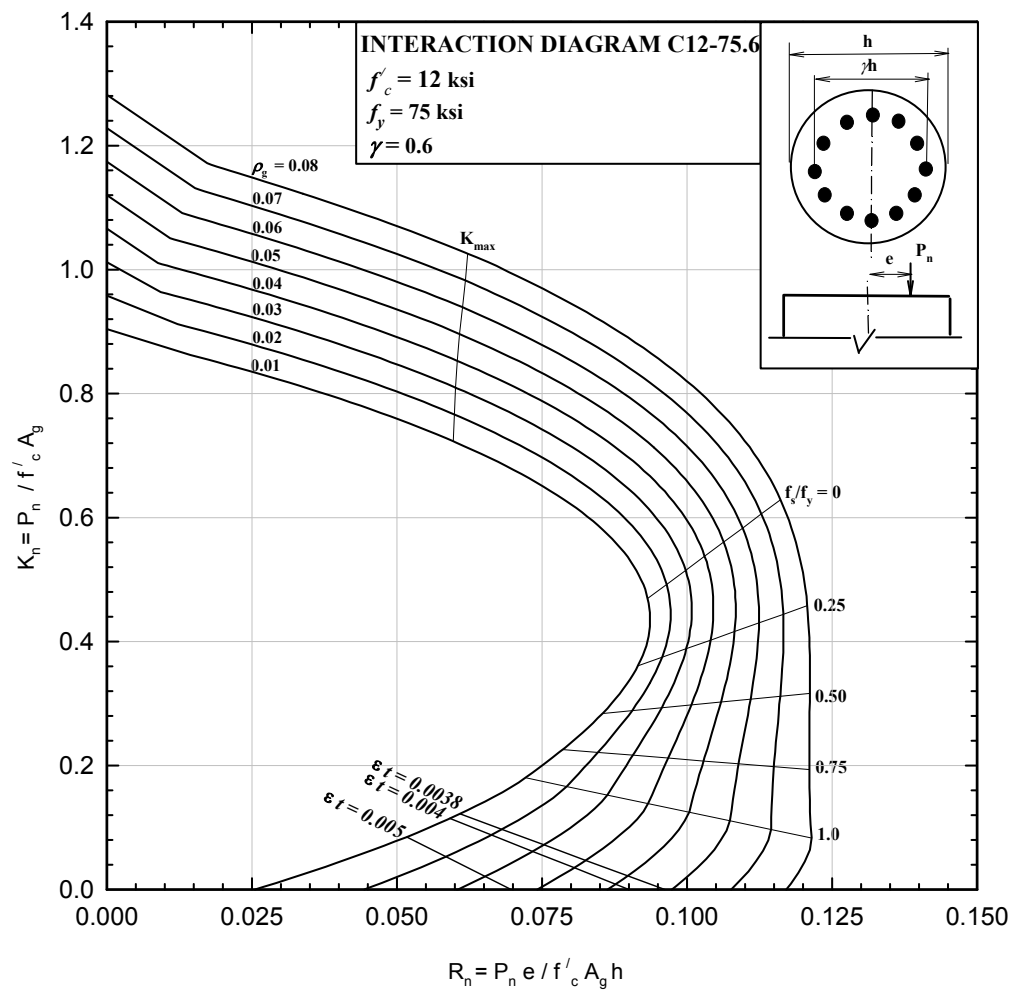


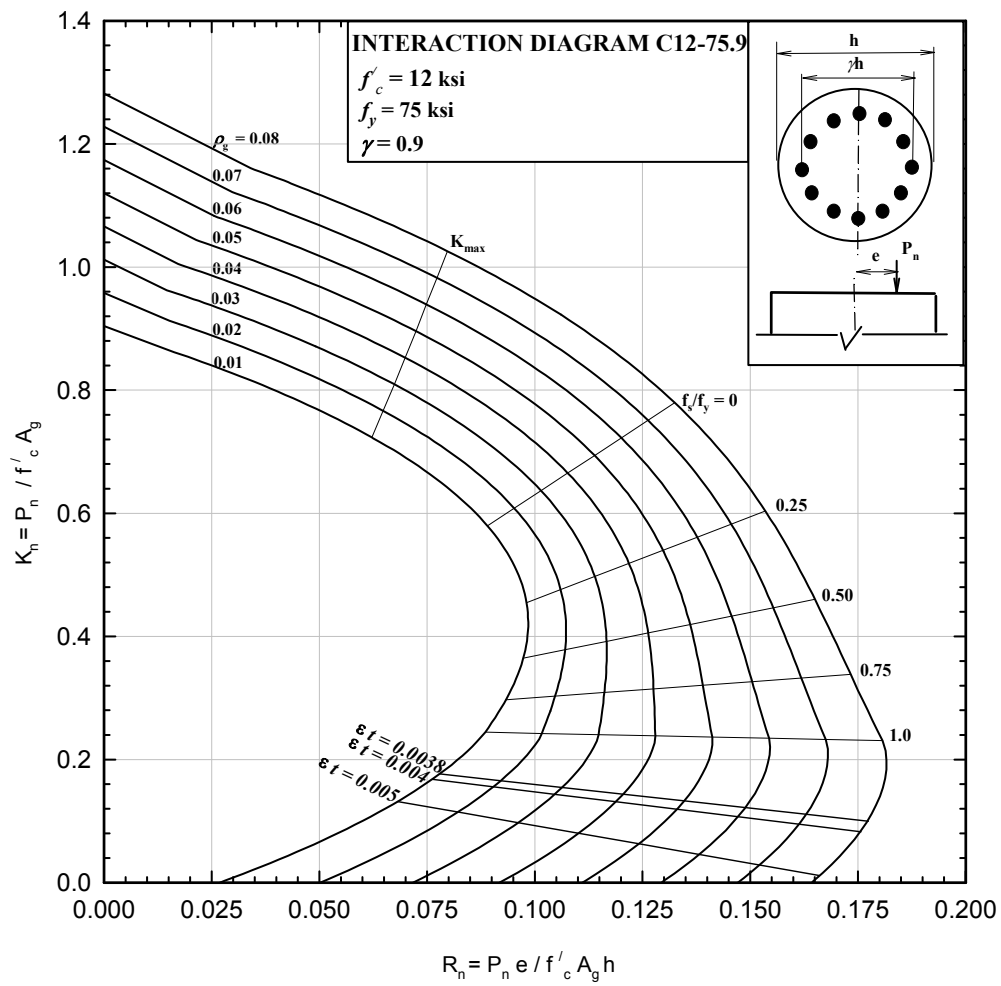
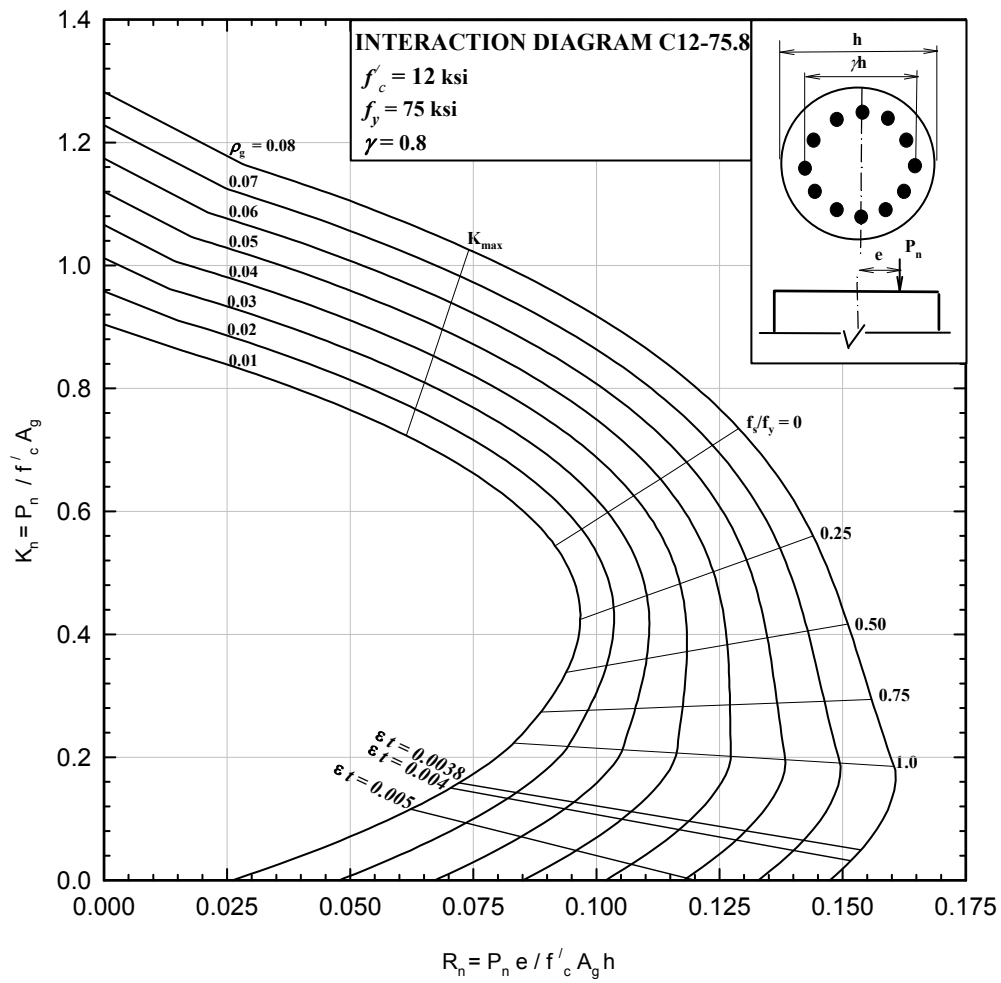


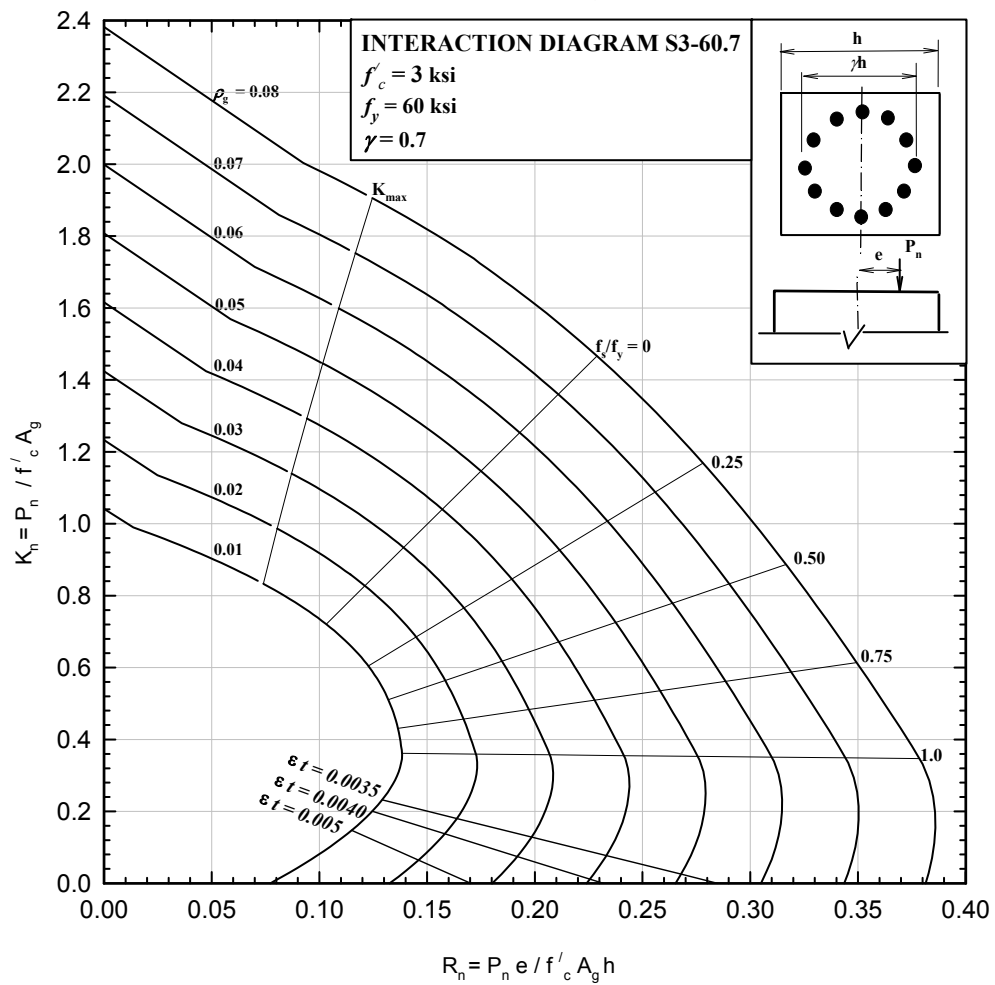
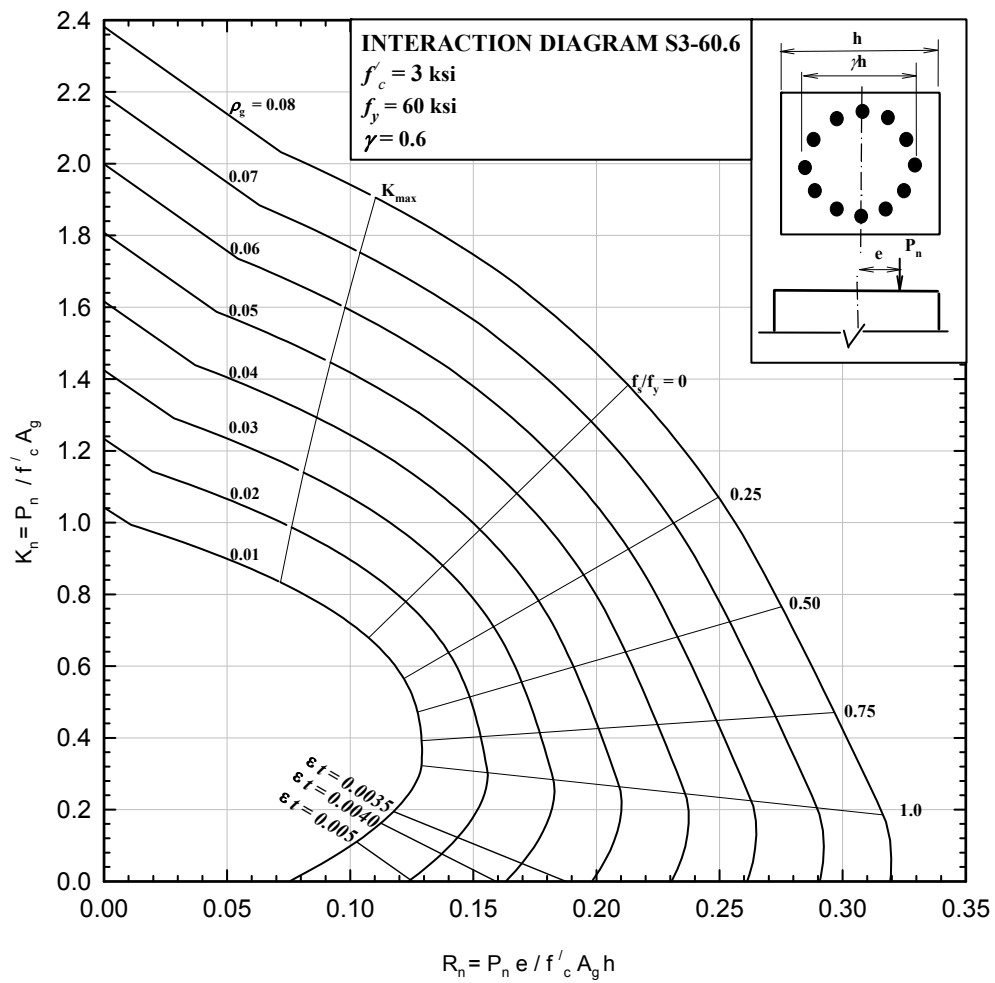


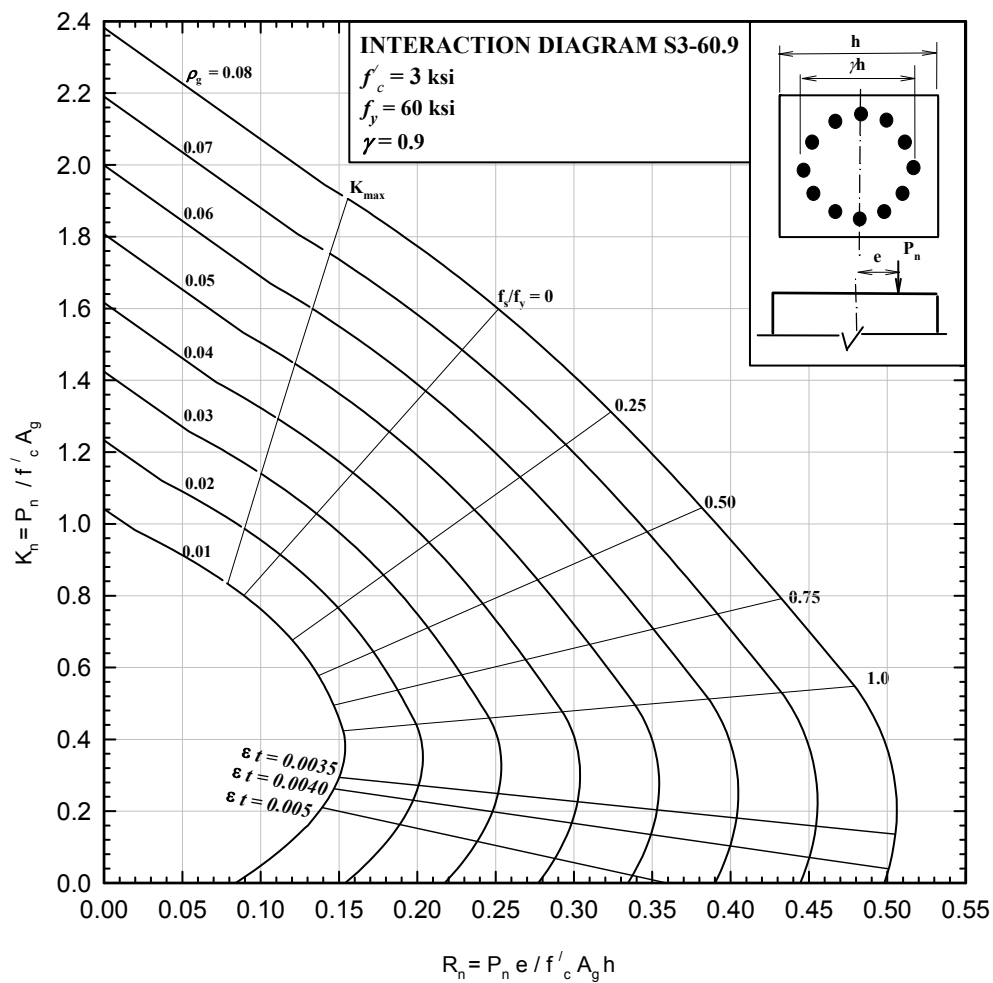
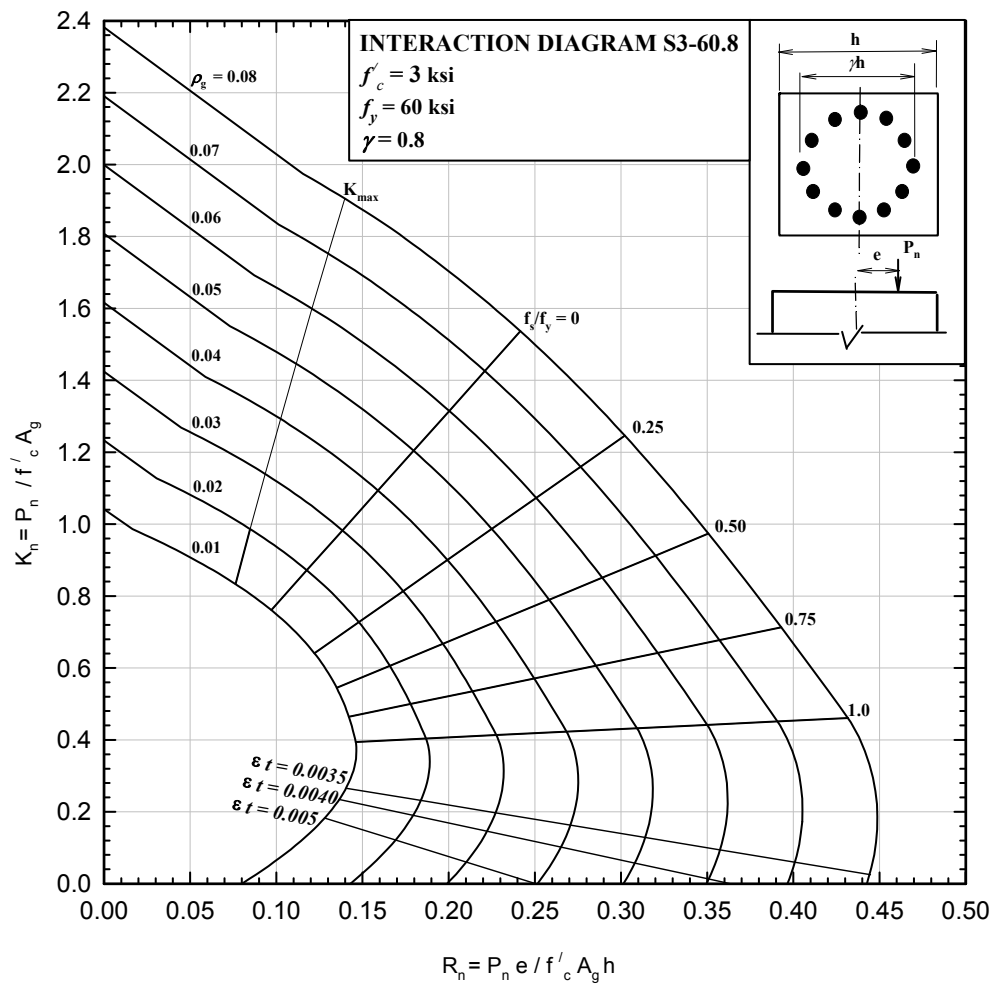


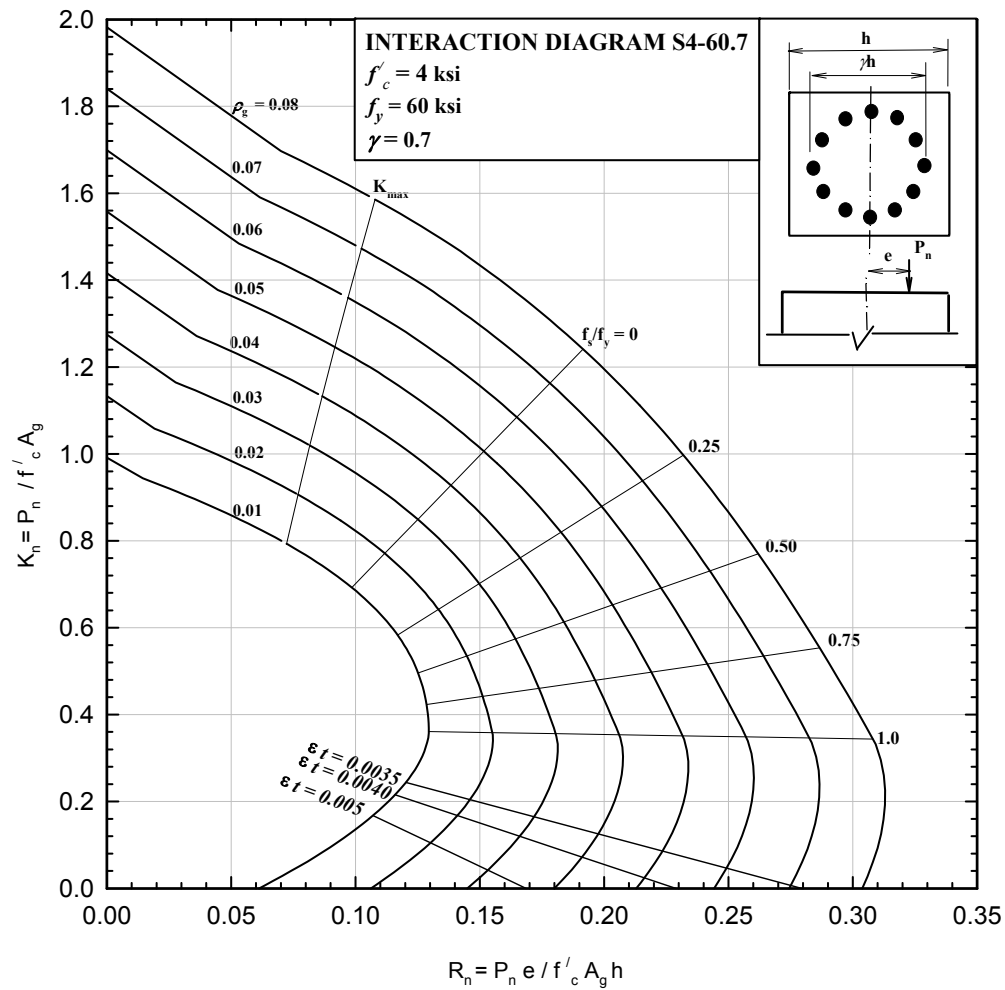
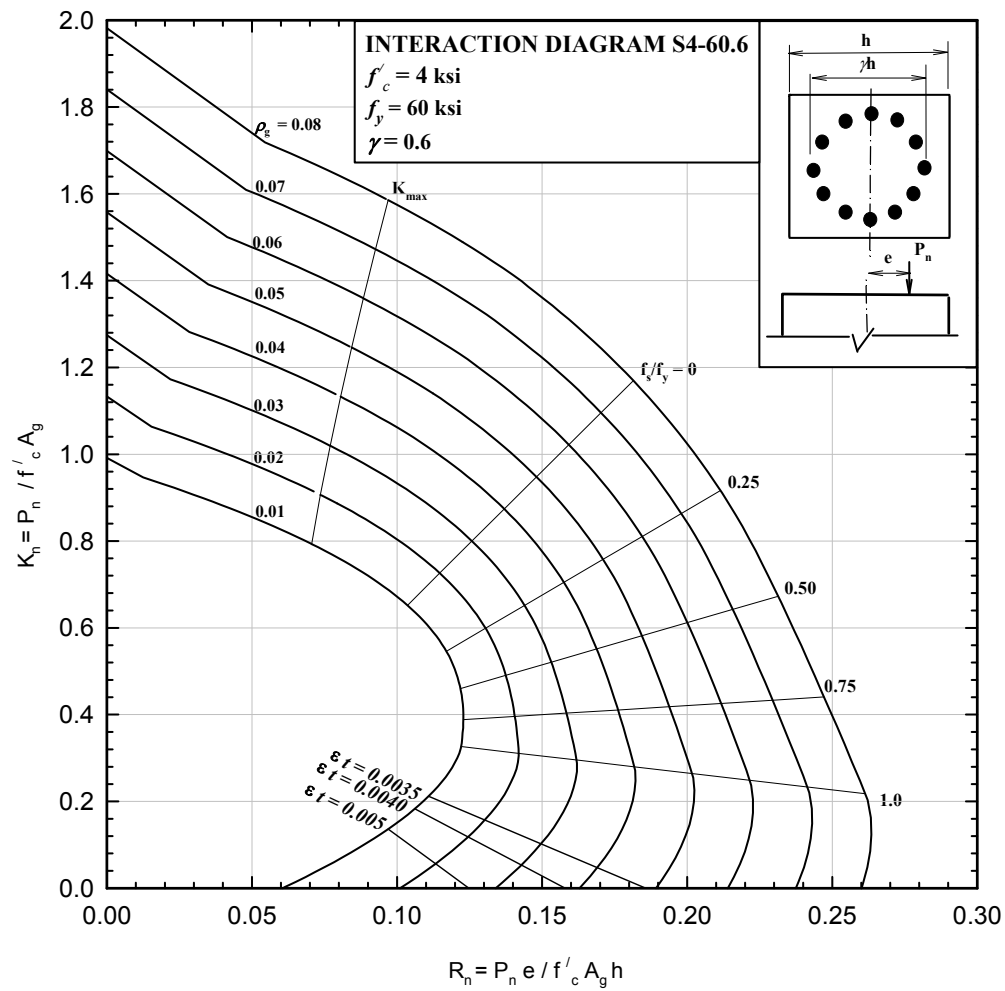


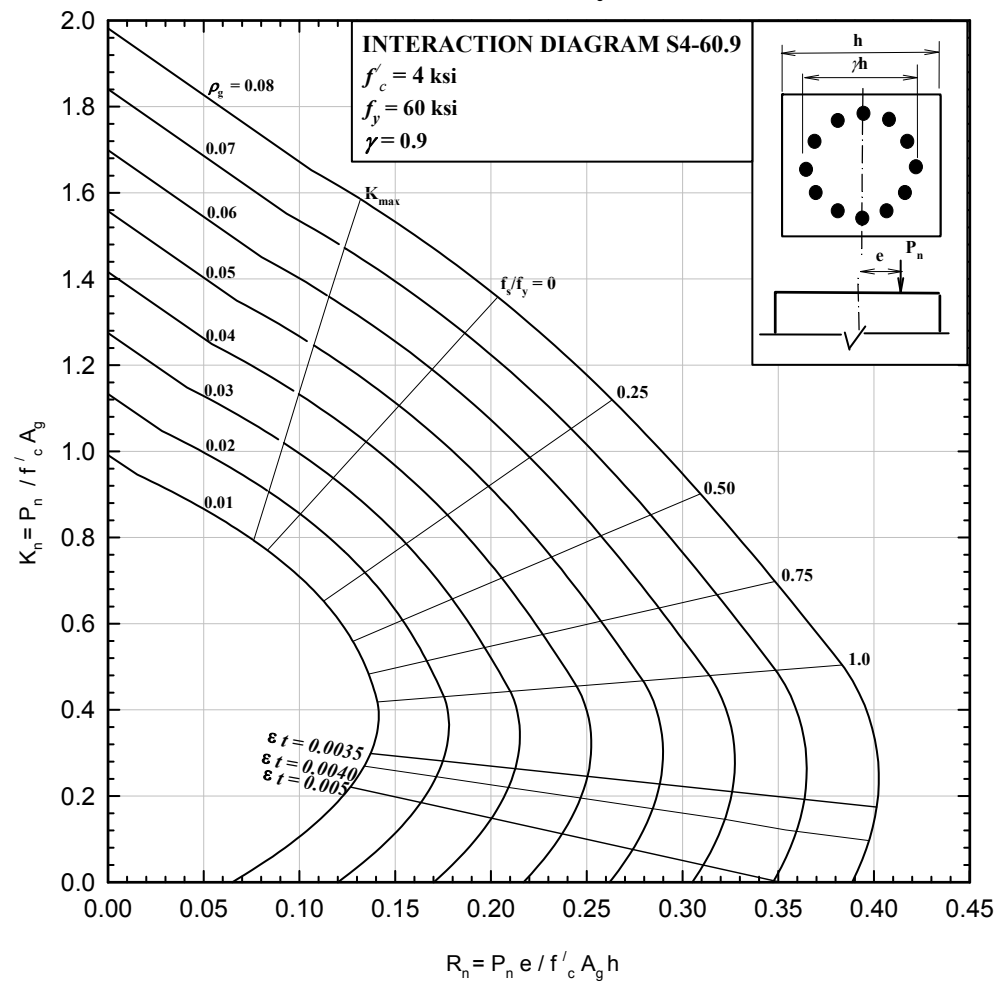
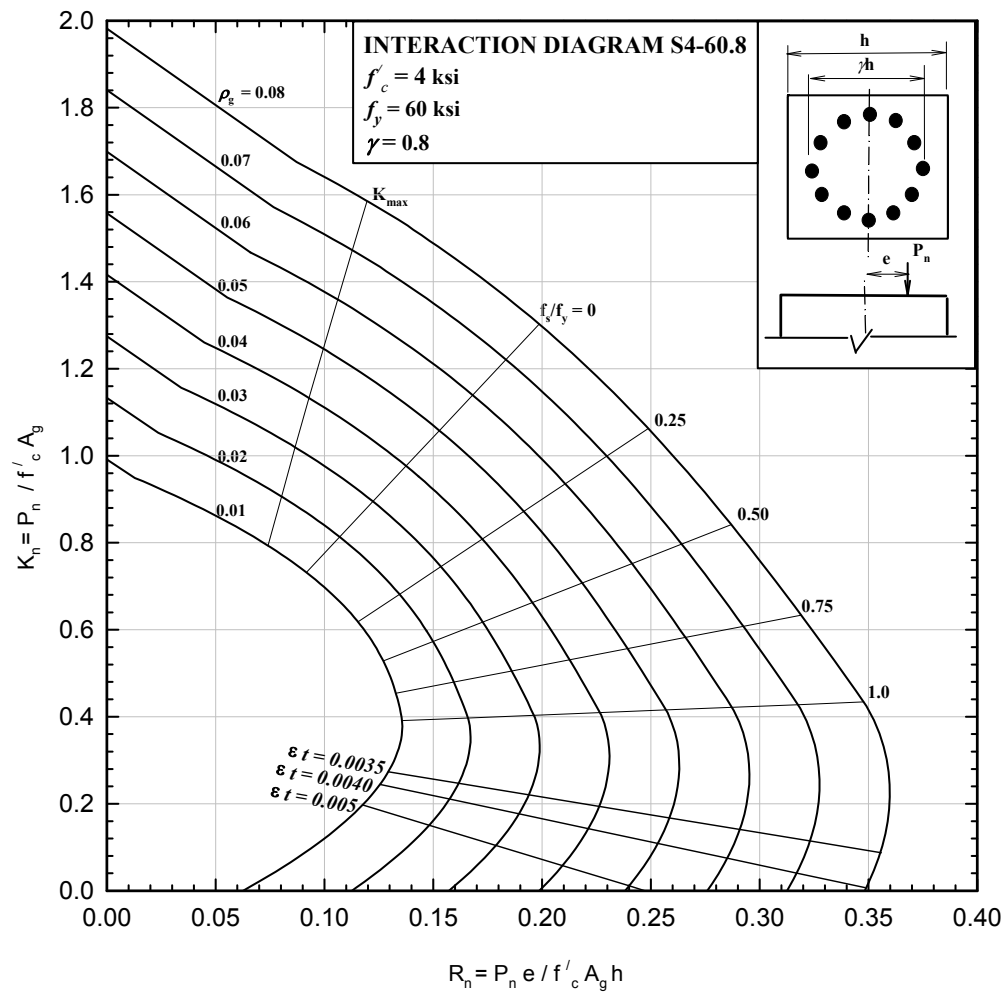


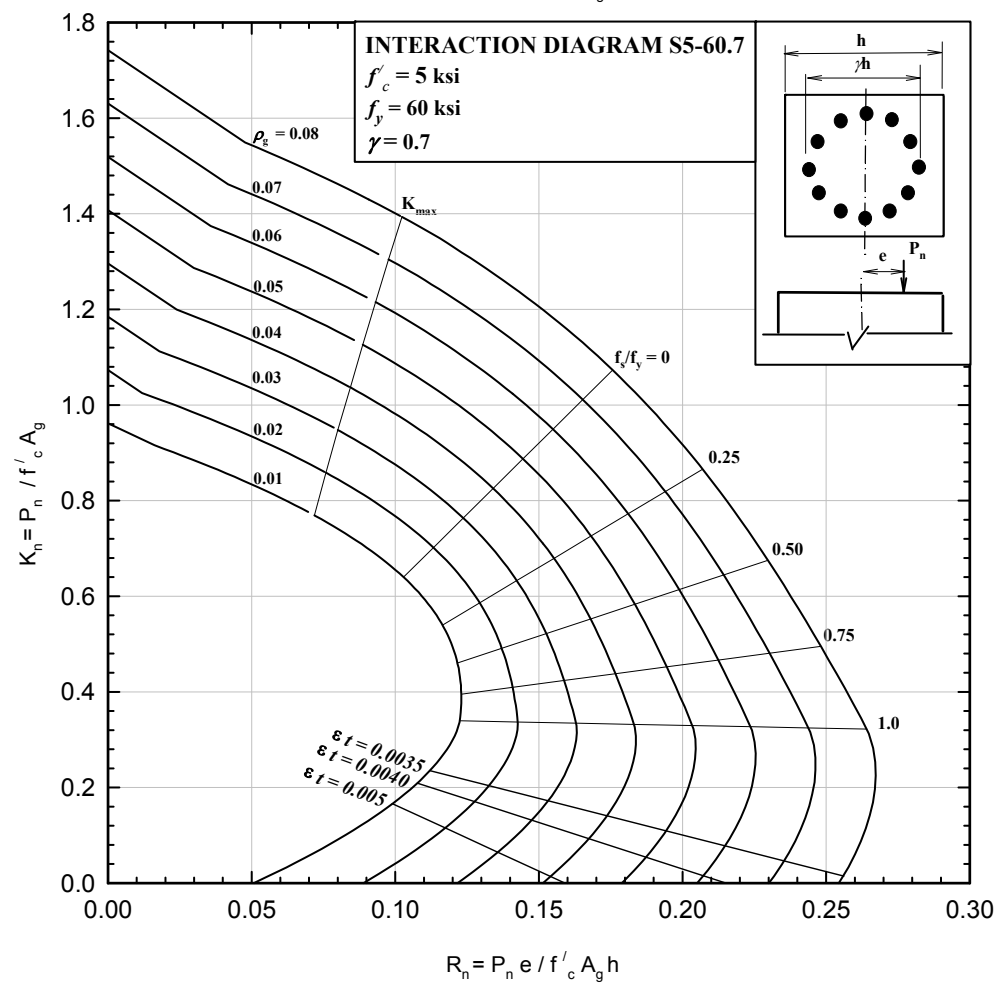
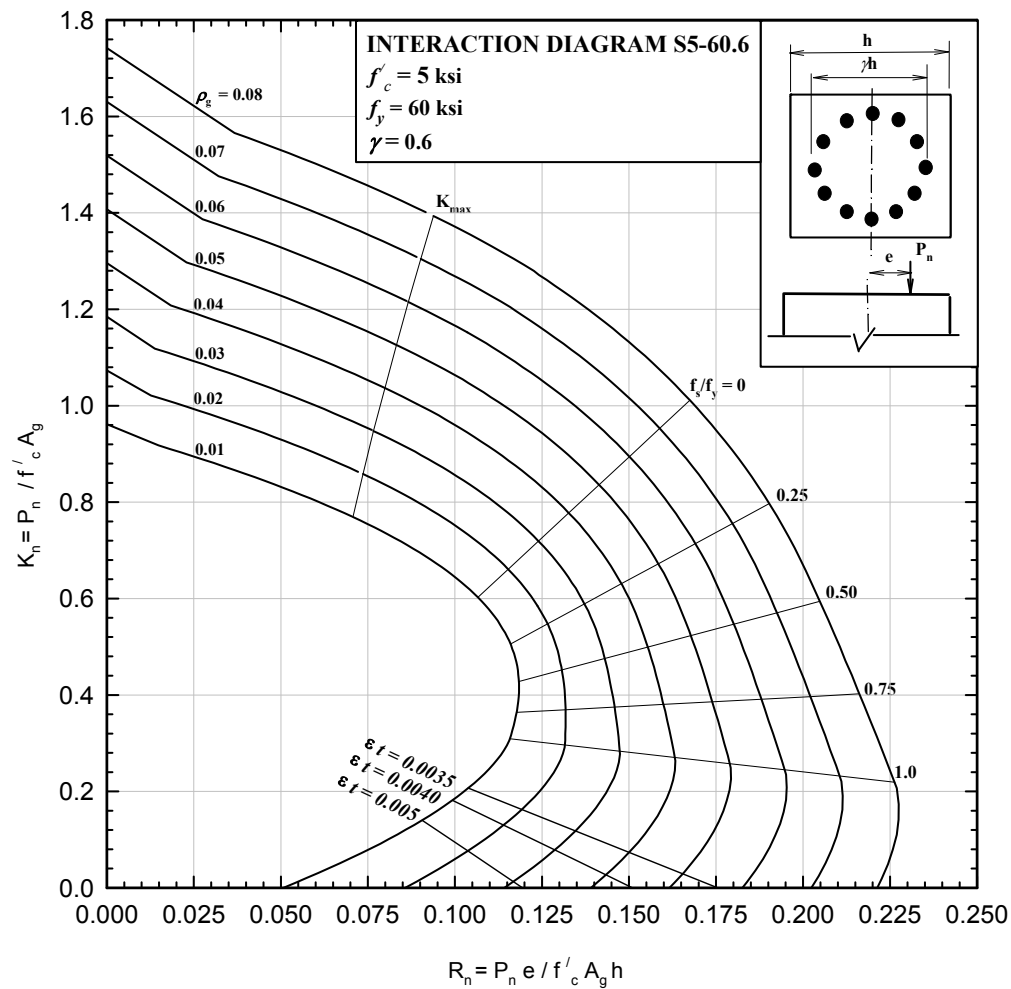


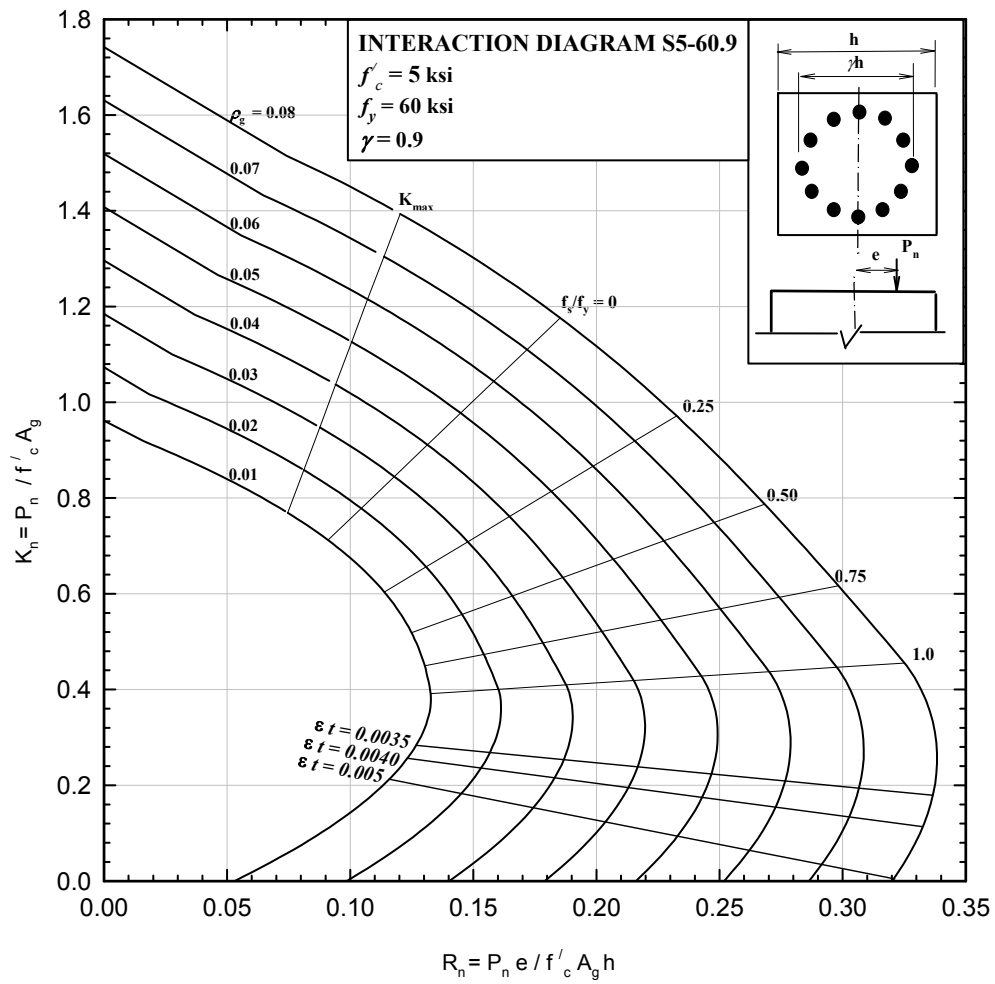
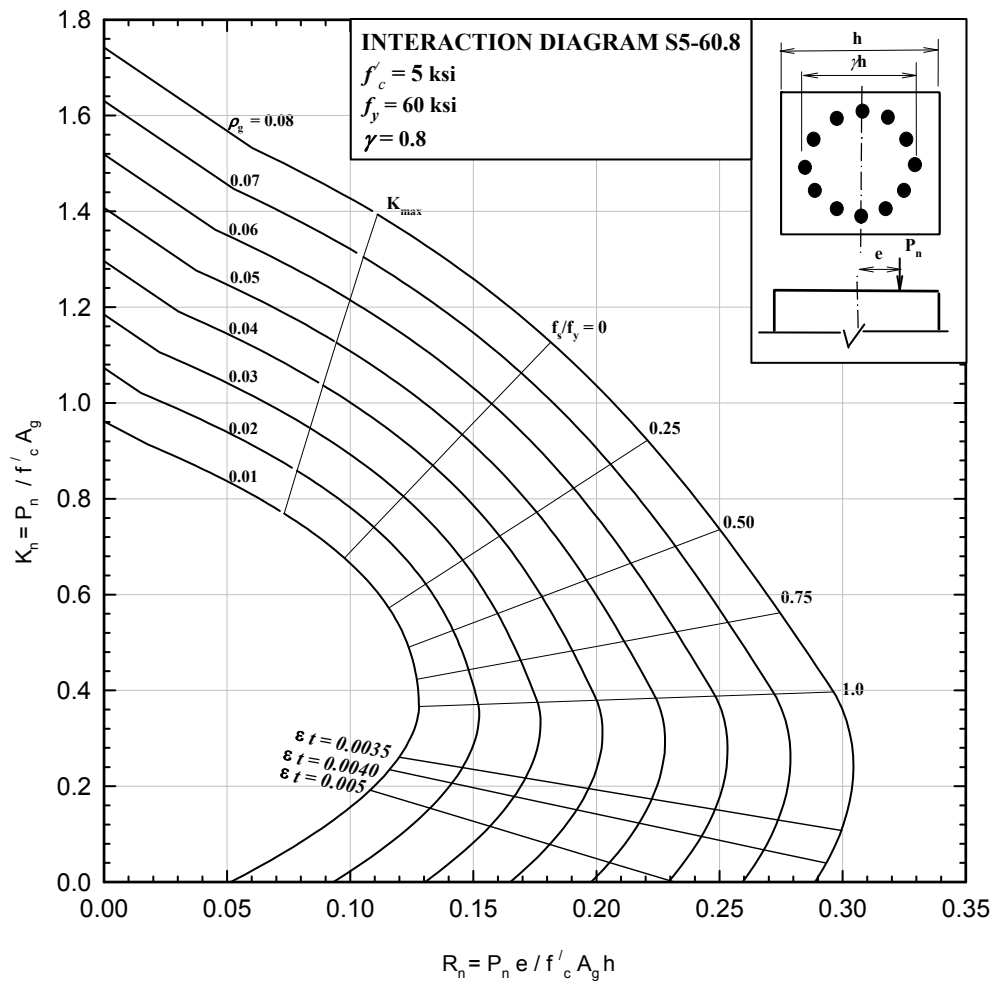


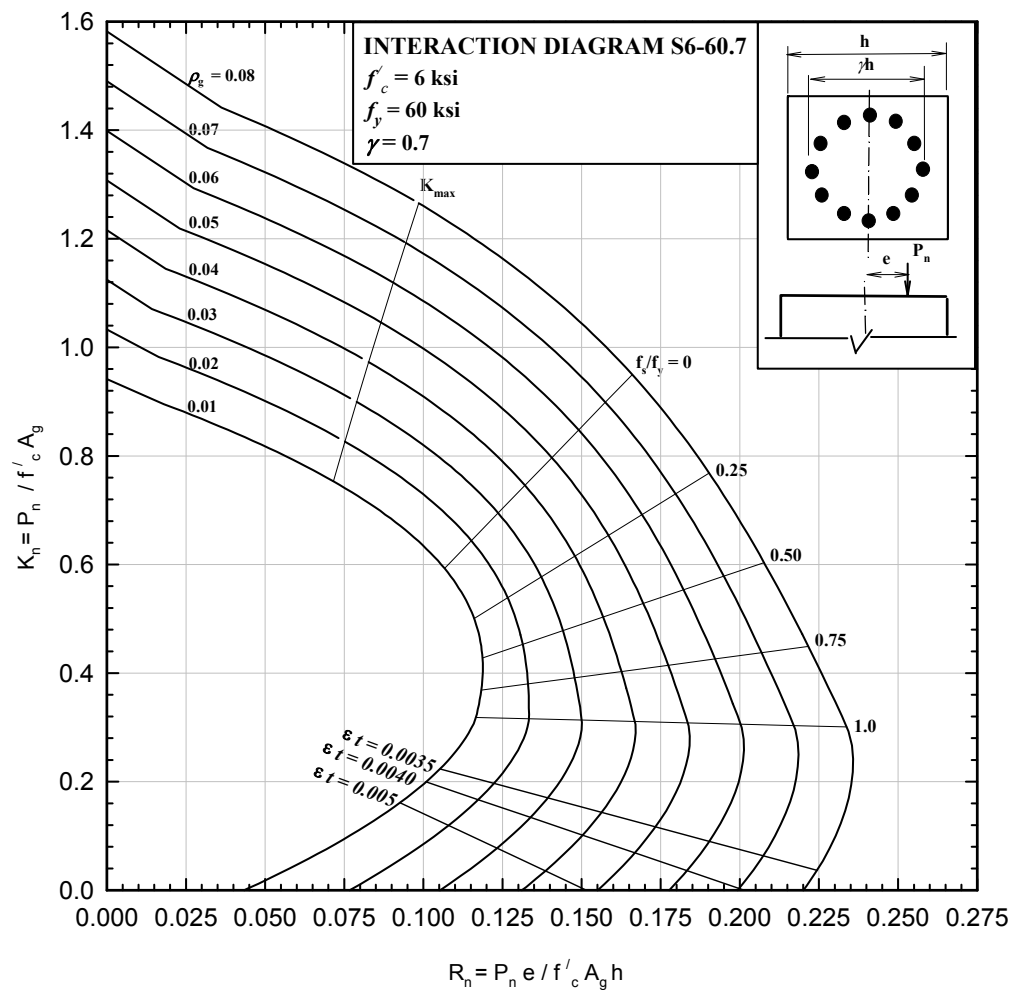
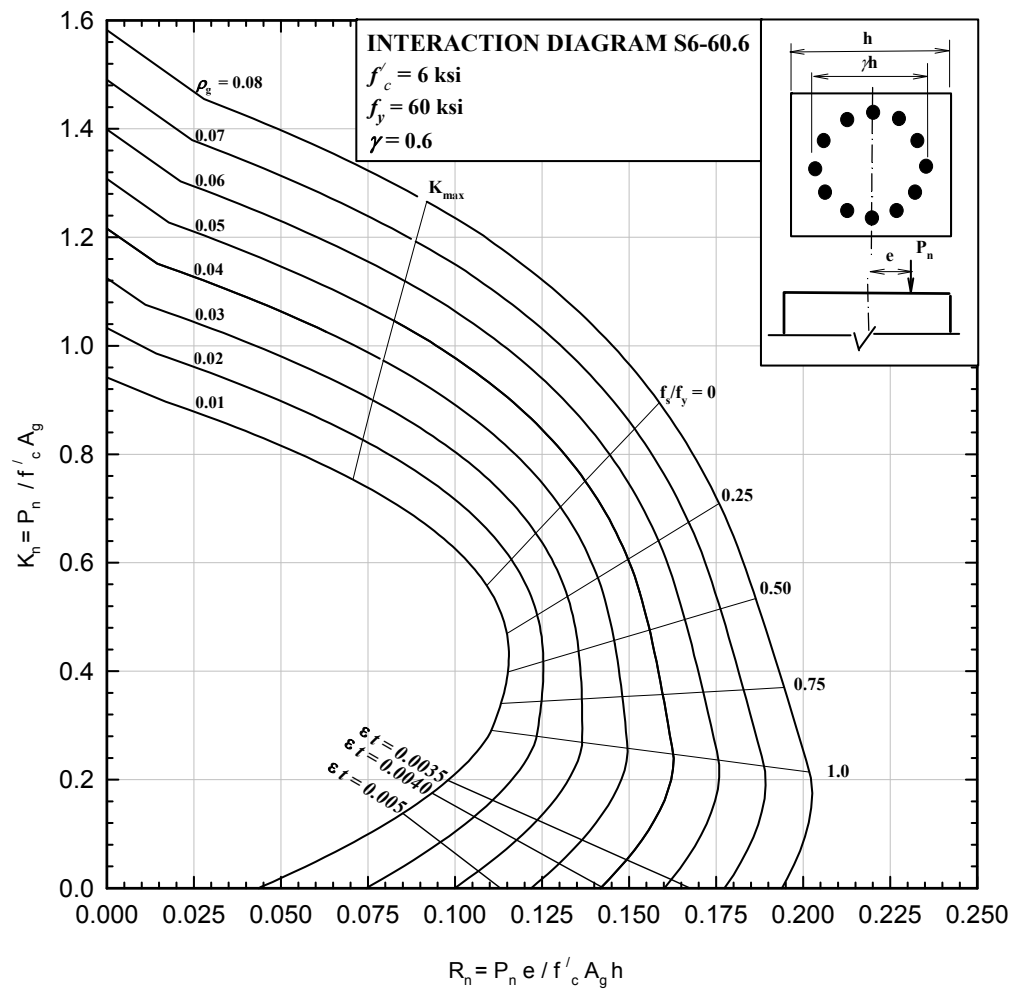


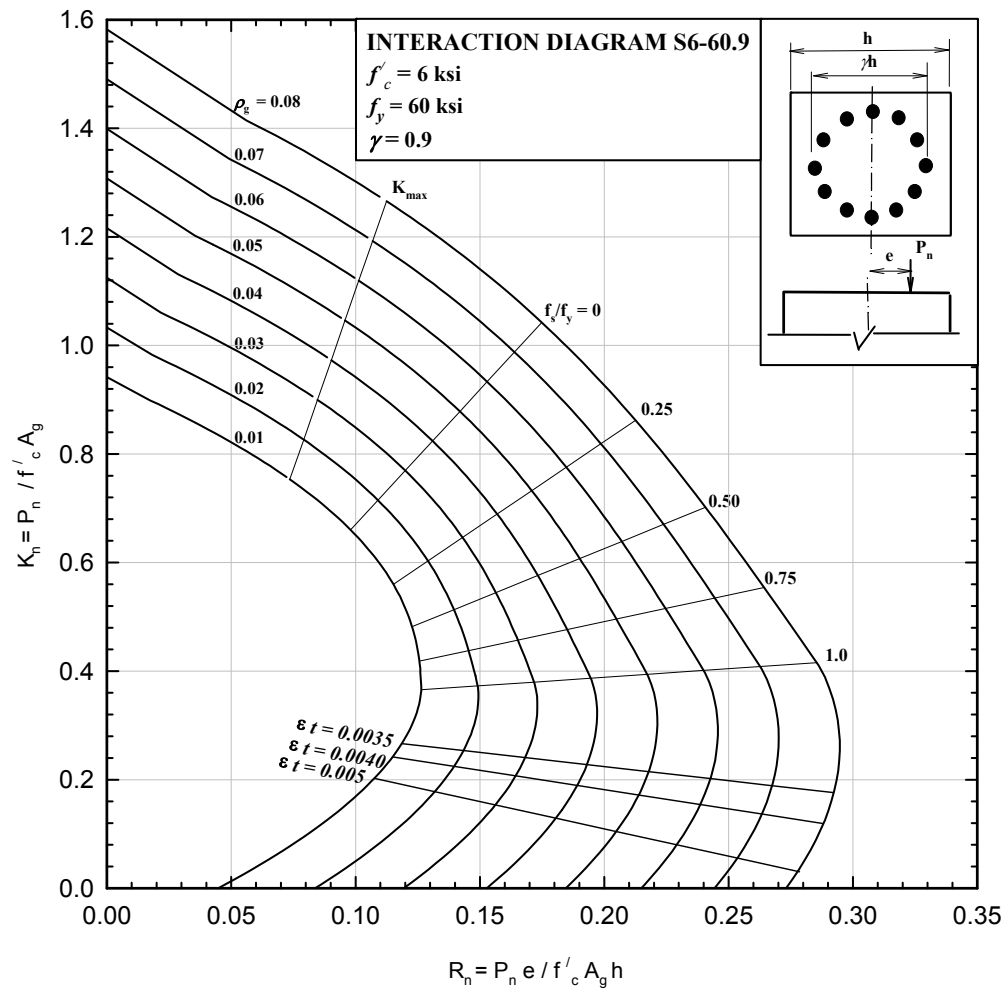
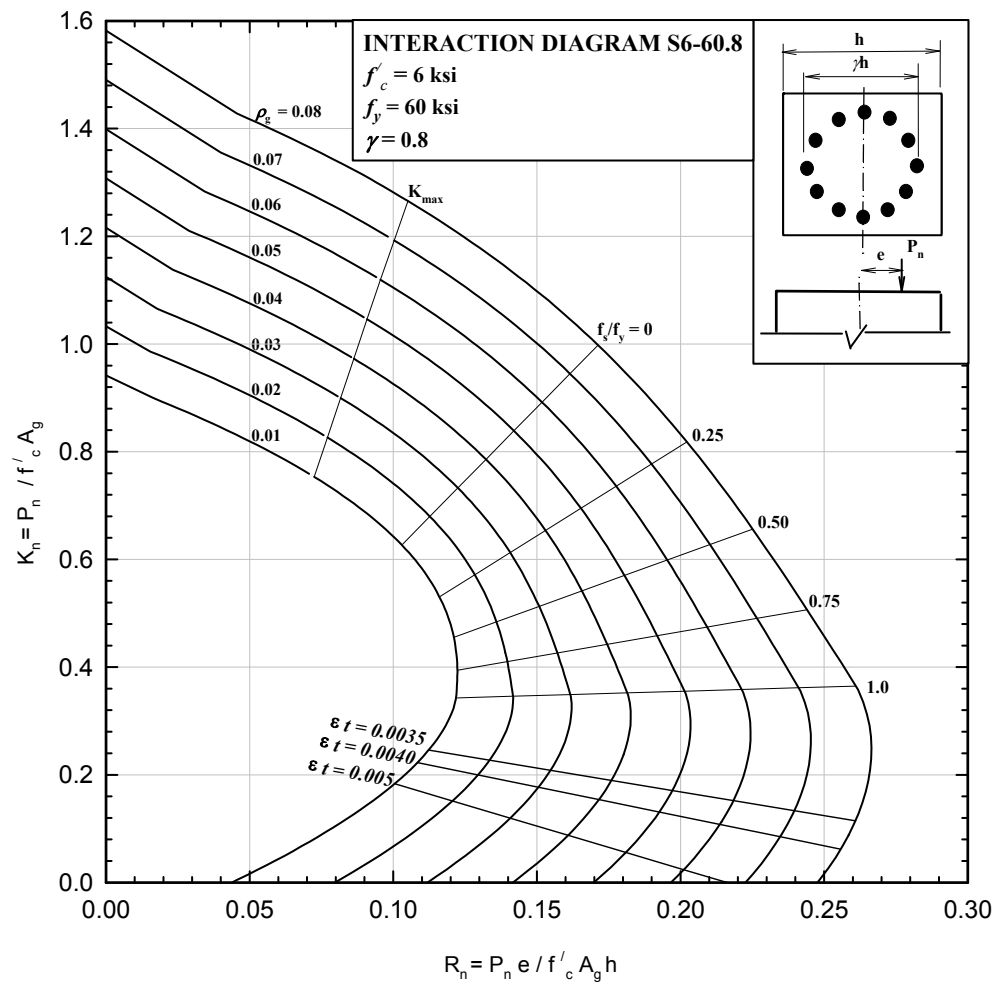


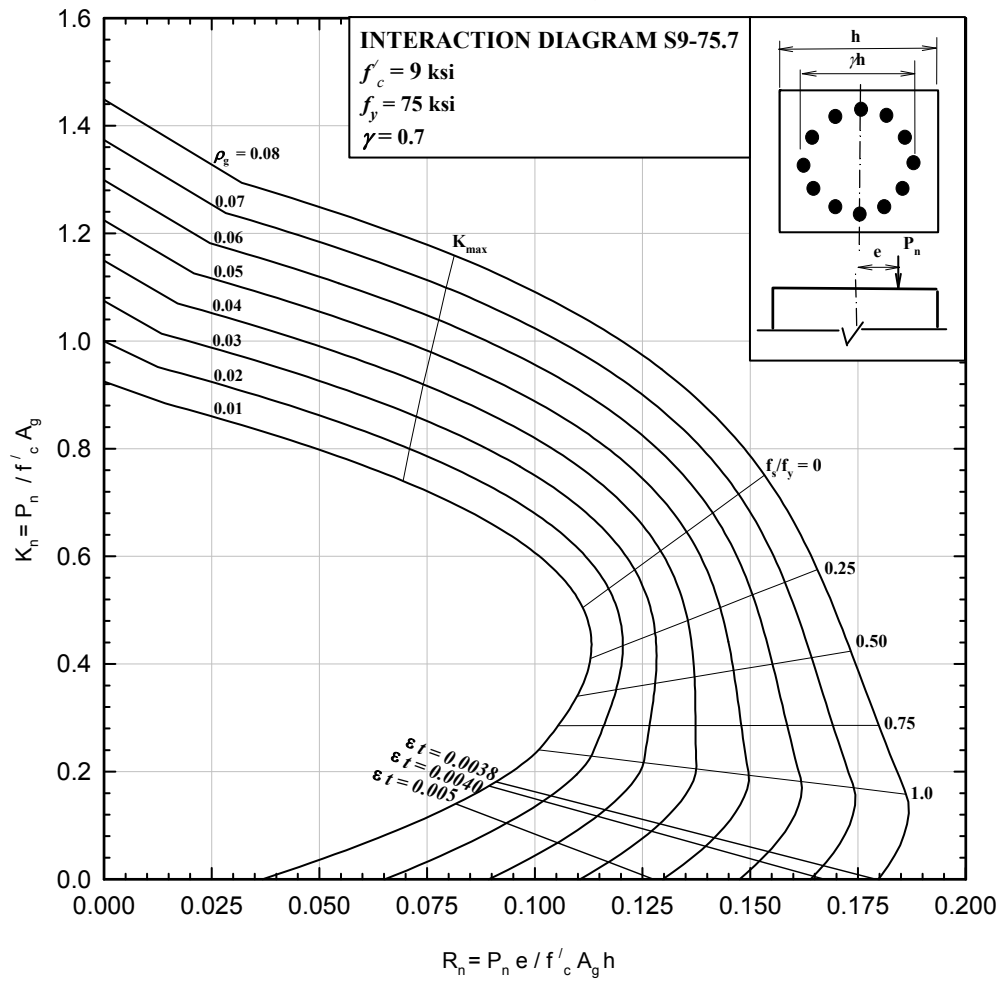
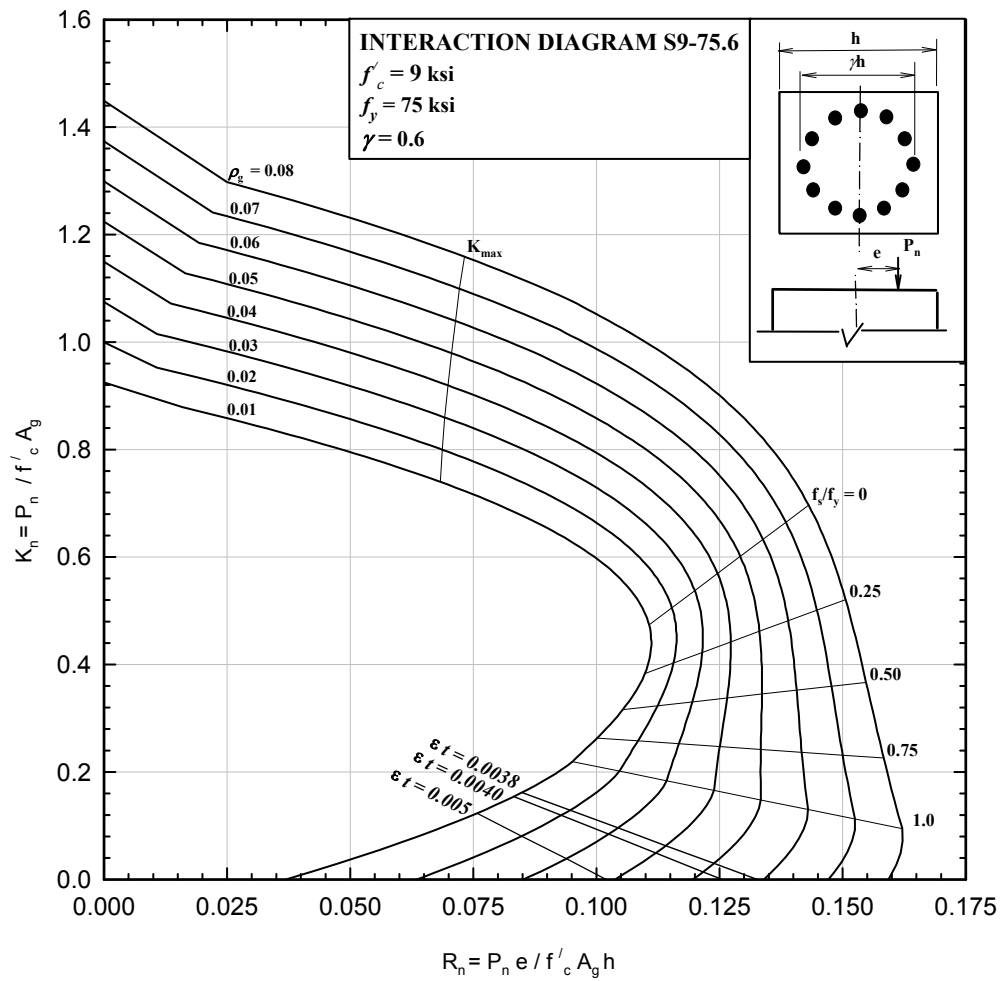


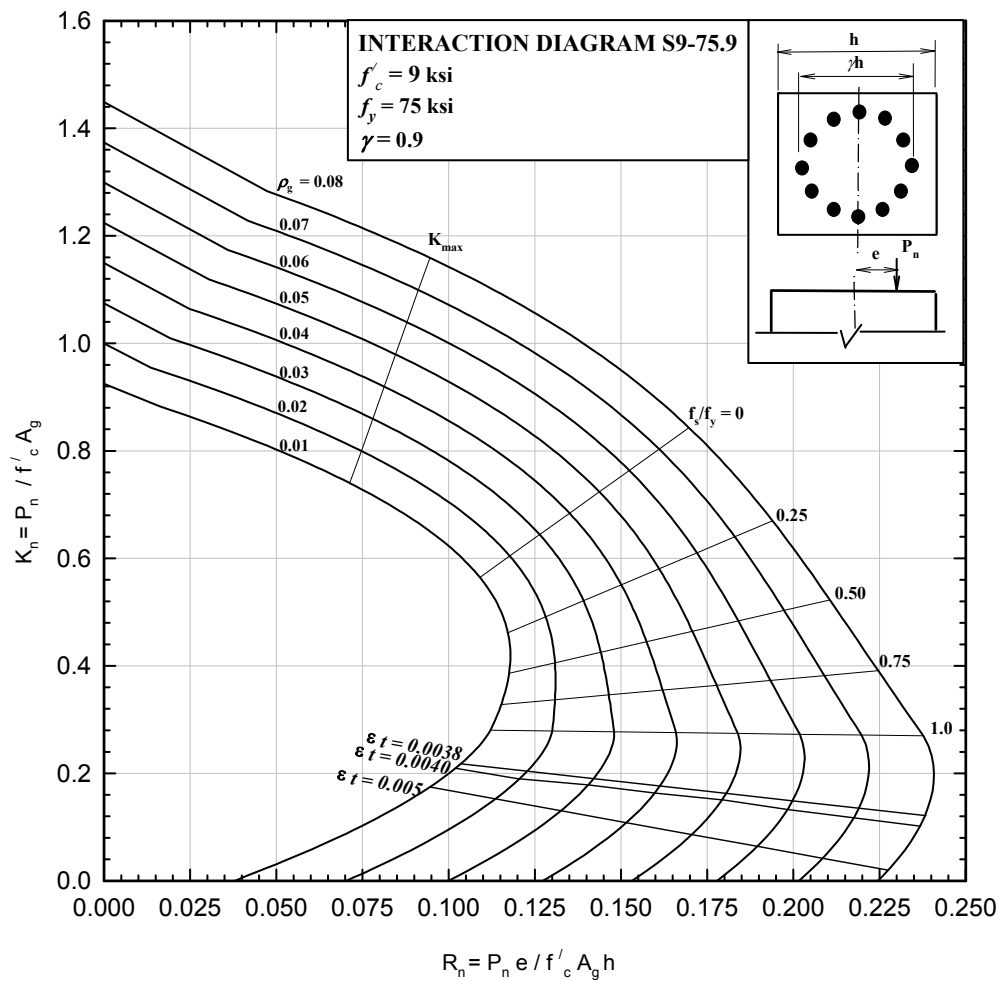
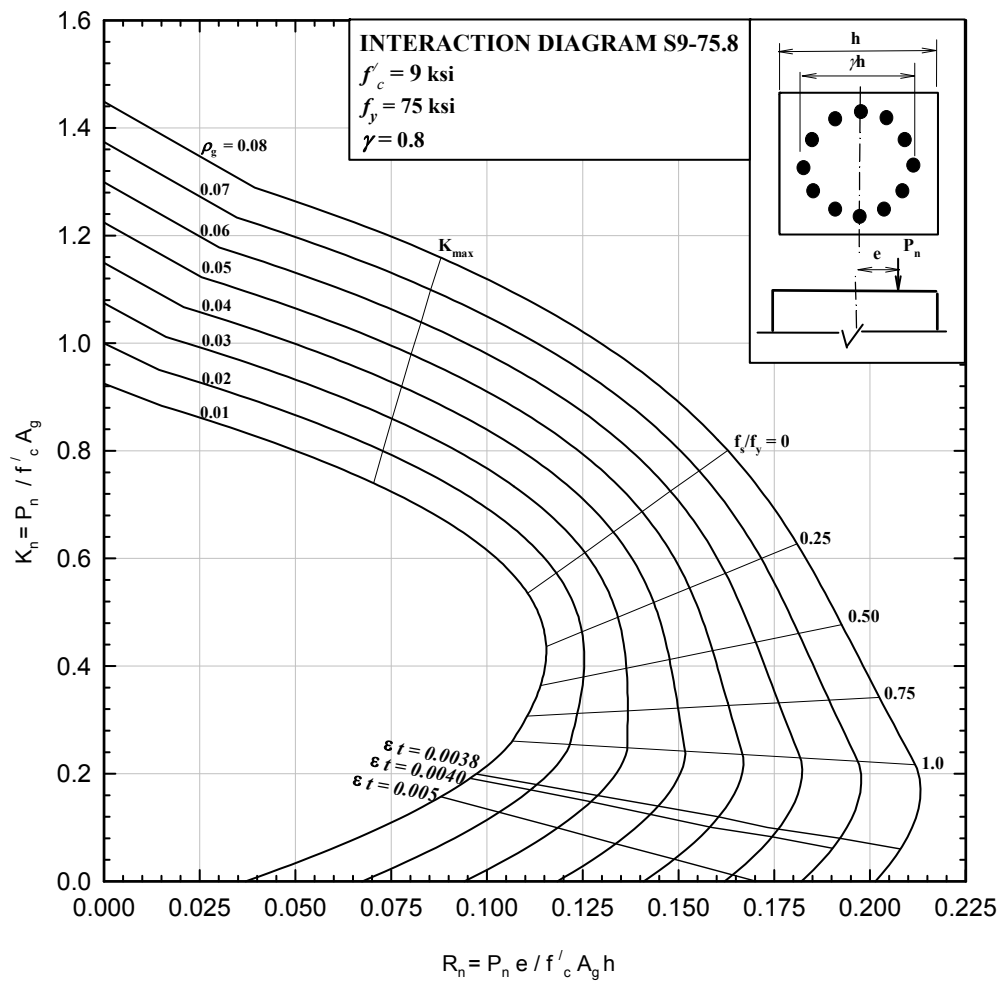


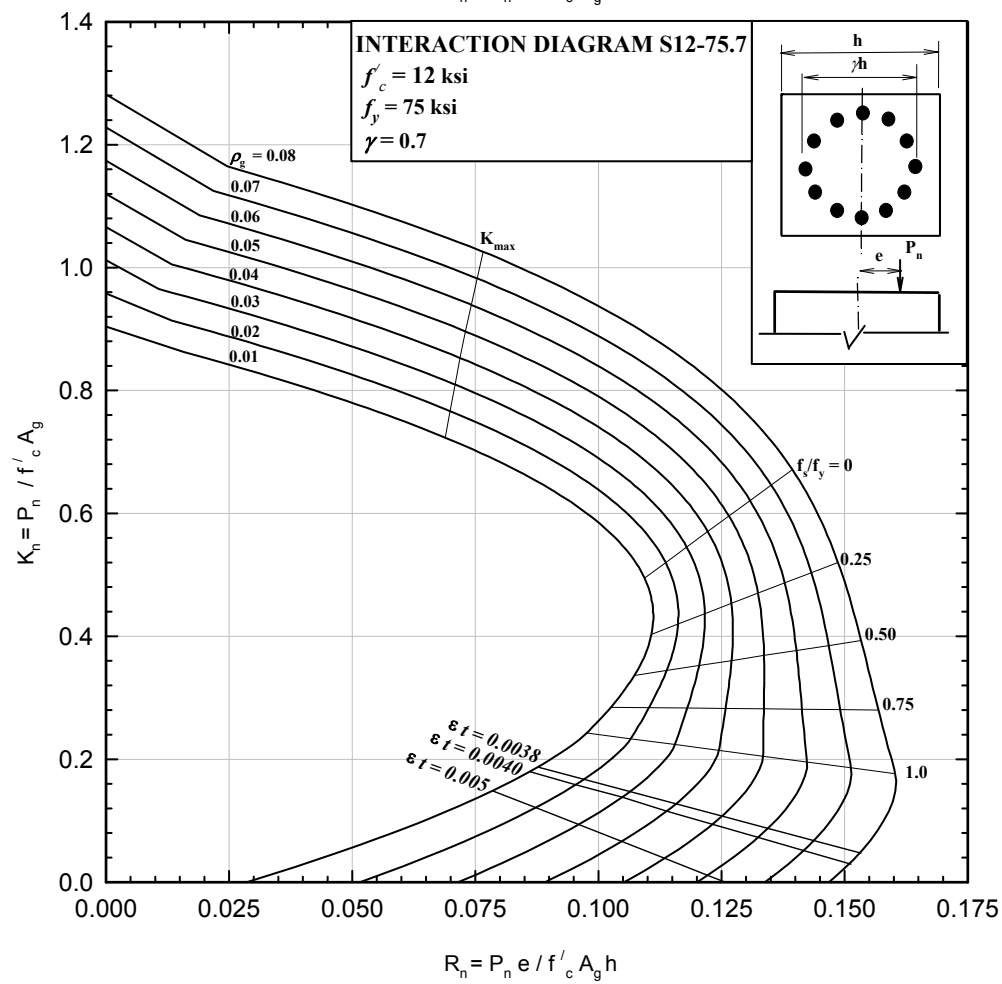
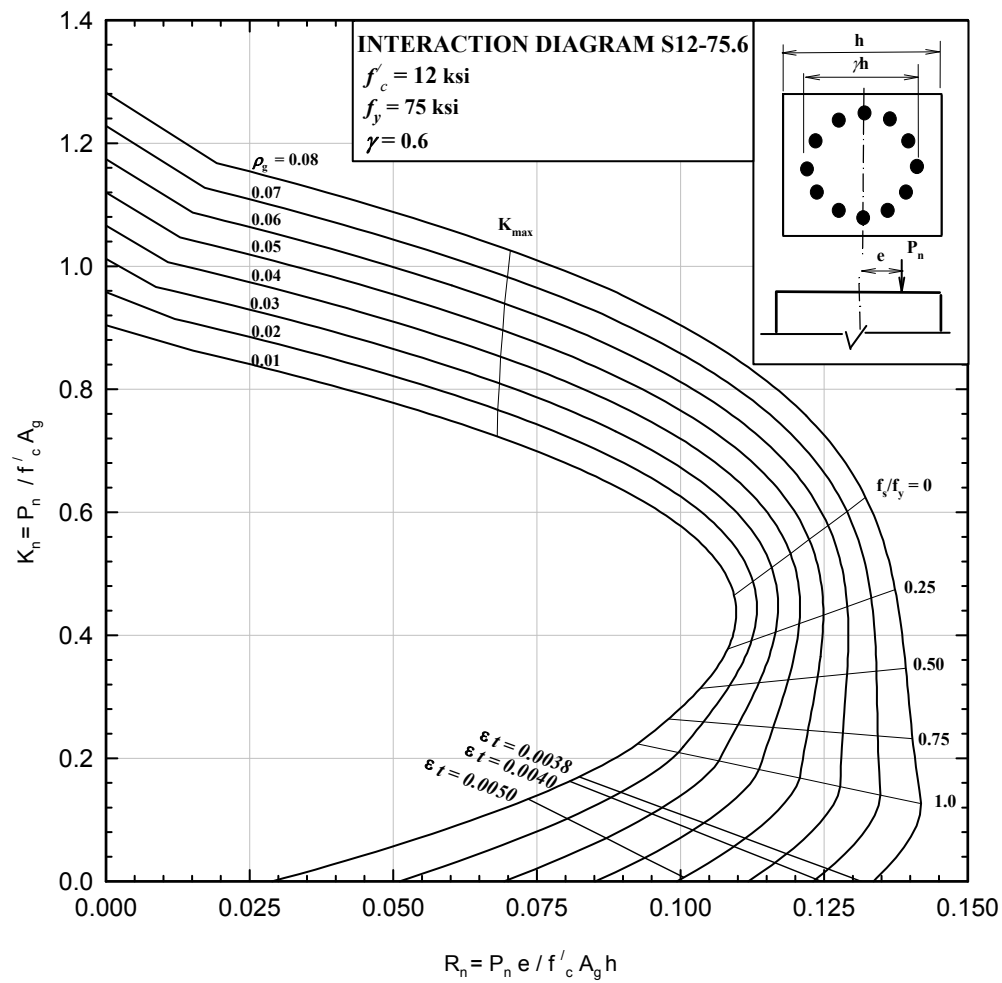


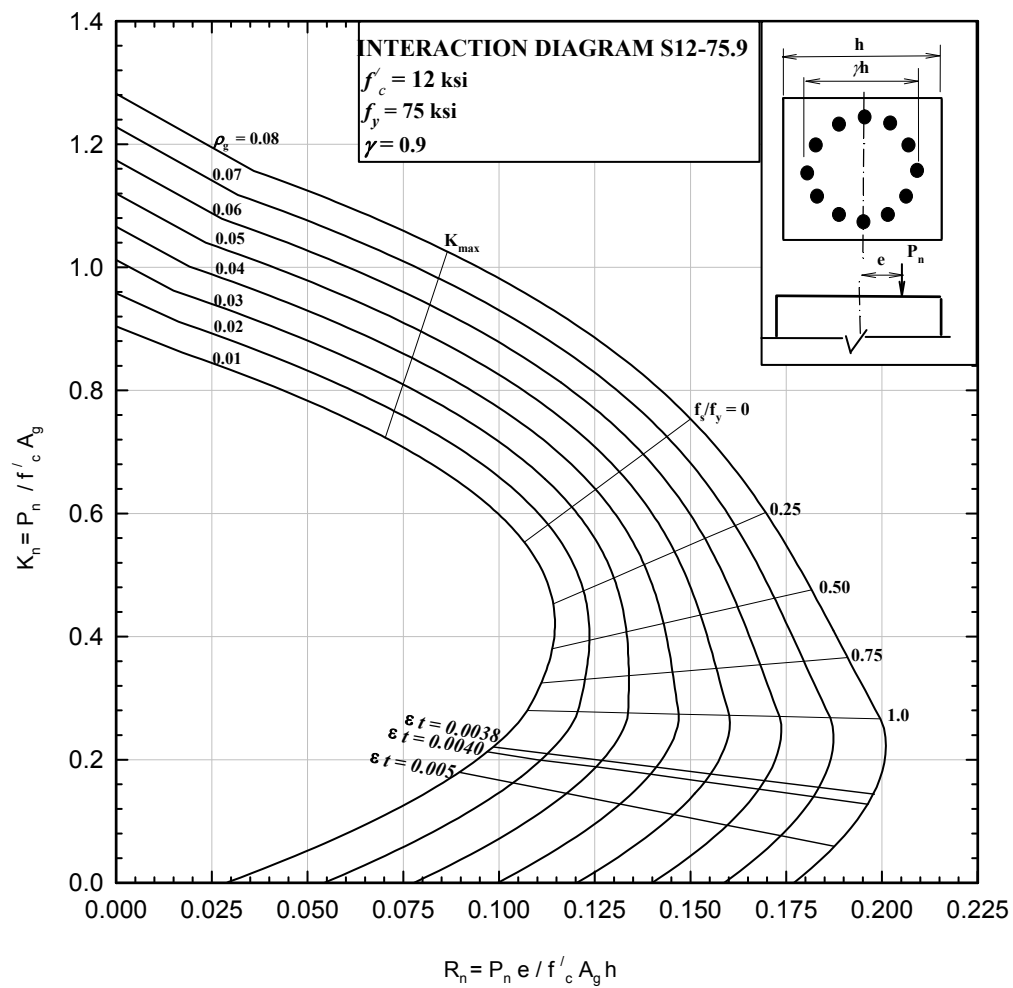
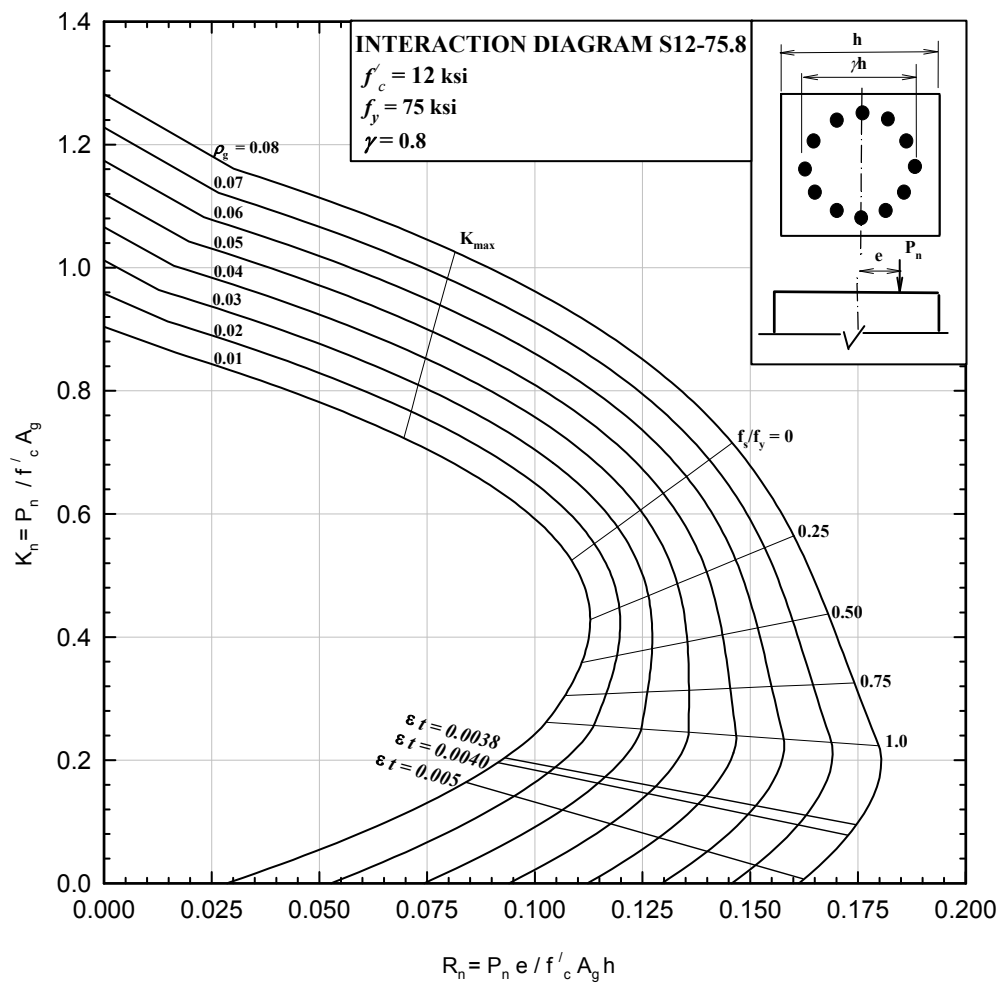














American Concrete Institute
Always advancing

38800 Country Club Drive
Farmington Hills, MI 48331 USA
+1.248.848.3700
www.concrete.org

